Table 22-11. Orbit Average Incident Radiant Fluxes on Surfaces of an Earth-Oriented Cube for Various Circular Orbits. β is the angle of the Sun out of the orbit plane. See Fig. 22-17 for orientation.

Cold Case—W/m²

500 km	$\beta = 0$			$\beta = 45$			$\beta = 70$			$\beta = 90$		
Surface Orientation	Solar	Albedo	Earth IR	Solar	Albedo	Earth IR	Solar	Albedo	Earth IR	Solar	Albedo	Earth IR
Zenith	418.2			295.8			143.1			1.3		
Nadir	30.4	79.1	186.8	44.7	66.3	186.8	143.1	42.7	186.7	1.3	15.2	186.7
Sun	0.8	24.5	58.3	630.7	23.7	58.1	123.76	19.0	58.1	1,317.0	16.5	58.0
Anti Sun	0.8	24.5	58.2		17.5	58.2		8.3	58.2			58.0
± Ram	287.3	24.6	58.0	226.0	20.6	57.9	143.1	13.3	57.9	1.3	5.3	58.0
700 km	$\beta = 0$			$\beta = 45$			$\beta = 70$			$\beta = 90$		
Surface Orientation	Solar	Albedo	Earth IR	Solar	Albedo	Earth IR	Solar	Albedo	Earth IR	Solar	Albedo	Earth IR
Zenith	418.2			295.8			143.1			1.3		
Nadir	41.3	74.6	176.4	62.0	62.6	176.4	143.1	40.9	176.3	1.3	18.8	176.2
Sun	0.8	21.3	50.8	661.0	21.4	50.6	1,237.6	18.1	50.6	1,317.0	18.2	50.6
Anti Sun	0.8	21.3	50.7		14.6	50.7		6.3	50.7			50.5
± Ram	299.7	21.4	50.5	238.4	18.0	50.5	143.1	11.8	50.5	1.3	5.8	50.5

Hot Case—W/m²

500 km	$\beta = 0$			$\beta = 45$			$\beta = 70$			$\beta = 90$		
Surface Orientation	Solar	Albedo	Earth IR	Solar	Albedo	Earth IR	Solar	Albedo	Earth IR	Solar	Albedo	Earth IR
Zenith	450.6			318.7			154.2			1.4		
Nadir	32.7	123.9	224.6	48.2	98.9	224.7	154.2	59.6	224.6	1.4	19.7	224.4
Sun	0.9	38.4	70.1	679.5	35.4	69.9	1,333.4	26.5	69.9	1,419.0	21.4	69.8
Anti Sun	0.9	38.5	70.0		26.1	70.0		11.5	70.0			69.8
± Ram	309.5	38.6	69.7	243.5	30.8	69.7	154.2	18.6	69.7	1.4	6.8	69.7
700 km	$\beta = 0$			$\beta = 45$			$\beta = 70$			$\beta = 90$		
Surface Orientation	Solar	Albedo	Earth IR	Solar	Albedo	Earth IR	Solar	Albedo	Earth IR	Solar	Albedo	Earth IR
Zenith	450.6			318.7			154.2			1.4		
Nadir	44.5	116.9	212.2	66.8	93.5	212.2	154.2	57.0	212.1	1.4	24.4	211.9
Sun	0.9	33.3	61.1	712.2	31.9	60.9	1,333.4	25.2	60.9	1,419.0	23.6	60.8
Anti Sun	0.9	33.4	60.9		21.7	61.0		8.7	61.0			60.8
± Ram	322.9	33.5	60.7	256.9	26.8	60.7	154.2	16.4	60.7	1.4	7.5	60.7

	Cold	Hot	Units
Solar Constant	1,317.0	1,419.0	W/m²
Albedo at $\beta = 0$	0.22	0.32	
Albedo at $\beta = 45$	0.26	0.36	
Albedo at $\beta = 70$	0.34	0.44	
Albedo at $\beta = 90$	0.49	0.59	
Earth IR	217.0	261.0	W/m²

the total energy density, or flux, incident on a surface, and the spectral content of that flux. The flux at 1 AU is called the *solar constant* and is calculated at the Earth's average distance from the Sun. As we change distances from the Sun the flux changes as the square of the distance. For deep space missions, or missions to other planets, the solar flux can vary dramatically.

The spectral content of the solar flux emulates the black body (perfect emitter) distribution for a source at 5,780 K. This concentrates the energy in the visible wavelengths with significant amounts near visible in the ultraviolet and infrared wavelengths. This concentration of energy at certain wavelengths allows the selective absorption of the energy (i.e., surface properties).

Table 22-11, Fig. 22-16◀, Eq. 22-19

Reflected Sun (albedo) is a function of orbit and orientation. The solar energy reflected by a planet, albedo, is also a significant source of radiant energy when the spacecraft is near the planet. Think about riding on a spacecraft in near earth orbit and staring down at the Earth directly below. You are only a few hundred miles above the surface but the surface is thousands of miles across and completely fills your view. From higher orbits, the Earth begins to fill less of our view until the albedo at geosynchronous orbits is negligible. The Earth diffusely reflects from 25% to 55% of the incident solar energy depending on the Earth surface properties. Land, sea, clouds, and ice all have different reflectance. So, when on the sunlit side of the Earth, surfaces facing the Earth can have incident energy from 25% to 55% of the

direct solar flux. The spectral content of the reflected solar energy is approximately the same as energy directly from the Sun. However, it arrives as diffuse energy and not as parallel rays directly from the Sun.

Computing the albedo incident on the various surfaces requires knowledge of spacecraft orientation and location relative to a sunlit Earth. Surfaces facing away from the Earth will receive no solar energy reflected from the Earth.

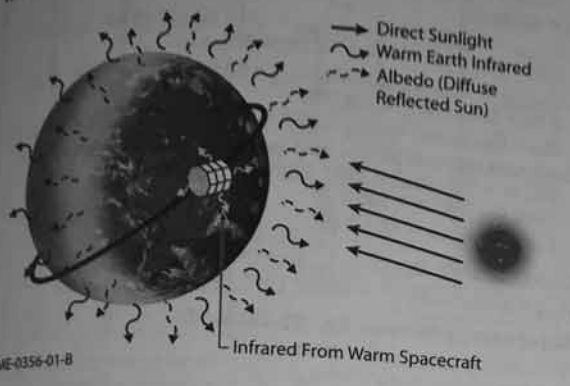


Fig. 22-17. Incident Energy on Spacecraft Surfaces and Radiated Energy from Spacecraft Surfaces.

Planetary infrared energy is a function of the planet's temperature and the spacecraft's orientation to the planet. Planets, moons and other large space bodies can also provide infrared heat sources to spacecraft. These sources are warm compared to space because they have absorbed energy from the Sun and re-radiate it as infrared energy. Most of these sources can be treated as uniform in temperature if they are rotating in the sunlight or have an atmosphere to spread the heat over the surface of the body. The Moon is a notable exception since it does not have an atmosphere and takes 28 Earth days for one revolution. Lunar designs must consider the wide temperature variation on the Moon's surface and resulting wide range of Lunar infrared energy as the spacecraft moves through its orbit. Infrared energy from a warm body in space is transferred just like a warm wall on a cold night or the heat from a fireplace. When you are close to the wall or fireplace, you can easily feel the heat. As you move away from the source the heat flux becomes less and less until it is negligible. The quantity of heat incident on a surface depends on the temperature of the source, and the geometric relationship between the surface and the source. A surface that remains normal to the source throughout its orbit will get a constant infrared flux. A surface pointing away from the source receives none.

Table 22-11 summarizes the incident energy values for several orbit conditions of altitude and β angle considering the accepted range of variation for solar and Earth physical properties, to produce the "hot and cold" range of incident fluxes.

Absorbed Energy; Surface Properties (α/ε)

The incident energy on each surface can be absorbed or reflected by the surface. The surface properties that define the absorbed energy are the, absorptance, (α for solar energy) and emissivity (ε for infrared energy). α is the surface property which defines the percent of incident solar energy that is absorbed by the surface. Solar energy is primarily in the visible part of the electromagnetic spectrum and is absorbed differently from energy in the infrared part of the spectrum. The percent of energy absorbed or emitted at infrared wavelengths is called emissivity. Emissivity is used to determine both the incident infrared energy absorbed and the infrared energy emitted by a surface at the surface temperature. Table 22-14 lists some properties of typical surface finishes and coatings.

22.2.4 Applied Radiant Heat Transfer and Temperature Control in Space

A typical thermal control problem must consider complex geometries and radiant thermal environments. Figure 22-17 shows a spacecraft in orbit, identifies the sources of radiant heating and shows a physical geometry. Following the spacecraft through its orbit and collecting all the radiant energy on each surface for the range of orientations is a task best left to computer programs. However, with a few reference tables (e.g., Table 22-11) and some interpolation, much of the information needed to do thermal analysis and design can be estimated if specific data does not exist. These estimates and assumptions can be "tested" analytically to see if they are critical drivers in meeting requirements. If they are, they can become the high priority items for precision definition, or become requirements placed on the rest of the system. Figure 22-18 shows a flow diagram to illustrate the analysis and design process. Table 22-12 gives an example with numerical values.

Spacecraft and mission information to be collected and computed:

1. A preliminary baseline mission profile is identified. This sets the thermal environment (heat sources for the spacecraft exterior). Variations are expected within the estimated limits. A major change in the mission profile could present a significantly different thermal environment for the flight system and therefore may require a very different design.
2. Location and orientation: (Trajectories, orbits, attitude) are known or baselined; radiant properties of heat sources (solar intensity, planetary reflectance and planetary temperature) are known:
   - Trajectory elements
   - Orbital parameters altitude, eccentricity, inclination
   - Orientation with respect to planet and Sun is known
   - Heat source characteristics are known
3. Power dissipations must be known or estimated for the spacecraft equipment (heat sources for the spacecraft interior)

Table 22-12. Example of a Single Node Thermal Analysis and Calculation. See flow diagram in Fig. 22-18.

INFORMATION:

1. **Mission profile:** circular Earth orbit, inclination 21.5 deg; 2 yr life
2. **Location & orientation:** 500 km alt., spinning cylinder with spin axis always perpendicular to the orbit plane
3. **Internal power dissipation:** 100 W with 200 W spike for 10 min each orbit
4. **Temperature requirements for internal average:** 20 C ± 20 C
5. **Configuration:** cylinder $L = D = 1$ m, ends closed, cylindrical surface 75% covered in solar cells, 25% access panels with very small instrument apertures

ANALYSIS:

6. **Compute thermal incident energy to the surfaces of a cylinder in the mission orbit:** (Note: the ends of the cylinder are closed and insulated so they will not play a significant part in the energy balance.)

   The 500 km circular orbit inclined 21.5 deg to the equator will experience a beta angle range of 0 to 45 deg. This translates into a time in sunlight of 62% to 68% out of the 95 minute orbit period. We can use the incident flux values from Table 22-11 if we make some adjustments for spacecraft geometry and surface orientation to Earth and Sun.

   - ***Direct Sun:*** max and min time in Sun as orbit plane Sun vector changes.
   - ***Cylinders normal to the orbit plane*** will present a constant area to the Sun that varies as the cos β
   - ***Albedo:*** reflected solar energy from the sunlit side of the Earth. Nadir surface fluxes can be used, but use surface area = ½ the total surface area
   - ***Earth IR:*** radiant energy from the warm Earth. Nadir surface fluxes can be used, but use surface area = ½ the total surface area

7. **Compute all absorbed energy:**

   β = 0 (cold)

   $Q_{solar} = SA_p[\%\ \text{Sun time}]\alpha_{avg} = 1317\ \text{W/m}^2 \cdot 1\ \text{m}^2 \cdot 0.62 \cdot (0.85 \cdot 0.75 + 0.9 \cdot 0.25) = 704.3\ \text{W}$

   $Q_{albedo} = Q_{incident\text{-}albedo} A_{IR}\alpha_{avg} = 79.1\ \text{W/m}^2 \cdot \pi/2\ \text{m}^2 \cdot (0.85 \cdot 0.75 + 0.9 \cdot 0.25) = 107.2\ \text{W}$

   $Q_{IR} = Q_{incident\text{-}IR} A_{IR}\varepsilon_{avg} = 186.8\ \text{W/m}^2 \cdot \pi/2\ \text{m}^2 \cdot 0.9 = 264.1\ \text{W}$

   β = 45 (hot)

   $Q_{solar} = SA_p[\%\ \text{Sun time}]\alpha_{avg} = 1419\ \text{W/m}^2 \cdot (1\ \text{m}^2 \cdot \cos 45°) \cdot 0.68 \cdot (0.85 \cdot 0.75 + 0.9 \cdot 0.25) = 588.4\ \text{W}$

   $Q_{albedo} = Q_{incident\text{-}albedo} A_{IR}\alpha_{avg} = 98.9\ \text{W/m}^2 \cdot \pi/2\ \text{m}^2 \cdot (0.85 \cdot 0.75 + 0.9 \cdot 0.25) = 134.0\ \text{W}$

   $Q_{IR} = Q_{incident\text{-}IR} A_{IR}\varepsilon_{avg} = 224.7\ \text{W/m}^2 \cdot \pi/2\ \text{m}^2 \cdot 0.9 = 317.7\ \text{W}$

   Compute internal power and add to absorbed energy:
   100 W + (10 min/95 min) 100 W = 110.5 W (orbit avg)
   Total Q_{in} cold = 1,185.5 W
   Total Q_{in} hot = 1,150.6 W

8. **Compute total Q_{out} :**
   $Q_{out} = \pi D L \cdot 0.9\sigma T^4 = 16.0 \times 10^{-8}\ T^4$ W

9. **Solve energy balance Eq. (22-22) for *T*:**
   Cold avg temperature = 18 C
   Hot avg temperature = 17.7 C

10. **Meets requirements of 20 C ± 20 C**

    Note that the changes in the orbit % time have been compensated by the changed projected area for the direct solar absorbed energy and the "cold" orbit average prediction is actually warmer. At this point re-doing the calculations with maximum and minimum surface properties will yield a wider range of possible temperatures.

11. **Change areas and finishes:**
    The surface of the "access panels" is available to add reflective coatings (low α) or reduce radiation capability (low ε) and result in lower or higher orbit average temperatures

4. Operational and non-operational temperature requirements are established
   - Average temperature limits
   - Allowable temperature gradients
   - Mission unique requirements (e.g., cryogenics)
5. Select simplified geometric shape: cube, sphere, several flat surfaces with an operational orientation representative of the mission flight system (spacecraft)
   - Spinning spacecraft might use a sphere or cylinder
   - Inertial fixed or planet oriented spacecraft, use a cube or flat plates

This is a lot of information to collect, but necessary to perform the analysis. Now we can start thermal analysis and design.

Steps in Doing a Thermal Analysis (Continuing to follow Fig. 22-18)

6. Compute the thermal heating environment: Move the geometric shape through the mission profile with the established orientations relative to the heat sources. For each surface and for each incident heat source, identify the maximum, minimum and average incident heat flux over time, for each mission phase. This involves a lot of averaging over time and area. Programs are available to compute all the geometry and time averages for this step.
   - It is often helpful to imagine yourself on the various surfaces moving through the mission orbit or profile, directed in the expected orientation and visualize the heat source. This can provide an intuitive check on the output of a computer generated set of incident heat sources.
   - Except for special cases (biological or radioactive heat sources), *all energy into a spacecraft arrives by radiant energy.* Even the electrical energy arrives as radiant energy on the solar arrays before being converted to electrical energy and then dissipated as thermal energy.
7. Compute the absorbed environmental heat, Q_{env} on each object or surface by selecting surface radiant properties (α for a solar source and ε for an infrared source) from the following:

Table 22-12, Fig. 22-17◀, Eq. 22-19◀

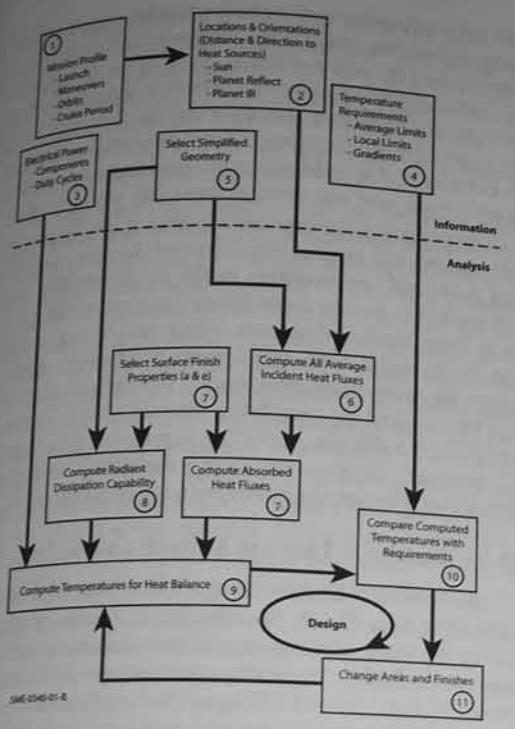


Fig. 22-18. Thermal Analysis and Design Process.

$$Q_{env} = \alpha S\left[A_p + RA_R\right] + \varepsilon IR\, A_{IR} \qquad (22\text{-}20)$$

where S is the solar irradiance at the planet; R is the percentage of solar irradiance diffusely reflected from the planet; IR is the irradiance of infrared energy from the planet; A_p is the projected area toward the sun; A_R is the area exposed to diffusely reflected solar energy from the planet; and A_{IR} is the area exposed to the infrared energy, emitted from the planet.

8. Compute the total IR dissipation capability of the spacecraft geometry by summing all the external surfaces

$$Q_{out} = \sigma T^4 \sum_{1}^{n} \varepsilon_n A_n \qquad (22\text{-}21)$$

where ε_n is the emissivity at surface n, and A_n is the radiating area at surface n.

9. Use the steady state heat balance equation for an isothermal object, include the electrical power from Step 3 and compute the steady state temperature for the object.

$$Q_{env} + Q_{in} = \sigma T^4 \sum_{1}^{n} \varepsilon_n A_n \qquad (22\text{-}22)$$

where Q_{env} is the heat absorbed from environment loads, and Q_{in}, is the internal power.

10. Compute temperatures for the maximum and minimum heat loads using the maximum and minimum surface properties, and simultaneously using the maximum and minimum internal power dissipation.

11 Solve for T and compare with the requirements.

9a. Compare the temperature results of all these cases and evaluate the adequacy of a passive thermal design.

10a. Adjust the surface properties so the steady state temperature, for the average heat load, is near the middle of the required temperature range.

11a. Iterate results using the available independent variables (e.g., areas, finishes, power dissipation).

Constructing a Multi-node Thermal Model of the Spacecraft

A single, isothermal temperature calculation will rarely be adequate to analyze and design the thermal control subsystem. Even fairly simple spacecraft require the construction of a mathematical thermal model. A thermal model is a collection of isothermal nodes which represent locations in the spacecraft that should each operate as an isothermal unit. For example, a thick structural metal base plate could be a single node since it would readily conduct heat throughout the plate and not support a temperature gradient across the plate. The plate will change temperature over time but the plate will be a uniform temperature at any given time. An avionics box mounted on the structural plate will not necessarily be the same temperature as the plate since it may have electrical power dissipated inside and the conduction path from the box to the plate is finite. Therefore, the avionics box is a separate isothermal node in the thermal model. The collection of nodes in a thermal model can range from one to thousands. The number will be a function of spacecraft size, complexity, and the resolution required of the thermal design. For an unpressurized spacecraft there are only two ways to transfer heat from one node in the model to another node in the model: conduction and radiation. The set of isothermal nodes and all the couplings which interconnect them with heat transfer paths form the "internal" thermal model. See Fig. 22-19.

Conduction Heat Transfer Couplings are calculated by determining the mechanical path that must carry the heat from one node to another and applying the conduction properties of the materials in the path. A collection of series and parallel heat transfer paths can be combined the same way a network of electrical resistors can be combined to represent a single conduction coupling between two nodes in the model.

The heat transferred between the nodes by conduction depends on the temperature difference between the nodes and the coupling from one node to the other. A sample calculation that represents this, shows conduction from a heat dissipation element to an avionics box, to a base plate on which the box is mounted (Fig. 22-19).

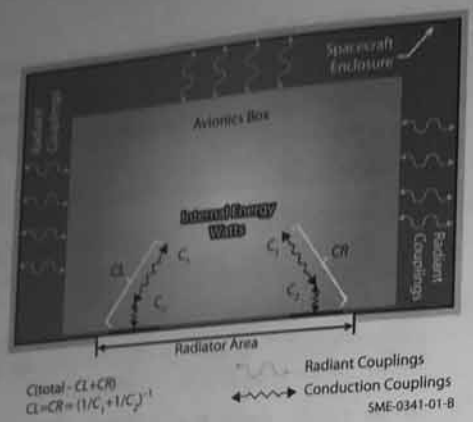


Fig. 22-19. Heat Transfer Inside Spacecraft.

For each leg

$$C_L = C_R = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2}} \qquad (22\text{-}23a)$$

$$C_{1,2} = C_L + C_R \qquad (22\text{-}23b)$$

Heat transferred from 1 to 2 will be:

$$Q_{1,2} = (C_L + C_R)(T_1 - T_2) \qquad (22\text{-}24)$$

Radiant heat transfer couplings between the isothermal nodes are determined by using the geometric view factor relationships between all the node surfaces and the emissivities of each surface. The radiant coupling will be a function of: surface area of each node, geometric view factor of one node surface to another node surface, and the IR emissivity of each surface. The heat transferred will be driven by the difference between the absolute temperature of each node raised to the 4th power. If two 1 m^2 surfaces are close and facing each other, their geometric view factor is 1.0. With a high emissivity on each (0.9) and a temperature difference of 1 deg K near room temperature would transfer 4.6 W from one surface to another. For small temperature differences the heat transferred is roughly linear. The heat transferred from surface 1 to surface 2 can be computed using an approximate radiation coupling as follows:

$$Q_{1,2} = \varepsilon_1 A_1 \times \varepsilon_2 A_2 \times F_{1,2}\sigma(T_1^4 - T_2^4) \qquad (22\text{-}25)$$

Where $F_{1,2}$ is the geometric view factor from surface 1 to 2 defined as the percentage of diffuse energy that leaves surface 1 and strikes surface 2.

22.2.5 Thermal Model for Analysis and Design

The thermal model is used as the primary tool for analysis and design of the thermal control subsystem. The "internal" thermal model is radiatively coupled to the environment by connecting all node surfaces that view outside the spacecraft to a thermal sink at 0 K. Now we can apply the thermal loads to the model. These consist of the average absorbed environmental loads (solar and infrared) to the various nodes that have external surfaces and adding the average dissipated electrical power directly into the relevant nodes. Solving the resulting set of simultaneous equations will result in a predicted temperature for each node that balances the total heat transfer matrix for the spacecraft. Transient temperatures can be calculated if we add mass and specific heat properties to each node, and use thermal loads that vary with time.

The surface properties are adjusted to get the predicted temperatures to meet requirements. Known variations and statistical uncertainties in the environmental loads are applied to see the temperature response to the changes in these forcing functions. The conduction couplings can be changed with appropriate mechanical requirements for component attachment. The radiant couplings can be changed by specifying internal surface coatings and radiation shields. Areas that require a narrow operating temperature range can have a dedicated external radiator and a heater controlled by a temperature sensor.

22.2.6 FireSat II Thermal Analysis and Design

The FireSat II thermal design follows the process described in Sec. 22.2.4. The FireSat II mission profile is set by the orbit parameters and the orientation with respect to the Earth. The orbit parameters are: a circular orbit of 700 km, inclined 55 deg to the Earth's equator. The flight dynamics associated with this orbit show that the angle between the solar vector and the orbit plane (β angle) can vary from 0 to ± 78.5 deg (orbit inclination + equatorial tilt). This geometry translates into a range of possible shadow conditions from no shadow over the entire orbit (when β angle exceed ± 60 deg) to the maximum shadow when the solar vector is in the plane of the orbit (β angle = 0 deg.) At the $\beta = 0$ condition, FireSat II will be in the Earth's shadow 35% of the orbit.

FireSat II requires that the surface that contains the instrument aperture be oriented to nadir so that it always points down to Earth. This nadir vector is the yaw axis for the spacecraft. The instrument requires a radiator surface with low environmental loading to provide an efficient way to cool the detector to its operating temperature. Since the Sun is the major environmental heat source, we can designate a surface that will be on the side of the spacecraft which is perpendicular to the orbit plane and shadowed by the spacecraft. The normal to this surface is the pitch axis. When the Sun crosses the plane of the orbit (moves from a positive to a negative β angle) the spacecraft must perform a 180 deg yaw maneuver to keep the instrument radiator on the shady side of the spacecraft. Operationally, this will occur every 45 days. The velocity vector passes through the forward and aft panels of FireSat II and is perpendicular to the yaw and pitch axis, and is therefore also the roll axis. This orientation is shown in Fig. 19-3.

The radiant heat sources that will fall on the spacecraft surfaces in this mission profile will be: the direct Sun, the solar energy diffusely reflected from the sunlit side of the Earth, and the infrared energy emitted from the warm surface of the Earth.

Power dissipation for FireSat II is 200 W total. Of this, 65 W is used by the instrument and 15 W of the instrument power must be dissipated by a radiator, not to exceed 238 K (–35 C). The location and sizing of this radiator is discussed later. Except for the cold radiator, the instrument and the rest of the spacecraft have normal temperature limit requirements of –10 to +40 C.

We now have the geometry and requirements sufficiently defined to perform a simple thermal analysis. The first step is computing the thermal heating environment. Since the diameter and length of FireSat II are nearly identical, it is possible to use a simple cube as a surface representation to collect the relevant heating rates for the analysis. Moving the Earth oriented cubical representation of the spacecraft geometry through the orbit causes the Sun to rise and fall on various surfaces. Also, the surfaces change their geometric relationship to the sunlit portion of the Earth. An Earth oriented spacecraft will usually have constant Earth IR heating over the orbit. Using the orbit average incident fluxes from Table 22-11 we can sum all the orbit average heating for FireSat II keeping the solar and albedo separate from the Earth IR.

For our hot condition, we use the β angle which results in no shadow time combined with the high end of values for the heating sources. For the cold condition, we use the β angle which provides the maximum shadow time for each surface combined with the low end values for the heating sources.

Using Table 22-11, the rate at which energy is absorbed for each surface is:

$$Q = [\alpha(S + S_A) + \varepsilon S_{IR}]A \qquad (22\text{-}26)$$

where S_A is the albedo flux.

With the surfaces mostly covered by solar cells we will assume the effective absorptance is 0.85 (some energy is removed as electrical energy) and an emmisivity of 0.9.

Adding up all the energy for the 6 surfaces on the cube yields: 1,378 W for the cold case 2,173 W for the hot case. When we add the internal power, the total energy into the system varies from 1,578 W to 2,373 W.

The energy leaving the system does so by radiation. Again, assuming solar cell properties gives a total radiation capability from the surface area of all 6 sides:

$$Q = A_S \varepsilon \sigma T^4 \qquad (22\text{-}27)$$

Summing the internal power with the 6 surface heat energy inputs and outputs is equivalent to linking them together with infinite couplings. This gives the thermal average for the spacecraft surface temperatures and the average temperature for an isothermal spacecraft.

The energy balances when the temperature is –5 C for the cold case, and +24 C for the hot case. Table 22-13 shows the details of these calculations.

We can now look at how to tailor the thermal coatings to meet our specific FireSat II requirements. Referring to the average incident flux tables for the β angle limits for FireSat II, we note that the nadir facing surface and the surface which is always on the shaded side of the spacecraft do not have enough incident energy to generate power for the spacecraft. These surfaces are available for thermal control coatings and other spacecraft equipment. This allows us 2 of the 6 sides on our estimating cube for thermal control coatings. These

Table 22-13. FireSat II—Temperature Calculations—First Estimate. From incident energy to temperature at thermal balance.

Cold 700 km $\beta = 0$													
Surf. #	Descrip.	Area (m^2)	Inc. Solar	Inc. Albedo	Inc. Earth IR	α	ε	Absorbed Solar + alb	Absorbed Earth IR	Elec. Power	Q_{in}	Rad Area/ abs Area	Temp (°C)
1	Zenith-Y	1	418.2			0.85	0.9	355.5	0.0	0.0	355.5	1	16
2	Nadir-Y	1	41.3	74.6	176.4	0.85	0.9	98.5	158.751	0.0	257.3	1	–7
3	Sun-P	1	0.8	21.3	50.8	0.85	0.9	18.8	45.7038	0.0	64.5	1	–84
4	A-Sun-P	1	0.8	21.3	50.7	0.85	0.9	188	45.603	0.0	64.4	1	–85
5	Ram-R	1	299.7	21.4	50.5	0.85	0.9	272.9	45.4509	0.0	318.4	1	8
6	A-Ram-R	1	299.7	21.4	50.5	0.85	0.9	272.9	45.4509	0.0	318.4	1	8
	Total	6	1,060.6	159.9	378.8	0.85	0.9	1,037.4	341.0	200.0	1,578.4	6	–5
Hot 700 km $\beta = 70$													
Surf. #	Descrip.	Area (m^2)	Inc. Solar	Inc. Albedo	Inc. Earth IR	α	ε	Absorbed Solar + alb	Absorbed Earth IR	Elec. Power	Q_{in}	Rad Area/ abs Area	Temp (°C)
1	Zenith-Y	1	154.2			0.9	0.9	131.0	0.0	0.0	131.0	1	–48
2	Nadir-Y	1	154.2	57.0	212.1	0.9	0.9	179.5	190.8	0.0	370.3	1	19
3	Sun-P	1	1,333.4	25.2	60.9	0.9	0.9	1,154.9	54.8	0.0	1,209.7	1	119
4	A-Sun-P	1		8.7	61.0	0.9	0.9	7.4	54.9	0.0	62.3	1	–86
5	Ram-R	1	154.2	16.4	60.7	0.9	0.9	145.0	54.6	0.0	199.6	1	–23
6	A-Ram-R	1	154.2	16.4	60.7	0.9	0.9	145.0	54.6	0.0	199.6	1	–23
	Total	6	1,950.1	123.8	455.3	0.9	0.9	1,762.8	409.8	200.0	2,372.6	6	24

All surfaces solar cells –200 W internal power; meters, watts, °C, $\sigma = 5.67 \times 10^{-8}$ W/m^2-K^4

surfaces, not needed for solar cells, are valuable areas. The nadir surface contains the instrument aperture and other spacecraft equipment which may require external mounting, (e.g., communication antennas). Let's assume that of our 2 m^2 available for thermal coatings, 0.5 m^2 is required for other uses. The average spacecraft temperature can be lowered by using coatings to reject as much incident heat as possible while maintaining a high heat radiation capability. With a low α/ε coating on 0.75 m^2 of the nadir and shadowed sides the total energy into the system becomes 1,379 W for the cold case, and 2,034 W for the hot case. The resulting cold and hot orbit average temperatures are –14 C and +12 C. If we cover the available 1.5 m^2 of our working surfaces with insulation to minimize their radiation capability, the average cube temperatures are raised to –3 C and 26 C. Our estimation calculations have shown that we can set the average temperature with passive coatings. The thermal coatings can now be further refined for the specific geometry of the spacecraft using actual surface areas and equipment to set the average temperature.

The requirement for a local cold condition is satisfied as follows. The element to be controlled at a low temperature is isolated from coupling to the spacecraft interior and strongly coupled to an external radiating surface. The external surface location is selected for minimum incident radiant energy loading. For a physical representation of this problem, refer to Fig. 22-22 in Sec. 22.2.8 under "radiators". The cooled element is the "avionics box" which, in this application, is well insulated from thermal coupling with the spacecraft enclosure. It is strongly coupled to a special radiator which is the surface that dissipates the net 15 W of power at a temperature of –35 C.

Start with the location of the radiator to compute its size. The side of the spacecraft that is always shadowed is the only surface that can meet the requirement for a radiator that can dissipate 15 W at –35 C. A surface at –35 C can only dissipate 180 W/m^2. All surfaces on FireSat II, except the shadowed side, have more than that in absorbed environmental energy at one or more β angles. The size of the radiator on the shaded surface is computed as follows. Lower the required temperature to –40 C to allow for some margin. Look up the maximum incident flux on the shadowed surface from Table 22-13. The maximum solar albedo is 34 W/m^2 and the maximum Earth IR is 61 W/m^2. Choose a coating to minimize solar absorption and maximize radiation dissipation. A white paint with $\alpha = 0.2$ and $\varepsilon = 0.9$ is a good choice. The absorbed environmental flux is $34 \times 0.2 + 61 \times 0.9 = 61$ W/m^2. At –40 C, the same surface can dissipate 167 W/m^2. Our net radiator capability is 101 W/m^2, and the required 15 W should therefore need about 0.15 m^2 of radiator area. Typically, the radiator is slightly oversized and has a controlled heater to set the exact temperature.

22.2.7 Thermal Hardware, Tools and Techniques

There are many tools and techniques that can be used to design the thermal subsystem and meet requirements. The art of thermal design lies in choosing and applying the possible solutions while also meeting system level cost and mass constraints. Most common hardware elements used for spacecraft thermal design and control are:

- Coatings (surface finishes)
- Insulation and isolators
- Heaters & their controllers
- Louvers
- Heat pipes

22.2.8 Thermal Control Components

Surface Finishes

In spacecraft thermal designs, wavelength-dependent thermal control coatings are used for various purposes. Solar reflectors such as second-surface mirrors and white paints or silver or aluminum-backed teflon are used to minimize absorbed solar energy, yet emit energy almost like an ideal blackbody. To minimize both the absorbed solar energy and infrared emission, polished metal such as aluminum foil or gold plating is used. On the interior of the vehicle, if it is desired to exchange energy with the compartment or other equipment, black paint is commonly used. Thus, the existing state of the art uses a rather wide variety of coatings which have wavelength dependent thermal properties. The problems of in-space stability, outgassing, and mechanical adhesion to the substrate have been resolved for most coatings. There are many fully qualified coatings, so development and qualification of a new coating for a new design is normally unnecessary.

The two primary surface properties of importance are the IR *emissivity*, ε, and the solar *absorptivity*, α. Table 22-14 shows the range of properties available for some common surface finishes. Two or more coatings are sometimes combined in a checkerboard or striped pattern to obtain the desired combination of average absorptivity and emissivity if it cannot be obtained in a single material.

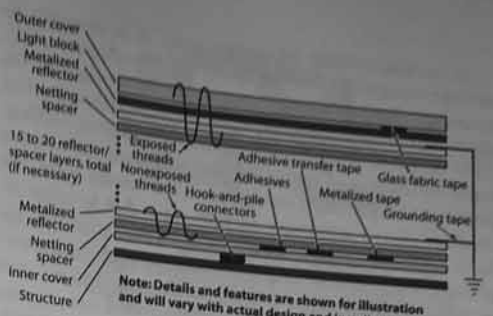
Thermal control finishes are affected in orbit by charged particles, ultraviolet radiation, high vacuum, and the contaminant films that deposit out on almost all spacecraft surfaces. The general result of these processes is an increase in solar absorptivity with little or no effect on infrared emissivity. This is normally undesirable from a thermal control standpoint because spacecraft radiators must be sized to account for the substantial increase in absorbed solar energy that occurs due to degradation over the mission. These radiators, which are oversized to handle the high solar loads at "end-of-life," cause the spacecraft to run much cooler in the early years of the mission, sometimes necessitating the use of heaters to avoid under-temperatures of electronic components. The degradation is, therefore, a problem not only because of the solar load, but also because of the change in load over the course of the mission. The stability of coating properties is important in order to both limit maximum temperatures and minimize heater-power requirements.

Table 22-13◀, Fig. 22-19◀, Eq. 22-27◀

Table 22-14. Properties of Common Finishes.

Surface Finish	B.O.L. α (Beginning of Life)	ε
Optical Solar Reflectors		
8 mil Quartz Mirrors	0.05 to 0.08	0.80
2 mil Silvered Teflon	0.05 to 0.09	0.66
5 mil Silvered Teflon	0.05 to 0.09	0.78
2 mil Aluminized Teflon	0.10 to 0.16	0.66
5 mil Aluminized Teflon	0.10 to 0.16	0.78
White Paints		
S13G-LO	0.20 to 0.25	0.85
Z93	0.17 to 0.20	0.92
ZOT	0.18 to 0.20	0.91
Chemglaze A276	0.22 to 0.28	0.88
Black Paints		
Chemglaze Z306	0.92 to 0.98	0.89
3M Black Velvet	~0.97	0.84
Aluminized Kapton		
½ mil	0.34	0.55
1 mil	0.38	0.67
2 mil	0.41	0.75
5 mil	0.46	0.86
Metallic		
Vapor Deposited Aluminum	0.08 to 0.17	0.04
Bare Aluminum	0.09 to 0.17	0.03 to 0.10
Vaporized Deposited Gold	0.19 to 0.30	0.03
Anodized Aluminum	0.25 to 0.86*	0.04 to 0.88*
Miscellaneous		
¼ mil Aluminized Mylar,	unstable	0.34
Beta Cloth	0.32	0.86
Astro Quartz	~0.22	0.80
MAXORB	0.9	0.1

* Anodizing and similar surface treatments must be carefully controlled in order to produce repeatable optical properties.



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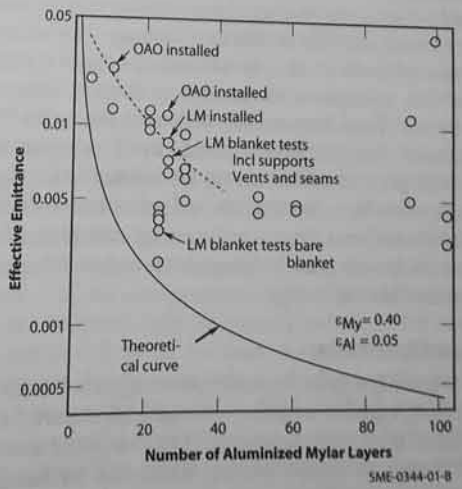
Fig. 22-20. Composition of a Typical MLI Blanket. Mulitlayer insulation blankets are made of fairly sophisticated layers of low-emittance films with low conductivity between layers. (Courtesy of NASA)

Insulation

Multilayer insulation (MLI) and single-layer radiation shields are among the most common thermal control elements on spacecraft. MLI blankets are used either to prevent excessive heat loss from a component or excessive heating from environmental fluxes or rocket plumes. Most spacecraft are covered with MLI blankets, with cutouts provided for radiator areas to reject internally generated waste heat. MLI blankets are also typically used to protect internal propellant tanks, propellant lines, solid rocket motors, and cryogenic dewars. Single-layer radiation barriers are sometimes used in place of MLI where a lesser degree of thermal isolation is required, since they are lighter and cheaper to manufacture. For applications requiring insulation under atmospheric conditions, foam, batting, and aerogel materials are generally used because MLI is not very effective in the presence of a gas.

Multilayer insulation is composed of multiple layers of low-emittance films with low conductivity between layers, as shown in Fig. 22-20. The simplest construction is a layered blanket assembled from embossed, thin mylar sheets (¼ mil thick) with a vacuum-deposited aluminum finish on one side of each sheet. The embossing results in the sheets touching only at a few points, thereby minimizing conductive heat paths between layers. The layers are aluminized on one side only, so that the mylar can act as a low-conductivity spacer. Higher-performance construction uses Mylar film metallized on both surfaces (aluminum or gold) with silk or dacron net, Tissuglas paper, or "Super-Flock" whiskers as the low-conductance spacers.

Figure 22-21 illustrates theoretical and experimental data for embossed aluminized (one surface) Mylar insulation versus number of insulation-blanket layers. The emittance for a multilayer blanket theoretically varies inversely with one over one plus the number of layers. However, in practice, simply increasing the number of layers past a certain value will not improve performance. As the number of layers increases, radiative heat transfer becomes small compared to conductive "shorts" between layers and other losses. Thus, a point of diminishing returns is reached. Considering these trends, about 25 layers are usually sufficient to obtain a minimum overall conductance value.



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Fig. 22-21. Effective Emittance vs. Number of Aluminized Layers. This figure illustrates theoretical and experimental data for embossed aluminized (one surface) mylar insulation versus number of insulation-blanket layers. Note that the emittance for a multilayer blanket theoretically varies inversely with one over one plus the number of layers.

Table 22-14, Fig. 22-21, Eq. 22-27◀

In well-controlled laboratory tests, it is possible to achieve values of 0.005 or lower for the *effective emittance*, ε^*. However, when a blanket is configured for spacecraft application, experience has shown that an ε^* of 0.015 to 0.030 is typical for medium-sized area applications. As the size of the application increases, the relative performance generally increases. This results from the smaller relative influences of heat leaks due to edge effects, seams, or cable penetrations. For very-large-area applications with minimal penetrations, the ε^* laboratory performance approaches 0.005 at 30 layers. Performance data from cryogenic tankage and controlled calorimeter tests typically show performance of ε^* down to 0.002. Control of discontinuities through the design and fabrication of insulation joints and penetrations is crucial to the problem of reducing the effective emittance of multilayer blankets. Compression of a blanket section dramatically increases heat leakage, as well. Small area blankets show high effective emittance along with considerable manufacturing variation. Very small blankets used to wrap propellant lines typically have effective emittances ranging from 0.05 to 0.30, making low emittance surface finishes a simpler alternative for internal propellant lines. External lines that are exposed to sunlight require MLI blankets to avoid overheating caused by the high absorptance/emittance ratios typical of low emittance surface finishes.

Missions involving planetary landers, rovers, or atmospheric probes usually require insulation that performs well in the presence of an atmosphere. While MLI may still be required to protect the hardware during interplanetary cruise, foam, batting, or aerogel materials are generally added to provide insulation after atmospheric entry and landing in situations where ambient pressure is expected to be high enough to substantially degrade MLI performance. If atmospheric pressure and gravity are low enough, simple trapped gas spaces alone may be sufficient to limit heat loss to the surroundings. The underlying principle behind all of these insulation types is to trap gas within volumes small enough to eliminate convection effects. Total heat transfer is thereby limited to what can conduct through the low conductivity insulation material and gas and radiate across the insulation through a process of multiple absorptions and emissions within the insulation material structure. To complicate MLI design further each layer must be grounded to reduce the chance of electrostatic discharge.

Conduction Isolators

It is usually a good thermal control approach to couple everything together to get an average temperature for the spacecraft internal components. However, there are often local temperature requirements which call for isolation. This can provide very low temperatures or efficient response to the addition of heater power. Radiation isolation is accomplished using MLI. Conduction isolation usually involves custom isolators. These increase in performance as their cross sectional area decreases, their length increases, and the conductivity of the material decreases. For extreme isolation requirements, long slender standoffs made of non-metallic material may be needed. This can be a systems design challenge when also providing adequate mechanical support.

Developments in composite materials have enabled many options in the design of conduction isolation (or enhancement).

Radiators, Thermal

Most spacecraft waste heat is rejected to space by *radiators*. These occur in several different forms, such as spacecraft structural panels, flat-plate radiators mounted to the side of the spacecraft, or panels that are deployed after the spacecraft is on orbit. Whatever the configuration, all radiators reject heat by IR radiation from their surfaces. The radiating power is dependent on the emissivity of the surface and its temperature. The radiator must reject both the satellite waste heat plus any radiant heat loads from the environment or other spacecraft surfaces that are absorbed by the radiator, as shown in Fig. 22-22. Most radiators are therefore given surface finishes with high IR emissivity (> 0.8) to maximize heat rejection and low solar absorptivity (< 0.2) to limit heat loads from the Sun. Typical finishes include quartz mirrors, silvered or aluminized teflon, and white paint.

The external surface finishes of the radiator couple it to space. Because these surfaces are also exposed to external sources of energy such as sunlight and Earth-emitted IR, their radiative properties must be selected to achieve an energy balance at the desired temperature between spacecraft internal dissipation, external sources of heat, and reradiation to space, as illustrated in Fig. 22-22.

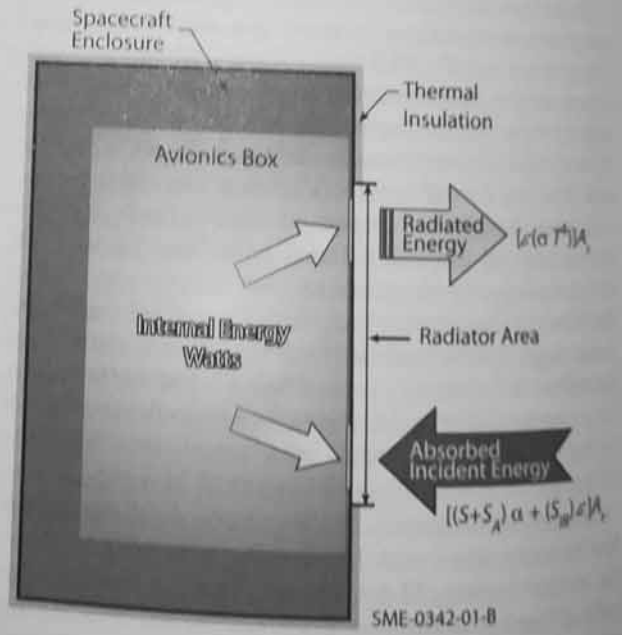


Fig. 22-22. Radiator Energy Balance. Note that we must select radiative properties of the radiator surface to achieve an energy balance among spacecraft internal dissipation, external heat sources and reradiation to space to obtain the desired temperature. A_r is the area of the radiator.

Table 22-14◀, Fig. 22-22, Eq. 22-27◀

The radiating power of a radiator is a strong function of temperature. The total heat leaving a radiator surface is given by the Stefan-Boltzmann law:

$$Q = A\varepsilon\sigma T^4 \qquad (22\text{-}28)$$

where A is the surface area, ε is emissivity, T is the absolute temperature, and σ is the Stefan-Boltzmann constant (5.6705×10^{-8} W/m^2-K^4).

The T^4 term results in a large increase in radiating capability with temperature. Referring back to Fig. 22-15, the radiating power at 50°C is about twice that at 0°C. At cryogenic temperatures the effect is even more pronounced, with a 70 K radiator having only 1/306th the heat rejection capability of a room-temperature radiator. This makes cryogenic radiators extremely sensitive to environmental heating and heat leaks through insulation and supports, and leads to special design considerations.

Most spacecraft radiators reject between 100 and 500 W of spacecraft internally generated electronics waste heat per square meter. The upper end of this range is typical of a radiator that runs at a fairly high temperature (say 40°C) and experiences relatively modest heat backload from the environment or other spacecraft surfaces. The lower end of the range represents a radiator running below room temperature in LEO, where environmental backloads can be substantial. The actual sizing is determined by a thermal analysis that considers the desired operating temperature, worst-case satellite waste heat, environmental heating, and radiative and conductive interactions with other spacecraft surfaces (See Table 22-15).

Heaters

Ideally, thermal control of a satellite or component would be achieved using only passive techniques, such as surface finishes. Unfortunately, orbital and seasonal variations in environment and component heat-generation rates, along with degradation of surface finishes over time, can drive temperature variations in a passive design to ranges larger than some components can withstand. Because of this, heaters are sometimes required to protect components from cold-case environmental conditions or to make up for heat that is not dissipated when electronic box is turned off. Heaters may also be used with thermostats or solid-state controllers to provide near-temperature control of a particular component. A final common use for heaters is to warm components to their minimum operating temperatures before they are turned on.

The most common type of heater used on spacecraft is the *patch heater*, several of which are shown in Fig. 22-23A. It consists of an electrical resistance element sandwiched between two sheets of flexible electrically insulating material, such as Kapton. The patch may have one circuit, or more than one, depending on whether redundancy is required within the patch. Redundancy is typically required on spacecraft systems since heaters do fail. Sometimes the redundancy is provided within the patch and sometimes it is provided by using

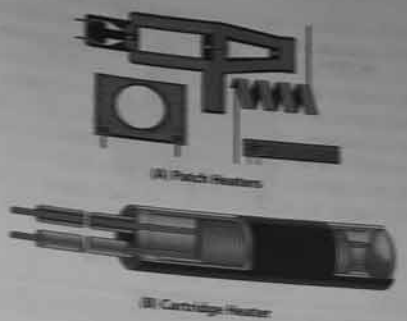


Fig. 22-23. Types of Heaters. Patch and cartridge heaters are used on many spacecraft to meet the heating needs of different types of equipment. These heaters are very important to the successful operation of key components, so we often use redundancy to increase reliability of these devices.

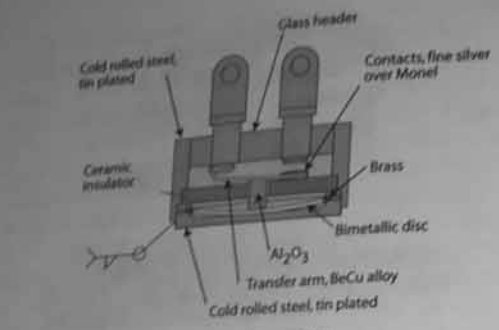
two separate patches. The patch heaters shown in the figure illustrate the custom shapes to which these heaters may be made. In most instances, however, a simple rectangular patch of some standard dimension is used.

A *cartridge heater* is another type that is often used to heat blocks of material or high-temperature components such as hydrazine-thruster catalyst beds. Such a heater is shown in Fig. 22-23B and consists of a wound resistor enclosed in a cylindrical metallic case. A hole is typically drilled in the component to be heated and the cartridge potted into the hole. Another attachment technique involves the use of a clamp or small bracket to hold the heater. These heaters are typically a quarter inch in diameter or less, and up to a few inches long.

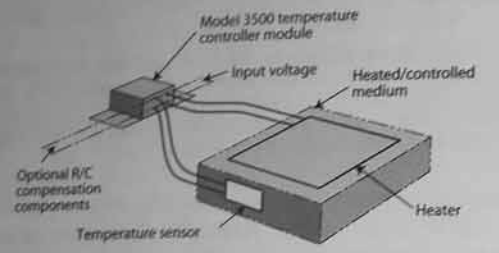
Almost all heater systems have some sort of switch or control. This typically involves a relay that is commandable from the ground to enable or disable power to the heater, a fuse to protect the spacecraft from a short circuit, and, most commonly, a thermostat or solid-state controller to turn the heater on and off at predetermined-temperatures. More sophisticated satellites sometimes use their on-board computer to monitor temperatures and turn heaters on and off as appropriate, using relays.

The simplest arrangement involves only the heater, a fuse, and a ground commandable relay to turn the heater on and off. This arrangement is typically used for heaters that are activated only for special events, or for heaters that can be left on all the time. A typical application is heating up the catalyst beds on hydrazine thrusters to 100°C before the thruster is fired. (Firing the thruster with a low initial catalyst-bed temperature decreases the catalyst life.) The heater is commanded on, the catalyst-bed is heated, the thruster is fired, and the heater is turned off until the next maneuver, all under ground control.

Historically, the most common control device is a mechanical thermostat, such as the one shown in Fig. 22-24A. These typically consist of a small hermetically sealed can containing a switch driven by a



(A) Elmwood Thermostat



(B) TAYCO Solid-State Controller

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Fig. 22-24. Common Control Device. Mechanical thermostates are typically used to control operation of heaters on a spacecraft.

snap-action bimetallic actuator. The temperature at which the thermostat clicks on, known as its *set point*, is fixed for any given thermostat. The engineer can select from an array of standard thermostats available from the manufacturer to get a set point close to what is desired, or a custom device can be ordered. In addition to the set point, the *dead band*, or the difference between the temperatures at which the thermostat turns on and off, is important. Thermostat dead bands typically range from 5 C to 15 C. A smaller dead band reduces the temperature swing of the device being heated, but increases the number of cycles on the thermostat itself, thus decreasing its reliability. Small dead bands also increase the chance of "dithering," in which the thermostat rapidly cycles on and off. This is a failure condition that can cause the set point to drift lower, resulting in an under-temperature of the component being controlled.

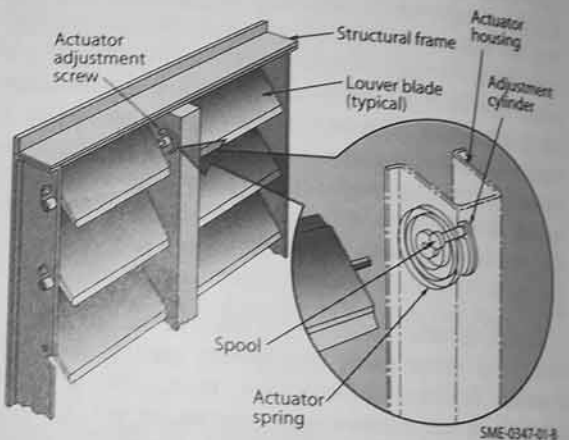
A solid-state controller, an example of which is shown in Fig. 22-24B, replaces the mechanical switch with an electronic device that has a higher reliability and life expectancy. These are used extensively on the DMSP, the Hubble Space Telescope, and the ISS. They employ a temperature sensor that can be located either at the controller or at a remote location, as desired. Another advantage of solid-state controllers is that extremely tight dead bands (<0.1 C) are possible for very precise temperature control, such as required on the Space Telescope. Optical systems, some sensors, and electronic frequency standards often require precise temperature control, which cannot be achieved with a mechanical thermostat.

Table 22-14◀, Fig. 22-25, Eq. 22-27◀

A number of military and scientific satellites use on-board computers to control heaters. Such systems read the temperatures from telemetry sensors placed throughout the vehicle and send signals to turn relay-controlled heaters on and off as required. This allows enormous flexibility since the control set points and dead bands can be adjusted on orbit by uplinking new tables to the spacecraft computer.

Louvers

Louvers are active thermal control elements that have been used in different forms on numerous spacecraft. While most commonly placed over external radiators, louvers may also be used to modulate radiant heat transfer between internal spacecraft surfaces, or from internal surfaces directly to space through openings in the spacecraft wall. In general, a louver in its fully open state allows the rejection of six times as much heat as it does in the fully closed state, with no power required to operate it. Thus, louvers find applications where internal power dissipation varies widely as a result of equipment duty cycles.



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Fig. 22-25. Louver Assembly Schematic. The most commonly used louvers are the "venetian-blind" type that are typically opened to allow heat to radiate away from the spacecraft and closed to keep heat from escaping.

The most commonly used louver assembly is the "venetian-blind" type shown in Fig. 22-25. These louvers consist of four main elements: blades, actuators, sensing elements, and structural elements. The louver is placed over a high emittance, low absorptance spacecraft radiator to modulate the flow of radiant heat from that surface. Blades, which are driven by the actuators, are the louver elements that give variable-radiation characteristics to the radiator surface below. While closed, louvers shield the radiator's view to space, but while open, they allow radiative coupling to space. The radiating characteristics of the radiator can be varied over the range defined by these two extreme positions

The actuators drive the blades according to the perceived radiator temperature. In most louver designs, a bimetallic spiral spring drives each louver blade independently to ensure maximum reliability. Thus a single

Table 22-15. Characteristics of Flight-Qualified Rectangular-Blade Louver Assemblies. Here you can find typical characteristics for three louver assemblies that can be used to estimate mass and size.*

	OSC	ATK	Starsys
Blades	3 to 42	Various	1 to 18
Open Set Points (°C)	Various	0 to 40	−20 to 50
Open/Close Differential (°C)	10 or 18	10 or 18	14
Dimensions (cm)			
Length	20 to 110	27 to 80	8 to 43
Width	36 to 61	30 to 60	22 to 40
Height		6.4	6.4
Area (m^2)	0.07 to 0.6	0.06 to 0.5	0.02 to 0.2
Weight/Area (kg/m^2)†	3.2 to 5.4	~4.5	5.2 to 11.6
Flight History	Multiple	XTE, Stardust	Multiple

* This table contains representative values from past louver designs. Contact manufacturer for additional design possibilities or values for specific designs.

† Weight without sunshield.

‡ The Starsys design is a slightly modified version of a JPL louver design that has flown on the indicated spacecraft.

point failure is associated with one blade, not the entire assembly. The actuator spring drives the blade angle as determined by the underlying radiator temperature. A strong conductive path between the actuator and radiator is therefore sought to minimize the temperature gradient between them.

Louver assemblies have been designed for operation in both shadow and sunlight. Two design approaches that have been followed for operation in sunlight are the use of a Sun shield to prevent direct solar illumination of the louver, or the modification of the louver assembly for high temperature operation if it is directly exposed to sunlight. Characteristics of louvers offered by some principal vendors are shown in Table 22-15.

Heat Pipes

A *heat pipe* uses a closed two-phase fluid-flow cycle to transport large quantities of heat from one location to another without the use of electrical power. The heat pipe can be used to create isothermal surfaces or to spread out heat from a localized source uniformly over a larger area. One-way (diode) heat pipes have been tested and flown, as have *variable-conductance heat pipes*, which maintain a constant temperature under varying heat load conditions. Since the driving mechanism is capillary pumping, a relatively weak force, traditional heat pipes may be susceptible to severe performance degradation when operating under gravity or acceleration, so planning is needed to facilitate ground testing. Loop heat pipes and capillary pumped loops are more advanced cousins of the basic heat pipes that can provide constant temperature operation with varying heat loads under gravity or acceleration.

To illustrate how a heat pipe works, consider a simple horizontal heat pipe in equilibrium with an isothermal environment, as shown in Fig. 22-26. The liquid in the wick and the vapor in the vapor space are at saturation. If heat is applied to the evaporator, raising its temperature, liquid in the wick evaporates (removing some of the added heat), which depresses the meniscus in the evaporator since less liquid is present there. This process also raises the local vapor pressure, since it must be in saturation with the heated liquid in the wick.

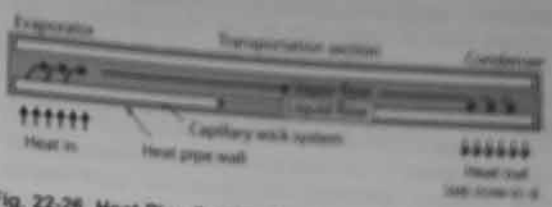


Fig. 22-26. Heat Pipe Schematic. Heat pipes are very efficient in moving heat from one place to another on the spacecraft. See text for a description of operation.

The difference between the increased curvature of the meniscus in the evaporator wick and the unchanged meniscus in the condenser wick causes a difference in capillary pressure sufficient to pull liquid from the condenser wick toward the evaporator. This replenishes the liquid in the evaporator wick. At the same time, heated vapor flows from the evaporator to the condenser, which is at a lower pressure. When this vapor comes in contact with cooler condenser surfaces, it condenses. This cycle is shown schematically in Fig. 22-26.

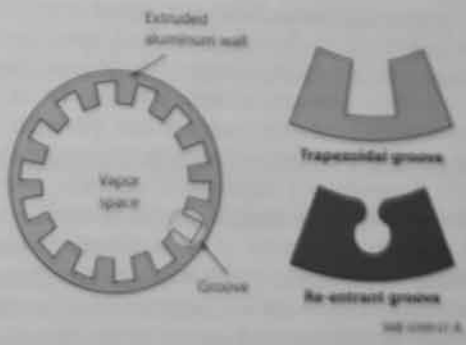


Fig. 22-27. Grooved Heat Pipe. This is an example of the most common wick/envelope design which consists of axial grooves in the wall of extruded aluminum tubing.

Elements of this most basic, constant-conductance, heat pipe consist of a working fluid, a wick structure, and an envelope. The most common wick envelope design consists of axial grooves in the wall of extruded aluminum tubing, as shown in Fig. 22-27. This class of wick is very susceptible to gravity effects during ground testing, but is relatively inexpensive to produce and very consistent. Heat-transfer capability is moderate, but sufficient for many applications. Grooves are typically rectangular or trapezoidal in shape, but more complex shapes such as

the "teardrop" or "keyhole" have been extruded with difficulty. Different working fluids can be used depending on the temperature range of interest. In most spacecraft applications, ammonia is the working fluid of choice. Laptop computers, on the other hand, typically use water in heat pipes used to cool high-power chips.

The most obvious application of a heat pipe is one requiring physical separation of the heat source and sink. If a heat pipe is used, it is not necessary to mount all hardware to be cooled directly on radiator panels, or to use relatively inefficient conductive couplings. A closely related class of application is that of the thermal transformer. A small high-powered box can be mounted on one side of a radiator with integral heat pipes; the heat generated will be spread out over a large radiator surface. This approach also permits more efficient use of available "real estate"—the area available for a radiator is seldom centered symmetrically about the heat source, facing the optimal direction.

Heat pipes have also been used to reduce temperature gradients in structures to minimize thermal distortion. The telescope tube of the Orbiting Astronomical Observatory had three ring-shaped heat pipes to minimize circumferential temperature gradients. The ammonia/aluminum heat pipes worked throughout the eight years of mission life.

The diode heat pipe was first proposed as a means of connecting a device to two radiator panels on opposite sides of a spacecraft, with the understanding that at least one of the radiators would be free of any direct solar load at all times during the orbit. The diodes would couple the device to the cold radiator, while preventing heat from leaking back into the system from the radiator in the Sun. This type of thermal design problem—in which heat from a temporarily warm radiator or from a failed refrigerator must be kept from leaking back into the system—is an obvious application for a diode heat pipe.

The variable-conductance heat pipe can be used to control the amount of active radiator area, providing reasonably good temperature control without the use of heaters. This is particularly attractive if electrical power is limited, and this type of design has been flown on a number of satellites. However, if the application requires maintaining a component at virtually a constant temperature, feedback control (at the expense of some heater power) may be employed. A sensor on the baseplate of the device to be controlled can be routed to an onboard computer, and whenever the temperature drops below the desirable range, heaters on the variable conductance heat pipe reservoirs are activated, causing the control gas to expand and block off more of the radiator area. If the temperature rises above the range desired, power to the reservoir heaters is reduced, increasing the active radiator area. This concept usually requires less power than using heaters directly on the box or system to be controlled.

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23 Space Logistics and Manufacturing

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Ground/User Segment Constraints on Design; Manufacturing and Testing Processes; Future LEO Communications Constellations

23.2 LEO Monolithic vs. Distributed Architectures
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23.7 Intangible Factors in Manufacturing
Manufacturing Efficiency; Perpetual Optimization

As part of the engineering of a space mission, we need to consider the construction of the system. In Chaps. 9 and 10, we discussed orbits and constellations of spacecraft. Sections 23.1 and 23.2 discuss the trade between a single spacecraft and a constellation of spacecraft in order to meet the same mission objective. The logistics associated with this decision will impact the strategy for manufacturing the spacecraft, launching the system into space, and establishing communication with the ground. Sections 23.3 and 23.4 will cover the standard approach for the manufacturing of flight hardware, including the validation that we have built the right system and the verification that we have built the system right. There is significant effort associated with the integration and test of a spacecraft, and the design must take this process into consideration. Section 23.5 discusses how building a number of spacecraft for a constellation will impact the design and approach to manufacturing, concluding with some interesting details about the construction of the Iridium constellation. Alternate approaches to space manufacturing and how modularity, adaptability, and reconfigurability can be designed into an architecture, leading to significant cost savings are described in Sec. 23.6. Finally, Sec. 23.7 concludes this chapter with a discussion of some less measurable factors, such as culture and motivation, that play into manufacturing, again pointing out that early consideration of assembly, during the design phase, is critical to efficient implementation.

23.1 LEO Communications Constellations

Stuart Eves, *Surrey Satellite Technology*

As will be appreciated from Chaps. 8 to 10, satellites in LEO have relatively limited coverage footprints on the surface of the globe by comparison with their cousins' higher altitude orbits. Bearing in mind this footprint limitation, a system designer wishing to achieve a reasonable level of communications performance is automatically driven towards a constellation involving multiple satellites. Traditionally, both satellites and their launch vehicles have been expensive, and this raises obvious questions about the financial wisdom of constructing multi-satellite communications constellations in LEO—would smaller numbers of satellites in higher orbits not represent a more logical investment? And yet the most prolific satellite series in history, the Russian Strela-1 system, is a communications constellation which over its lifetime saw the launch of some 350 or so

relatively short-lived satellites. And in the 1990s, Iridium, Globalstar and Orbcomm all invested large sums in the creation of LEO communications systems. See the book website for a more extended discussion of the historical development of LEO constellation.

The explanation behind the apparent contradiction relates to the user communities that these satellite systems were endeavoring to serve, and the locations of those users on the surface of the Earth. These user communities were either mobile, with small, low-power hand-held receivers, or (in the case of the Strela-1 system), espionage agents who presumably had no desire to advertise their presence by erecting a satellite dish on the roof! In most cases, such terminals will not be "cooperative," (in the sense that the user will not necessarily be able to ensure a clear line of sight to the satellite, or use a highly directional antenna to track the satellite as it moves across the sky). In order to establish a satisfactory link budget to such an uncooperative terminal, it is necessary to ensure that the *Effective Isotropic Radiated Power* (EIRP) from the satellites is sufficient to overcome these limitations. Specifically, the system designer must make certain that the free space path loss (which is dictated by the range between the transmitter and receiver) does not render the system infeasible.

23.1.1 Ground/User Segment Constraints on Design

In order to make valid comparisons between candidate LEO communications systems, it is also necessary to consider the constraints that the user segment places on the design of the satellite. In its simplest terms, this means assuming a common data rate requirement for each system, and ensuring that the satellites deliver an equivalent EIRP at the surface of the Earth, so that the link budget equations can legitimately be compared.

It is for this reason that direct comparisons between systems such as Iridium, Globalstar and Orbcomm are complex, since they were designed with rather differing EIRP and data rate assumptions. (Iridium's power and data rate requirements were higher than either of the other two systems, and debates about their relative merits tend to end up in rather subjective areas, such as the relative voice link quality at 2.4 and 4.8 kbps.)

However, assuming that the constellations **are** being required to provide equivalent levels of service, it is the capabilities and utilization of the user terminals that will ultimately drive the system design. For ease of use, hand-held mobile terminals cannot generally be expected to track the satellites across the sky—satellite engineers tend to assume that users of their systems are "uncooperative" in this regard! This dictates that the terminals are equipped with relatively low gain broad beam antennas, and this usually drives the system towards comparatively low frequencies and more limited data rates than can be achieved through fixed-dish GEO-based systems (despite the physical proximity of the LEO satellites).

It is also reasonable to assume that most users will not be operating in an "open sky" environment. The satellites will thus be expected to make communications links to users in cluttered urban environments, where obscuration by building and trees is likely, and multipath effects are more of an issue. Communication system designs tend to work to assumed elevation angle limits, and these elevation angle limits are frequency dependent. Relatively low frequencies such as UHF can be made to work down to elevation angles of about 5–10 deg elevation, whereas higher frequencies may well require elevation angles as high as 20–25 deg to work successfully. The difference that this makes to the effective size of the coverage footprint of the satellite is significant, and hence frequency choice is one of the most important system design trades that must be performed.

Constraining the elevation angles for a system in LEO obviously shortens the effective communication periods with a given point on the ground. The result is that, as well as requiring more satellites in the constellation to provide continuous coverage, the chances that the system will need to "hand-over" a user from one satellite to another during a communication session is also increased, imposing a greater computational load on the system.

Another obvious constraint on the system design is the RF power that can be emitted by the handset. There is a natural reluctance to avoid "frying" the brains of handset users, and although most of us can think of several people to whom we would happily give such a handset, the health and safety authorities in most countries tend to take a dim view of microwaving the user's neurons. By comparison with modern mobile phones, the original Iridium handsets look uncomfortably large and heavy, and it should be remembered that any communications system will need to conform to user expectations, in terms of size and weight, in order to be a commercial success.

Users are also "uncooperative" in the sense that they tend to cluster into densely populated urban areas. The communications load on the satellites is consequently highly variable around the orbit, and it is clearly necessary to make reliable estimates of traffic loads on the system at peak times before establishing the system capacity, and before sizing crucial parameters such as the orbit average power that the platform will need to supply to the payload. Clearly too, the maximum power consumption will coincide with maximum thermal dissipation requirements of the mission. The thermal control subsystem will thus need to accommodate both the shorter term variations in satellite power usage that occur from orbit to orbit, and also the longer term variations in illumination from the Sun that occur as the orbit planes regress with respect to the Earth.

It will be readily appreciated that, at any given time, the satellites in one plane of a multiple-plane LEO constellation will be experiencing a different pattern of lighting and eclipse conditions than the satellites in another. In a sophisticated system equipped with inter-satellite links, such as Iridium, the possibility exists to route the communications traffic around the network in such a way as to balance the loading on the satellites. More generally, in regions where several coverage footprints overlap, (e.g., over the poles in the case of the

Iridium system), it is possible to route the traffic through the satellite best-placed to handle the traffic.

More specialized users may also impose rather exotic requirements on the system designer. Military special forces and covert agents tend to prefer systems that offer low probability of interception and exploitation. This leads to techniques such as burst transmission, spread-spectrum and frequency hopping, and encryption being employed, and these will all impose constraints on the data rates that can be achieved by the system.

It is not only the user segment that will impose requirements on the satellite design. As discussed earlier, the anchor stations which form the fixed ground segment will need to be able to support the data rates generated by the users in a given region. This may be achieved by a steerable dish antenna, but such an antenna system will clearly only support one satellite at a time, and it is perfectly possible with large constellations, where footprints overlap, to have more than one vehicle in view. This can be solved by appropriate scheduling, which will limit the contact time, (and hence the effective region), that can be serviced by the ground station, or by installing multiple antenna heads. (Satellite to fixed ground station links are typically at higher frequencies than the satellite to user links in order to accommodate the required data rates, with the result that using wide-beam antennas to communicate with more than one satellite simultaneously is not a realistic solution.)

23.1.2 Manufacturing and Testing Processes

Manufacturing and testing of communications satellites clearly has to accommodate real-world constraints, such as time and cost. Particularly in a competitive commercial environment, there is a significant driver to complete satellite construction and testing quickly. A similar pressure potentially exists to maintain the pace of construction especially if you have missions that are intended to be launched on a regular schedule, as was the case with the Russian systems described earlier. In both cases, there is a strong incentive to maintain a common hardware design and establish a "production-line" approach to the satellite construction. In general, this reduces costs through economies of scale, and spreads the non-recurring design costs over a larger number of satellite platforms (See Sec. 23.5).

As the largest constellation constructed in recent times, the Iridium constellation provides an interesting example of the procedures that can be used. Each of the Iridium satellites were constructed on a gimbaled, "island" platform, to which each of the subsystems were delivered for integration. The pace of integration was greatly facilitated by the gimbaled support structure, since a major component of the time-budget for assembly integration and test activities is often the handling time that is necessary to move the satellite into an attitude where a component can safely be added.

For unique, single-satellite production programs, there is often little heritage data on either the design or the expected outputs of the test program. By contrast, the test program for a large constellation can exploit the experience gained with satellites that have already been completed. The test data acquired from early satellites from the production line will clearly need to be compared carefully with the design specifications established by the system engineers to ensure compliance, whereas later satellites can simply be shown to be compliant if they exhibit similar behavior to their predecessors under the same test conditions.

Some pre-launch testing is intended to confirm "workmanship", and these basic functional tests will be required for all satellite systems in the foreseeable future, due to the costs associated with both the satellite hardware and the launch. In the future, mass production of identical satellites for large constellations may permit a "batch-testing" approach, where only some of the satellites are tested before launch. It is also possible that the exigencies of a conflict situation would mean that "responsive space" satellite hardware would be assembled and launched without some of the usual time-consuming tests.

There are, however, some tests, such as mass-properties measurements, where the aim of the test is to establish physical parameters which should not vary significantly from satellite to satellite in a constellation. Measurements of the satellite's moments of inertia and center of mass are made to allow the calibration of momentum wheels, thrusters, and other elements of the Attitude Control System, and any slight variation from one satellite to another (due to slight variations in the wiring harness, for example) should lie well within the performance envelope of the devices used to control the satellite's pointing. It is reasonable to assume, therefore, that for constellation satellite designs, mass-properties testing is only likely to be required on the first vehicle in the series.

Another factor that influences the system design is the availability, of on-orbit spare satellites. As has been mentioned previously, in the case of Iridium, the system design involved a single spare satellite for each of the orbit planes. This offers the opportunity to replace a failed satellite relatively quickly, and without the requirement for a separate launch. It should be noted that the length of time required to achieve this repositioning is dependent on the maneuver capacity of the satellite, and drift times of a couple of weeks would not be unusual, given the likely fuel constraints on the vehicle. Some gaps in coverage are thus likely to persist during this period while the in-plane relocation takes place. In the future, it has been suggested that the advent of responsive launch capabilities would provide the opportunity to insert replacement satellites into a constellation within this sort of time frame. In such a scenario, it might be seen as preferable to keep the spare satellites on the ground, and launch them directly into the slot where they are required. One of the important factors influencing the choice of strategy is the financial cost to the service provider of the downtime in terms of lost revenue and possible contract penalty clauses.

Iridium used an imaginative approach to constellation maintenance when they needed to replenish more than one plane via a single launch. The approach which they

adopted was to launch a batch of new satellites into a parking orbit in the same orbital plane as one of those requiring new satellites, but at a lower altitude. Some of the recently launched satellites were then immediately boosted into their operational orbit locations. Due to the higher orbit regression rate resulting from their lower altitude, the remaining satellites in the parking orbit drifted in right ascension relative to the operational constellation. When they were beneath the required plane, they were then boosted to the correct operational altitude.

A word of caution is also appropriate here. The commercial pressure to be "first-to-market" with a satellite-based mobile communication led to a very rapid pace of procurement with all the LEO communications constellations, with the result that much of the satellite production was complete before any significant degree of in-orbit experience had been gained with the hardware. Hence, in contrast to more conventional programs (where lessons learned with earlier satellites in a series can be used to refine the design of the later missions) there was a significant risk of introducing systematic design faults into all the satellites in the series.

23.1.3 Future LEO Communications Constellations

Future LEO communications constellations are likely to follow Iridium's lead and incorporate inter-satellite links. These may be LEO-LEO links, but there are also options to use LEO-GEO links for some of the connections, especially for those regions where the LEO satellites are not in view of a suitable ground station. Such systems could potentially make use of so-called "dark-beams" on the GEO satellites, i.e., those beams with fewer terrestrial users for which the charges are generally lower.

To avoid creating a future debris risk, future LEO communications will follow Iridium's lead and make provision for disposal maneuvers causing the satellite hardware to re-enter the atmosphere within 25 years of the end of the mission.

One innovation which would require regulatory changes would be the merger of terrestrial and space-based mobile systems, allowing users to operate via ground-based infrastructure when suitable hardware is within range, and switching to satellite-based links when in more remote regions. This type of system could also prove beneficial in time of crisis, for example, if any or all of the ground structure became unavailable or overloaded with traffic.

23.2 LEO Monolithic vs Distributed Architectures

Stuart Eves, *Surrey Satellite Technology*

23.2.1 Pros and Cons of Monolithic vs. Distributed Architectures

As the capabilities of small satellites increase, the option to use a constellation of several inexpensive spacecraft as an alternative to a single, large expensive one becomes increasingly attractive.

However, the decision to launch a constellation of satellites rather than one large vehicle impacts all aspects of the system design process, and traditional design processes must change if satellite constellations are to achieve their full potential.

This section will show how areas such as sensor design, platform design, orbit choice, on-orbit operations, communications, redundancy, testing, and launch are all impacted when the baseline objective is to launch multiple satellites.

From a design perspective, it is clearly most cost effective to create a single satellite design, and then vary the concept of operations of the individual spacecraft in the different orbit planes of the constellation. The alternative, (designing bespoke vehicles for each orbital plane) potentially constrains the deployment and replenishment operations, and necessarily involves significantly more expense. But proponents of large satellites would argue that a bespoke design makes the most efficient usage of a satellite's capacity, and that a modified concept of operations, constrained by the satellite's orbit, is also an inefficient concept of operations.

It is the task of the system designer to make an appropriate choice between these two competing views, taking account of changing technical capabilities.

Coverage Requirements

The coverage required from a system is an obvious place to start. If the service is required over a limited area of the globe, and can be satisfied by relatively infrequent passes, there is evidently little incentive to move to a constellation solution.

The applications of LEO satellites are becoming more global in terms of their coverage requirements. As a result of the increasingly globalized economy, systems are used to collect or transfer data over many locations of the globe, rather than simply satisfying an individual governmental focus. This trend is expected to increase as more international and commercial entities enter the market. System design will also need to increase the area coverage rate (or duty cycle) of the satellite system. In the surveillance domain, applications such as monitoring climate change, deforestation, pollution, agricultural productivity, and water resources, all necessitate the collection of large data sets over wide areas. And increasingly, these are applications which benefit from very frequent revisits. Repeat coverage with temporal frequencies of days, rather than weeks or even months, permits a range of change detection products that can be used, for instance, to estimate crop yields, (and hence predict commodity prices), or in cases of extreme climactic conditions (such as droughts), anticipate potential famines and plan disaster prevention operations.

Not all disasters can be predicted in advance, of course, and a high revisit frequency is also clearly a major advantage when a region is affected by a major

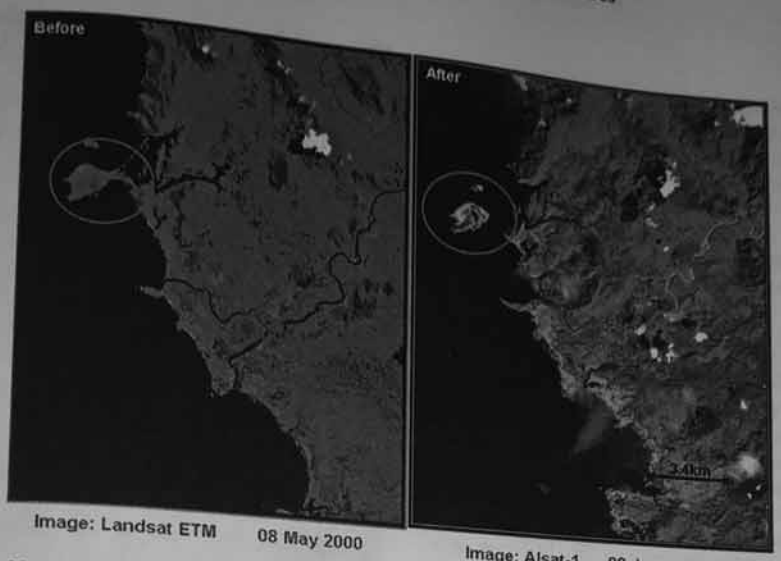


Fig. 23-1. DMC Image of the Indian Ocean Tsunami Shown in Comparison to a Landsat Image of the Same Scene Taken Before the Disaster.

catastrophe such as an earthquake or major tsunami. (An example of the utility of a constellation in this circumstance is illustrated in Fig. 23-1.) The change detection product illustrated was created using timely data over the region affected by the Indian Ocean tsunami from the Disaster Monitoring Constellation (DMC). This imagery was compared with earlier data from the Landsat satellites in order to highlight the areas of greatest change, and hence the regions most in need of disaster relief operations.

Covering such large regions with an individual, monolithic satellite in LEO is potentially problematic and distributed constellations appear increasingly attractive, especially if the temporal requirements call for daily global surveillance coverage from a passive imaging sensor (i.e., one which can only operate on passes when the target region is illuminated).

Clearly, one of the first issues that has to be addressed is how many satellites will be required to meet this coverage requirement. Some estimation techniques are potentially helpful at this point.

The need for adequate spatial or radiometric resolution means that surveillance satellites, especially imaging missions, are rarely deployed above 750 km. At such altitudes (and assuming a fairly "generous" roll capability of around 45 deg) each satellite in a constellation can be "responsible" for a swath of about 600 km on the surface of the Earth.

If it is assumed that the satellites are in near-polar Sun-synchronous orbits (a convenient supposition for many current surveillance systems), then the North-South passes over the equator are close enough to 90 deg for the following approximation to be useful.

Dividing the Earth's circumference (40,074 km), by the assumed swath width (600 km) gives a value of about 67. This is the number of sunlit passes per day required to cover the equatorial region. A satellite in LEO will normally complete between 14 and 15 orbits per day (at an altitude of 750 km, the value is approximately 14.5), and dividing 67 by 14.5 gives a figure of 4.6. Since satellites come in integer amounts, the minimum number of satellites required to provide this coverage pattern is 5. (See Fig. 23-2)

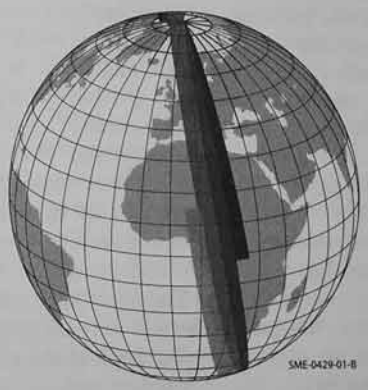


Fig. 23-2. A Five-Satellite, Coplanar, Near-Polar Orbit Coverage Pattern Illustrating How Successive Passes Provide Adjacent Coverage Swaths on the Earth's Surface.

Obviously, though, these swaths must be regularly spaced in order to avoid gaps in the coverage pattern. This can be achieved most simply by spacing the satellites evenly around a single orbital plane, which has the advantage that the coverage is provided at a constant local time of day. In this case, clearly, the Earth-central angle is 360/5 or 72 deg apart, which means that the satellites are not in line-of-sight contact with one-another.

Not only is a shared launch usually the simplest means of achieving the required coverage, it is also

potentially the cheapest. All 5 satellites can be launched on a single vehicle, reducing the cost of the constellation, provided they all have propulsion capability and require a common inclination value. We will explore the implications in Sec. 23.2.2.

23.2.2 Satellite and Sensor Design for Distributed Architecture Missions

Sensor

Swath requirements and sensor design are clearly related to the number of satellites in the constellation. For example, in the case of the Disaster Monitoring Constellation (DMC), and the RapidEye commercial remote sensing constellation, the chosen configuration was 5 satellites in a common Sun-synchronous orbit plane, with the satellites at 72 deg separation. The goal in both cases, was to achieve daily global access opportunities delivered in different ways because of the differing area coverage and resolution requirements. In the case of DMC, the need was to monitor large geographic areas, so the satellite platforms were designed to stare at nadir, and their sensors were equipped with a 600 km swath, 32 m resolution, imaging capability. By contrast, RapidEye required higher spatial resolutions of approximately 6 m, so the instantaneous swath of the sensors could only be 80 km. In order to provide the opportunity for global daily revisit with this system, the satellites have a roll capability of up to 45 deg to allow the 80 km swath to be directed anywhere within ±300 km from nadir.

Further beneifts of constellation operations can be derived if satellites are capable of cueing one another. In order to provide the area coverage rates needed to establish effective cueing, a high resolution surveillance sensor is required. Hence many SAR satellites, (and some optical satellites) usually trade off special resolution and area coverage. For example, the SSTL-300 design is equipped with both wide area multispectral sensors capable of delivering swaths of over 200 km at about 15 m resolution, and a high-acuity sensor with a baseline resolution of 1.4 m and a swath of 12 km. In practice, this cueing is most frequently achieved by comparing two images collected on different dates, identifying the regions of change, and then using this information to cue high resolution sensors to identify exactly what these changes are. Clearly, the high revisit frequencies available from constellations allow changes to be identified considerably more quickly, including variations that occur from day to day.

Platform Design

Another area where constellation satellite design is different from that of monolithic satellites is the need to select an appropriate structural configuration for multiple launch scenarios.

In the case of monolithic, large satellites, some designs have made efficient volumetric use of the available space within the launch vehicle shroud. (The Hughes series of cylindrical, spin-stabilized GEO comms satellites provide one example of this, and the Russian Strela LEO missions mentioned in the previous section, are another.)

Since most launch vehicles were designed to carry larger satellites, modern small satellite designs are seldom constrained by mass limits, and rarely fill the available volume on their own.

Multiple launches of small satellites are becoming more common. The "box-body" designs that have routinely been employed, (largely for simplicity), in recent years have not been volumetrically efficient. The challenge for distributed systems is to find a design that both meets the needs of the mission and at the same time fits well inside the launch vehicle shroud, as seen in Fig. 23-3.

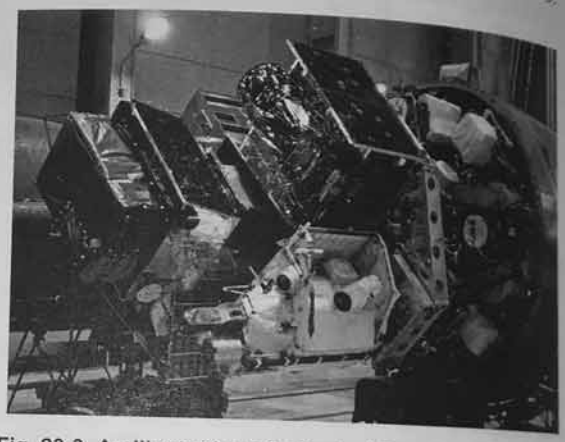


Fig. 23-3. An Illustration of a Typical Small Satellite Launch Configuration, Illustrating the Difficulties Involved in Trying to Make Volumetrically Efficient use of the Fairing.

One such design is the SSTL-300 imaging mission, which has a heptagonal cross-sectional design. (As well as providing the opportunity to use a rare word like heptagonal) The rationale behind this design is to provide greater opportunities to launch these missions as part of a constellation. The satellite would be mounted on its zenith-pointing face for launch, with its camera pointing up rather than down. It can be seen from the illustration in Fig. 23-4, that 3 such satellites make a reasonable approximation to a circle, and hence make good use of the space available. The mounted solar cells comprise 4 of the body facets, with others being used as radiator surfaces, or surfaces on which to mount the star cameras (since they point in the anti-Sun direction).

It will be noted that, unlike the earlier designs, (like TopSat) where the optical axis of the missions were oriented "horizontally" at launch, (although, of course, the launch vehicle pitches during flight), the SSTL-300 has its optical axis oriented vertically at launch. The aim of this design change is to minimize the potential loading on the camera optics, since it is clearly possible to exert fairly extreme loads on the very sensitive camera designs during launch. This is especially true of the small satellites that are typically used to build up distributed constellations, since they have comparatively little plat-

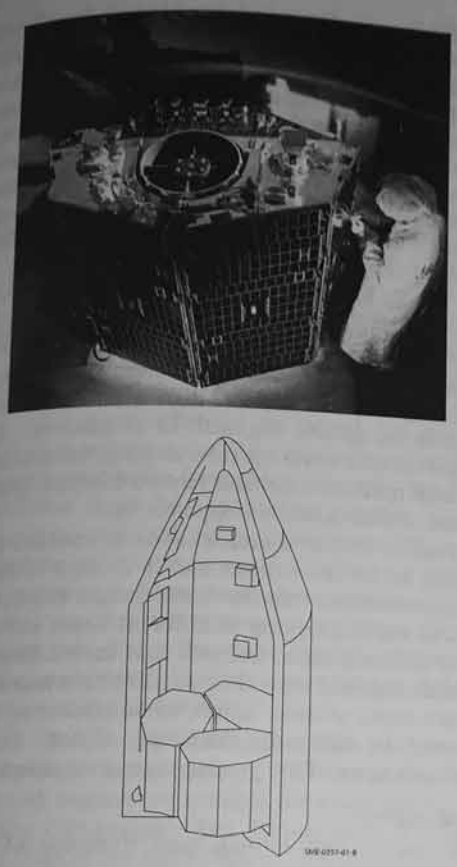


Fig. 23-4. Illustration of the SSTL 300 (Top) and the Envisaged Launch Configuration for 3 Satellites (Bottom).

form structural mass between the launch vehicle and the camera which can absorb some of the vibrational and acoustic loads that are imposed during lift off.

One of the obvious requirements that is likely to be placed on the designer of a distributed system is the ability of the satellites to be launched on more than one launch vehicle. Apart from the obvious commercial advantages of being able to negotiate on price with more than one launch supplier, the likelihood is that a distributed system of satellites is very likely to require multiple launch vehicles in order to reach orbit in an acceptable timeframe, and the turn-around time between launches for a single class of launch vehicle may not be sufficient. This was especially true for the Iridium constellation, where more than 70 satellites were launched on a range of different vehicles.

The technical implication of this approach is the need for a satellite design which avoids structural resonances over the range of frequencies that correspond to the different vibrational modes potentially imposed by the different launch vehicle options. This is obviously something of a challenge, but the situation is helped by the fact that small satellites are comparatively robust, with proportionately fewer resonant modes than their larger cousins.

In those cases where launch vibration concerns remain, (as with the case of TopSat, which had a sophisticated, and potentially vulnerable three-mirror off-axis optical design), the system designer may elect to resort to using a "soft ride" system. Mechanical engineers often like to preserve the mystique of soft-ride systems, describing them as "magic" rubber washers. In practice the "magic" is generally concerned with selecting an appropriate rubber material that will dampen out the vibrations of the launch vehicle, rather than one which will turn the launch platform into a trampoline and make the situation far worse!

Another design consideration for multiple-plane constellations is the fact that, over the system lifetime, a satellite's given plane will have:

A) At any given point in time, a different pattern of activity compared to satellites in other planes of the constellation

B) Over a period of time, differing patterns of activity from month to month

Both these sources of variation arise from the fact that, the satellites in different planes will be oriented differently with respect to the Sun, and that these planes will regress relative to the solar vector. This will affect the times when an optical surveillance system can operate. An active surveillance system, such as a radar or a communications system, will also have differing capacities over time depending on the time-variant illumination conditions of the orbit plane, which will dictate the power generation capacity, and hence the duty cycle of the system. (It should be noted that most active systems will also have a duty cycle limit imposed by the need to cool the payload components. Here too, the ability to radiate waste heat will ultimately be constrained by the angle that the orbit plane makes to the Sun.)

Another area where distributed systems differ from monolithic missions is in terms of their fuel budget allocations. Like all satellites, members of a constellation will need to perform regular station-keeping maneuvers to maintain altitude, but there are additional constraints arising from the need to keep the members of a constellation at the desired relative separation.

In general, satellites in different locations within a constellation will experience differing perturbations, leading to the need to establish a "key" satellite with respect to which the other satellites will maintain station. Obviously, though, this "key" satellite will have to perform fewer maneuvers than the remaining members of the constellation, with the result that their propellant usage will be different over time. For a short-lived mission, this may not matter unduly, but for a longer-lived constellation, a CONOPS which varies the "key" satellite from time to time, in order to distribute the station-keeping requirements more evenly, is a wise policy.

If satellites in a constellation are deployed on a single launch vehicle, then the nominal situation is that they will be deployed around the nominal orbit height, with

some satellites moving "prograde" around the orbit plane, and some moving "retrograde" relative to the key satellite, (which is assumed here to be close to the nominal design altitude of the constellation). Clearly those satellites with the greatest drift rates following deployment will have to perform the largest maneuvers in order to establish a common orbit period with the key satellite. Immediately, therefore, the key satellite will be "fuel rich" relative to the satellites that are endeavoring to keep station with it, and hence there is often a case for switching "key" satellite fairly early in the mission.

Indeed, a flexible strategy at this stage is quite likely to be a pre-requisite if the propellant management is to be optimized. Launch vehicles do not always deliver their cargo to exactly the correct altitude and eccentricity. In that case, the initial key satellite may need to be altered in order to minimize the propellant expenditure required to configure the satellites as intended.

A flexible strategy also potentially allows the oldest and most fuel depleted satellites to be "husbanded" toward the end of their missions. A satellite that is almost out of fuel, (save for the de-orbit allocation which is now increasingly viewed as essential), can remain as a useful member of the constellation if the other satellites treat it as the "key" and maneuver with respect to their aged cousin!

Another constellation-related propellant requirement is the potential need to reconfigure the constellation to optimize performance in different potential modes of operation. Hence, satellites in a common orbital plane may be operated relatively close together in a peacetime scenario to collect collocated contemporaneous data of specific locations. In the SAR domain, this could allow the generation of 3-D terrain information products through interferometry, and in the optical domain, could involve the collection of instantaneous stereo pairs, or contemporaneous sampling of the bi-directional reflectance distribution function (BRDF) using hyperspectral sensors. In order to operate in these modes, the satellites clearly need to be relatively close together.

The same co-planar assets could be used at wider separations, (Earth-central angles up to about 40 deg), to cover the same region with a small time interval between the collections. With appropriate rapid processing and cueing, this would allow the first satellite, (possibly operating in a wide-area mode), to cue the second, (possibly operating in a more focused high-resolution mode. Alternatively, the satellites could simply generate two very similar products and then apply change-detection processing to the results to identify the moving targets or other changes in the scene.

Finally, the satellites could be used at maximum separations around the orbit plane, (a mode perhaps most likely in a crisis situation), since this mode of operation would maximize the area coverage rate of the constellation and minimize the "maximum-wait-time" between passes for a user at a given location. Usually, unless the number of satellites in the constellation is large, this mode of operation would result in the satellites being separated by more than 40 deg, which would place them "over the horizon" with respect to the other satellites in the system.

All these modes are viable only if the satellites have been previously designed with mode-changing maneuvers in mind. The total number of such reconfigurations is unlikely to be high, and the propellant expenditure does not have to be excessive if the drift rates, (and hence the propellant usage), are low and the maneuvers are in-plane, but they will not be possible at all if there is insufficient propellant capacity designed into the satellite.

23.2.3 Communications and Inter-Satellite Links (ISL)

Much has been written about constellation design, but an equally important element of a distributed system design is the ground segment. In no small part, this is because the area coverage rates and data volumes associated with multiple satellites require adequate provision in terms of data downlink opportunities.

As noted earlier, a constellation of satellites can be operated in different orbits and in a range of different collection modes. This inevitably means that the satellites will require tasking at different times; will collect data at different rates; and will thus have different data downlink requirements. Inter-satellite links may serve to alleviate some of these issues, while others may be best addressed by the judicious siting of both fixed and mobile elements of the ground segment architecture.

Commanding

The use of a high-altitude asset, (typically in GEO), to relay commands to a LEO satellite in near real time is clearly a far more cost-effective proposition if that LEO satellite is just one of a number of assets that can be serviced by the GEO platform. Previously, GEO data relay has been largely reserved for very high-end platforms, but as the UAV community has demonstrated, the use of high-altitude relay satellites to perform both commanding offers significant benefits. As an example, the Inmarsat satellites are capable of supporting 500 kbps data rates through satellite-borne terminals similar to their terrestrial BGAN system.

It can be demonstrated that a single satellite in GEO can see a LEO surveillance asset at some point on each of its orbits, with the result that for most systems, the longest gap between commanding and having a suitable imaging opportunity, (a sunlit opportunity for optical sensors), is on the order of 45 minutes.

In a comparatively dense constellation, the satellites may be separated by in-plane Earth-central angles of less than 40 deg, and this would provide the opportunity to establish continuously available LEO-LEO command links around the orbit plane. More realistically, the constellation may consist of a more limited number of satellites in several different orbit planes. One approach to improved commanding timeliness in a more sparse constellation is to equip the satellites with a LEO-LEO ISL system which establishes links between the satellites when they come within view of each other close to the

poles. This is the approach adopted by the SAR-Lupe constellation of 5 satellites, which operate in three different, near-polar, orbital planes. Clearly some phasing of the satellites is required to ensure that they pass close to the pole at suitable times, and hence periodically come within view of other members of the constellation. This phasing must be conducted without compromising the coverage pattern, and hence will involve changes to both the orbit right ascension and true anomaly.

Data Return

In the future, the timeliness offered by a surveillance constellation will need to be matched by an architecture for the data-return communications that support it. There is little point in offering hourly revisits over the target region if the opportunities to download the data only occur once or twice per day. The most attractive option, if suitable inter-satellite links exist, is to relay the data in near-real-time from the surveillance satellite in LEO. This could be achieved via an indigenous ISL system hosted on the surveillance constellation but the inter-satellite separations in-plane would be constrained if there was a need to maintain continuous long-range ISL links for moving data around the globe, (and the surveillance constellation size would probably have to rival that of the Iridium network to be feasible).

The Iridium network itself represents a possible alternative communications medium, but the data rates that this network support are suitable for voice communications rather than imagery data. Hence, even if equipped with the space-borne equivalent of an Iridium handset, a member of a LEO surveillance constellation would struggle to deliver information in a timely fashion. This conclusion may change with future generations of LEO communications networks, which are likely to support higher data rates. The advent of greater on-board processing capabilities on future surveillance satellites may also help by reducing the size of the image files which need to be transmitted.

In the short-term, therefore, the most appropriate ISL adjunct to a surveillance constellation would appear to be a data relay asset in GEO, focused primarily on the delivery of timely, thin-route commanding information. In part, this is because a satellite in GEO can focus its coverage on the specific region of operational interest, and hence can transfer commands to satellites about to access the region of interest. In the longer term, as ISL capacity improves, it will also be possible to retrieve data from those assets which have just accessed the crisis area in a far more timely fashion.

If an ISL capability is not available, a suitable network of ground stations at varying longitudes and latitudes is the next-best alternative. Ideally these ground stations should be linked either via the internet or via satellite communications to allow them to route commands and return data in the minimum possible time. This sort of architecture has been demonstrated to work successfully with the DMC constellation.

For a military surveillance constellation, there could be a number of different user communities. In practice, the most critical one is the in-theater, war-fighting component, but the strategic intelligence community, the homeland defence community, allies and the system training requirements are all likely to impose additional collection requirements on the system. It should be possible, therefore, to estimate the regions of the globe where the surveillance system will be most active, and then to configure the ground-station network to provide downlink opportunities as soon as possible. Clearly, for the warfighting scenarios, it is somewhat more difficult to plan in advance where the capacity will be most needed. It is suggested that this is the most appropriate role for mobile ground stations, which can be sited in response to a particular crisis, and can therefore ensure that any given satellite in the constellation does not reach its memory capacity and then start to miss imaging opportunities.

23.2.4 Reliability, Redundancy, and Survivability

One of the most subtle aspects of designing at the constellation level concerns the redundancy philosophy. Traditional satellites are typically dual-string redundant in order to guarantee a reasonable probability of meeting the individual satellite's design lifetime. This calculation clearly changes if the satellite assets in question are intended only as a short-term adjunct to a military surveillance capability that is targeted on a particular, short-term conflict or crisis situation, and here the calculations probably favor a single-string approach.

But how is the debate resolved when there could be a mix of in orbit assets and spares on the ground using the same basic design? The answer is to look not at the redundancy of a single satellite, but at the probabilities associated with the constellation as a whole. When multiple satellites are involved, the overall performance degradation resulting from an individual failure is modest, and in a scenario where surge launches are envisaged as a routine part of the system, this would suggest that a far more relaxed approach can be taken to design reliability and the testing procedures.

The debate is complicated by the fact that a military surveillance constellation is likely to be targeted during time of crisis. The failure statistics are less likely to be independent if the cause of failure is hostile enemy action, but again there would appear to be grounds for thinking about satellites as conventional assets where attrition is expected, rather than as grand-strategic assets where failure is to be avoided at all costs. Some mitigation of this *conditional probability problem* may be achieved if the satellites in the constellation are fitted with threat warning systems—essentially the ability to *cry for help* or shout a warning in the event that they come under attack. The advantage of such an approach is that the other members of the constellation can then take precautionary measures, (e.g. rapid maneuvers) intended to reduce their vulnerability and so improve their availability statistics.

Table 23-0, Fig. 23-4◀, Eq. 23-0

Again, the utility of ISL's becomes apparent, but this time for monitoring the platform telemetry and relaying any messages from the threat warning sensors, rather than for interacting with the sensor payload.

Somewhat ironically, one advantage of traditional, large, stove-piped programs is that they typically have bespoke weaknesses and vulnerabilities which do not necessarily read across into other large, stove-piped programs. A potential aggressor is thus presented with the challenge of finding a weakness in several different, separately designed systems. This fact is often overlooked by enthusiasts for "network enabled" capabilities, who tend to overlook the fact that by creating a large number of assets all working to the same standard, they have potentially also created an opportunity for an aggressor to target a significant proportion of their capabilities. (This was arguably one of the principal reasons for the Allies' success in WWII, where the German Enigma machines were cracked, allowing all their most secret communications to be exploited.)

It is strongly recommended, therefore, that one element of constellation spacecraft that **should** be redundant is the command and control subsystem, incorporating a second, "crisis mode" for communicating with the satellites that is only engaged in the event that the primary mechanism is compromised.

Another rather ironic advantage of large monolithic programs arises from their comparatively pedestrian production cycles. If, despite the pre-launch testing, a particular system has a systematic weakness, a traditional monolithic production process is likely to reveal that weakness following the launch of the first satellite in the series, allowing those on the ground to be modified accordingly. By contrast, a rapidly launched distributed system could potentially launch several satellites in a relatively short period of time, with the result that a significant proportion of the satellites in the constellation could be affected by the same design error.

Counter-arguments in favor of distributed systems are the fact that the non-recurring design costs of the system can be spread across multiple missions, and the experience that the builders of the system gain through building multiple copies of the same design. Traditional monolithic systems may take more than a decade to reach fruition, with the result that an engineer in the traditional space industry might only work on two or three satellite designs in an entire career. By contrast, relatively short-lived, and quick-to-build, satellites provide the engineers who work on them valuable experience which helps to avoids errors in both design and assembly. This is, in effect, an example of the traditional commercial advantage of economies of scale being applied in western space industry for the first time. The commercial LEO comms systems mentioned in the previous chapter were some of the first to benefit from these advantages, and the RapidEye constellation of LEO surveillance satellites represents a more recent example.

By comparison, the Russian space industry has been aware of the advantages of mass production for many years. Although comparatively unsophisticated, the Strela communications satellites referred to in the previous chapter are the most populous distributed constellation in the history of space. Over 350 satellites based on this design were launched over a period of more than 20 years before finally being replaced by a more capable system.

For those used to a more rapid pace of technological change, it will be appreciated that maintaining access to the parts required to build a near-identical satellite some 20 years later is an achievement in itself. The driver for doing so is the need to preserve continuity of service, and this is dictated in part by the design and deployment of the ground terminals—which is the subject of the following section. In many cases, however, the desire for data continuity and the re-use of existing technology is outweighed by the desire for data improvement. Here constellations of shorter-lifetime, rapidly replaceable satellites clearly have the edge over large monolithic systems. The later missions in the constellation can employ more up-to-date technologies and thereby deliver better performance. The DARPA F6 concepts envisages a subtly different concept of "distributed" operations, by spreading the functions of a larger monolithic satellite not across a constellation, but across a cluster of co-orbiting sub-satellites equipped with short-range inter-satellite links. In the F6 concept, it is envisaged that new technologies can be introduced to replace elements of the cluster without the requirement to change all of the hardware.

Get More

23.2.5 Ground Segment

Many of the advantages that are associated with a distributed space segment also apply to the terrestrial systems which interact with them.

While it is conventional to have a widely dispersed user community consisting of multiple terminals, (especially for satellite communications systems), the command and control segment of many space systems has been largely "monolithic" in character, consisting usually of one or two ground stations invariably in fixed locations.

While this model is generally sufficient for communications systems, (especially traditional ones operating through geostationary satellites which have very large, fixed, communications footprints, allowing continuous contact between the satellite and the ground), it is far less satisfactory for low Earth orbit systems.

In LEO, a satellite's communications footprint is typically only a few thousand km across, and the laws of orbital dynamics dictate that a satellite can only spend a small percentage of its time over a given point on the Earth's surface.

For real time communications services, (and especially safety-critical communications to manned spacecraft), intermittent communications with the ground is either undesirable or unacceptable, and so multiple ground station networks were established to support such systems. These networks involved stations at multiple longitudes, and at a variety of latitudes, in order to facilitate more frequent contact between the space and ground segments. Ideally the aim was to ensure that con-

tinuous contact could be maintained, but over unpopulated areas such as the southern oceans and Antarctica, this was not always perceived to be a cost effective solution, and some periods of outage were tolerated by the LEO communications networks.

The precise configuration of the ground station network will depend on the coverage that is required, and the orbital configuration that is selected to best meet that coverage need. But there are some specific advantages to a distributed ground station network that will potentially enhance the performance of all LEO systems that are not equipped with inter-satellite links. These advantages are:

1. More frequent access to the satellites in order to send commands to the on-orbit assets, improving the timeliness of the system
2. More frequent opportunities to download data from the mission, again increasing the end-to-end timeliness of the system
3. Improved speed of checkout of the system, using multiple ground stations. This advantage is of particular utility for the sorts of distributed space systems envisaged by the Operationally Responsive Space community
4. More frequent opportunities to monitor telemetry from the satellites, providing more responsiveness in the event of an anomaly on the satellites
5. Importantly, the increased opportunities to download data from the satellites means that onboard storage capacity can be re-used several times per day, greatly increasing the output from the system and its revenue generating capability
6. In a military context, faster response times for reporting from threat monitoring packages mounted on the satellites

Conclusions

Distributed satellite constellations offer multiple advantages to the system designer, and while they do incorporate certain vulnerabilities, (similar to those encountered in terrestrial computer networks), the advantages appear to significantly outweigh these disbenefits.

Just as today we would find it hard to imagine life without the world wide web, in the future it is anticipated we will be equally reliant on the "space wide web" of distributed satellite systems.

23.3 Spacecraft Manufacturing, Integration and Test

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Much of the focus of *Space Mission Engineering: The New SMAD* is the "what" of space systems—what these systems are able to sense and communicate back to mission operations and end users, quantifying what their reliability will be, and what these systems might cost. This section focuses on the "how"—how individual space qualified hardware components are assembled together, integrated in configured items and then tested as complete spacecraft before being delivered to launch integration (Chap. 27).

An overview of the process of how spacecraft are manufactured from components and then integrated into their final configuration is shown in Table 23-1. This process is fundamental to understanding how space systems are built, as well as the impact of the underlying type testing theory. The methodologies and processes described here apply to space hardware used for both monolithic and distributed/fractionated systems (Secs. 23.2 and 23.5).

In addition, the timeline required for the integration and testing of a spacecraft's flight hardware and flight software can drive the schedule and ultimate cost of space systems. The testing and qualification of flight software is addressed in Chap. 20, and its availability is a critical element of how soon a full up spacecraft can perform its integrated system testing and enter into vehicle-level environmental testing.

23.3.1 Qualification Testing of Space Hardware

Flight hardware for space systems is qualified for space through environmental and operational testing of hardware designed for space applications. The design of this hardware uses materials that will survive in the applicable environments and selecting controlled manufacturing processes to produce this hardware. This regime of qualification testing of space hardware applies to individual piece parts such as electronic components and graphite composite boom structures, to functional components such as reaction wheels and space thruster motors to the individual payloads and spacecraft bus that comprise a flight vehicle.

Qualification of space hardware is critical given, at best, a limited ability to service space hardware on orbit and the environmental stress imposed by launch and relevant space operating regimes and required operating life times of space systems. Unmanned spacecraft are exposed to vibroacoustic launch loads that would kill humans, radiation environments around Jupiter and Saturn many times greater than those in Earth orbit, temperature ranges from +100 C to –100 C transitioning in and out of Earth eclipse and must not only survive but operate reliably and effectively for a decade or two for some missions.

Mission designers and engineers involved in the development of space systems need to understand the interaction between the availability and production of space qualified hardware and feasibility of a given space program. The time and cost required to manufacture and test space flight hardware can and will strongly influence a space program's cost and schedule. These span times and costs are directly driven by the precision manufac-

Table 23-1. An Overview of Spacecraft Manufacturing, Integration, and Test. Delivery of a qualified spacecraft to the launch site involves effective coordination and planning by multiple staffing functions and relies on a global supply chain.

Step	Comments	References
1. Review of Governing Program Documents and Plans	Identify applicable system and lower level tests, verification activities, environmental effects, permissible materials affecting the space segment	Chaps. 6 & 7, Secs. 23.4, 26.5
2. Develop Spacecraft Designs to Implement Required System Functions; Work with Subcontractors and Global Supply Chain	Work within ascribed program budget and schedule to generate required documentation such as drawings, equipment specifications, software development plans, testing plans in coordination with suppliers of space hardware and relevant software; use/adapt space qualified components and units; meet interface requirements with ground and user segments and other supporting systems	Chaps. 14–22, 26–29
3. Manufacture, Assemble and Test Component-level Space Hardware; Develop and Test Unique Test and Support Equipment	Procure subcontracted components and elements; order and purchase space qualified materials and components; develop and perform necessary component qualification tests; procure, develop and validate necessary ground support equipment; produce and signoff flight components; assemble and (qualification/acceptance) test configured items; document and deliver configured items to Integration and Test	Sec. 23.3, Fig. 23-9, Table 23-7
4. Perform Spacecraft-level Integration and Testing	Assemble and interface subsystems and units with flight harnesses; evaluate interface compatibility across subsystems and elements; load core spacecraft executive software and test device drivers; stimulate system with test vectors; perform closed loop testing; execute comprehensive system tests and environmental tests; prepare and package spacecraft for shipment to launch site	Table 23-4, Fig. 23-8

ture of components for flight, specialized technologies required for testing these components and subassemblies and the often unique facilities needed for manufacture, integration and test of space systems.

Given the impact of qualifying hardware for spaceflight, the theory of type testing provides the basis from which space hardware is qualified for space. From the hard won lessons of how to reliably design and build aircraft and launch vehicles from subsystems comprised of multiple components performing individual functions for manned systems, analogous processes are used for manufacturing, integration and test of spacecraft flight hardware. What is unique to space hardware, in comparison to aircraft and launch vehicles, is the time span in which a spacecraft may be initially proposed versus when it actually is launched, a magnitude reduction in production volumes, no on-orbit serviceability, the stress of launch/space environments and required system operation on a 24/7/365 basis for multiple years.

In order to reduce the risks of on-orbit failures and improve flight hardware reliability, we use the theory of type testing. The theory of type testing is the fundamental principle that drives the production of space hardware.

As shown in Table 23-2, type test theory depends on preparing and controlling complete and exact engineering data (drawings, specifications, and procedures). If the engineering data controls the hardware construction completely, all items built to the same data are equivalent and the results of any single-item test are valid for all items that are built using the same data. In particular, if a representative article *(type test article)* passes a sequence of *qualification tests*, all other articles built to the same engineering data should also pass. In other words, the design is *qualified*. Thus, production of space hardware will focus on making sure subsequent articles are identical by directly controlling engineering data (*configuration management*), using material qualified for space and employing the necessary controlled manufacturing processes (under *statistical process control*). Once the design is qualified, subsequent articles built from the same data and controlled manufacturing processes are then subjected to less severe *acceptance tests* that certify the workmanship of these articles.

Table 23-2. Theory of Type Testing. Type test theory is the basis for qualification testing.

Logic	Conditions & Conclusions
Assertions	• Engineering data is complete and exact • Engineering data completely controls manufacture • Manufacturing processes are under process control • Materials used in manufacturing are space qualified and exhibit minimal lot variability • All items manufactured to the same engineering data and using the same materials and with the controlled manufacturing processes are identical
Therefore	Results of a qualification test for one article are valid for all subsequent articles

With the expense, schedule impact and potential risk exposure of qualifying hardware for space, the decision to use new materials, mission-unique components or modified hardware is generally not taken lightly for a space program. Whether because of obsolescence or unavailability of specific components, changes to mission requirements or size/mass/power constraints, a program will need to decide what must be modified or changed with respect to a previously qualified design and actively manage the qualification process for the new article or re-qual (delta qual) of modified designs.

Table 23-2, Fig. 23-4◀, Eq. 23-0

Table 23-3. Space Hardware Qualification Approaches. Depending on the mission type, heritage of a spacecraft's hardware, program cost and schedule constraints, different methods are used to demonstrate that the hardware is qualified for space

Qualification Method	Method Unique Characteristics
Dedicated Qualification Hardware	A specific set of qualification components is constructed and **qualification tests** are performed at the relevant **component qualification levels**. This set of components or a second set of hardware qualification components is assembled into the qualification spacecraft and tested as a spacecraft at **spacecraft qualification levels**. Usually dedicated qual H/W is not approved for flight given cost, risk and schedule constraints. For programs building multiple identical spacecraft, a dedicated qual unit may be cost effective.
Qualify the First Set of Flight Hardware	The first set of flight components is tested at **component protoqualification levels** or **protoflight levels**, then assembled into a S/C which is tested at **spacecraft protoqualification levels/protoflight levels**. This spacecraft is then launched. This is the **protoqual or protoflight concept**.
Qualify by Similarity	Demonstrate that components and the environments they will be subject to are identical to previously qualified hardware. Similarity basis uses type testing, configuration management, conformal material lot sampling, statistical process control. Programs will still expose components, assemblies and S/C to a testing regime but at **acceptance levels**.

An example of a delta qual might be to test an existing thruster motor to run for double its current qualified operation time because of in-stock availability of previously built thrusters versus the cost and schedule associated with developing and qualifying a new thruster (and the expense of building and acceptance testing the new thrusters). In the case of a new qual, using a new processor chip in a qualified Spacecraft Control Processor (SCP) unit because of the on-orbit computation load imposed on the SCP by a new payload or modified mission requirements may trigger not only a qual of the new chip but also a delta qual of the SCP. In either of these cases, a qualification test program must be developed, agreed to and properly funded.

Qualification testing of all flight hardware (and software) is needed to show that a specific spacecraft design is suitable for space [DoD, 2004]. These tests certify that the hardware and software work together properly and that the hardware can and will survive and operate in the prescribed environment. The qualification program for a space system will test each component not already qualified for space as well as the complete spacecraft. The program may also include additional functional and/or environmental testing of selected assemblies or subsystems. Table 23-3 lists three different approaches to qualify hardware for a spacecraft design.

Qualification testing of a component checks how it performs *(functional test)* and testing its ability to survive the anticipated environment (*environmental test*). Functional testing of each component under qualification is powered and exercised by its own special-purpose test set during test. Environmental testing exposes a component to the relevant environmental stresses it will see that can affect its operation. These include vibration loads analogous to in-fairing launch loads, multi-g shock loads similar to launch vehicle separation and thermal vacuum that simulates operation in space. If the component must survive nuclear weapons effects, it may also undergo a flash X-ray test, which simulates a prompt radiation dose. Sometimes, it may be tested to assess its electromagnetic compatibility (EMC) with the rest of the spacecraft to determine its susceptibility (e.g., its sensitivity) to any emitted electromagnetic frequencies from a payload or another part of the spacecraft. Figure 23-5 shows a typical component-level qualification sequence.

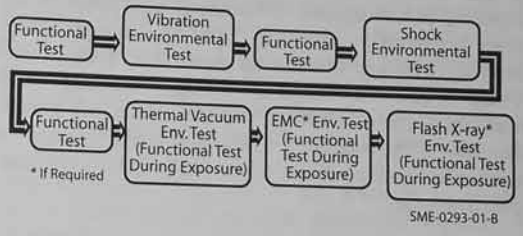


Fig. 23-5. Component-Level Qualification Testing Flow. A component is qualified through a series of functional and relevant environmental tests.

Going back to type test theory, qualification testing focuses on both design and process verification. These qualification tests are conducted to demonstrate that the component design, its manufacturing process and materials selected repeatably produce flight articles (generally identified as hardware configured items or HWCIs) that meet specification requirements. These HWCIs include items such as deployment hinges, solar array assemblies and solid state data recorders—an item, at the assembled level, that has sufficient capability to perform a specific set of functions and that can be tested as a single unit.

After testing has qualified a unit or HWCI for use and operation in space, additional tests, known as acceptance testing, will be performed on any similar/identical articles designated for flight. Acceptance testing is performed on a unit (e.g., software, per batch or lot of manufactured articles) prior to delivery to Spacecraft Integration and Test. Acceptance tests are conducted to demonstrate the acceptability of each HWCI to specification requirements and provide quality-control assurance against any workmanship or material lot deficiencies.

Acceptance testing is applied to stress screened items in order to cause incipient failures due to latent defects that may be in the parts, materials, and workmanship that comprise the unit. Figure 23web-4 (see Sec. 23.4.5.1 on the web) shows the difference between expected temperatures, acceptance test level temperatures and protoqual/qual temperatures for a unit/HWCI.

As an example, Table 23web-1 identifies qualification and acceptance test levels for specific tests applied either at the unit/HWCI level or at the spacecraft/space vehicle levels [DoD, 1999]. These test levels and exposure durations differ because units arrive for vehicle-level assembly having been exposed to and passed their specified component/unit level tests. Engineers should also note that application of these same unit test levels to the integrated space vehicle is not recommended and can involve the very real potential of damaging the fully integrated vehicle.

Figures 23-6 and 23-7 show the magnitude of the typical qualification and acceptance vibration levels and device shock to which components are tested to versus the actual expected load levels imposed by an Atlas-Centaur upper stage on an attached spacecraft. For example, at the spacecraft/launch vehicle interface ring, the expected power density spectrum is equivalent to 2.7 times the force of gravity (2.7 g). Applying an acceptance environment that is 6 dB above this expected load would be ($2.7 \times 4 = 4.3$ dBg + 6 dBg = 10.3 dBg) or an equivalent load of 10.75 g. This is considerably greater than any of the measured vibrational loads imposed on an attached spacecraft and would likely damage the integrated vehicle.

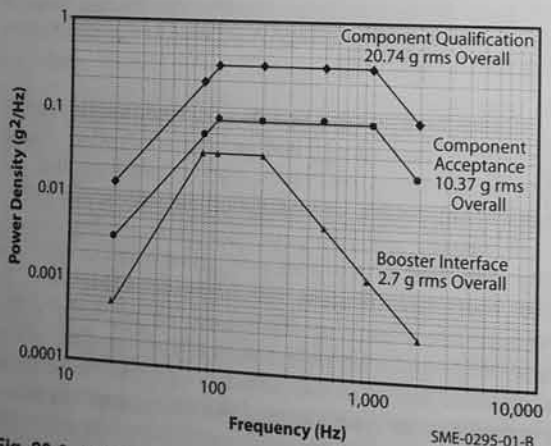


Fig. 23-6. Vibration Levels for Components on the Atlas-Centaur. Component qualification vibration levels are specified by a power density spectrum which is higher than the expected spectrum.

23.3.2 Supply Chain Considerations and Availability of Space Qualified Hardware

Sometimes there is an extended period from when a space mission is initially proposed to the launch and operation of a system in space. As an example, the Chandra X-Ray Observatory (Chandra or CXO), originally referred to as the Advanced X-Ray Astrophysics Facility, was initially proposed to NASA in 1976. Initial work began in 1978 and Chandra was ultimately launched on STS-93 in 1999. In comparison, LCROSS (Lunar Crater Observation and Sensing Satellite) was proposed and selected in early 2006 as a secondary payload to the Lunar Reconnaissance Orbiter, readied for launch in late 2008 and subsequently launched and impacted the Moon in 2009.

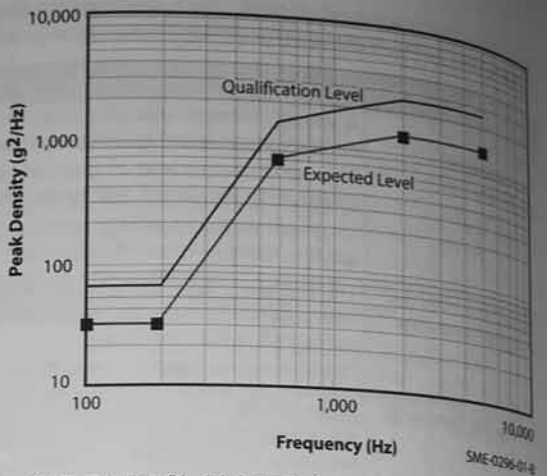


Fig. 23-7. Device Shock Levels for an Atlas-Centaur. Qualification levels are selected to be greater than the expected design levels.

Given the wide variability between the different types and classes of spacecraft as well as their available funding and mission timelines, all space programs are ultimately dependent upon the vendors, suppliers and subcontractors that provide space qualified hardware for flight systems. For some systems such as spacecraft launched into geosynchronous orbit with communication payloads (geocomm sats) or cubesats that are secondary payloads to other missions, a steady demand for production and launch of these systems enables a worldwide supplier base. Here, a geocomm sat may be ordered by a satellite services provider and be ready for launch in less than three years.

In the case of Chandra, multiple design configurations were developed and driven by its launch from the Space Shuttle—initially, the satellite was planned to be operated in LEO and serviced on-orbit similar to the Hubble Space Telescope. Other options considered included robotic servicing and orbit raising and maneuvering by the Orbital Maneuvering Vehicle before a final configuration was selected and built which eventually inserted Chandra into an elliptical orbit of 133,000 km by 16,000 km by use of an Inertial Upper Stage booster rocket once released from the shuttle cargo bay. For Chandra, over two decades passed between when it was originally proposed to when it actually launched.

Whether for a one of a kind spacecraft such as Chandra or Hubble or for production builds of geocomm sats, decisions are made whether to use existing spaceflight

qualified hardware or to develop new hardware specific to the mission at hand. Some of these selections may be driven by cost and available schedule—in the case of a commercial geocomm sat derived from platforms such as Boeing's 702 or Lockheed's A2100, designs based on previously flown equipment such as propellant tanks, reaction wheels, and solar array panels can be used and are qualified by similarity.

Other decisions are driven by availability and change —memory chips used and tested in previous flight builds may no longer be produced forcing selection and potentially qualification of different memory chips (change driven by obsolescence) or that chips used in heritage designs may be susceptible to memory upsets in a different orbit regime (change driven by space environment).

Designers and engineers must play a direct role in selecting the appropriate hardware and thus the relevant producers of this hardware when developing a space system. Related to and part of a space program's risk management plan, decisions on which hardware configuration is used in a system need to be documented and potentially tracked if risks are identified in terms of part or component availability, space operational issues or non-technical problems due to regulatory regimes such as ITAR/EAR. Mission designers and system integrators must be aware of the types of problems than can occur related to equipment availability and supply chain issues throughout the development, manufacture, and integration and test of a spacecraft and manage these problems accordingly.

Engineers and supply chain personnel must also verify that vendors supplying spacecraft hardware have satisfactory quality assurance programs. This will involve performing test surveillance, certifying vendors' programs and periodically auditing their performance. Test surveillance includes certifying relevant test equipment and procedures, witnessing tests, approving test records, and reviewing results.

Since test equipment used in the production of flight hardware is in contact and directly affects hardware intended for space, the design, manufacture and calibration of all test equipment used on flight programs must be managed as well. All information used to develop and build the test equipment such as mechanical and electrical assembly drawings, parts lists, wire lists, panel photographs, and calibration test procedures must also be treated as controlled engineering data.

Quality assurance established at both the supplier and system integrator must maintain data records for all formal tests and document and track any testing failures. Often quality assurance personnel keep all failed parts and record failure and anomaly results, so that lessons learned from failures can be used to correct design weaknesses. In addition, while formal procedures govern all tests, troubleshooting may need to deviate from approved procedures. For such troubleshooting operations, quality assurance must approve and maintain records of the exact steps involved.

23.3.3 Space Hardware Manufacture and Assembly

We differentiate between the component and unit level configurations for space hardware versus the integration and testing of a spacecraft bus and overall space vehicle/satellite. At the component/unit level such as processor chips, solar cells, reaction wheels or solid state data recorders, these pieces of a spacecraft perform specific, individual functions (e.g., process data, generate electrical power, spin to control attitude or store collected data in space).

These components must operate in concert with other devices (and relevant software) to provide the necessary spacecraft-level functions and capabilities—generate and distribute power, process and store data on-board, hold operational spacecraft attitudes and handle on-orbit disturbances, communicate to/from ground stations, keep and maintain safe operational temperatures for all components, and perform real-time control of spacecraft operations. Here we focus on the manufacture and assembly of the lower-level hardware and address spacecraft-level integration and test in the following section.

Needless to say, different classes of hardware are manufactured and assembled with different materials and processes. Structural piece parts which are used for the primary load bearing elements of a spacecraft may be machined from metal billets or formed from composite materials on tooling and then assembled together using fasteners and/or bonded together. In contrast, digital electronic units such as the spacecraft processor, payload interface units or solid state recorders are built from discrete electronic chips which are soldered to printed circuit boards which are then assembled together in machined housing frames to form these digital electronic boxes.

For other components such as multi-layer insulation (MLI) blankets, sheets of thermal management material such as aluminized Kapton and scrim are measured, trimmed and assembled together by sewing and/or bonding at the edges. Devices such as space thrusters are constructed from components (e.g., rocket nozzles, titanium tubing, and catalyst beds), assembled with tooling and then subjected processes such as brazing to enable the thrusters to operate for extended periods on-orbit at temperatures over 1,750 C.

Fundamentally, the objective in component/unit level manufacturing and assembly is to produce flight hardware with sufficient functionality that can be tested and measured. In the case of electronic boxes, these are connected to test sets to verify the integrity of signals into and out of the boxes, measure their power demand and thermal dissipation under differing operational conditions and then subject them to flight environmental qualification/acceptance testing. A space thruster would be hot fire tested in a testing chamber to determine its thrust levels and to meet its acceptance performance criteria. Reaction wheels would be connected to wheel control electronics, mounted on lazy susans and then exposed to both test and operational duty cycles to verify

Table 23-4. Design Verification Tests. These tests demonstrate functionality and verify component, subsystem and spacecraft level operation.

Test Category	Example Tests	References
Structural Tests: Verify primary structural members handle flight operational loading conditions	• Static loads test • Modal survey	Sec. 22.1
Deployment Tests: Verify deployment and operation of devices stowed for launch	• First motion • Full extension deployment • Restowage	Secs. 22.1, 25.1, 25.2
Separation Tests: Demonstrate complete separation of components from attachments when released	• Release • Rates of separation	Secs. 22.1 & 22.2, 25.4, 27.8
Antenna Testing: Measure generated electromagnetic fields and actuator/pointing control	• Pattern tests • Closed-loop tracking	Chap.16, Sec. 21.1
Attitude Control Testing: Verify attitude control responses and mode transition behaviors	• Closed-loop testing	Secs. 19.1 & 19.2, 22.1

wheel performance to demonstrate that when a wheel is commanded to rotate counterclockwise, it does so with sufficient accuracy and performance—i.e., it meets spec.

Underlying all of the manufacture and assembly of space hardware is sufficient information from designers in the form of part and assembly drawings, approved part and material documents (PAPLs and PAMPLs), testing procedures, and any relevant test fixtures. These data are the relevant engineering data necessary to move forward with the manufacture of components. For space programs subject to the development phases identified in Chap. 3, close coordination between design and manufacturing is needed to insure a smooth transition from component level design into actual production.

While many of the tables related to the design of individual subsystems in Chaps. 18 through 22 identify a final "iterate and document" step, design changes late into a component's or unit's development can significantly lengthen the overall production of flight hardware. As an example, if a digital processor unit requires a specialized chip such as an ASIC, late design changes can delay the release of the ASIC design to a foundry—where the production cycle time for a custom ASIC may span anywhere from 12 to 24 months. And this delay propagates into a delay of qualification testing of the overall digital processor unit, which, if it is the main spacecraft control processor, may then significantly delay integration of the spacecraft.

For components which are identified as long lead items such as ASICs and control moment gyroscopes, their production cycle times may end up driving the schedule (and thus overall cost) for a whole program. The potential impact of major long lead items will likely be managed as part of the whole program's risk management activity and possibly impact unit design and manufacturing to provide specific steps where risk reduction activities such as early design releases of engineering data may be scheduled and budgeted to check out and possibly test the compatibility of the design with its manufacturing and assembly processes.

At the component and subsystem levels, design verification tests (Table 23-4) can be performed to demonstrate functionality of components and verify operation of both components and subsystems. Structural tests such as static load testing apply a proof load to the flight structure and modal surveys verify the natural frequency, damping factors and mode shapes of the spacecraft. Deployment tests insure the proper deployment of appendages such as solar arrays and booms. Closed loop tests are used to quantify and verify performance of algorithms such as sun tracking on either solar array drive gimbaling or software state transitions due to changes detected such as low battery charge or loss of communication with a simulated ground station.

23.3.4 Spacecraft Integration and Test

Spacecraft bus and space vehicle-level integration and test (I&T) encompass the activities that occur between completion of assembly and testing of unit hardware and software and launch site processing of the spacecraft with the launch vehicle. Fundamentally, this is where the pieces of space hardware are put together and along with the developed flight software, tested as a whole system to make sure that, as an integrated whole, the spacecraft correctly responds to commands and will survive launch to operate as intended in space.

The essential elements that comprise I&T include: the HWCIs and SWCIs that are to provide the basic functions of the spacecraft, all necessary testing apparatus and facilities, relevant documentation (i.e. test & verification plans, approved integration and test procedures, handling documents, and automated test sequences), trained I&T staff and available/on-call support personnel as well as critical I&T management tools (i.e. I&T flow, schedule and budget, test floor and configuration control software). How any and all of these are applied on a given flight program depend directly on the class and type of space system being integrated, prior experience on similar or identical builds and the maturity of the HWCIs and SWCIs under integration.

I&T occupies an unique place in the development and fielding of a space system. Most often, it bears the brunt of delivery delays for both hardware and software and must execute test procedures during early integration

Table 23-4, Fig. 23-7◀, Eq. 23-0

with a combination of available EM and flight units that together may generate incorrect telemetry responses to valid spacecraft commands and which have to be tracked down and understood. Specialized test facilities such as acoustic chambers and thermal vacuum chambers may be reserved years in advance by a program as designated in the I&T Plan but conflicts can happen because of schedule slips and end up affecting multiple systems going through I&T. Problems and delays during I&T can be extremely expensive both in terms of time and cost—having a whole I&T crew wait for an essential unit to be reworked can negatively impact a system's schedule and cost.

I&T is also a unique phase in a space system's life cycle because it is usually the first time that all of the "pieces" of the flight part of a system come together to determine how well they operate together and their effectiveness as a total system. Opportunities for the integrated and tested spacecraft to interface with elements of the ground segment (Chaps. 28 and 29) may allow for compatibility testing and validation of many end-to-end system functions. Additionally, I&T is the time and place for execution of the test and verification plans developed during the early phases of a program as discussed in Sec. 23.4 and critically close the link with the system level requirements as developed in Chap. 6.

Planning for Integration and Test

Given the criticality of I&T to a space system, how and where a spacecraft is to be integrated and tested is usually extensively described when a system integrator submits a proposal to develop and deliver the spacecraft. Availability of relevant environmental testing facilities, readily tailorable test procedures, suitability of electronic hardware for system-level test sets may also be called EGSE (electronic ground support equipment)) and prior experience with integrating a given payload or payload suite are all included in the proposal submittal along with the I&T processing flow and schedule. An example of an I&T flow is shown in Fig. 23-8.

The first two steps in the flow define spacecraft bus-level I&T activities. This level of integration and testing is performed prior to integration of the payload. Bus integration normally starts with delivery of an assembled structure or structural frame where the propulsion subsystem may or may not already be integrated into the structure. Liquid propulsion components are assembled to the structure first because the fields used to braze them together are not compatible with electronic components. The spacecraft wiring harness can go on before or after the propulsion components.

Electronic assembly of the spacecraft commonly starts with electric power and command and data handling subsystem components. Integration of the remaining subsystems depends more on the particular spacecraft design. Although the subsystems may have been preassembled and functionally tested, it is common to integrate HWCI one at a time and test subsystems after they are fully integrated together on the spacecraft.

After bus-level integration and testing has completed, integration of the mission payload onto the bus defines vehicle-level integration of the space system. Interface compatibility testing may be performed prior to integrating the payload onto the bus or can be conducted with EM and/or payload simulator hardware and software.

I&T Execution

All operations to be performed on a spacecraft during I&T must be planned and executed under control and quality surveillance. These operation sequences and the data to be collected during each sequence are documented in I&T procedures. Records must be maintained —either or both in hardcopy and electronically—and the results of each integration and test step reviewed before proceeding to the next operation.

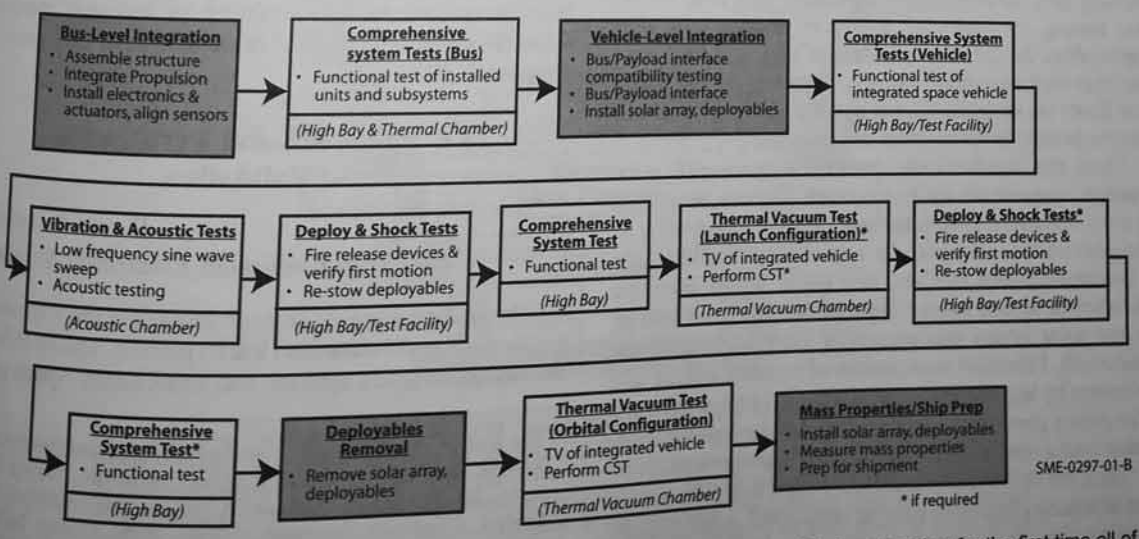


Fig. 23-8. Spacecraft Integration and Testing Flow. Integration and test of a spacecraft brings together for the first time all of the hardware and software required for an operational system and makes sure that everything works correctly as specified.

Table 23-4◀, Fig. 23-8, Eq. 23-0

To functionally test a spacecraft, an I&T crew normally uses a test set comprised of one or more computers with attached information displays for command generation, telemetry decoding and logging, and issuing automated test sequences. The information displays typically consist of an event screen that shows all signal activity and page displays devoted to particular subsystem parameters. The test set also provides stimuli and data-measurement equipment for end-to-end testing of the spacecraft and payload subsystems.

A complete spacecraft functional test may take several days to run and involve subroutines for each subsystem or functional group. Such a test is called an *integrated system test (IST)* or a *comprehensive system test (CST)*. Subroutines are used that test the specific functions of each subsystem, so that only affected subroutines during regression testing need to be run if a subsystem anomaly was detected and subsequently fixed. A reduced version of this functional testing suite may be defined and used at selected points in the integration flow.

During each CST, the test crew records all anomalies or out-of-tolerance measurements and formally resolves each discrepancy. Anomalies which result from operator error or malfunctioning test equipment and which do not damage the spacecraft are easy to resolve. But spacecraft malfunctions demand thorough investigation. If design errors have caused the problems, the design must be corrected and retested. Test rules normally require rerun of an integrated system test (or all affected subroutines) during regression testing or system configuration changes.

A CST timeline normally follows the same operational sequence and transitions involved in going from the launch event to on-orbit operations while also matching the applied test environment: vibration, shock, and thermal vacuum. Similarly, the spacecraft's physical configuration during system-level environmental testing flow is matched to the operational sequence with 1) stowed solar arrays and deployables during vibration testing, 2) performing first motion release/separation tests after vibration testing, 3) deploying and/or physically removing deployables during thermal vacuum testing. Often, test facilities such as thermal vacuum chambers impose physical limits on an articles' size and the ability to perform particular tests at the relevant environment.

First, the mechanically integrated spacecraft goes through temperature cycling to verify proper assembly. A comprehensive system test and performance testing of the payload then take place in a special test facility. (For a communications satellite, this might be an anechoic chamber or screen room.) Separate performance testing of the solar arrays precedes their integration with the spacecraft. *Vibration tests* consist of acoustic testing and vibration by low-frequency sine waves. The system must also pass a *pyroshock test firing* of the deployment ordnance-and a check of the mechanical deployments.

Once the deployables are re-stowed, a thermal vacuum test of this configuration may be conducted. After demonstrating the deployments again, the I&T crew then removes the deployables so the spacecraft will fit into the thermal vacuum (TV) chamber. The spacecraft is then thermal vacuum tested in its orbital configuration. The thermal vacuum cycle activity serves as the final gate to showing readiness for shipment to the launch site. A minimum number of error-free hours (200 is typical) is required of this final run. An ambient comprehensive system test and mechanical verification series may complete the qualification of the integrated space vehicle. *Integrated system tests* (shorter functional tests) supplement this sequence between each environmental exposure and during TV testing. This shortened version of a CST is often used after an environmental test to assure that no failures occurred during the screening activity.

I&T Drivers

As noted earlier, the class and type of space system being integrated, prior experience on similar or identical builds and the maturity of the HWCIs and SWCIs under integration all directly impact the execution of the I&T function on a space program. For Class A programs that require at least single fault tolerance, design life of over 5 years [DoD, 1986], the largest qualification margins are imposed along with additional requirements specific to documentation and at the lowest feasible risk levels. Imposing and managing to these requirements increases the cost and schedule for I&T.

Also, for multi-spacecraft builds as described in Sec. 23.5, the ability to directly influence the vehicle design and developing and verifying the I&T process can greatly reduce the per-unit cost. Use of lean manufacturing methods such as continuous process flow and reducing operation cycle times [Womack et al., 1990] as applied to the production of the Iridium constellation have shown their feasibility and efficacy for space systems (Sec. 23.5). Other approaches to larger volume production, such as modular space vehicle systems where groups of common modules such as spacecraft buses and sensor payloads are built in batches, stored and are subsequently integrated together for a rapid response capability may also foster the ability to leverage lessons learned/knowledge gained from repetition of I&T procedures with block builds.

23.4 Space Mission Verification and Validation

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This section presents an overview of the space mission verification and validation (V&V) process, and its role in the design of space systems and subsystems. Space pro-

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Table 23-4◀, Fig. 23-8◀, Eq. 23-0

gram execution and management of cost, schedule, and risk, are guided by definition and implementation of an affordable verification and validation program that incorporates available (or implementable) analysis, simulation, and test methods and facilities. This section gives a step-by-step approach to such a program, particularly as it applies to first article or single article space missions.

An effective V&V program must be an integral part of the up-front planning process and continue through deployment and certification of the on-orbit operational system. It requires a level of planning that spans all levels of the product hierarchy (see Sec. 14.4), demanding coordination among all suppliers and developers in the chain. Responsibility for this coordination lies with the prime integrator, who establishes and enforces general policy and flow of verification events, communicated through formal documentation such as the System Engineering Management Plan (SEMP), System Verification Plan (SVP), Software Development Plan (SDP), Qualification Test Plan (QTP), Acceptance Test Plan (ATP), Software Test Plan, System Safety Plan, EMC Control Plan, and Parts, Materials, and Processes (PM&P) Control Plan. [See Blanchard and Fabycky, 2006; Buede, 2010; EIA, 1994; INCOSE 2007; MITRE, 1998 and AIAA, 2010].

Before proceeding to more specific discussions of the space system verification and validation program, it is important to establish a clear distinction between these two types of activities. *Verification* is a formal process that assures a system or product will meet written requirements, in the specified mission environment. For verification to be considered complete all verification products must pass *sell-off* by the receiving agency. *Verification products* are frequently embodied as Requirement Verification Reports (RVRs) that show compliance to specific requirements in the system or product *compliance documents*: specifications, standards, interface document. Methods of verification, inspection, analysis, test, and similarity are described later in this section. *System validation* is the process for assuring that the system will provide needed services in the mission operational environment.

23.4.1 Space System Verification

Formal connection of verification products to the specification, design, and build of any system can be generally represented by the filled V model as shown in Fig. 23-9. Specific implementation of the scheme is governed by the *System Engineering Management Plan (SEMP)*, which defines tools and products used to generate and capture requirements, design disclosure, error budgets and allocation documents, models, hand-offs to and from manufacturing and Integration and Test (I&T), major simulators, legacy resources, and verification products. It also stipulates rules and protocols to be observed in linking products of the various teams responsible for generating and maintaining them. The SEMP spawns other plans (listed above), including the System Verification Plan (SVP). Initial release is required at System Requirements Review (SRR) and maintained through Test Readiness Review (TRR).

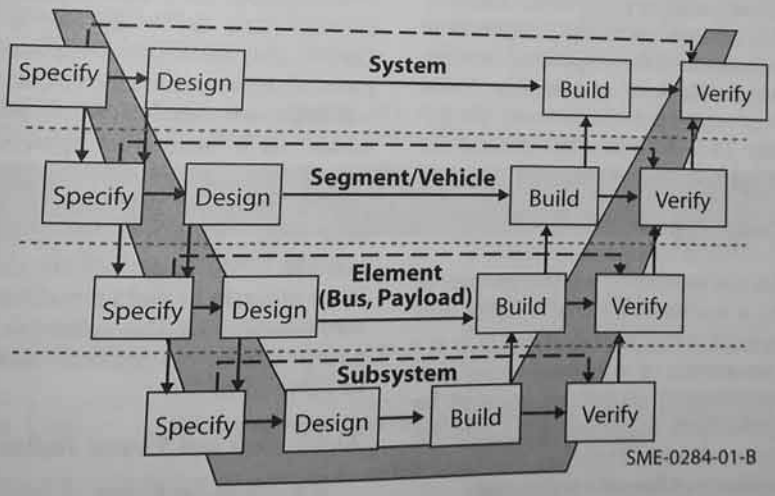


Fig. 23-9. The Filled V Model of System Development. Applicable to a wide variety of systems, the Filled V Model provides a template for architecting the relationships (*schema*) between development products generated over the engineering and manufacturing development phase of a program. It shows how verification depends on the requirements and builds of a specific level of assembly, as well as the relation with verification products above and below in the hierarchy. Requirement trace and decomposition (a design function) are covered in Chapter 6. Note that the hierarchy extends down to lower levels than shown here: component/unit; subassembly; piece-part; wafer; foundry. All must be addressed to achieve a complete space system.

Table 23-4◀, Fig. 23-9, Eq. 23-0

Table 23-5. Steps in Space System Verification Planning.

Step	Comments and Required Information	Ref
1. Capture Requirements in Hierarchical Set of Specifications.	• System spec and SEMP released at SRR • Standards and external interfaces defined • First layer of hierarchy captured in requirement database, schema baselined down to subsystem level	Sec. 6, Chaps. 14, 27, 28, 29
2. Create Verification Cross Reference Matrix (VCRM).	• VCRM preliminary at SRR • System spec VCRM baselined at SDR	Sec. 23.4.2, Table 23-6
3. Map Verification Methods to Available Assets—Assess for Effectiveness, Affordability, and Schedule.	• Draft Verification Plan at SRR • Analysis assets: e.g., heritage simulators and models, program-developed models • Test assets: e.g., thermal vacuum chamber, vibration table, fixtures, acoustic chamber, wind tunnel, optical bench, EMI test facilities, RF ranges	Secs. 20.2.4, 23.3, 23.4.4, 23.4.5
4. Identify Gaps and Update: *- Requirements baseline* *- Verification methods (VCRM)*	• At PDR, preliminary Verification Plan down to unit level, baselined at system level • Trade technical, cost, schedule constraints • Preliminary Integration & Test Plan at PDR	Secs. 11.2, 23.4.3, 23.4.5, 23.5.4, 24.17
5. Plan and Track Closure of Verification Events. Periodically Re-evaluate for Effectiveness in Verifying Requirement Compliance. Hold Test Readiness Reviews (TRRs).	• Verification Plan and I&T Plan baselined down to the unit level well before CDR • Link verification records to requirement database • Test Requirements Document (TRD) with quantitative pass/fail criteria, including measurement error allocation	Secs. 6.3, 23.4.4 Fig. 23-9
6. Conduct Physical and Functional Configuration Audits (PCA and FCA): Unit/Component, Subsystem, Element, Segment. System Sell-off.	• PCA establishes qualified drawings, part list, assembly lines, or suppliers • FCA assures verification of all requirements, archives approved deviations/waivers • Roll up for system sell-off to customer	Sec. 20.1; [MITRE, 1998] [DoD, 1985]

Verification planning is an activity performed at every level of the hierarchy across hardware and software products, and their integrated configurations (See Fig. 20-10 in Sec. 20.2.4). It is important that the *System Verification Plan* (SVP) stipulate a roadmap that takes best advantage of program assets and team capabilities, without over-constraining verification planning at the lower levels of the hierarchy. Another strategy increasingly important in cost constrained environments is that of pushing verification to the lowest practicable level of assembly. Initial release of the SVP is required at *Preliminary Design Review* (PDR). Table 23-5 describes the process for developing a space system verification plan.

23.4.2 Verification Methods

Formal verification is a process by which a product or service (or, *end item*) is evaluated for conformance with written requirements in its compliance documents. It is a rigorous, objective assessment of end item capability to meet each unique requirement, sometimes assessed for groups of related requirements, documented in a *Requirement Verification Report* (RVR). There are at least four widely recognized methods of formal verification:

Inspection: visual inspection of *end item* (EI) and/or its drawings, or possibly an associated *end item data package* (EIDP).

Analysis: definitive qualitative or quantitative pass/fail assessment of a validated mathematical model of the end item, usually at both BOL and EOL.

Test: a regimen exercising the end item, or a validated physical representation thereof, in such a way as to collect data used to directly determine pass or fail.

Similarity: comparison of features of the end item with those of another end item confirmed to be similar, and meeting equal or more stressing requirements.

In addition, some users separately add a special category of test, *demonstration*: a regimen exercising features and functions of the end item, or a validated physical representation thereof, successful completion of which implies compliance for a group of related requirements. Each method is appropriate to the type of requirement being verified, subject to cost constraints and the program's risk posture. Greater certainty in the verification method usually translates into greater expense, hence the desirability of defining methods early in the program to avoid misunderstandings or cost overruns downstream. This definition is typically captured in a *verification cross reference matrix* (VCRM), as shown in Table 23-6.

23.4.3 Real and Virtual Builds

Implicit to the notion of verification is a specific *end item* that is being evaluated for compliance. Since many requirements relate to performance or robustness over mission life, it is not always practical to directly perform an evaluation of the as-built end item prior to delivery and deployment. It is therefore necessary to evaluate a surrogate judged to be similar to the end item, or at least to possess similar characteristics in the area of interest

Table 23-5, Fig. 23-9◀, Eq. 23-0

Table 23-6. Sample Entries for a Verification Cross Reference Matrix (VCRM). There is a unique VCRM for each specification or interface, usually appended at the end of the document.

Unique ID	Para. Number	Text	Analysis	Similarity	Inspect	Test
FIRE1370	3.2.1.8	Space vehicle pointing knowledge shall be ±75 μrad, 1-σ over mission life.	X			
FIRE2040	3.2.2.4	Dimensions of the space vehicle shall not violate launch adapter envelope specified in FIRESAT-ICD-7100.			X	
FIRE2890	3.3.1.5	Materials used to integrate subsystems shall be on the Program Approved Material and Part List.		X		
FIRE3020	3.3.2.2	Radiated emissions shall not exceed 20 dB uV/m between 1MHz and 200MHz.				X

for this particular evaluation. Thus is introduced the generalized concept of build as represented in the Filled V Model (Fig. 23-9).

For the purposes of verification, there are two categories of build: real and so-called "virtual" builds. In the category of *real builds* naturally falls not only the *end item deliverable* (EID) itself, but also qualification units (items built from the same production paper and tooling), mock-ups, prototypes, and others. For software items, it includes code running on emulators embedded in test beds, high level programming language compilations on alternate processors. In most cases, the verification depends on a similarity argument. The greater degree of departure of any real build from the end item, the more uncertainty exists in the verification activity using it.

All type testing, particularly qualification, rests on this concept (Table 23-2). If a type test article passes a sequence of qualification tests, all other articles built to the same engineering data (see Sec. 23.3) would also be expected to pass. In other words, the design is qualified; succeeding builds of this design need only verify proper workmanship [Reeves, 1999]. Alterations to any features of a qualified design must themselves be qualified, since the similarity assessment has been invalidated—a process known as delta qualification. Any of the following changes can require a delta qualification, unless a compelling similarity argument can be made: part or material substitution, change in fabrication tooling or facility, new or revised requirement, change in operating environment or duration, and others.

Virtual build is a more abstract category, including source code listings, logical and arithmetic descriptions (state machines, formal statements, combinatorial expressions, and diagrams the like), closed form mathematical descriptions, and computer models. In this case the build possesses characteristics, similar to the end item, that are specific to a verification type. Table 23-7 describes the application of virtual builds to the four verification methods. (Note that all ground based testing involves a certain amount of simulation and simplification, including environments, interfaces, time spans, and even operational states. The art and science of this discipline will be addressed later in this section.)

Table 23-7. Virtual Builds in End Item Verification. This illustrates several types of virtual builds and their role in different verification activities for end items at various levels of the space system hierarchy.

End Item	Verification Method	Virtual Build	Application
Attitude Control Subsystem	Analysis	MATLAB model	Statistical assessment of performance across parameter variations in tolerance, temperature, radiation, and life
Ground Terminal Software	Similarity	High level programming language code	Variable type compatibility confirmation, based on heritage code
Spacecraft Fault Management	Test	Hardware-in-the-loop test bed	Exercise defined fault scenarios to validate response of chains
Motor Drive Electronics	Inspection	Schematic	Assure proper grounding, isolation, filtering, and shielding as specified in ICD
Motor Drive Electronics	Analysis	SPICE model (net list derived from schematic above)	Statistical assessment of performance and derating across parameter variations in tolerance, temperature, radiation, and life
Thermal Control Subsystem	Analysis	SINDA model	Verify energy balance and temperature limits over life and environments

Table 23-8. Sample of Modeling Methods Commonly Used for Space Mission Verification.

Model Type	Example / tool	Typical Validation Event / Data	Correlation Parameter(s)
Circuit Analysis Model (closed form)	Steady-state analysis	Breadboard	Voltage, current
	Small-signal analysis	Breadboard	Gain vs. frequency
Circuit Analysis Simulator (non-real time)	SPICE	Breadboard	Voltage, current, gain vs. frequency
	VHDL	Brassboard	Timing, logic
Mechanical Stress Analysis	NASTRAN™	Loads test	Stress and strain
Control System Block Diagram	Typically s- or z-domain	Hardware/Software-in-the-loop (HIL/SIL) test bed; mass simulator	Natural frequency, damping ratio, gain, closed loop error
Control System Simulator	MatLab™		
Thermal Model	SINDA™	Thermal balance test	Temperature
Reliability Block Diagram	Lambda Predict™	MIL-HDBK-217 database; life test	Time to failure; accelerating factors
Mass Properties Model	dBMass™	Mass balance test; spin balance test; weighing	Bulk mass, center of gravity, moment of inertia
Dynamic Simulator (non-real-time)	I-DEAS Vibro-Acoustics™; SYSNOISE™	Modal survey; Sinusoidal vibration	Modes, q-factor
Spacecraft Simulator	Custom	Test bed	Demonstration
Link Budget	Tables, graphs, spreadsheet	RF test range; Pointing and alignment test bed	Antenna gain vs frequency, alignment error, temperature, lobes
Error Budget (various)	Monte Carlo Simulator; spreadsheet	Optical bench test; guide-to-hit test; channel characterization	Alignment angle, radiance, in-band noise
Orbit Simulation	STK™	Orbital data from other spacecraft	Altitude vs. time, period, ground track
Phenomenological Model	Custom	Ground truth testing; celestial observation	Radiance, humidity, background noise, absorption

23.4.4 Role of Modeling and Simulation

Model Selection and Management

Modeling is the art and science of creating virtual builds (or, models) of systems of varying complexity and levels of integration. Proper selection and use of models during program development can do much to control cost and risk. In certain instances there is advantage to creating models during concept development for one-time use to support *rapid prototyping*. This can also be effective in other time-critical situations such as trade study and anomaly resolution efforts. But the more frequent application of models is in completing virtual builds in support of design validation and verification of requirements that must be met over the life of the program. In this sense, the methods are predictive, since it is usually impractical to directly confirm compliance with such requirements prior to committing flight assets to service. This is a challenge somewhat peculiar to space missions, demanding extensive ground test programs representing a significant proportion of life cycle cost.

Many requirement categories are common to space mission operations, hardware, and software, and are supported by modeling methods and tools standard in the industry. A few examples are given in Table 23-8. Standardized tools affordably provide a framework within which specific models can be built with manageable risk.

Table 23-8, Fig. 23-9◀, Eq. 23-0

These modeling methods have been shown to provide fairly accurate predictions of system response to operational environments and operating conditions, when used in the intended application. As design matures, details are added to reflect the product specifics (e.g., corrections discovered during test). It is important to track versions of the model and the modeling tool used to support various program milestones such as design reviews and sell-off activities.

This discipline is no less important for management of custom models developed by a specific program or organization. Modern space mission programs often manage evolution of model fidelity using a documented *Model Management Plan*.

Models fall into one of several general categories, or combinations thereof:

- ***Physical***—hardware realization of end item configuration to the fidelity necessary to achieve objectives at a particular point in system development
- ***Functional***—rudimentary emulation/simulation responding to commands, time-outs, changes in operating state, transmission mode; may represent a range of degrees of fidelity
- ***Descriptive***—mathematical representation of EID physical features and its external stimuli

- ***Behavioral***—mathematical representation of relationship between EID inputs and outputs, and its external stimuli, usually much less complex than a descriptive model
- ***Empirical***—involves the use of curve fits, look-up tables, and other data-based media to predict excitations and responses (e.g., response of standard structural elements and configurations to shock produced by various pyrotechnic sources along with measured attenuation vs. distance from the source)
- ***Statistical***—statistical representation of EID responses to inputs and external stimuli; also applies to statistical representation of range of values for parameters in descriptive, empirical, and behavioral models (stochastic)
- ***Closed Form***—expressed in a set of equations, state machine description, table, or graph
- ***Finite Methods***—represented by a large number of simple finite entities representing a description of the EID, most commonly a *finite element model* (FEM) grid

More and more, complex systems are being architected through descriptive behavioral models which graphically define, specify, construct, partition and document the functional and physical architectures and designs of systems/software under development. The *Universal Modeling Language* (UML) and its *System Modeling Language* (SysML) extension are the commonly accepted standards to render these models and related system views, such as the *DoD Architectural Frameworks* (DoDAF). These tools are used during V&V to hierarchically capture test cases and standard mission scenarios [INCOSE, 2007].

Model Validation

Significant effort is typically expended on a space program to confirm validity of predictions made by models. *Model validation* (also called *model correlation*) is a formal process of demonstrating the ability of a virtual build to respond to stimuli in a way that imitates what the EID would do, to a quantifiable degree of error. It is important for analyses conducted throughout a space program's life to account for error introduced by the modeling process. This is usually embodied in a term, or set of terms, referred to variously as *model uncertainty factor* (MUF), uncertainty margin, or contingency, depending on the discipline involved. See Akaike [1983]; Blair, et al. [1994], Draper [1995]; Food, et al. [2002]; Kaplan and Bremmer [2010]; Madigan and Raftery [1991]; Raftery [1992]; and Yunis [2008].

As development matures from initial sketches and concepts, through preliminary and detailed design, prototype, integration, and flight build, an explicit objective of the program is to reduce these uncertainty terms in the various budgets maintained on the program. Target values are established in advance, indicating readiness for major program milestones, such as design reviews. These are achieved through a series of dedicated model build cycles and tests of the type given in Table 23-8. A generalized process for evolving model maturity and reducing uncertainty over a space program life cycle is given in Table 23-9. Ultimately, the model and the end item must respond to test case conditions, sequencing, and stimuli with sufficient similarity as to justify use of the model to predict mission performance. These predictions form the basis for many verification products.

Simulators and Emulators

A word should be said about the types of facilities and other resources used for system simulation. The majority of space missions do not include exo-atmospheric human activity (i.e., "unmanned"). Flight equipment prototypes of varying fidelity are integrated and checked out using dedicated *test beds*. In modern systems, extensive software and firmware content obliges designers, systems engineers, and test planners to conduct *software-in-the-loop* (SIL) testing. At the least, this involves exercising interfaces and sequences from pre-defined operational scenarios. SIL and SIL-like tests start early in the development cycle, at low levels of assembly, through the use of breadboards and flight-like prototypes. Embedded software and firmware unit testing is conducted through the use of emulators. An *emulator* is a device that executes instructions in the same way as the flight processor and its core peripherals, usually at a much reduced clock speed.

For systems involving moving assemblies, it will further involve *simulators*, also of varying degrees of fidelity. Development testing uses real-time software simulations connected through test drivers to circuits designed to replicate the hardware interface to the subsystem. Dynamic subsystems dedicated to thrust vector control, spacecraft attitude control, and attitude determination, often extend the simulation to include physical loads, actuators, and sensors, to accomplish highly representative *hardware-in-the-loop* (HIL) testing for purposes of system verification and model validation. Sensor subsystems often employ simulated mission subjects during development and integration, such as point sources, flood sources, black body radiators, and other types of apparatus. Communication systems employ RF ranges, RF signal "hats," and data simulators, to name a few methods.

Today's space systems are designed and deployed in the context of existing surface-based and space-based systems. Interactions within the subject mission and with these existing systems should be exercised early and often, utilizing prototypes and test beds to proof interfaces and operations. As end item deliverables become available, they are substituted for the initial and intermediate prototypes (usually development or articles that have already gone through unit or subsystem qualification). These are also used during crew training and *man-machine interface* validation for ground and operation stations, extending the test regimen to include *man-in-the-loop* methods. Finally, where exo-atmospheric human operations are to be a part of the mission, dedi-

Table 23-9. Steps in Space System Model Generation and Validation.

Step	Comments and Required Information	References
1. Identify Operating Characteristics of System to be Modeled: Performance, Environments, Duration.	Life and environments usually fixed by mission profile defined in CONOPS. Performance parameters are driven by specifications and technical budgets, with an eye to maximizing parameter count per model	Secs. 4.1, 6.2, 29
2. Select Modeling Approach Addressing Selected Characteristics.	Includes physical modeling of system and environments. Simulation limits addressable parameters, but offers flexibility in range and number of test cases, lower material cost and delay, plus ease of upgrade and maintenance. Method selection critiera: sufficiency for defined parameter space, cost, maintenance, reuse, test case accommodation	Secs. 5.4, 23.4.5
3. Define Mission Cases to Meet Modeling Objectives.	Mission cases are typically chosen to represent stressing and nominal conditions from system CONOPS or specification	Secs. 5.5, 29, 30
4. Build Model and Determine Uncertainty Terms.	Tailor model complexity to address parameters of interest within constraints of program resources. Use technical budgets and standards to define uncertainty terms	Secs. 6.3, 11.3
5. Devise Test Cases for Correlating Model to End Item.	Test cases are chosen to optimize correlation of model parameters to those of end item. Approximate mission cases as closely as practical within resource constraints	Chap. 22, Table 23-9
6. Subject Model and End Item (or surrogate) to Test Case Environments and Excitations.	Surrogates (or, prototypes) are commonly used in early to middle phases of system development, when models are often immature. Model objectives are among entry criteria for test case runs	Chap. 22, Table 23-9
7. Correct Model and Reduce Uncertainty Terms.	Divergent responses are reconciled through updates to model. Resultant uncertainty terms are updated using a blend of statistical methods and industry standards	Secs. 6.3, 23.4.4
8. Iterate Until Uncertainty Terms Satisfy Budget Objectives.	Objectives are established by program phase in the SEMP, again based on industry standards and technical budgets	External

cated *vehicle simulators* are used in extensive operational simulations in collaboration with the other system elements and external systems.

23.4.5 Introduction to Space System Testing

23.4.5.1 Environmental Testing and Defect Screening

It is rarely practical to exercise an end item or test article in the full mission environment and duration before committing it to space deployment and fielded service. (One counter-example is the Pioneer Jupiter series, which encountered Jupiter in 1973, path-finding the Solar System grand tour nearly six years before the 1979 launches of Voyagers 1 and 2. Lessons learned were used to modify equipment on the subsequent spacecraft, especially for radiation hardness.) More commonly, proving suitability for deployment and use in the space environment requires subjection of the flight article to ground test conditions that represent stresses of transportation, launch, and space flight. Table 23-10 describes a number of test regimens used to qualify and accept space flight equipment. Practice has shown that these methods can be used to promote timely qualification through accelerated life testing.

Accelerated life testing, as used here, is a quantitative approach to predicting life of a fielded space asset, by subjecting that article and similar ones to stresses exceeding those expected to be experienced during the mission. Traditional methods of reliability testing, qualification, and acceptance, have seen innovations such as test-to-failure and *highly accelerated life testing* (HALT) enter as viable alternatives or supplements [Meeker and Hahn, 1985]. Suitability of these latter methods depends on inherent product robustness, technology maturity, and other considerations [Weibull, 2007].

In addition to demonstrating hardware and software robustness, a primary objective of space hardware environmental testing is to demonstrate ability to perform as required while operated on orbit, after experiencing the rigors of ground processing and transportation, launch, atmospheric flight, and orbital maneuvering and deployment (see Sec. 23.3). It is therefore desirable to subject units, subsystems, payloads, and vehicles to environmental levels, durations, and sequences similar to those expected for the mission. The intuitive construct of this approach is to *Test Like You Fly* (TLYF) [White and Wright, 2005] built on methods usually associated with system validation. Ground testing results are used to characterize, and confirm predictions for, such parameters as post-launch alignment and settling for line of sight (LOS), integrity of moving mechanical assemblies, RF and optical transmission, and thermal control and monitoring. While predictions and measurements are used for final vehicle verification and sell-off, characterization provides data to be used during on-orbit operations. For example, parameter variation under well-chosen temperature conditions will be used to calibrate data for each specific build of the vehicle and payloads during mission life. See the book website for a more extensive discussion of both environmental testing and qualification and acceptance testing, for space and ground systems.

Get More

Table 23-10. Environmental Tests Used for Spaceflight Hardware and Equipment.

Test Name	Description	Assembly Level
Radiation Total Dose	Test characterizes changes in device performance after exposure to total ionizing radiation dose predicted for mission, typically at highly accelerated rate and at elevated temperature	Electronic piece-part
Enhanced Low Dose Rate Sensitivity (ELDRS)	Test characterizes changes in device performance after exposure to total ionizing radiation dose rates more closely approximating that predicted for mission	Electronic piece-part
Radiation Prompt Dose & SEE	Radiation stresses simulating single event phenomena: high-energy protons, neutron bombardment	Electronic piece-part
Electrical Stress Test	Application of electrical stresses above normal use or rating, usually high voltage (Variants: hi-pot test, di-electric withstanding)	Electronic piece-part Cable, printed circuit
Static Load Test	Application of static forces to a UUT, enveloping static and quasi-static acceleration loads predicted to occur in flight	Sub-assembly Subsystem Element
Burn-In	Sustained high temperature operation of UUT over pre-defined time span, watching for failures or unfavorable parametric trends	Electronic piece-part Hybrid micro-circuit
Thermal Shock	UUT exposed to abrupt temperature transitions between extremes	Electronic piece-part Hybrid micro-circuit
Seal Leak	Fill with pressurized gas and immerse in liquid to reveal leaks	Hybrid micro-circuit Unit
Salt Spray	Spray mist of 5% salt solution on UUT in a chamber	Unit
Humidity	UUT exposed to 90 to 95% relative humidity at elevated temperature	Electronic piece-part Unit
Thermal Cycle	Subjects UUT to rapid temperature transitions between defined extremes under normal atmospheric conditions, dwelling at extremes for hours at a time	Unit Element
Thermal Vacuum	Subjects UUT to rapid temperature transitions between defined extremes under near-vacuum conditions, dwelling at extremes and other plateaus for hours at a time. Explicitly simulates on-orbit conditions and operations using conductive and radiative coupling	Unit Element
Random Vibration	Simulates pseudo-random vibration spectrum predicted for launch and atmospheric ascent	Unit Element
Sine Sweep Vibration	Excitation of a UUT structure involving the application of fixed sine wave forces with brief dwell time, stepping through the applicable frequency range to identify resonances and other responses	Unit Element
Acoustic	Subjects UUT to pseudo-random acoustic loads representative of those seen during launch and early ascent	Unit Element
Pyro-Shock	Exposure to transients simulating those resulting from sudden application or release of loads associated with deployment, staging/separation, impact, and release events	Piece-part Unit Subsystem Element
Electromagnetic Interference (EMI)	Encompassing one or more regimens involving: measurement of conducted and radiated emissions from operational UUT; and/or subjection of operational UUT to conducted and radiated noise to assess ability to operate in such environments	Unit Element
Electromagnetic Compatibility (EMC)	Monitoring of critical circuit EMI safety margin (EMISM) during simulated runs in operational configuration	Subsystem Element Segment

23.4.6 Space System Validation and Final Certification

System validation is the process for assuring that the system will provide needed services in the mission operational environment. System validation involves the exercise of simulators, emulators, and development hardware and software, using mission operators, infrastructure, and procedures, through progressive levels of integration and fidelity as the program matures. Activities focus on exposing system elements and subsystems to interfaces (including users and operators), data interchange, available heritage system elements, and event sequences representative of the operational mission and environment. Detailed mission scenarios (or, threads) from system *concept of operations* (CONOPS) are enacted to capture incompatibilities that may have escaped the requirement derivation process. System validation is distinct from both system verification and from model validation described earlier in this section. It is also distinct from the process of

requirement validation, shown in Chap. 6, though that process is often incorporated as an early part of the system validation plan. Validation is the final line of defense against previously undiscovered mission hazards related to sneak paths, software bugs, and missed hand-offs.

A commonly used method for planning space system validation testing follows a simple mantra: *First Day, Best Day, Worst Day, Last Day*. Each scenario represents a standard mission operational thread or set of threads, with actual commands and messaging, operator displays, communications links and relays, preferably at projected durations. First Day threads emulate launch day operations, and post-launch deployment and initialization for the spacecraft. This is almost always the most hazardous part of the mission, with success hinging on real-time decisions by ground crews, as well as flight and ground processors. It is essential that fault response scenarios be enacted multiple times in the months leading up to this mission phase (also falls under category of Worst Day).

Best Day scenarios, more often referred to as *day-in-the-life* (or, DITL, pronounced like fiddle), represent the operations for which the mission was desired in the first place. For many missions, this is simply continuous operation, with occasional interruptions for calibration or maintenance activities, such as orbital station keeping. More complex missions involve intricate autonomous or operator-directed orbital rendezvous and docking, real-time tasking, planetary encounter, surface roving, sampling and other *in situ* measurement. In such cases, it is crucial that potentially debilitating phenomena are accounted for, such as celestial body eclipse, communication lag, Doppler effects, solar activity disruptions, and other inherent aspects of the mission. A subset of these scenarios is typically run with on-orbit assets after launch, as a part of final *system certification* for operational use.

Equally critical is the exercise of anomalous scenarios (Worst Day). Here the system equipment and operators conduct normal operations which are interrupted by the occurrence of a fault. This fault can be a hardware failure in space or on the ground, *single event effect* (SEE), or even natural or man-made catastrophe. The aggregate space system must demonstrate ability to respond to and correct the fault, and return to normal (or pre-defined degraded) operations.

Finally, all space missions are required to provide the means for retiring space and ground assets, as part of mission termination (Chap. 30). For space assets in particular, this is critical to preserve availability of orbits and orbital stations that are free of debris or other hazards. Interplanetary missions are required to assure elimination of materials that could harm as-yet-undetected life forms. Exercise of realistic operational scenarios must demonstrate ability to de-orbit or otherwise safe the asset in question, even in the presence of faults. In some cases, mission planners desire the capability to go into dormant or follow-on mission bridging states, and this also will need validation.

Table 23-10◀, Fig. 23-9◀, Eq. 23-0

23.5 Multi-Spacecraft Manufacturing

Wade Molnau, *General Dynamics*
Jean Oliveri, *SpaceX*

Historically, satellite manufacturing, integration, and test has been a crafted and arduous process. Each spacecraft is essentially unique, and is manufactured and tested appropriately. Commercial satellites, while alleviating some of the major impedances to fast and efficient satellite manufacture, have fared only slightly better. The advent of commercial constellations of satellites forces us to seek and develop completely new strategies. We need to incorporate ideas and methods from other industries into satellite supply-chains to meet the cost and cycle-time requirements needed to make space systems compete effectively with their terrestrial counterparts.

This chapter describes a few of the vital changes that need to be addressed to manufacture and test multiple satellites efficiently. These methods and strategies apply to the whole satellite supply chain, and to piece-part, assembly, subsystem, and spacecraft levels. Section 23.3, which details methods used to manufacture and test single satellites, will be augmented here for multiple satellite systems.

Small satellite systems (< 10 spacecraft) may not fully benefit from the methods presented here, but some points will be applicable. Designing manufacturable satellites and associated production systems requires up-front investments in time, money, and capital. Each program needs to trade the benefits of these methods with anticipated investment costs. As the number of satellites grows, the benefits and the usefulness of these methods increases.

We begin with a short description of the goals and challenges of manufacturing, material, test, and launch processing organizations. The majority of the chapter then concentrates on four phases of the manufacturing and test of the spacecraft constellation: (1) creating the manufacturing vision, (2) influencing the design, (3) developing and verifying the process, and (4) producing the spacecraft.

23.5.1 Challenges for Manufacturing, Material, Test, and Launch Processing Teams

The challenges presented to the manufacture of satellite constellations are extremely different from those traditionally conceived for spacecraft. These challenges, however, are not much different from those for other commercial products. Spacecraft for large constellations are designed and built to cost and schedule goals—just like cell phones or cars. Table 23-11 compares a few of the challenges facing the manufacture of a constellation of satellites, while Table 23-12 shows manufacturing issues as they pertain to key metrics—quality, time, and cost.

For instance, time to market often is not only a goal, but a competitive requirement for viability. Time becomes an overriding factor. This drives not only the overall time to market, but the time to complete each system element.

Table 23-11. Typical Constellation Manufacturing Challenges. Note the significant difference between traditional space vehicle manufacture and the newer manufacture process.

Key Metrics	Traditional	Multi-Satellite
Quality	Build, Test, Rebuild, Retest	Build, Verify, Ship, Shoot
Spacecraft Cost	66,000 $/kg	19,800 $/kg
Cycle Time (Integration)	225 days	24 days
Cycle Time (Build to launch)	18 months	2 months

Table 23-12. Constellation Manufacturing Issues. The quality, cycle time, and cost issue relating to the production of a constellation of satellites are different than those encountered while producing single satellites. While quality requirements are different—but still stringent—cycle time and cost issues are now paramount.

- **Six-Sigma Quality is Required**
  - Satellites are not field serviceable!
- **Time is Critical**
  - For Commercial Ventures → First to Market
  - For Government/Military → Leverage the Market
- **Money is a Limited Resource**
  - Fixed Budgets, Commercial Pricing, Continuous Profit Improvement

For manufacturing, this translates to drastically reducing the cycle time required to produce a satellite. We employ lean methodologies to minimize time and cost inefficiencies in the design, build and deployment of our systems.

Quality, as with all spacecraft, is still imperative. Satellites are not field serviceable, and simple defects can render a satellite unusable. The traditional approach places emphasis on high-cost and time-consuming quality assurance methods that check and recheck, test and retest, and verify and re-verify the hardware. Instead, we employ *six sigma* methods to verify that the processes are performing with high-quality results. Robust designs using in-control processes will produce high-quality products—without associated high costs and long cycle times. Six-sigma methods are a cornerstone for lean manufacturing methods (see sidebar).

To meet the additional cycle-time and quality requirements, manufacturing personnel must participate concurrently during the creation of the design. They are responsible for influencing the design and tailoring the design into a final, producible product. Finally, manufacturing must effectively integrate the product during production.

In summary, the challenges presented are to recreate the value-added or useful elements of space hardware design and manufacture while eliminating the non-value-added activities. We must add elements for quick, efficient, and high quality execution. We must drive out non-value-added or superfluous activities and requirements.

Material Challenges

Traditional manufacturing drove spacecraft parts to the costliest and slowest-delivery grade—*S-level*. Unfortunately, these parts are not inherently better than commercial parts—they have just been screened for various properties to improve their overall expected performance or failure rates. The parts are mostly produced on commercial part lines and even from commercial part batches.

Our challenge is to use the *right* parts for the mission—not the highest level parts available. We need to target commercial parts, replacing them only when required. We need to discourage military grade or S-level parts, but allow them in situations where they *must* be used. A simple comparison of part costs, as shown in Fig. 23-10, depicts the drastic difference for S-level, Mil-spec, and commercial parts. Significant cost savings can result from selecting lower-cost parts when applicable. Significant time savings also can result from selecting lower grade parts. Figure 23-10 shows typical lead times for various parts and the significant advantage of using Mil-spec parts over S-level parts. Furthering this comparison, we can receive commercial parts typically even quicker—often in less than a week.

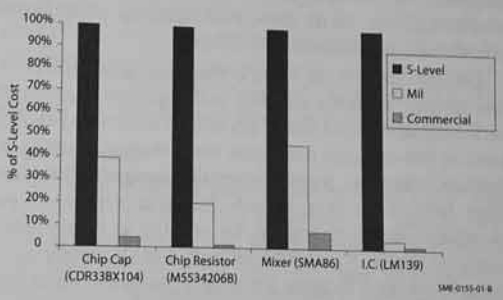


Fig. 23-10. Representative Parts Price Comparison. Commercial parts represent a significant cost savings over Mil-Spec and S-level parts.

Parts selection needs to consider producibility, radiation, out-gassing, and other requirements. We need to strongly consider parts that support efficient manufacturing methods. We should target parts that use the standard process flows—for instance, automated placement and mass reflow for electronic assemblies.

Finally, parts selection needs to be a concurrent engineering activity. All stakeholders—purchasing, production, design engineering—need to have their concerns taken care of. Success in the material area heavily influences overall program cost and schedule performance.

Test Challenges

The test function drives the overall cycle time of spacecraft delivery. This holds true not only for traditional satellites, but for constellations as well. The key is to design test as a process—this includes minimizing

Six Sigma Manufacturing

Six Sigma began at Motorola in 1985 as a means to measure and improve performance in all phases of its business. The ultimate target is virtually perfect execution.

Sigma (σ) is used to designate the distribution or spread about the mean or average of any process or procedure. A simple definition of the *measure* six sigma is 3.4 defects per million opportunities (DPMO), or 99.9997% perfect. Sigma level improvement is not linear. In fact, improvement rates escalate rapidly as the base sigma level slowly increases.

3 Sigma = 66,810 DPMO = 54,000 postal letters lost per day
6 Sigma = 3.4 DPMO = 35 postal letters lost in a year

Six Sigma *methodologies* are a set of tools and procedures to improve processes. Fundamental methods include problem solving, process control and process characterization. They can be applied to any process from engineering to business. Six Sigma *programs* are structures within companies for managing and maintaining continuous improvement. [Harry, 1997 or Harry et al. 2010]

Lean Six Sigma (LSS) began to become popular in 2003 as the marriage of lean and six sigma. What could be better than using lean to drive out waste and non-value producers and six sigma for optimizing value-added processes? [George, 2002]

cycle time, maximizing throughput, and achieving high-quality results.

The purpose of the overall test program is to verify the adequacy of the design and assembly processes. The requirements to verify these two elements—design and processes—are inherently different.

Design verification, more commonly called *qualification testing*, requires detailed and thorough analysis of performance capabilities. This still is a time-consuming process that includes functional test, vibration and shock, thermal vacuum, electromagnetic compatibility, and other tests. This may include multiple iterations, and essentially involves testing the satellite to greater than expected levels. We should cut back and ultimately eliminate these qualification tests after we understand and proof the design. We can achieve significant savings in time and money by relinquishing the need for full-up testing on every unit.

Conversely, *process verification* continues throughout the production of the spacecraft. These tests only verify the performance of the production system and include only simple performance checks. The control of the individual processes throughout the production cycle ensures the high-quality results. Complete testing of the spacecraft would be redundant and non-value added—and not consistent with the principles of lean production. Figure 23-11 shows some key differences between traditional spacecraft production and lean production.

In regards to components, subassemblies, and subsystems, it is critical that they achieve their qualification testing and product testing as early as possible. We need to detect quickly issues with design or process. We can integrate high quality and well-understood components, subassemblies, and subsystems with no additional integration level testing—with high confidence. At integration, we need to verify only incremental processing—usually only shipping and connection.

Table 23-12◀, Fig. 23-11, Eq. 23-0

In summary, test provides possibly the greatest opportunity for significant cost and cycle-time improvements. Intelligent selection of reduced testing, clear distinction between design and process verification, and performance of lowest-level testing greatly enhances the cost-effective, high quality, low cycle-time production of spacecraft without significantly increasing risk.

Launch Processing Challenges

After finishing integration and test, we package and ship the satellites to the launch site for processing and preparation for launch. Traditionally, this has been a complex and time-consuming segment of the satellite delivery process. The requirements for building a multi-satellite constellation include quick and efficient processing of satellites for launch. These requirements may also force the use of multiple launch sites and launch-preparation areas, further driving the need for standardized and simplified launch processing.

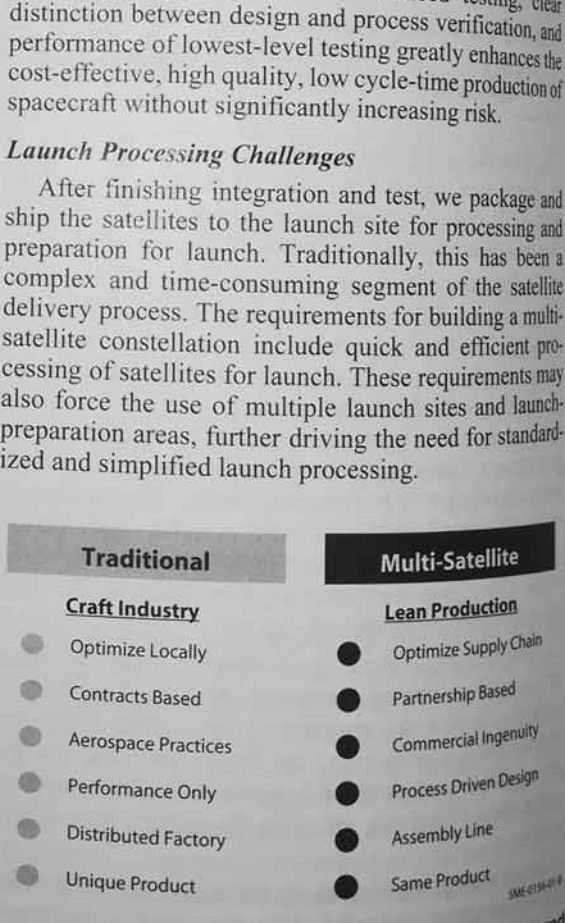


Fig. 23-11. Lean Satellite Production Principles Compared to Traditional Craft Production. Lean production methods originated in the automobile industry. These methods can be applied to the production of satellites as well. See Womack, et al. [1990] for in depth discussion of lean manufacturing.

We need to consider launch-site processing from the start of the program. We should target decisions and trades that support simplified processing. Changes and processing are much easier to handle while at the integration factory. This is where the facilities and skill levels are the most available. Launch-site processing typically takes place at a remote location—often not owned or operated by the satellite manufacturer—that makes processing more difficult.

The spacecraft are transported to the launch-site processing area via protected environment or container. The container environment must provide for temperature, humidity, vibration, and shock protection during the shipping process. In order to support the streamlined processing of the satellites upon their arrival, the shipping process is subject to basic process control. We deem the process to be successful as long as the required parameters—temperature limits, humidity limits, vibration levels, and shock levels—are not exceeded. If these parameters exceed their limits, we need to perform in-depth analysis or additional testing at the launch-site. But, if none of the parameters are exceeded, the satellite can continue with the streamlined process flow that contains little, or preferably no, testing.

Figure 23-12 depicts a streamlined launch-site process flow. The processing includes all the steps needed to prepare the spacecraft from shipping condition to flight condition. Launch team members receive, inspect, load onto the launch dispensers, fuel, package, and transport the space vehicles to the launch vehicle. They charge and monitor the flight batteries, which may be installed at the launch site, when the spacecraft are mounted on the launch vehicle.

We remove all difficult tasks that can be driven to earlier processing. We minimize hazardous operations. For instance, we can replace deployment ordnance with other mechanisms. We should minimize or eliminate test operations. If the shipping process was in control (e.g., no limits exceeded on shipping sensors), we need no test verification of the process. Well-designed tooling and process development can make the remaining processes quick and efficient with little threat of damage to the spacecraft.

23.5.2 Creating the Manufacturing Vision

One of the most important factors relating to a successful production program is the creation of a manufacturing vision. More and more customers today want the "better, cheaper, faster" solutions to their product needs. We should create the manufacturing vision based on the customer's and the market's expectations for the product. The entire project team must embrace this vision. All parties involved with manufacturing must share the manufacturing vision and we must also include everyone else involved with the program from the early design phases through production and test. We must create manufacturing goals to support the vision. Creating manufacturing goals for quality, cycle time and cost must directly relate to the manufacturing vision. For example, if there is a manufacturing vision to produce enough satellites to populate a constellation in a single year then the manufacturing goals must reflect this vision. These goals must be in the form of quantified short cycle times, quantified quality levels and quantified costs. Once the manufacturing goals are in place, the next step is to execute these goals and pursue the manufacturing vision. We must constantly measure the progress towards meeting these goals and share them with the team. We must evaluate progress and make necessary changes to achieve the stated goals. Changes may occur in the design or in the production and test processes. Changing the design and processes to meet the manufacturing goals is an iterative process. It is a give and take scenario where the sum of the changes equals achieving the manufacturing goals.

Establish Goals

We must establish goals in the areas of cycle time, quality and cost. In today's competitive environment, these areas are everything. Many times multiple companies have similar ideas and the only differentiating characteristic between their products is which one makes it to the consumer market first and captures the majority of the market share. Missing the time to market goals will cost a company market share, revenue and brand equity. Therefore we need to know up front when our product must be ready for our customer and then determine our cycle time goals to support this overall schedule.

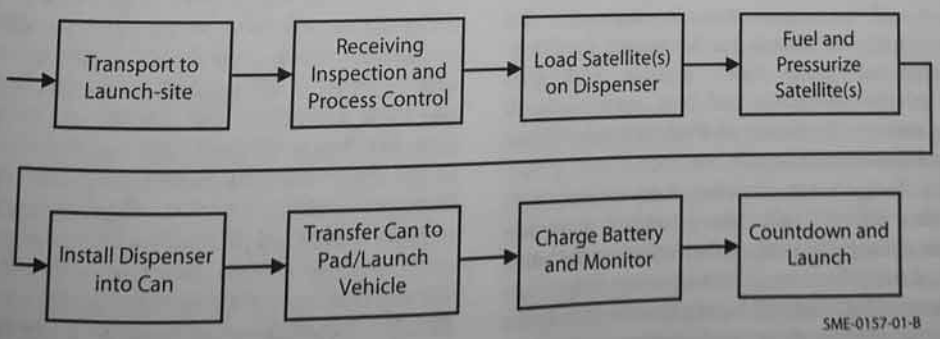


Fig. 23-12. Streamlined Launch-Site Process Flow. Traditional launch-site processing can take several weeks. Streamlined processing can reduce the overall cycle time (transport to launch) to less than two weeks.

Table 23-12◀, Fig. 23-12, Eq. 23-0

As stated previously, one of the unfortunate aspects of building satellites is that they are usually not field serviceable (Hubble Telescope is the most prominent exception). If there is a quality problem after we launch the product into space, we have to live with the degraded performance or, in an extreme case, accept that the product does not meet our mission requirements. We must establish quality goals so that we can design our product accordingly. When we choose components for our product and select processes to build and test our product, we determine our resulting quality levels. As we choose components and processes we need to calculate our cumulative quality predictions and track them to our goals. We will then have the visibility to see what components and processes have a positive or negative impact on the overall product quality. With this data we now make informed trade-off decisions. We can determine which components or processes to change to increase the product quality. Often quality drives cycle time. If a product is designed with low quality levels then we should expect to perform more rework and additional testing. This in turn adds cycle time. Hardware designed around robust processes which exhibit low defect rates result in products with fewer defects which leads to less test time, less rework and lower costs.

Most customers desire low-cost solutions. More and more of the traditional "cost plus" projects are being proposed as fixed price contracts. The days of cost-plus contracts and cost overruns being absorbed by the customer are quickly disappearing. Today the contractor carries the burden of performing to an agreed-upon fixed price. Because of this, the contractor must know the cost goals and understand how they are going to perform to these cost goals. Performance to cost goals directly relates to performance to cycle time goals and quality goals. The addition of unplanned cycle time and of effort required to correct quality problems result in increased cost. In general any unplanned work results in additional cycle time and in additional costs.

Involve Manufacturing Early

Manufacturing early involvement may not always be the accepted way of doing business. Early manufacturing involvement allows the design team to be informed regarding the effects of their choices on downstream manufacturing and test operations. It empowers the design team to make informed decisions and predetermine the expected cycle time, quality and cost performance of the assembly and test operations. It allows management to be aware of what to expect when the product reaches assembly and test.

Getting the design team to accept concurrent engineering can be a difficult task. Many times a company needs a culture change for everyone to embrace the early participation of manufacturing. Management must support concurrent engineering and must pay attention to the analysis results created by the manufacturing representatives. The manufacturing and test participants on the concurrent-engineering team must embrace the manufacturing vision and possess the tools to influence the design to meet this vision. They must show the other members of the design team that their participation is value-added and that early team involvement will save the program time and money in the long run. Everyone on the project team must share the manufacturing vision in order for it to become reality.

Select Parts and Processes Effectively

As was discussed earlier the sum of the project's individual process cycle times (both design and manufacturing/test) must support the overall cycle time goals and the manufacturing vision. Cycle time is inherent in the design. We predetermine the cycle time to assemble and test a product when we choose parts and processes for a particular product. Therefore, during the product design phase we must give appropriate detail to the proper selection of parts and processes. Overall the design team must realize that a design consists of parts and processes. Often the choice of a particular part will dictate the use of a particular process. This relationship is what gave rise to the phrase "pick a part, pick a process." Often design engineers choose parts that meet a limited set of criteria from which the design engineer is searching. They often do this without knowing the downstream impacts that these part choices have in assembly and test.

It's important to have manufacturing and test representatives involved in the part selection process. They can perform analyses to show how certain parts may require additional process steps which have lower process yields leading to more defects, higher cycle times and ultimately higher costs.

An electronic assembly example of this type of analysis is shown in Table 23-13. We can extend this analysis to satellite design and manufacture. Part X and part Y are equivalent in the areas of form, fit and function; therefore, we would assume that the parts are interchangeable. However, the parts vary greatly when we address associated processes, process yields and overall cost. Part X is an 84-lead *ceramic quad flat pack* that must have the leads formed, trimmed, and tinned. The part must then be loaded into a tray for presentation to the part placement (pick and place) equipment. Form, trim, tin and tray placement are non-preferred processes. We must perform these processes in-house, which contributes to increased process time for that part type. The low process yields associated with these non-preferred processes drive this part to contribute 0.273 *defects per unit* (DPU) which leads to increased cycle time and costs associated with rework and potential test failures. Part Y is an 84-lead plastic *quad flat pack* that comes to us formed, trimmed, tinned and loaded on tape and reel to be presented to the pick and place machine. This part uses only preferred processes which leads to lower process cycle times, lower defects per unit and lower rework costs. All of these factors are reflected in the lower total cost of Part Y. Therefore, when selecting a part the design team must look at the downstream impact of their part selection and include the total cost impact of the part in their design tradeoffs.

Table 23-13. Part/Process Cost Relationship.

Part	Package	Part Cost	Processes Used	Process DPU	Process Time (Hr)	Assembly Cost	Rework Cost	Total Cost
Part X	84CQFP	$25.00	1. Lead form/trim 2. Lead tin 3. Pick & place —tray 4. Oven reflow	0.273	0.05	$3.20	$40.95	$69.15
Part Y	84QFP	$17.50	1.Pick & place —tape 2. Oven reflow	0.00306	0.002	$0.128	$0.459	$18.00

By relating parts to processes we can determine the expected time to manufacture and test a product, the associated quality level and the overall cost. Once we make these estimates we can perform "what if" analyses to see if other part/process combinations will make attaining the cycle time, quality and cost goals more realistic. We perform this iterative process early in the design phase, so we can include many design tradeoffs between all of the disciplines.

Parts have many attributes that influence the effectiveness of manufacturing processes. Some of these attributes fall into the categories of cost, interchangeability, availability, reliability and simplicity. Early part-supplier involvement is critical to ensure acceptable levels in all of these categories. The suppliers need to know what our cost goals are for the parts being supplied. The supplier also needs to understand our product design and how the supplier part interacts with our design. The supplier can work with us to determine the appropriate variables to control to ensure that the part is interchangeable and reliable. The supplier also needs to understand our production schedule and our material management philosophy in order to make sure that their parts are available to us when we need them.

Part cost directly impacts product cost. It also has an impact on process development and process execution. Processes used to assemble high-cost parts are usually designed with additional controls and verifications. Typically, this leads to higher stress levels for the people doing the assembly work and more engineering involvement for process verification. Use of higher cost parts drives the desire to perform more rework rather than scrap hardware in the event of unwanted process variations.

Part interchangeability is critical in a product design regardless of the production quantity. The ability to remove and replace a component quickly translates into less non-value-added time. Interchangeable parts have the same form, fit and function for all parts. The supplier of interchangeable parts must use process control to insure that the parts coming into the assembly area are truly interchangeable. If a part is not truly interchangeable, assemblers will have to spend additional time to alter mechanical characteristics of the part to make it fit or perform additional tests to adjust for the varying performance of the part.

Part simplicity directly relates to part cost, interchangeability, availability and reliability. Typically the more simple the part design, the lower the cost. A simple part will have less variables associated with its design, therefore, it will positively impact the component reliability and interchangeability. A simple part will require less production set up time and have fewer variables to cause scrap, and thus, should be more available for use.

Part availability is critical to maintaining product flow in the assembly area. We never want to stop the assembly process due to parts unavailability. Many times the risk of part unavailability is controlled by creating material inventories. However, there can be a great deal of cost associated with carrying part inventories. Typically, we must allocate valuable factory floor space to material storage near its point of use. An alternative to storing material on the factory floor is to store it in a stock room and move the material to the assembly line when we need it. Both of these scenarios result in material being stocked at the assembly location and increase the risk of having large quantities of parts that may become obsolete or require rework/retrofit if there is a design change. The best alternative to either of these scenarios is to have *"just in time (JIT)"* suppliers. Working with JIT suppliers requires less stocking activities and hence less chance of having vast amounts of product that has to be reworked or retrofitted when there is a design change, but can add risk. The supply chain information —especially deliveries, cycle times, and quality—must be well understood to make JIT plausible. If not well understood and reliable, JIT is very risky.

Once material is ready for production it is time to use our manufacturing processes. In the typical aerospace manufacturing environment, processes were developed after the design, so the product could be manufactured and tested to meet the customer's needs. Before the implementation of concurrent engineering this meant that the design was "thrown over the wall" to manufacturing and manufacturing did "whatever it takes" to build, test and deliver the hardware to the customer. Typically, this meant that the processes were highly flexible, required specially skilled workers, lacked good control and were seldom characterized. Today with the advent of *concurrent engineering*, we design in the controlled and repeatable manufacturing processes from the beginning and the manufacturing output is highly predictable.

Concurrent engineering has brought about a change of attitude regarding manufacturing processes in the aerospace industry. No longer do we perform processes just

because "that is the way we've always done them." High-quality processes are characterized, controlled and repeatable. Many of these processes have been borrowed from the commercial manufacturing world. The mass reflow of electronic assemblies is an example of this exchange. Mass reflow—the mass soldering of parts on printed circuit boards using a single, controlled thermal cycle—was developed to efficiently produce electronic products for delivery to consumers. Although slow in gaining acceptance, we now use it for spaceflight hardware and other high reliability applications.

The concurrent manufacturing engineer should never be afraid to ask "why" we design or process something the way we do. An example of this is to look at the integration of deployables on a satellite. In the past, certain deployables have always been integrated with the satellite in the vertical position. This was done to minimize the amount of stress that could be preloaded into the deployable mechanisms. Changing the satellite from the horizontal position to the vertical position and back to the horizontal position requires time, space, additional tooling and creates a lot of risk associated with handling the satellite. With today's modern analysis tools much of the stresses associated with preloading the deployable mechanisms in the horizontal position can be modeled and the risks analyzed. If the risks are determined to be low enough then the satellite plane change from horizontal to vertical can be eliminated. This will save time, space, and tooling and also reduce handling risk.

Typically, satellite manufacturing has been a low-volume process. To date, many companies have built or are proposing multi-satellite constellations. These multi-satellite projects drive manufacturing to create production lines for assembling and testing satellites. The "assembly line" concept of these satellite production lines has roots in the automotive industry. Many of the new satellite production lines take mass production assembly line technology one-step further and implement many of the *"lean" manufacturing* principles currently used in the automotive industry. Some of these "lean" manufacturing principles include JIT suppliers, supplier partnering, concurrent engineering, pull production*, and statistically controlled processes.

Manufacturers in other industries have taken the "lean" manufacturing philosophy to the next step—*"agile" manufacturing*. "Agile" manufacturing incorporates all of the principles used in "lean" manufacturing with the addition of flexibility. An "agile" manufacturing line can support multiple product lines on the same assembly line. It also can adapt and respond to product changes which may be directed by changing customer wants and needs. Agile manufacturing is a future possibility for satellite manufacturing.

One of the guiding principles behind lean and agile manufacturing is waste elimination. We must evaluate all of the processes associated with assembly and test in terms of value-added activities and non-value-added activities. During process development and implementation, we should maximize the value-added processes and the non-value-added activities should be minimized. Reduction of the non-value added process steps will drive waste elimination.

Reduction in the number of inspection points is a good example of waste elimination. Satellite production programs have typically had a high content of quality inspection points. Inspection points are examples of non-value-added operations. In the past, multiple inspection points were required because the assembly processes were highly variable and uncontrolled. By institutionalizing concurrent engineering, processes are now designed into the assembly flow that are controlled and repeatable. Because the output of these processes are predictable, we no longer need multiple inspection points. Therefore, we eliminate waste by removing the non-value-added inspection steps.

Another non-value-added recurring process is test. Because test is non-value-added, we need to minimize it to eliminate waste. Although some testing is required, we should minimize redundant testing. One of the benefits of using controlled and repeatable processes is that the output from these processes is predictable. Therefore, we don't have to test something that has been tested earlier if we know that the processes used after the initial test have not injected variability into the product. A good example of this principle is the previously-mentioned minimization of launch-site processing. We test the spacecraft prior to shipping it to the launch site, and don't test it at the launch site unless shipping indicators or sensors show that it exceeded a shipping process limit.

Test is just another process. During the early phases of design, we should view the assembly and test efforts as processes that must be controlled and repeatable. Similar to satellite manufacturing processes, we should establish goals for all test processes. These goals should relate to cycle time, quality (predicted test yields), and cost. We should question any non-value-added steps and, if possible, eliminate them. If a test continually exhibits a 100% yield rate then we need to determine if the test is necessary.

The reality is that test yields are not always 100% and finding defects at higher levels of integration can negatively impact cycle time and cost. The goal of performing a test is to find the defect at the lowest possible level of integration. This reduces the amount of time and effort required to get to the problem area and fix it. Even after tests have been performed there are instances when the quality of the lower level hardware may still be suspect at the higher levels of integration. Therefore, it is important to design in accessibility at the higher levels of integration. We may need this accessibility to repair a defect that has passed undetected through a lower level test operation or to incorporate late design modifications prior to sealing the satellite. Design for accessibility needs to be a goal of the program vision because it doesn't happen without conscious effort. For a detailed discussion of process design and production trends, see Shunk [1992].

Table 23-13◀, Fig. 23-12◀, Eq. 23-0

*** Pull Production Systems**

Traditional satellite manufacture and mass production in general is based on a push system. In other words, the material and work-in-process is generally pushed or processed through the system. The emphasis is on processing the hardware through the production line as quickly as possible. Unfortunately, this can create pockets of inefficiency and bottlenecks (throughput limiters). These problems require excess factory space for inventory of material and product, and can create excessive defects and cycle times.

Lean production utilizes a pull production methodology. Material and work-in-process is not processed until upstream stations are ready. Downstream processes do not create excess inventory, because the production system will not allow production to continue unnecessarily. The system is tuned for a continuous process flow, with necessary materials being delivered Just In Time in small quantities.

23.5.3 Influencing the Design

Success of production depends on the manufacturing/production disciplines being involved throughout the product design. Through concurrent engineering, we develop the product involving various disciplines of the trade. For satellite design, this includes such expertise as electrical, mechanical, software, test, industrial, and reliability engineering; it also includes other less-technical but equally important functions such as contracts, finance, procurement and scheduling. The manufacturing role, in particular, acts as the main interface between many product teams due to their direct responsibility of ensuring a producible product. All experts must get involved with the design early and continually to the extent they can influence the product to meet their respective goals or to compromise with others. At the very least, if the experts can't reach an acceptable term, they can highlight the issue very early in the program so that they can continually work it into an acceptable term.

To influence the design, the program team must first establish a common vision for the enterprise. This common vision serves to create a working platform that allows the supply chain to collectively communicate. It provides the central theme to which the program will focus regardless of where anyone's portion of the business is located. The common vision creates the need for goals and guidelines in each concurrent design area within the program. Overarching goals must be established to provide incentive for the program; they should be difficult to attain, yet attainable within the program life.

To supplement the goals, the team must establish specifically defined rules and guidelines to bring the goals to fruition. The vision, goals, and guidelines provide the program with a standard, central direction that, in turn, provides the program with the opportunity to be efficient and united. The intention of a common vision is not to strip individual company identity, rather to converge on certain business ideas that alleviate challenges due to diverse mindsets, backgrounds, and corporate cultures.

Designing to goals and guidelines sounds simple but may be difficult in practice; few programs have these clearly outlined at the inception of the program. Implementing goals at the beginning of the program is where they are most effective. DARPA studies have shown that approximately 70% of the cost of the system is fixed by the end of the idea/conceptual design phase (Fig. 23-13).

A concept becomes feasible through refinement and, in turn, producible through other iterations of refinement. If we initiate the guidelines of design at the conceptual level, it is less costly and less painful to iterate to a feasible design. The cost to correct problems occurring from a previous program phase increases by an order of magnitude, approximately (Fig. 23-14). A simple design change during the conceptual design phase will cost 10 times during breadboard phase and 100 times during engineering model phase.

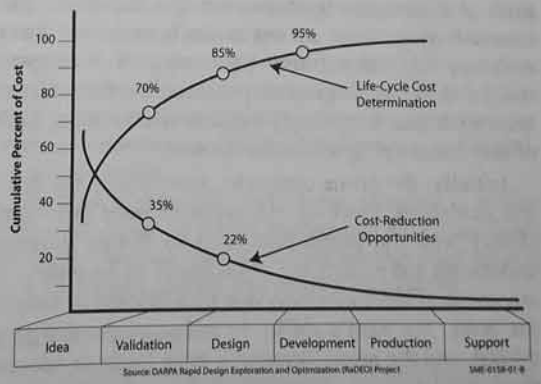


Fig. 23-13. Project Phase/Cost Assignment. The ultimate costs of a system are greatly influenced by decisions made early in the product life cycle. Based on model fit to empirical data.

The feasible design should already be largely configured to the goals and, therefore, iterating to the production level should be relatively smooth. The iteration process is smoother because the team knows what is expected through the long-term goals.

Table 23-13◀, Fig. 23-13, Eq. 23-0

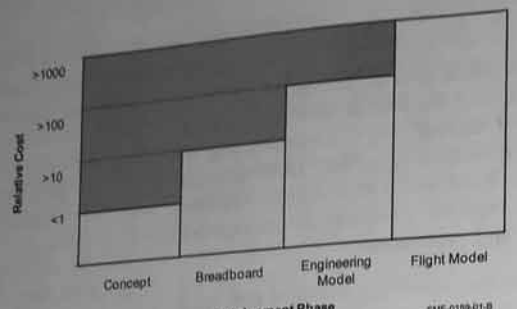


Fig. 23-14. The Relative Cost of Modification / Corrective Action. The relative cost of fixing a design problem escalates rapidly as the design matures.

Production goals and guidelines should include all production-related information for the product. This information encompasses the expectations for both parts and processes as these two ingredients form a product. It also provides structure for the product to grow properly since it addresses quality, capabilities, production rates, facility requirements, procurement, cost, warranty, manufacturability, and testability. For example, The design dictates product quality. If the product has a 2-sigma design, the best we can expect is a 2-sigma product.

Knowing the expectations initially not only assists the product to mature into production, and assists the product team to develop. Having these goals gives the team a purpose to commonly focus their attention. Successful teamwork resolves issues seemingly otherwise unresolvable. Communication becomes clearer and less conflicting because the team uses common terms to explain the common goals. A good team forms to produce a working product without relying upon functional titles and positions. Additionally, trust forms within the team. Yet, trust and competition are not mutually exclusive. Team members may and will form partnerships with competitors for different projects or products, but this practice must not hinder the primary business relationships. Lack of trust causes program inefficiencies.

Initially, the prime contractor must derive the goals and guidelines; however, the suppliers must then work closely with the prime contractor to further develop, understand and modify the expectations as necessary. In any event, if team members don't understand or believe the goals, the team's ability to influence the design is diluted, and the prime contractor must begin to educate them about the goals. Furthermore, it is especially important to derive the goals so they are meaningful. For example, assembly time of a high volume, high complexity product (indicative of the constellation satellite industry) typically comprises less than 35% of the overall cycle time while production test comprises the remaining 65% [Olivieri, 1997]. Using the Pareto Principle* approach we find that the greatest opportunity to influence the design is in production test.

Table 23-13◀, Fig. 23-14, Eq. 23-0

• The Pareto Principle

If we rank order contributors to a problem from greatest influence to least influence, the *Pareto Principle* states that only a few of the contributors will comprise the bulk of the influence. In general, 80% of the contribution comes from 20% of the factors. Pareto analysis is a technique for ranking items according to their contribution and analyzing the top contributors, or the "vital few." The other 80% of the factors that only influence 20% of the contribution are the "trivial many."

Design influence is most effective through mathematical analysis. Personal bias and tradition are not the basis for influence. When we select a part for a product, we inherently select the process as well. This means that predicting behaviors (e.g., quality, reliability, cycle time, assembly process) of the product is possible. We do this by collecting data from suppliers but where data is not available we either run an experiment or locate like-product data. Key predictions include, but are not limited to: *defect per unit*, *defects per million opportunities*, statistical tolerance and variability assessments, *design to unit production cost*, solids modeling, cycle time, process modeling and mapping, design of experiments, and simulations.

Having the data equates to having the knowledge to influence design. The only way we achieve key goals in areas such as quality, manufacturability, and profit is if they are designed in.

Influencing the design is an enterprise challenge. Awareness of our up and down stream product map helps us make smart decisions for the overall program. Locally maximizing cycle time, quality, assembly ease and other areas of importance, does not always provide the same benefit to the overall program. Standardization of analysis tools, metrics, goals, and guidelines help to relate design impacts to the supply chain.

23.5.4 Process Development and Verification

Quality, and the resulting efficiency of manufacturing, is never an accident. It is the result of good design practices and sound process development methods—with the addition of management and execution diligence. Significant process development and verification activities must take place to manufacture high quality products. For satellite manufacturing, we need to apply these methods at all levels—suppliers, sub-tier suppliers, payload assembly and test, space vehicle integration, and launch operations.

In the manufacturing context, we define a process as any activity that changes or touches the product or requires resources. The objective is to use simple, uncomplicated processes that yield high quality results with short cycle times. Very few processes meet these general criteria without development. Therefore, sound

development methods are key. The end goal is to have controlled and repeatable high quality processes during the production phase of the program.

Development Methods

We can apply structured development methods to process and product development. Many methods exist, and most have merit. Here, we use a *four-phase process development*, a.k.a. *process characterization*, model. Other useful and powerful models include DMAIC (Define Measure Analyze Improve Control) and DFSS (Design for Six Sigma). We should first apply a development model to critical processes and then extend it to other processes as necessary.

The process characterization model consists of four phases.

Phase 1: *Process Definition*—Map the process to understand the variables and characteristics involved.

Phase 2: *Process Capability*—Establish current level of performance of the process. Does the process perform as needed?

Phase 3: *Process Optimization*—Investigate the variables to determine which variables drive the process output. Determine the best levels for these variables to provide optimal process output.

Phase 4: *Process Control*—Monitor the process and its important variables to determine when its performance has changed or is out of control.

We iterate the cycle to further improve the process as necessary. (See Fig. 23-15.)

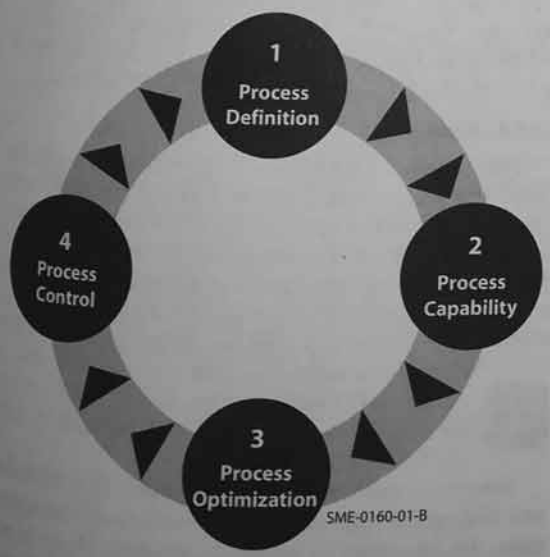


Fig. 23-15. Process Characterization Four Phase Model. We iterate this cycle to continually improve the product or process.

We characterize the process to identify and remove sources of variation. Ultimately, variation is the cause for product quality problems. We use statistical tools, such as *Design of Experiments* and *Statistical Process Control*, to reduce the variation, which improves quality and subsequently reduces cost and cycle time. See Montgomery [2000] for detailed discussions about statistical tools.

Cycles of Learning

In addition to the iterations caused by characterizing the production process, we may need additional cycles of learning when developing satellite constellation production systems. Varying techniques are available, depending on the subject matter.

For instance, we use simulation techniques to model and predict performance. Discrete event simulation packages can model processes, factories, and whole supply chains. We can use these models to determine resource requirements, facility size, inventory schemes, and delivery schedules. We can explore proposed changes to pre-established baselines without risk.

A second source of cycles of learning is the use of prototyping. Prototyping allows for physical evaluation—appearance and dimensional—while changes are easier and less expensive to make. Production changes are extremely costly, and potentially devastating. We can potentially avoid them through early prototyping. The use of *Dynamic Test Models* and *Engineering Development Models* are advanced functional examples of prototypes.

A third source is the use of *pathfinding activities*. The more complex a system and its logistics, the more beneficial pathfinding activities become. They revolve around building a scale representation, usually 1:1, of a product and using it to simulate all handling, manufacturing, and logistical activities. For satellites, this includes practicing and iterating every process, from subsystem receipt to integration processing to launch processing. The resulting benefits of pathfinding include handling streamlining, mechanical checkfit verification, documentation improvement, process development, tooling improvement, facility development, and logistical simplification.

In short, any activity that we can simulate or realize before we create an actual product can have significant payback. These sources of learning are important tools to gain significant process and product knowledge.

23.5.5 Production

All of the previous phases—creating the manufacturing vision, influencing the design, and process development—lead to the final phase—production. All previous goals are to make this phase easy to execute. Failure or lack of attention in the previous phases becomes readily apparent here. They surface as schedule slips, exceeded budgets, and poor quality.

Executing the overall plans and goals established in the earlier phases is paramount. While many production issues are similar for building one satellite or one hun-

dred, some areas are significantly different. These areas include supply chain management and production floor management.

Supply-Chain Management

A supply-chain focus is key to the successfully executing the production phase of a multiple satellite program. The supply chain is defined as all of the suppliers and activities involved with producing a satellite. This involves the major suppliers—bus, payload, and antennas—and the subtier suppliers.

Communicating and coordinating across the supply chain is critical. Fundamentally, all the suppliers need to have a common vision and sense of direction. In effect, the major suppliers become partners. Openly communicating goals, expectations, and problem solving is essential. To facilitate communication, we need to seek a common language and culture through common training and team building. The resulting program culture can be built on the strengths of each partner (Fig. 23-16).

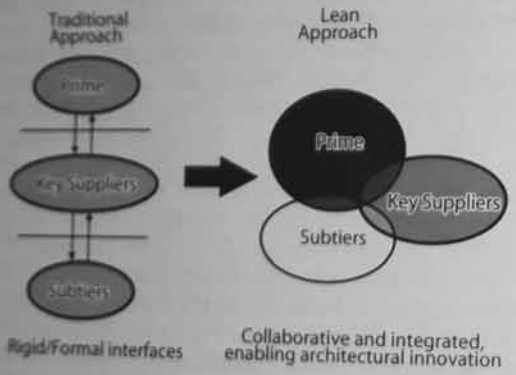


Fig. 23-16. Lean Supply Chain Model. Creating a collaborative and integrated supply chain enables lean performance while traditional approaches create boundaries and limitations.

Concurrent engineering across company lines must occur. All company entities must share the key design rules or characteristics early in the process and maintain them throughout the program. We must carefully define and fully understand the interfaces and handoffs. We should develop and use common tooling and processes. For instance, shipping containers carrying subassemblies to final integration should support integration activities without a tooling transfer. They should seamlessly integrate into the factory. Finally, common metrics across the supply chain allow for meaningful summation of data, so managers can judge the overall system health. These metrics should center around quality, cycle time, and critical parameter performance.

Production Floor Management

Performance and executing the production phase uses principles similar to any production environment. In addition to supply chain management, consistent performance takes lean manufacturing principles of production floor management, including good processes, qualified workers, and timely information.

Short cycle, well understood processes and [illegible] processing is fundamental to lean manufacturing. Simulation or other flow design tools help to determine optimum production flows that reduce the risks of production delays. The flows and processing should include *Just In Time* (JIT) concepts to minimize *Work In Process* (WIP) inventory and improve quality and cycle time. JIT hinges on delivering material items when the flow needs them in a pull fashion. We don't move items until they are ready for processing. Less hardware is in process, thereby reducing the potential for damage and reducing overall floor space requirements.

Qualified workers that understand the processing of space hardware are still required for satellite production. While still required, their roles and specific skills are very different than traditional programs. Highly specialized and narrowly skilled positions now become broader and more generalized. The design attributes resulting from a successful multi-satellite design team will require less intensive processing. In addition to being less difficult, the processing needs to be more repeatable. We can train or certify workers for each repeatable operation, so they can typically monitor their own work. We minimize quality inspections. Instead, the individual or the team is responsible for performing each operation correctly and safely.

The capture and use of timely production data is a third production floor management challenge. Producing multiple satellites with short cycle times requires us to collect, use, and archive data differently from traditional methods. Paper, the traditional documentation medium, is too cumbersome, untimely, and not available to multiple disciplines. Constellations based on a large number of satellites require volumes of data be available to multiple users. We need factory systems for work instruction and process planning; transmittal of product data between suppliers, integrators, and users; and for factory data collection. Factory data collection includes calibration, as-built, WIP tracking, and quality data and/or information. Using common tools for these functions across the supply chain would be an added advantage.

23.5.6 Summary

These methods apply to multiple satellite constellations and were applied to the Iridium program, with outstanding cost and schedule results. Extensive constellation reliability and performance data has been collected on the Iridium system and its satellites. The system is still operating in 2010, even though every satellite has exceeded its 5–8 year design life. Some operational satellites have been in use for over thirteen years at the time of this publication. See the book website for a description of the Iridium Satellite Assembly Process.

Get More

Smaller constellations, non-communication payloads, and high-value missions may use subsets of these practices. In all cases, thoughtful selection of parts and processes and skillful application of concurrent engineering practices are paramount.

23.6 Alternative Approaches to Space Manufacturing

Jaime Esper, *NASA GSFC*

Current NASA space systems are mainly driven by performance and reliability to ensure science requirements are successfully met. Exacting the most performance is necessary, if one is to be assured of only a handful of flights in one's lifetime. Reliability is of course a way to ensure that time wasted is kept at a minimum, and that precious data is acquired (relatively) on schedule. But what if flights were more frequent and opportunities for participation in meaningful science increased? If failure were an option we would not hesitate to take the risks necessary to advance the state of art in supporting the acquisition of volumes of specialized data. If space system costs were reduced, then failure may not be seen as wasteful. Of course, spectacular failures are often the result of launch vehicle or spacecraft malfunctions. So what if both the launch vehicle and the spacecraft were "acceptably" inexpensive? To achieve this, it is necessary to examine the way space systems have been designed, manufactured, assembled, tested, flown, and operated. In other words, we must find alternative approaches to space system manufacturing and implementation. At the same time, we must examine what type of missions may be candidates for this new paradigm, and an architecture developed that clearly establishes some boundary conditions on which to operate. This game-changing application approach also needs to be placed into context with past experiences, to extract what works, and what needs improvement.

23.6.1 Historical References for Alternative Approaches

There are a number of examples of programs and spacecraft that tried to incorporate ideas to reduce the cost of flying space missions, either by providing a multi-mission spacecraft, or by judicious application of programmatic approaches and principles to run a production line. A few examples from the NASA side include the *Multimission Modular Spacecraft* (MMS) and the *Small Explorers Program* (SMEX). Other organizations have also experimented with alternative cost-effective approaches, including Freja, the joint Swedish-German micro-satellite for atmospheric research, and the US Iridium communications production. We will examine each of these in more or less detail, to highlight the areas where the aerospace community has advanced ideas on alternative approaches that attempt to reduce cost, and at the same time advance the state of the art in space manufacturing.

The Multi-Mission Modular Spacecraft (MMS)

The MMS was a standard, 3-axis stabilized spacecraft design conceived to accommodate the largest number of missions, whether in low Earth or in geosynchronous orbits [Bartlett and Cepollina, 1975]. Specifically, 4 design reference missions were envisioned: Sun, Earth, astronomical observations from low-Earth orbits, and Earth observation from geosynchronous orbits. The motivation for developing and implementing the MMS concept was not much different in 1969 than it is today: the desire to get more for less, or at least, to be able to go the extra mile with a constrained budget. The key to the concept was modularity, and the ability to incorporate mission-unique hardware with minimal impact to the established bus design. Cost savings were expected through maximum use of *standard components*, design of a set of *standard subsystem modules* with application to the reference missions, and leveraging the Space Shuttle for "orbit re-supply and retrieval of modular subsystems and instruments," thereby extending the usefulness of a particular mission. Shuttle servicing potential required a clear perspective on interface simplicity. This called for maintaining a minimum number of electrical and mechanical connections at the servicing interface, preservation of thermal interface integrity, one structural and thermal design for all reference missions, and maximum use of previously flown components, including the use of NASA standard components. A basic premise of the MMS design included the use of *interface standards*, permitting module exchange between spacecraft. This would also allow for re-design of internal subsystem configurations in order to implement required performance upgrades. In addition, backward-compatibility would permit the re-fitting of operational spacecraft with subsystems produced at a later date [Falkenhayn, 1988]. The baseline MMS configuration contained three standard modules: attitude control, communications and data handling, and power subsystems. Additional modules were considered mission unique, and included configurations for antenna, solar arrays, and propulsion. The standard subsystem modules were supported by a thrust-axis load bearing structure, outfitted at either end by transition and base adapters. An exploded view of MMS is shown in Fig. 23-17.

The MMS program made important contributions to I&T processes and procedures [Greenwell, 1978]. In the early days at the NASA Goddard Space Flight Center (GSFC), complete prototype spacecraft were produced and tested to environmental levels greater than those expected under operational conditions, in order to demonstrate design margins of safety. After proving hardware so tested would not be degraded, GSFC developed the *protoflight* concept. Instead of building an entire copy for testing alone, protoflight hardware was built and tested to prototype levels but for durations normally used to test flight hardware. This saved the cost of producing hardware intended for testing alone. Protoflight testing was usually applied to the first unit in a series, but not to the production units. Testing at the module level was favored over component or spacecraft system (sans the payload) testing. This was a reasonable approach, as it was envisioned that modules would be produced by different vendors and integrated already

Table 23-13◀, Fig. 23-16◀, Eq. 23-0

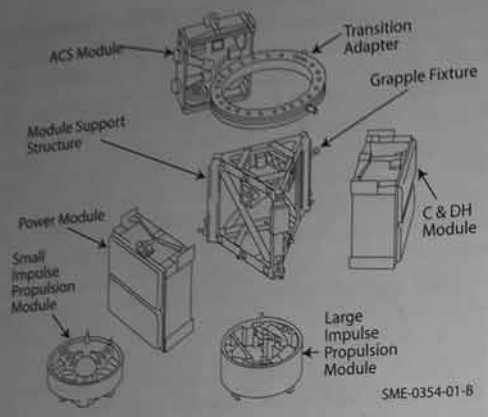


Fig. 23-17. The Multimission Modular Spacecraft. [Bartlett and Cepollina, 1975].

pre-qualified into a flight system on the ground, or as replacement units for orbiting spacecraft. Testing at the payload or observatory level was still executed, but that was the responsibility of the project end-user. Many concepts developed during MMS were incorporated into the defacto standard for spacecraft testing contained in Goddard's *General Environmental Verification Specification* (GEVS) for STS and ELV payloads, subsystems, and components, and applied in spacecraft servicing such as SMM and Hubble Space Telescope (HST).

The Small Explorers Program (SMEX)

SMEX was a NASA program started in 1988 to provide frequent flight opportunities for focused and relatively inexpensive space science missions. The program executed at GSFC included a small team that worked across several mission boundaries, with overlapping mission developments. This staggered approach allowed satellite launches once every 12 to 18 months. Structures, thermal designs, and attitude control systems were designed per mission needs, whereas other subsystems were taken from an inventory of spacecraft designs that were developed for the initial 3 SMEX missions (SAMPEX, FAST, and SWAS). The architecture used standardized instrument and operational interfaces including a distributed system (at the subsystem level) with standard hardware and software interfaces. To this end all subsystems/instruments communicated through a MIL-STD 1553/1773 bus [DoD, 1978] (low rate) and RS-422 (high rate) serial port. Initial SMEX spacecraft relied primarily on the GSFC Wallops Flight Facility (WFF) and the Deep Space Network for ground station coverage.

The "SMEX-Lite" architecture was developed to optimize the spacecraft and mission design process in order to shorten development, integration, and test times. Subsystems were broken into functional slices, covering a single basic function. This minimized the rippling effects of changes within the system box. Functional slices had to be easy to interface and transparent to each other, behaving akin to an early implementation of aerospace *plug-and-play* (PnP) components. Most software development activities were planned using a PC representation of each of the functional slices. This concept resulted in the definition of an open architecture for mission design which included the electronics, sensors/actuators, software, solar array, mechanical system, and ground operations system.

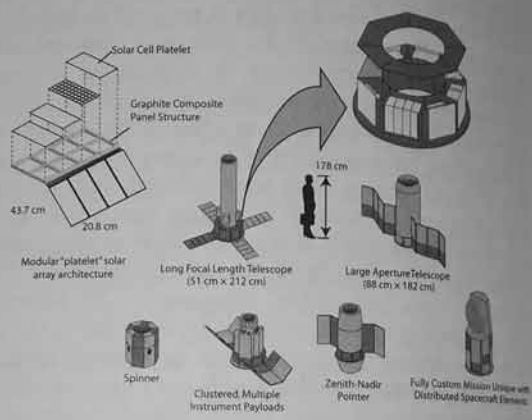


Fig. 23-18. SMEX-Lite Architecture Flexibility.

Industry standards were used for interfacing the functional slices (e.g., PCI bus, MIL-STD 1553). Reliance on programmable devices allowed for rapid prototyping and flexibility of changes to flight circuitry. The use of a standard card size allowed for one mechanical/thermal box design. The solar array design used a modular "platelet" approach, consisting of full string cell configurations mounted on a composite mini-panel. The reduction in spacecraft electronics lead to the adoption of a standard mechanical bus configuration: a monolithic aluminum casting piece, with electronic boxes directly incorporated into it. Ground system testing was to use standard TCP/IP protocols. There was only one SMEX-Lite spacecraft built (TRIANA), and although it was never flown, the SMEX-Lite concepts survived the test of time and have been implemented in numerous NASA and non-NASA spacecraft and flight missions.

The SMEX-Lite modular architecture flexibility, including its structure and modular (sliced) subsystem architecture, promised to enable multiple instrument configurations and mission implementations. Figure 23-18 shows this potential assortment of layout and applications.

Iridium

The Iridium satellite constellation was first placed operational in 1998. This experiment on global communications and data coverage initially found itself in trouble, and not too much later in 1999 its parent company

Iridium LLC went into bankruptcy. With new management, Iridium Satellite LLC is now a viable provider of services to industry and government organizations. The most relevant aspect of this venture for the discussion at hand is that Iridium represents the first mass-production example of commercial satellites in the world.

From a general industry perspective, the production line approach uses similar principles already described: units are assembled concurrently, with standard components that are integrated as the line progresses. In a way, the next project is underway before the previous one is complete. Only the initial units are flight qualified, and the rest are verified by similarity. The one drawback to this approach however, is when it comes time to upgrade the existing system. As the units are identical, all components come together to form a complete system module. Design flexibility may be traded against system flexibility, and the module level of integration. The success of these constellations relies on system-level redundancy, rather than on module-level redundancy, where modules may be interchanged among satellites of different capability, but with standard interfaces. This philosophy is carried into the current thinking behind modular, adaptive, reconfigurable space systems.

The Swedish Small Satellite Program

The Swedish Small Satellite Program has been effective in utilizing tight budgets achieving excellent results to that effect through streamlined project organizations where responsibility is pushed to the lowest possible level, and piggy-back rides [Lundahl, 1998]. The Freja satellite was dedicated to aurora research, and included simple, in-situ instruments and an UV imager. Freja-C was based on its predecessor, and in a sense followed the production-line approach. Astrid, the first Swedish microsatellite, carried instrumentation to study upper atmospheric plasma, with a total mass of 27 kg. Astrid-2 again followed the successful paradigm of limited production run, and was also dedicated to upper atmospheric plasma research. The important point to make here is that the Swedish program started with a basic approach of simple satellites achieving important results, and used serial compatibility with previous mission designs (backward compatible).

23.6.2 A New Architecture for Alternative Space Mission Implementation

Since late 2002 GSFC has been taking a new look at space mission approaches, incorporating experiences from past successful spacecraft and programs. In particular, there are 2 progressively refined, yet common system architectures: the *Modular Reconfigurable Rapid* (MR^2) architecture [Esper et al. 2004], which in many ways remains the same basic approach adopted by the DoD Operationally Responsive Space (ORS) Office today, and a slightly new variation that incorporates adaptive systems into the mix: *Modular Adaptive Reconfigurable Systems* (MARS) [Esper, 2005]. These are the result of government and industry experience and collaboration that extends as far back as the beginning of space flight.

MARS is a system-level architecture that comprises all elements of a space mission life cycle, including ground and space segments. MARS establishes a set of basic guidelines upon which technologies can be developed. Exactly what it means for a technology to be MARS-compatible will be explained. It should be realized however that any static definition of an architecture that has "adaptive" and "reconfigurable" in its term would simply be counterproductive, or at the very least, transient. So this definition is intended to evolve over time.

The defining terms offered for MARS may be thought of as initial, or boundary conditions, in the traditional sense of the term. These boundary conditions, which factor both programmatic and technical concerns, are used to develop a particular solution domain containing a set of technologies. Again, technology evolution and programmatic concerns can necessitate the changing of these initial conditions. This is one key result of the 20/20 hindsight obtained from MMS. There are neither standard boxes nor standard spacecraft. Even standard interfaces, a key element of MARS, are allowed to evolve over time. Insofar as interfaces are known to evolve at a slower rate than technologies though, they qualify here for the term "standard," with a caveat.

So what are the basic premises used to set the stage for development of MARS technologies and systems? They are summarized below. But first, a few definitions are in order.

Modular: MARS systems contain selectable mechanical, electrical, and software components that may be used and re-used in modular or quantized numbers. Modular (or quantum) components used in a system must be capable of evolving to incorporate advances in technology, and they must accept standard interfaces and plug-and-play principles (e.g., Personal Computers). Collectively (and individually) MARS modular components and systems must result in intelligent units, referring to their ability to assemble into larger components or systems on the ground or on-orbit. A module can have varying levels of integration, from the chip to the card, box, subsystem, system, and system of systems. As an extension of the above, a module is capable of carrying a function either by itself, or as a conglomerate system.

True today as it was 40 years ago, plug-and-play interfaces are not yet implemented in space flight systems. The ability to recognize components autonomously as they are added to a modular system is a key in realizing schedule savings and streamlining integration and test. This is important for astronauts on a Moon or Mars base, or for cost-savings on the ground.

Adaptive: This term is used along the lines of the classic control systems definition. In simple terms, an *adaptive control system* is one that is capable of learning from its environment, and acting accordingly. An adaptive MARS system would permit reconfiguration of its

mechanical (including thermal), electrical, or software characteristics to changing requirements, whether they are precipitated through an *a priori* definition (**"I did not intend for the system to do this, but now that you mention it, by design it can adapt"**), or autonomously required on the field based on previously unanticipated events.

This concept is entirely new, and had not been entertained by the MMS nor the SMEX or other system-level architectures. Rather, due to technological limitations MMS relied more on standard subsystems designed to accommodate the largest number of missions *a priori*. Missions that fell outside of this design envelope were better served by one-of-a-kind approaches.

Reconfigurable: The system must be capable of morphing in order to apply to a host of missions. It also must be easy to produce, integrate, test, and launch. In addition, it must be capable of operating alone or as a collective part, physically detached or attached. Reconfigurable systems are considered at varying scales, from electrical components and MEMS, to whole spacecraft, or pieces of a Moon base. It differs from adaptability in the sense that reconfiguration is expected based on a set of known boundary conditions. A system may be reconfigurable, yet not be capable to adapt to changing requirements in certain cases. However, a system that can adapt is always reconfigurable in that particular instance. MMS can best be described as a reconfigurable-only system, within the limited set of missions for which it was conceived.

System: Normally, one should not have to define what a system is. Yet, there may be as many definitions as there are engineering disciplines. The MARS architecture allows all as it should: a system takes on as many incarnations as there are modules. One could consider an atom as being the most elemental module (true, one could go even smaller), and indeed it is a system composed of protons, electrons, and neutrons (the simple model). For practical purposes, a system is an entity that carries out a specific function, whether it opens and closes tiny MEMS "Venetian blinds" (variable emissivity coating), distributes power over an entire spacecraft, or coordinates construction tasks among multiple robots and humans. The important thing to remember is that the MARS architecture is not limited to small or large systems: it can be as broad or as narrow as required. This broad application of the MARS architecture stands in stark contrast to the MMS design philosophy of only modularizing 3 key subsystems: attitude control, communications and data handling, and power subsystem. **Levels of modularity or integration are then used in MARS to provide the reconfigurability to satisfy a broad range of predefined mission classes, or the adaptability to adjust to unexpected mission requirements.**

The basic design methodology for MARS spacecraft and systems is summarized as follows. Note that these are only broad suggestions in what is hoped to be a reasonable set of principles to follow. However, alternative concepts are not only welcome, but also expected as the space community finds the best implementation approach for the evolving times.

- MARS spacecraft and systems should take advantage of multi-billion dollar commercial standards for manufacturing, computing, and communications technology. This results in sustainable systems, where not only the government is the contributing source of funds to maintain technology relevance.
- The modular design architecture must be capable of advancing along with technology improvements. In other words, the module can contain technology relevant to its time, and is not fictiously constrained to some standard.
- Standardization is implemented at the interface, not at the subsystem or system level. Electrical interfaces should use commercial standards, such as Ethernet, Firewire, USB, Spacewire, CAN, and other well-established approaches. Of preference would be those electrical interfaces that lend themselves to a plug-and-play implementation. Mechanical and fluid interfaces should also be standard and flexible enough to accommodate various layout configurations. Specialized interfaces can be developed **only** if required for a particular set or sets of applications. Components should electrically attach via standard interfaces much as peripherals attach to a computer. Each peripheral would need to come with its software driver. The driver is either pre-loaded in the operating system, or loaded as needed by the user. Mechanical and fluid interfaces represent a special challenge, since there is currently little use of standards, except in the launch vehicle industry or for space station applications.
- Choice technologies (not standard) may be incorporated, with a list reviewed and **updated** at regular intervals to maintain technological relevance.
- A modular open systems architecture should be encouraged throughout, whether in the software or hardware arena. This would encourage participation by a broad community, and would ensure minimal implementation lock-ins or bottlenecks.
- The flight software should be based on a layered architecture, with maximum re-use of infrastructure and application modules. No doubt, software technology has improved leaps and bounds since the MMS days. Today, software can be developed which is completely independent of the platform or operating system. One of the key factors in the success of the PC platform for instance, is the choice and backwards compatibility afforded between many different types of hardware and software, a concept partly borrowed by MMS and the early SMEX program. This is made possible by the BIOS (Basic Input/Output System) and the soft-

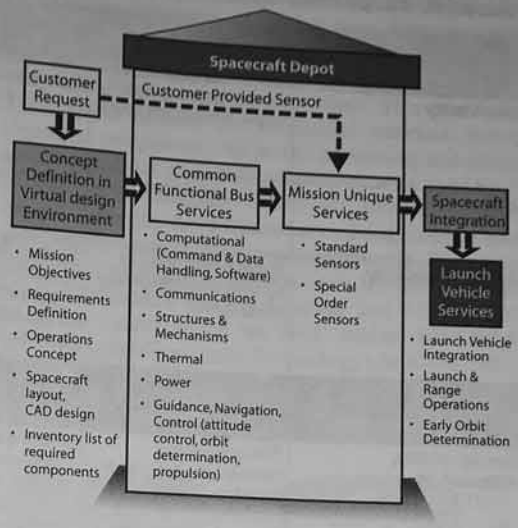
ware layer model. The system BIOS acts as an interface between the hardware and the computer operating system, on top of which the application layer resides. Each layer communicates to the layer below through a standard interface. This way neither the operating system nor the application layers "care" which the hardware host is. The *Open Systems Interconnect* (OSI) Reference Model also defines a modular approach to networking, with each layer (module) responsible for some specific aspect of the networking process. Earlier attempts at software development had more to do with defining standards for High Order Programming Languages [DiNitto, 1978], than concentrating in standard interfaces between highly portable software modules. From an operating system perspective, effort should be spent in ensuring proprietary implementations are kept to a minimum, to stimulate inter-operability among different providers. Whenever a proprietary operating system is used, assurance of compatibility with open code systems should be required.

- As much as possible, communications and information exchange should be done directly from the user to the spacecraft and back, through a distributed internet-like operation. The ensuing architecture calls for the use of spacecraft and space systems as point extensions on the Internet: i.e., each connected through a ground system Ethernet.

Table 23-14 shows a comparison of MMS, SMEX, and MARS concepts and operating principles. Note the similarities, but more importantly, the significant differences that result from 35+ years of technology evolution and hindsight. MMS and the early SMEX were important trailblazers for today's MARS systems and without them we simply could not have evolved to where we are today.

Process Flow

The reduction in design-through-launch life cycle and ensuing cost savings necessitate the existence of a well-established infrastructure. This infrastructure may be based on several approaches: from the pre-built system sitting on a shelf, to the (more desirable) one of having components in a warehouse ready to be integrated to suit. The latter approach lends itself to the greatest degree of flexibility, as modular components may be chosen all the way from the chip level to the box level for a truly unique application every time. Figure 23-19 shows the high-level process behind MR^2 spacecraft [Esper, et al. 2004]. First, a need is identified and a request for action is issued to the MR^2 team. The space system concept is next defined within a virtual design environment, where the mission objectives and the requirements are fine-tuned to develop an operations concept and a spacecraft design. The design also generates an inventory list of required parts and components, which can then be accessed at a *Spacecraft Depot* (SD), in much the same way one would fetch parts to build a project at a "do it yourself" store. In order to be effective, this "store" must contain a variety of select components defined to accommodate multiple mission applications. The length of time for this process depends directly on whether the application sensor (instrument) is sitting on a shelf, is provided by the user, or needs a special order.



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Fig. 23-19. MR^2 Spacecraft Process Flow—The Spacecraft Depot.

The physical location of the SD is a matter of choice and desired objectives. Implementations range from a distributed, virtual SD with components maintained at various manufacturer's sites, to a centralized warehouse. Its final implementation would depend on the need for quick access to all components, versus the market drivers of supply and demand. Clearly, quick-response systems would benefit from a centralized location. This model would not be much different from the one used by individual projects today, which acquire parts in quantities and store them as spare parts. The idea would be to extend this practice to creating a "super project" (in fact a *program*), capable of servicing more than just one mission need.

23.6.3 Boundary Conditions and Mission Applications

Now that several leading approaches to space manufacturing have been identified, and a promising new approach consisting of the sum-value of previous attempts has been defined and placed into historical context, it is important to establish the kinds of missions that can exact benefit from it. Clearly there are legacy approaches for missions that still require the traditional (and expensive) paradigm to space mission implementation. But there are a larger number of possibilities, so far underexploited, that can benefit from current alternatives.

Table 23-14. Comparison of MMS, SMEX, and MARS Concepts and Operating Principles.

Defining Characteristics			
Characteristic	**MMS**	**Early SMEX/SMEX-Lite**	**MARS**
Modularity	Implemented primarily at the subsystem level	Drop-down one level from the "box" to the "functional slice"	Uses **levels of integration**, where a module can be a MEMS device, electronic chip, card, box, subsystem, system, or system-of-systems
Adaptability	Not implemented	Not implemented	System **capable of learning** and acting either *a priori* or real-time from its environment
Reconfigurability	System designed to service a specific class of missions	System designed to service a specific class of missions (small explorers). Drop-down in modularity provides more flexibility than MMS	The system is **entirely morphable** in order to apply to a host of missions. Made possible by modular levels of integration and adaptability

Design Methodology			
Key Area	**MMS**	**SMEX-Lite**	**MARS**
Sustainability, Affordability	The supporting infrastructure did not develop as expected, requiring Shuttle flights, broad spacecraft servicing, and individual project buy-ins	SMEX-Lite provided significant advances in implementing industry interface standards; in particular the PCI bus for internal avionics card interconnects. It also provided a sound basis for realizing cost-savings through an open system architecture concept. Monolithic structure with tailored thermal designs	Leverage multi-billion dollar **commercial standards** for manufacturing, computing, and communications technology. This includes PnP interface definitions featuring device discovery, IP communications, and reconfigurable mechanical/thermal concepts that incorporate the latest advances in technology. Modular Open Systems Architecture for increased sustainability and affordability
Technology Relevance and Effectiveness	Relied on individual projects for technology development	Leveraged the existing Small Explorer Program to inject common hardware and interfaces across different missions	Relies on establishment of a technology or program series that can carry evolutionary changes in a sequential, phased manner
Standardization, Reliability	Primarily at the box or subsystem level	Modularity standards dropped down to the "functional slice" level. Some standard subsystems, common form-factors for electronic components, and aerospace industry standard electrical interfaces	Standardization at the **interface only**. Choice technologies, rather than standard that are updated regularly on a well established recurring approach
Spacecraft Operating System	One-of-a-kind, often proprietary implementation	Open architecture for software systems. However, tailored operating system	**Open code architecture** (expanding on SMEX work). Adoption of one or two compatible Operating Systems with interchangeable software modules
Flight Software	One-of-a-kind, often proprietary implementation	Structured software leveraging interface standards. Software re-use from mission to mission	Based on a modular **layered architecture**, with maximum re-use of infrastructure and application modules
Data Transport System	One-of-a-kind, often proprietary implementation	CCDS standards applied throughout. TCP/IP for ground testing	Distributed **internet-based/like** command and data flow

The legacy approach includes a rigorous systems engineering process to translate mission objectives into engineering practice and will result in spacecraft that, either because of the nature of the investigation, or because of their destination, require exacting performance and hence unique designs. Such is the nature of most planetary missions, or large systems with a complex instrument suite. Alternative approaches such as the ones described here can on the other hand satisfy more benign destinations, and are well-suited to the Earth-Moon system. Of course, a communications constellation in orbit around Mars can certainly use the Iridium experience, notwithstanding the greater expense in placing equivalent spacecraft in such orbits. The same can be said if a large number of microsatellites are launched to say, Venus. The system-level redundancy can be made to tolerate certain individual failures, so exacting performance may not be required of each individual unit. Only until a well-established program of recurring flights is implemented, we will have to differentiate between approaches for building spacecraft relatively inexpensively, versus those that cannot tolerate performance

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limitations (hence high cost). If an extended program is established, for instance, for flights to Venus, then the distinction becomes less clear, as the cluster of spacecraft may gather the same, or even more varied bits of information from the Venusian environment.

23.6.4 Point Design Discussion

From an implementation point of view, instantiation of MARS-compatible spacecraft (or equivalently flexible point designs) could make use of *form factors* to constrain a set of mechanical and electrical components all using specific interconnects. This may be better illustrated by borrowing an example from devices we all use extensively today. In the computer industry, the "full tower" and "mini tower" computer cases are intended to fit the *Advanced Technology Extended* (ATX) or Micro-ATX form factors, respectively. Each of these form factor specifications define **mechanical dimensions, mounting points, input/output panel, power, and connector interfaces** between the computer case, the motherboard, and the power supply. Furthermore, these cases would fit standard form factor drives and other peripherals with a set of defined fasteners. How many devices may be fitted within a case depends on its actual size or external form. So a case intended for desktop use is smaller than one intended for use under the desk (the vertical "tower"). If this computer example were to be extended to spacecraft, then we would conceptually have the equivalent "full tower" or "mini tower" system that apply to specific mission classes, at least for a finite period of time. This is because like computer form factors, spacecraft form factors would evolve. In fact, the ATX form factor itself was introduced by Intel in 1995 to replace the outdated AT form factor. Similarly, the "full tower spacecraft" may be replaced by a newer version, once its defining specifications become obsolete. The point is that module or *bus* (if we must use that "bus" word) standards, if specifically referring to form factors, may satisfy in a limited sense the technological needs of a time, but are not by all means intended in this architecture to constrain inevitable advances (i.e., modules are capable of evolving to accommodate advances in technology). As time progresses and changes do occur, these specification standards are indeed replaceable. It is important to stress that form factors can be used to constrain mechanical and electrical interfaces in a particular point design MARS-compatible spacecraft, but must not be intended for instance to constrain the modular components themselves. For this reason, the author prefers the use of "form factor" versus "module or bus standards," to ensure there is no confusion about its defining intent. Evidently, from an application perspective, a particular instantiation of a MARS-compatible spacecraft may indeed make use of particular form factors, as these may emulate the serial (yet more flexible) production of a telecommunications satellite with acronyms such as the "B series," or for that matter, the "model-T" from the automotive industry. As with computer form factors however, at some point the "B series" or the "Model T" will become irrelevant. Until that time, they may well service multiple missions.

Figure 23-20 provides an illustration of a MARS-compatible micro-satellite structure. This particular instantiation defines the mechanical interface among structural components, but leaves positioning and attachment of modular electrical components wide open based on equipment selection. In fact, the modular structure itself is internally reconfigurable, meaning that bay sizes may vary to accommodate component needs. The micro-spacecraft itself accommodates industry-standard electrical interfaces, including Ethernet and RS-422. This particular model series has been dubbed SMART-09 (Small Rocket/Spacecraft Technology), and is a collaborative effort between NASA and the US Department of Defense, Operationally Responsive Space (ORS) Office.

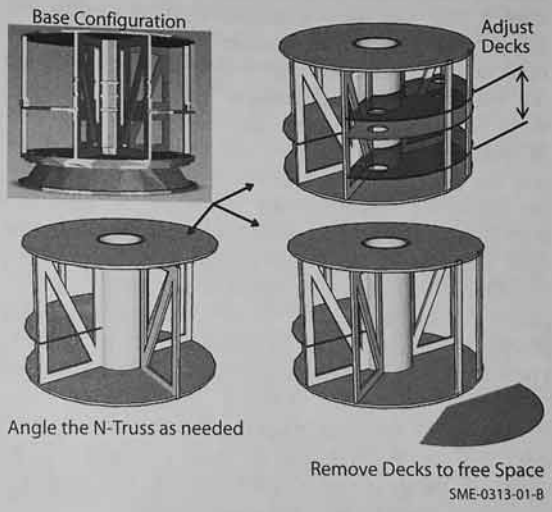


Fig. 23-20. SMART Features a Modular, Reconfigurable Structure.

23.7 Intangible Factors in Manufacturing

Markus Rufer, ***Scorpius Space Launch Company***

Intangible factors in manufacturing are by nature the kinds of things that do not come properly listed, sorted and labeled. Writing about them in a logical, organized way would therefore be very difficult, although daring to do so would be rewarded by beginning to convert them into tangibles. Covering all the intangibles is impossible, because we don't know how many of them exist. Their potentially significant impact on operational results is what prompted this writing. The purpose here is to induce an awareness of their existence and to help the reader develop a sensitivity to be able to detect them in a manufacturing world that is dominated by numbers and controls—meaning tangibles, of course.

The following is a fairly extemporaneous account of my thoughts and observations on this topic, supported by published historical data findable with common search engines. I want to acknowledge that throughout my years in manufacturing across various industries, I have greatly benefited from the works of Tompkins [1989] and Goldratt [1999, 2004]. Their books are not about intangibles per se, but Tompkins' exceptionally sharp account of all the essential requirements for successful manufacturing and Goldratt's enlightened theories have shaped my thinking about manufacturing in many ways. I appreciate their extraordinary contributions in this field and the inspiration they provided me.

This section takes a less conventional look at aerospace manufacturing and its specific challenges. In particular, it highlights a small selection of intangible factors not often discussed, but perhaps no less significant in their impact on manufacturing efficiency and cost. I will use some examples from the automotive industry for contrast, because cars are subjected to fairly open and unregimented market forces—rockets are not; not yet. Why is this distinction important? It actually wasn't important for a long time, because the main actor in space as well as the main customer for space flights was the government, a fact that has profoundly shaped the aerospace manufacturing landscape. As aerospace activities begin to shift toward space commercialization, open competition will begin to affect decision making. Drastic cost reductions for launches and spacecraft fabrication cannot be achieved by tweaking the entrenched practices. Reductions can be achieved only by applying a new mentality, innovative approaches, and the incorporation of advanced enabling technologies. New approaches to seriously reducing the cost for access to space will create advantages for those who dare to be the agents of change in this industry and who will challenge established relationships and organizations.

People with technical backgrounds are used to processing hard data, verifiable facts, and proven science. They prefer quantifiable, measurable numbers over qualitative attributes. For example, we cannot directly quantify the benefits of a worker's loyalty or creativity, so we usually end up discounting them. They are not purposely dismissed, but they don't readily fit a spreadsheet, a chart or a quarterly report and as a consequence, their impact on operational results is invisible or deemed coincidental. Companies that neglect human factors provide good examples confirming this observation. Their typically high personnel turnover is rarely ever identified as a problem and practically never addressed. The resulting costs can be significant, but there are no good tools available to capture them discretely.

Just because certain aspects of manufacturing are made of intangible factors does not mean they don't need to be recognized, understood and addressed. Corporate culture, individual attitudes, habits, inclinations, and motivations can play very significant roles in impacting operating results.

Table 23-14◀, Fig. 23-20◀, Eq. 23-0

23.7.1 Manufacturing Efficiency

Starting this segment with the history of rocket manufacturing would be appropriate here. However, by the time Robert Goddard launched his first liquid propellant rocket, the Ford Model T was already coming to an end after producing over 10 million vehicles. Lean manufacturing methods were already invented—though there were no fancy names for them yet. Unlike cars, which initially got a slow start with very few entirely hand built curiosities, rocket manufacturing started up rather quickly in large numbers as manifested by the German production of the V-1 and V-2 rockets, and then gradually decreased in numbers over the years.

Henry Ford and his right hand man Charles Sorensen were the first ones who successfully took the elements of a manufacturing system—people, machines, tooling, and material—and arranged them in a system for continuous manufacturing for the production of the Model T automobile. Such early systems already included the essential ideas of lean manufacturing, namely a streamlined flow of activities and materials as well as a focus on the elimination of waste. However, the capability to manage product variety, which required changeover, quick setups and small production batches, was not developed until the 1950s. Manufacturing has gone through a significant transformation since Henry Ford started his automobile production and great progress has been made in material sciences, processing methods, automation, warehousing and distribution of manufactured goods.

Numerous very useful books have been written about the subject of manufacturing and in particular about efficiency improvements. More recently, the knowledge about lean manufacturing methods and Six Sigma principles have become nearly indispensable for the professionals in the manufacturing field. As the continuous improvements in all aspects of manufacturing are producing positive results, the industry has become more focused on the measurable, directly cost-impacting elements of these methods. As a result, devising the processes and the tools for measuring the improvements in manufacturing efficiency have become a field of specialized expertise all of its own. Before shedding more light on how effects such as the large scale implementation of improvement programs can shape manufacturing environments, it must be stated that there is a set of essential rules that are the foundation upon which all further efficiency improvements must be based.

Basics

No manufacturing cost reduction effort can be successful unless it is built on the solid ground of these widely proven methods for improving efficiency:

Aim for Simplicity—minimize part numbers, part variety, assembly surfaces, simplify assembly sequences, component handling and insertion, for faster and more reliable assembly.

Standardize—on material usage, components, and aim for as many off-the-shelf components as possible to

allow improved inventory management, reduced tooling, and the benefits of mass production even at low volumes.

Rationalize Product Design—standardize on materials, components, and sub-assemblies throughout product families to increase economies of scale and reduce equipment and tooling costs. Employ modularity to allow variety to be introduced late in the assembly sequence and simplify *Just In Time* (JIT) production.

Use the Widest Possible Tolerances—increase the tolerance on non-critical components and thus reduce operations and processing times.

Choose Materials to Suit Function and Production Process—avoid choosing materials purely for functional characteristics; material choice must also favor the production process to ensure product reliability.

Minimize Non-Value-Adding Operations—the minimization of handling, excessive finishing and inspection will reduce costs and lead time.

Design for Process—take advantage of advanced process capabilities to reduce unnecessary components or additional processing, such as the porosity of sintered metal components for lubricant retention. Design in features and functions to overcome process limitations, such as features to aid mechanical feeding. Avoid unnecessary restrictions of processes to allow manufacturing flexibility in process planning.

Teamwork—promote concurrent engineering. Establish a product or project based development/manufacturing organization involving a formalized multi-disciplinary/departmental teamwork structure.

Include All Stakeholders Early in the Process—get the external contributors involved right at the concept stage. All activities must support a throughput oriented manufacturing approach.

Any organization that can implement and maintain these 9 rules to a meaningful degree is in good shape and has exploited a near full range of widely recognized and proven tools for efficiency improvements in manufacturing. Depending on the nature of the business, further lean principle tools can be implemented.

Mission Assurance Orientation

One key characteristic that may be used to distinguish aerospace manufacturing from other industries is its "mission assurance orientation." To highlight the contrast for this discussion I will call other industries' manufacturing approach "throughput oriented." This is not necessarily tied to specific production output, since a plastic cup manufacturer and a forklift manufacturer are both throughput oriented, even though their output is drastically different. The difference between them and aerospace manufacturing lies mostly in the mentality and the resulting approach to production related challenges. The real requirements (safety) and the perceived ones (political) for near 100% space mission success have elevated the quality standards so close to a zero defects mandate on every level, that the impact of cost lowering efforts are virtually offset. As a byproduct, there is latent confusion between striving for excellence and striving for perfection that often clouds policies by assuming that all defects are equal, and therefore, zero defects are best. Needless to say—the focus on zero defects is counterproductive and unaffordable. Further, the quality requirements originally devised specifically for piloted missions have gradually crept into the supply chains and their contributors of unmanned programs and added significant and unnecessary costs and production lead times. It must be stated here, that the perceived need for near 100% mission success not only drives cost up dramatically, but is also an impediment to technological progress. Choosing a throughput oriented approach vs. a mission assurance oriented approach is not an either/or proposition, but rather states the definition of the desired outcome and the necessary alignment of the supporting activities to achieve that outcome. The concept that a choice has to be made between high quality low-volume production and low quality high-volume production is a false one, as many industries are proving—the cars we drive represent just one good example. (Some of the low cost high-volume-production cars are the most reliable ones, while some of the high cost low-volume-production cars are some of the least reliable ones.) Certainly, economics of scale do not favor the typically low volume production in aerospace manufacturing for cost reduction opportunities, but that does not negate the impact of other approaches that enhance flexibility and resilience to manufacturing disruptions. **In a throughput oriented environment, all contributing elements are subordinated to the goal of achieving a disruption free manufacturing environment, which means that all processes are designed to accommodate that goal.**

In contrast, a mission assurance oriented environment aligns all activities in support of achieving near 100% mission success. The traditional process of building a launch vehicle or a spacecraft consists of a string of idiosyncrasies, using mostly single sourced components that are custom fitted on-line to unique subsystems. There are very few opportunities to benefit from such factors as process repetition, modularity, standardization, and learning. Launch vehicles are integrated mostly in a serial fashion in a product centered layout method in a dedicated environment with fixed production assets. There is often a high degree of interdependent steps and a low off-line content, making the process "throughput adverse." Such an environment is not capable of absorbing unexpected events very well and is therefore extremely vulnerable to disruptions. Moreover, as a consequence of the high content of interdependent processes, the disruptions have a highly amplified impact on cost and schedule, and they reverberate often throughout the entire manufacturing organization and its external suppliers.

In throughput oriented manufacturing operations, the need to get product out the door on time is a hard requirement. You cannot ship Christmas trees in January or Easter bunnies in May! This hard requirement creates important counter balancing input to the lean control systems in a manufacturing organization that tend to minimize material buffers (inventory). In the aerospace

environment, the "controls mentality" gets positive reinforcement from the perceived need for a near zero defects requirement, always in alignment with the near 100% mission success goal. Lean manufacturing systems aim at eliminating waste. Inventory is always a focal point, since it hides waste. Inventory is used to hide the imperfections in the system, which means that the level of inventory is a direct indicator of the level of imperfection in the system. For example, Toyota is running inventory free manufacturing systems, relying entirely on just-in-time deliveries of all needed components. However, Toyota neither single sources its parts, nor does it single source its labor (it is worth noting that unionized labor makes the labor content of a product single sourced), nor does it use hand made custom fitted parts. Toyota's system allows the buffers against disruption to be far removed from the factory, way down the supply stream. To be clear—Toyota running inventory free operations does not mean inventories are eliminated—it means either that the suppliers are carrying inventory for it (suppliers with internal inventory can react to a quality issue or production delay very quickly and can recover literally within hours) or that some suppliers are actually integrated into the factory, and their products are made right there on a just-in-time basis.

In the aerospace industry, the prime contractors are seeking to improve their manufacturing margins by using their considerable leverage to compel their suppliers to carry the burden of buffering their systems, which often exceeds the capabilities of small specialized companies. In addition, aerospace parts are mostly single sourced, custom made, fitted in place, and have long lead times, which moves the need to have flexibility and production buffers back onto the integration floor of the primes. As a result, material buffers become necessary, meaning inventory—the very thing lean methods are trying to eliminate! So, what makes perfect sense on paper, namely reducing material inventories at the prime contractor level and pushing the buffers down the supply stream can have costly consequences in the aerospace industry. Lean manufacturing with a fat inventory?

Actually, just having an inventory of the long lead custom made parts will make a difference. In the aerospace industry, the small, specialized contributors to the supply chain cannot afford to carry sufficient inventory of custom made parts for the large prime contractors—so when there is disruption (i.e., there is a part not conforming to specs) the supplier cannot recover fast enough to avoid disruption of the integration process.

Intangible Factors

Production efficiency improvement efforts and the resulting cost reductions must be quantified, and the essence of the generated data must be applied in subsequent production cycles in order to generate ongoing improvements. Since the implementation of efficiency improvement systems costs money, the results of their application must be measured against return on investment models and their predicted savings. Thus, an operation is required to have the talent and the capacity available for the implementation and operation of continuous improvement systems. Managers, by their very mentalities, are typically drawn to the controllable, quantifiable aspects of the operations they manage. Their jobs are defined to a degree by installing controls and systems, which is in contrast to leaders, who are typically interested in growth and innovation, relying on people's creativity and loyalty. Leaders want to do the right things—managers want things done right.

Managers (not leaders) usually do the hiring of the personnel required to implement and operate control systems, even if the mandate originates from the top of the organization. Following their natural inclinations, managers subconsciously hire new personnel who share their views and attitudes. One unintended consequence of the implementation of large scale improvement systems is a gradual shift in the personnel mix of an organization in favor of the "controls people" at the cost of the "creative people." Over time, this process can transform the culture of a corporation toward becoming more conformance, compliance and controls oriented; all tangible aspects.

As this shift often leads to the desired operating results in the near term, this positive feedback generates further consensus among the leadership about the beneficial impact of these methods. The consequences of this diminishing creative influence within the organization are most likely not recognized until their impacts are severe. Some of the most revered and world renowned companies have experienced a considerable upswing after installing lean methods and six sigma principles, only to fall behind in the global competition within just a few years after their impressive successes. (e.g., Motorola, in the field of cellular phone manufacturing.) That outcome occurred not because the manufacturing efficiency could not be sustained, but because of other factors, not the least of which was diminishing innovative power. The benefits of creativity and innovation are not directly measured as output per unit or in dollars and are therefore not easily captured in a system of controls. One rather perfidious consequence of this type of creeping transformation is that it compels a company eventually to compete on price alone as its products lose their superior performance and best-in-class status. (e.g., Maytag washers.) See more in Sec. 23.7.2.

Another aspect of the problem is that highly specialized industries such as aerospace are vulnerable to the effects of a monoculture of their own making. The agricultural definition of a monoculture actually works well here to make the point: **A farming system given over exclusively to a single product. Its advantages are the increased efficiency of farming and a higher quality of output. Disadvantages include a greater susceptibility to price fluctuations, climatic hazards, the spreading of diseases and the degradation of the soil over time.** In other words—short term optimization is achieved at the cost of long term damage. Unfortunately, launch vehicle manufacturing is subject to the same problem—short term optimization is always bought at the cost of long term damage. A single, homogeneous

aerospace culture without cross industrial diversity and routinely questioned standard ways of doing things has its price tag, which leads us to take a look at how the talent this industry needs is selected and hired. The well known trend of increasing shortages of aerospace technicians, engineers and scientists has not been addressed in a way that generates meaningful results, despite a twenty year (and counting) history of organized reporting about this troubling fact. The narrow technical and specialty oriented selection criteria for recruiting in our industry has most of the relevant talent rotating among a handful of employers who compete for the same or similar space programs, while considerable amounts of corporate knowledge has been lost in mass layoffs and retirement.

The long lasting, ongoing trend of permanently disappearing jobs from various industries and manufacturing companies who used to build all the things that the USA is not producing anymore, meets the ascending aerospace personnel demand that arises from years of neglect to attract sufficient talent for the industry. There has also been a long trend associated with the cyclical nature of business for the primes that trigger big layoffs, which causes many of those terminated to not return to the aerospace industry, further exacerbating the personnel problem. This is an opportunity for the much needed cross pollination of skills and ideas. There are countless scientists, engineers, workers, technicians, assemblers and toolmakers from other industries willing to offer their considerable range of transferable skills for the benefit of the aerospace industry. What is required now is for the aerospace companies to step outside their routines and look into other fields—to reach out and acquire the talents in those fields. Our industry cannot stay competitive unless it is innovation driven and risk eager. That same mentality is required to achieve real cost reductions—which will lead eventually and inevitably to a throughput oriented manufacturing mentality.

23.7.2 The Perpetual Optimization

Getting really heavy stuff all the way up into space is a hard thing to do. Making engines stronger and vehicles lighter is the way to tackle this challenge. Since we have been going to space for about 60 years, performance optimization is now in the DNA of every aerospace engineer. Mass reduction is almost always the dominant issue in most technical discussions and particularly when it comes to choices within the trade space. Despite the frequent mention of cost reduction requirements, the need for mass reduction seems to win the debate every time. The need for performance optimization has become the underlying culture in our industry, upon which most other decisions are made. No 80–20 rule here—unfortunately. As a consequence, the application of scarce metals, rare materials, exotic processes, and extremely specialized capabilities is perceived as a basic necessity to achieve the demanded performance. The impact on cost is nothing less than spectacular. Beyond that, the rare, the scarce and the exotic come from highly specialized sources that serve niche markets with little or no competition and who therefore dictate price, availability and production schedule.

Backing off even a small fraction from the optimum and stepping into "robust design," "large margin," and "throughput orientation" territory has enormous benefits. Even a minimal level of modularization of subsystems, and standardization of interfaces and fasteners will make a significant difference. Adding flexibility to the material flow, using multiple-use tools, movable production assets, wheeled setups, and cross trained personnel who work sliding scale and split shift schedules, are just a few measures that will reduce costly dependencies that are otherwise mandated by the demand for optimization.

Operations Managers and MBA'S

In technical companies, the production managers were traditionally operations people who were educated in the area of business concerned with the production of goods and services. Operations management training programs typically have included instruction in the principles of general management, manufacturing and production systems, plant management, equipment maintenance management, production control, industrial labor relations and skilled worker supervision, strategic manufacturing policy, systems analysis, productivity analysis and cost control, and materials planning.

Over the past twenty years or so, our most prestigious business schools have gradually added sophistication and emphasis to the instruction of the financial aspects of managing an organization. Today, an ever increasing number of companies employ operations managers with MBA centered backgrounds. Symbolically speaking and overstated to highlight the point—MBA's tend to think of assets as things you buy and sell; operations people tend to think of assets as things you maintain and utilize. (The former is usually related to eliminating jobs, the latter is related to creating them.) The need to achieve impressive quarterly results for the market makers is in direct conflict with the requirements for building and maintaining an efficient manufacturing company with lasting substance. For example, our Detroit auto manufacturers are intimately familiar with the Toyota manufacturing system and understand its superior performance. Theirs was a long term plan with long term investments for long term benefits—which is one reason why US car manufacturers were slow to implement it, since their focus is on measurably improved results every quarter instead of performance over a production model life cycle of 5–7 years. (Principle 1 of the 14-principle book The Toyota Way for employees: **Base your management decisions on a long-term philosophy, even at the expense of short-term financial goals.)** In aerospace, the same rules must apply. We need to understand and lay open the trades—what is the long term cost to achieve a short term benefit? Who pays for it and why?

The comments made earlier regarding operations managers and MBA's were not meant to imply that one is to be preferred over the other, but rather to highlight

some of their possible intangible aspects like background, education and mentality—and as a consequence, to understand what kind of people they would likely be hiring and how that may impact business results over time. In any case, it's worth giving some deep thought to whether we want to be in the business to make a killing—or to make a living. Aerospace manufacturing is particularly suited for the latter—not to mention that it has critical implications of national interest that cannot be addressed with a short term perspective.

Cost Cutting vs. Outperforming

American manufacturing companies have, over a long period of time, competed for international market share by pricing policies. As a result, cost cutting has established itself as the central theme in manufacturing. That strategy has worked fairly well because the United States itself is a very large market, and domestic deliveries have traditionally generated a large portion of company revenues. Smaller countries that had a difficult time gaining global market share based on pricing made up for it by innovation and specialization for niche markets. Fine precision mechanics, machinery, vehicles, chemicals, household equipment, optical instruments and machine tools are just a few examples of goods coming from smaller countries that have earned leading positions on the global manufacturing stage. To illustrate the impact of choosing between competing on cost vs. performance we might take a look at Germany, who at the time of this writing, is in the process of displacing China as the largest exporter in the world, selling mostly high-cost high-tech products, with many of them being the most expensive ones on the market in their respective categories. Germany is also the leading producer of wind turbines and solar power technology in the world; clean energy technologies that are expected to become some of the largest global economic sectors for decades to come. In the USA, the continuously increasing outsourcing for manufacturing has diminished our involvement in manufacturing engineering, which in turn has affected our R&D capabilities, since the two are tied in a feedback loop in most cases. The countries that are now manufacturing the goods that we buy have the long lasting benefit of gaining the expertise in manufacturing engineering and R&D, which creates the opportunities and incentives for young people to become engineers—who will in turn become tomorrow's technology innovators.

If existing trends continue, it's a matter of a relatively short time period until other nations will be in a position to outperform us in aerospace (not to be confused with undercutting us in price). We don't have to start working for $2/hour to compete with them. We have to outperform them—with innovation and ingenuity, which can be achieved only if we manufacture the components and systems domestically, so that we can grow superior R&D and manufacturing engineering capabilities again. This requires a consistent and long-term commitment to superiority in space as a national policy. Our policy makers must be educated about the extent to which our nation's security and well-being is dependent on broad-based and highly advanced manufacturing capabilities.

Back to the Basics—The Intangible Ones

Here are 5 basic rules or guidelines for the manufacturing manager that can help with keeping a high level of awareness of intangibles:

1. Manufacturing managers must fight vigorously the forces in the organization that demand quick fixes and short term optimizations. The alignment of the contributing factors to the main goal is paramount—not how well the individual measures produce immediate results. However, keen observers are aware that the annals of manufacturing history are full of records of such lost battles, as neither our political system nor our capital markets are used to anything but a short-term view. So here is an important tool to use: There are indeed shortcuts to be offered to the executives who demand them, but they each have a price tag and a payment due date, and it is the job of the manufacturing manager to present clearly the impact of such choices. Shortcuts never come for free, and the executives need to be informed of their long term costs. They must take documented ownership of these decisions because of their potentially profound impact on the well-being of the whole organization. Long term goals cannot be achieved with short term plans.
2. Always keep the 80–20 rule in mind. In manufacturing, you are the last in line. You are expected to make up for the time that was lost in concept development, design, engineering, and product definition. You will never have enough time to do what needs to be done, because the sales people sold a product that you don't have and promised a delivery date that you can not achieve—that's a universal law in manufacturing. Make sure you provide visibility on production progress (or setbacks) and realistic delivery dates to your sales people, so they can manage the customer's expectations accordingly and keep the executives appraised, so they can manage cash.
3. Always be alert to the fact that manufacturing has two main enemies: uncertainty and disruption. They dwarf everything else by comparison—there is your 80–20 rule. Uncertainty can be caused by such factors as financial constraints, environmental impacts, political or economic considerations. Disruption can be caused by such factors as unreliable supply sources, insufficient product definition, and deficiencies in the master schedule for production. Treat these two, uncertainty and disruption, as the main issues and everything else as noise, and you will be highly effective. Continuous production is the key. This word must be branded in your brain—CONTINUOUS.

Table 23-14◀, Fig. 23-20◀, Eq. 23-0

4. The elimination of waste (lean principles) and the reduction of defects (Six Sigma) are not an all-or-nothing package. They can be implemented stepwise in modules, and they can be mixed and matched to specific needs. The former improves efficiency, and the latter improves process. Not every manufacturing company needs to implement all the elements of a total continuous improvement system. More is not necessarily better; in fact, most companies will do fine with a small selection of basic controls. Highly regimented environments with optimized controls can reach a point of diminishing flexibility and reduced tolerance to disruptions. At this point, the cost savings with respect to lean operations can be outweighed by diminished resistance to cost-impacting disruptions.

   Front end, preventive solutions that are designed to reduce inefficiency (Lean manufacturing, Six Sigma) must be complemented with back end solutions (buffers) to reduce the impact of unplanned disruptions.

5. Consider the human factors—create an environment that is designed to match workers motivations, attitudes and expectations—factors that affect the outcome of a process. Make sure that equipment and environmental design, as well as training, create conditions that are attractive to personnel. Make your employees stakeholders. Create long term oriented incentives (short term incentives are mostly destructive to an organization). Hire people who represent individual differences across the human spectrum in every physical and mental dimension that is relevant for good system performance. Diversity is not a political term in a manufacturing operation; it is a way of enhancing creativity and fostering innovation. Attract creativity and inspire innovation—it's the engine that will propel a manufacturing organization to the forefront of opportunities. It is also the only sustainable force of the new economy.

Connecting the Dots with a FireSat II example

The speculation about how such thinking would apply to a FireSat II production program and its possible impact on the outcome in terms of cost, schedule and commercial viability leaves room for various scenarios. So here is one attempt to incorporate some of the discussed ideas:

The vision offered here for consideration includes forming an ad hoc cooperative (COOP) among the stakeholders of a FireSat II program—not unlike some healthcare provider partnerships, where a number of doctor-owned medical offices share the income from patients and insurance reimbursements. Surplus money goes into a profit pool that is used for rewards at the end of the year. The FireSat II COOP would include the teams that build systems or subsystems such as the bus structure, payload, bus sensors, avionics, command and data handling electronics, communications, the propulsion system, the launch, and ground controls. Traditional hierarchies would be eliminated, and teams would be horizontally connected and communicate by means of real-time video communication tools on demand. The team members are all shareholders in the COOP, and their efforts to under-spend the allocated budgets will be rewarded by distribution of surplus funds from the pool after each successful deployment and commissioning of a FireSat II. Mission failures will lead to cancellation of the rewards, and the profit pool will help support self-insurance funding for replacement costs. This arrangement of co-ownership also provides incentives for all participants to share resources and capabilities.

Basics—Apply a *design for manufacturing* (DFM) and *design for assembly* (DFA) mentality at the concept stage. Materials and process choices should also take logistics, availability (vulnerability to disruption), and other flow-affecting factors into account. A word about standardization seems to be of particular importance at this point. One of the biggest achievements in the field of standardization in modern history has to be the invention of the 40 foot shipping container. There is not a nation or entity in the world that has not adapted their environment to its use for the transportation of goods by road, by rail, by sea and by air. In aerospace, it seems to be an established standard that all hardware and software design is mission specific, and that every "trip" has to be designed from the ground up. Imagine you would need a different car for every trip you make! The blatant trivialization here is intended to make the point that at least some degree of standardization and modularity for "space shipments" will have to be implemented if commercialization of space is to be an achievable goal.

Throughput Orientation—Reduce the requirements catalog to the essential functions of the product. Design multi-functionality into your production environment, *not into your product*. The former reduces cost; the latter increases cost. Aim for modularity in the preliminary design phase. Obtain consensus on the product definition early to allow good production planning and coordination among the coop members. Defining the interface data for all involved parties and their products very early in the process will be central to achieving that goal. Interdependent processes and serial steps must be minimized. *Disruption* and *uncertainty* are both primary cost driving elements in a manufacturing organization and must receive constant attention.

Optimization—Apply a robust workhorse design standard as a baseline. Mitigate system redundancy requirements by using large margin design approaches that provide robustness in trade for optimal performance.

80–20 Rule—The decisions made in the first 20% of the program determine 80% of the overall cost performance. That means you have the *concept stage* and the

preliminary design phase to make an impact. Decisions made after this point are practically inconsequential on cost containment. In fact, just the opposite occurs—the later in the process design changes are implemented, the bigger and the more disproportional the impact on cost increases are. All members on deck for this one—no exceptions! Cost cannot be reduced or contained at the back end of the program; cost is designed into the product, and into the process that produces it.

Continuous Manufacturing—More important than actual production quantity is the requirement to maintain a continuous production mode. Continuous manufacturing demands a firm commitment to a build quantity and the provision of the necessary resources for the entire program to avoid disruptions. Single sourcing must be avoided or reduced by any and all means. The supply of unique components and scarce materials must be buffered by small inventories. Remember, all vendors are stakeholders, and whoever creates the program disruption spoils the pool for all!

Product Quality—The FireSat II is a low-cost unmanned mission design. An average projected 90–95% mission success rate would be desirable to create room for meaningful cost reduction opportunities. The product design and the quality requirements should be defined accordingly. Note: Higher quality is not "better" and lower quality is not "worse". Quality is designed into the product and the processes that make the product. The quality standard must be in alignment with the requirements of the end user—no more and no less. The missions could be self-insured by the COOP and the payload replacement costs of failed missions could be partially carried by the profit pool.

Business Model Innovation—Finally, as a business model for the FireSat II product line, one can imagine that offering the data services on a leasing basis could add a near term profitability dimension that would attract the involvement of financial institutions such as equipment leasing companies or lease/finance banks. This model could have a significant impact on the success of the commercialization of the space business in a larger context. Smaller entities who may not have the budget to buy a FireSat II would be leasing the services they need from the satellite operator. The availability of space equipment financing and space-based services leasing could become a profit proposition for the participating lenders. This approach also addresses the fact that obtaining funds for capital investment is usually much more difficult for typical customers to achieve than increasing existing operational budgets, which is where the FireSat II leasing program would fit.

The discussion of this section comprises only a small cross section of a wide range of cost impacting intangibles in manufacturing. It is worth noting however, that intangibles are not only difficult to quantify, but they are often the slowly creeping factors that can escape the attention of even the most alert manager, until their impact becomes severe. The purpose of this discussion is to help develop the senses and sharpen the alertness of the manufacturing manager to recognize intangible factors in the daily routine of looking over lists, charts, graphs and reports. They may not fit your spreadsheet—but they are no less important.

Table 23-14◀, Fig. 23-20◀, Eq. 23-0

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Table 23-14◀, Fig. 23-20◀, Eq. 23-0

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24 Risk and Reliability

24.1 Reliability
Definitions and Basic Concepts; Recent Spacecraft Reliability Experience; Design Process for Reliability; Software Considerations; Redundancy and Fault Tolerance; Human Factors Considerations; Reliability Testing

24.2 Space System Risk Analysis
Risk Assessment Throughout the Design Process; Risk Management; Identifying the Risks; Qualitative Risk Assessment; Quantitative Risk Assessment; Fever Chart Quantitative Risk Assessment, Probabilistic Risk Assessment; Expected Productivity Analysis; Summary

24.1 Reliability

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When a bulb in our desk lamp burns out, it is easily replaced. When the switch that controls the bulb fails, the replacement is not quite as simple but still within the capabilities of most mechanically inclined teenagers and even some adults. We expect a higher reliability of the switch than of the lamp because it requires more effort to repair a failure. When a spacecraft command receiver fails on orbit, it takes an extraordinarily long screwdriver to fix it. You get the general idea: the command receiver has to be much more reliable than the light bulb or the switch on the desk lamp. This need for very high reliability in all parts and subsystems of a spacecraft is the basis for including a *reliability program* in most space projects.

24.1.1 Definitions and Basic Concepts

Before describing the details of the reliability program, let us briefly discuss the meaning and metrics of *reliability* in the context of space missions. A common definition of reliability is "the probability that a device will function without failure over a specified time period or amount of usage" [IEEE, 1990]. If the phrase "without failure" is taken to mean "without failure of any kind" it defines *basic reliability;* if it is interpreted as "without failure that impairs the mission" it defines *mission reliability.* In spacecraft that employ extensive redundancy there can be a significant difference between these two reliability metrics. Mission reliability is the more important concept, and when we use "reliability" without a qualifier, it always means mission reliability.

Item Reliability

The elementary expression for the reliability as a function of operating time, t, for a single item subject to random failure, is given by an exponential distribution:

$$R = e^{-\lambda t} \qquad (24\text{-}1)$$

where λ is the failure rate and $1/\lambda$ is the Mean Time Between Failure (MTBF). Here, R is the probability that the item will operate without failure for time t (success probability).

In terms of absolute numbers, electronic components usually make up the majority of parts on a space vehicle. Because of the high quality of such parts used in spacecraft, their failure rates are usually random. For solar cells or subassemblies having moving parts (e.g., reaction wheels, solar array drives, or propellant valves), wear-out is a major failure mechanism, and these components have an increasing hazard (the failure rate normalized to the number of survivors). In such cases, the Weibull distribution, is used. One form of that distribution is

$$R = e^{-(\alpha t)^{\beta}} \qquad (24\text{-}2)$$

where α is called the *scale parameter* (with the same units and therefore somewhat akin to λ in the exponential distribution) and β is called the *shape parameter* (if less than 1, the failure rate *decreases* over time, if 1, the failure rate is constant, and the distribution assumes the same form as the exponential, and if greater than 1, then the failure rate increases over time).

Other distributions commonly used in reliability engineering are the Gamma, Normal (Gaussian), Lognormal, binomial (for discrete processes), and Bernoulli (also for discrete processes). Further information on the form and use of these distributions can be found in MIL STD 338B [DoD, 1998] or in the NIST/SEMATECH [2010], reliability handbook, as well as many industry standards and reliability textbooks.



The converse of reliability is the probability of failure. If there are only two outcomes: success and failure*, probability of failure, F, is given by:

$$F = 1 - R \qquad (24\text{-}3)$$

System Reliability

For a spacecraft subsystem or assembly made up of n non-redundant non-recoverable elements, all equally es-

* Methods of assessing mission success with degraded states, are described later.

sential for spacecraft operation, the *system* (or *series) reliability*, R_s, or success probability to time t (where t is the time since the start of operation), is

$$R_s = \prod_1^n R_i \tag{24-4}$$

where R_i ($i = 1, 2, \ldots n$) is the reliability of the individual elements. Where the reliability behavior over time is exponential, this becomes

$$R_s = \prod_1^n R_i \approx e^{-\sum_i^n \lambda_i t} \tag{24-5}$$

and λ_i the failure rate of the individual elements. Note that the right-hand equality holds true when the exponential distribution is used to represent reliability because the elements are believed to have a constant failure rate. Where a system consists of n elements in parallel, and each (i.e., any one) of these elements can by itself satisfy the requirements, the parallel (or redundant) reliability, $R_{s,p}$ is given by

$$R_{s,p} = 1 - \prod_{i=1}^n (1 - R_i) \tag{24-6}$$

where the reliability of the parallel elements is equal, say R_i. the above simplifies to

$$R_{s,p} = 1 - (1 - R_i)^n \tag{24-7}$$

In some cases, more than a single element is needed to satisfy the mission requirement (e.g., 2 out of 3 transmitters or power supplies), the reliability of such a "m out of n", where m is the minimum number of elements needed and n is the number that were on the vehicle at the beginning of the mission, the system reliability is given by

$$R = \sum_{k=m}^n \frac{n!}{k!(n-k)!} R_i^k (1 - R_i)^{(n-k)} \tag{24-8}$$

In terrestrial applications it is customary to distinguish between active and quiescent (dormant) failure rates, the latter being about one-tenth of the active rates. This reduction accounts for the absence of electrical stress when a component is not energized. However, in many cases, space components are derated (see Sec. 24.1.3) so that the failure probability due to electrical stresses even in the active mode is quite small. The distinction between active and quiescent failure rates is therefore much less important for spacecraft environments in systems with conservative design margins, with respect to power dissipation, the electrical stress induced by power is very low.

Series and parallel constructs are often combined in a satellite system and can be evaluated by combining the effects from Fig. 24-1 which shows the reliability predictions as a function of time for three possible system configurations each consisting of four subsystems:

1. ***Single System:*** A single string system consisting of the four subsystems placed in series. Eq. (24-2) and Eq. (24-4) would be used to calculate the system reliability.
2. ***System Redundancy:*** A dual redundant subsystem consisting of two strings, each consisting of the four subsystems in series; this is sometimes referred to as large scope redundancy or block redundancy. Eq. (24-2) and Eq. (24-4) would be used to calculate reliability of each string, and this string result would then be used in Eq. (24-6) to calculate the system reliability.
3. ***Partitioned Redundancy:*** A system in which each of the four subsystems is configured as a dual redundant system, and each of these subsystems is placed in series to calculate the system reliability. This configuration is sometimes referred to as a cross-strapped system or small scope redundancy. Eq. (24-2) and Eq. (24-6) would be used to calculate reliability of each dual redundant subsystem, and these results would then be used in Eq. (24-4).

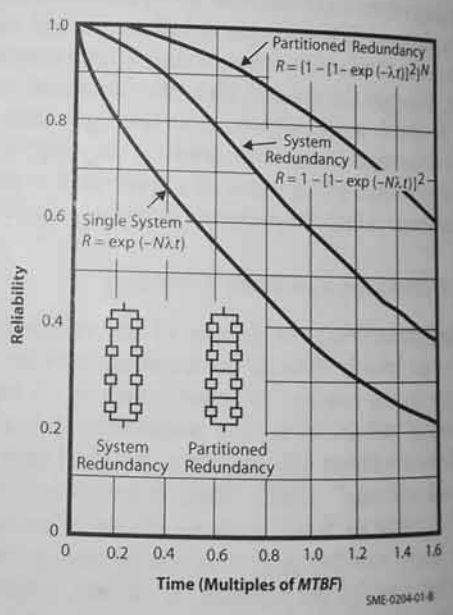


Fig. 24-1. Effect of Partitioning on Reliability, *t* is the Time from the Start of the Mission, *R* is the Mission Reliability or the Probability that at Least Essential Mission Elements will Survive.

The general equations resulting from the substitutions are also shown in the Fig. 24-1. The partitioned redundancy curve refers to the original (single) system being divided into quarters and then each quarter made redundant. The redundant and the quarter-partitioned curve involve (at least superficially) the same resources (in each case twice those of the single system), and yet the quarter configuration has a pronounced reliability advan-

Table 24-0, Fig. 24-1, Eq. 24-8

tage, particularly at the longer time intervals. It would therefore seem that redundancy of small scope would be preferred over large scope. A more detailed analysis would, (considering the reliability of the switching elements and other consequences of small scope redundancy for the specific system under consideration) be necessary to confirm or refute this general conclusion.

Reliability Predictions for Systems Composed of Electronic Components

Reliability Predictions for electronic components (the bulk of the satellite hardware) are performed by adding component failure rates to determine the aggregate failure rate of a higher level assembly or subsystem. This approach has been formalized in military standards such as MIL-HDBK-217F [DoD, 1991] and is a special case of series system reliability analysis for very low failure rate components. Table 24-1 shows typical values for some classes of electronic parts that are space qualified. The failure rates shown in the table are much higher than those for other handbooks for predicting electronic part reliability developed for the telecommunications industry (e.g., Telecordia RS-332 [2003], IEC TR 62380, [2004]). Non-space qualified parts have failure rates that are much higher (between 12 and 333 times).

The failure rates of the individual electronic components (sometimes called piece parts) are affected by factors such as temperature and application (i.e., electrical stresses placed on the part by the environment). Formulas are provided for adjusting the baseline failure rates in DoD [1991]. This method of reliability estimation may be used for comparison of design alternatives, when test or on-orbit incidents indicate insufficient reliability, or when a more expensive payload is to be incorporated (which increases the cost of failure).

As was noted at the beginning of this section, the method of adding failure rates is an approximation of a series system reliability and, as noted above, is valid for high reliability components. The following discussion shows the mathematical basis for this method. Where the quantity λt is small, the following approximation based on the Taylor Series expansion of the exponential function is frequently used

$$R = e^{-\lambda t} \approx 1 - \lambda t \tag{24-9}$$

Therefore

$$F = 1 - R \approx \lambda t \tag{24-10}$$

and for a system s made up of a number of n components in series, the failure problem becomes

$$F_s = \sum_{i=1}^{n} \lambda_i t \tag{24-11}$$

If the operating period, t, is the same for all components, the failure rate of the system becomes the sum of the failure rates of the individual components, that is

$$\frac{dF_s}{dt} \equiv f_s = \sum_{i=1}^{n} \lambda_i \tag{24-12}$$

where λ_i represents the failure rate of the ith component of the system.

Design Life

Design life is the intended operational time and an important vehicle-level parameter which is usually the key programmatic level system reliability requirement. De-

Table 24-1. Representative Piece Part Failure Rates (failures per 109 hours) from MIL-HDBK-217F.

Part Type	Failure Rate, × 10^{-9} per hour		Notes
	Space Flight Enviro.	Launch Enviro.	
Bipolar Gate Logic Array	0.9–19	17–200	Max 60,000 gates
Bipolar Microprocessor	7–27	60–215	Min 8 bits; Max 32 bits
MOS Microprocessor	12–47	70–250	Min 8 bits; Max 32 bits
MOS SRAM	2–11	24–75	Min 16 K, Max 1 M
Bipolar SRAM	2–8	30–75	Min 16 K, Max 1 M
Diodes General	1.3	170	
Transistors General	0.05	5F	
Transistors RF Power	165	900	
Resistors	0.01	1	Composition
Capacitors	0.1	10	
Relays	40	6,000	

sign Life determines the amount of consumables (e.g., control gas) that must be provided as well as the quality and test requirements for items subject to wear out, such as bearings and batteries. The mission reliability calculated at the design life is the *mission success probability*. Since this quantity is less than 1.0, the *expected life* is less than the design life. The customary measure of the expected life is the *mean mission duration* or *MMD*, given by:

$$MMD = \int T dR \qquad (24\text{-}13)$$

where T represents a horizontal (time) line in the shaded area of Fig. 24-2 and dR is the associated (vertical) increment in reliability. MMD is a frequently used figure of merit for spacecraft reliability improvement programs. When the spacecraft does not have any redundant elements, a plot of reliability vs. mission time will be the exponential curve, but when redundancy is provided the initial part of the reliability plot tends to be convex to the origin. This accounts for the shape of the curves in Fig. 24-2.

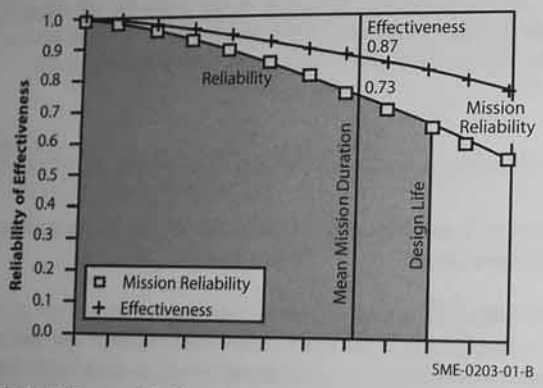


Fig. 24-2. Frequently Used Reliability Concepts. Design life is governed by wear-out and expendable stores. Mean mission duration is less than design life because failures can terminate a mission before end-of-life conditions are reached.

Degraded States

Most satellites perform multiple functions (e.g., weather observation and tracking ocean traffic), and performance of any of these functions may still be useful even when it is degraded. Under such circumstances the success criteria are no longer obvious. Should failure be defined as the event when the spacecraft fails to furnish any useful data, or should it be the point at which its performance deviates from any part of the specification? This problem can be addressed by establishing multiple reliability requirements. Assume that the specified weather observations are intended to use three frequency bands but that observations on even a single band are still useful. The success probability (SP) requirement may be formulated as shown in Table 24-2.

During the development of the satellite it may be found that it will be difficult to obtain a success probability for all three bands operational greater than 0.80 but that the single band mission reliability can be raised to

Table 24-2, Fig. 24-2, Eq. 24-16

Table 24-2. Hypothetical Reliability Requirement for a Weather Satellite.

Condition	Success Probability Requirement
At Least 1 Band Operational	0.95
At Least 2 Bands Operational	0.92
All 3 Bands Operational	0.85

0.99. If the requirements shown in Table 24-2 are part of a contractual document it may take lengthy negotiations to agree on the change, even though it may be technically quite acceptable. To avoid such situations, *mission effectiveness* rather than reliability may be specified. There are several definitions of mission effectiveness [DoD, 1998], but all aim at a single metric that represents reliability weighted by the operational capability level to which the reliability is applicable. In the above example assume that the 1, 2, and 3 band capabilities are assigned weights of 0.25, 0.35, and 0.4, respectively (it is desirable that the weights add up to 1.0). A simple mission effectiveness requirement may then be stated as $0.25 \times 0.96 + 0.35 \times 0.92 + 0.4 \times 0.85 = 0.902$. A similar calculation shows that raising the single band reliability to 0.99 will not compensate for reducing the 3 band reliability to 0.80, but that this reduction could be tolerated if the 2 band reliability is raised to 0.95 at the same time. Where there is a sound technical basis for specifying mission effectiveness it will generally reduce both cost and development time compared to specifying multiple reliability values.

Availability

In cases where failures are recoverable (e.g., radiation events, power transients, switchover to a redundant element, and software failures), two new parameters become important: recovery time and availability. Recovery is the time necessary to restore the vehicle to an operable state. Recovery can generally occur automatically with all mechanisms on the vehicle (often called "autonomous recovery") or with intervention from the ground control station. Usually, autonomous recovery can occur much more quickly (often in seconds).

Availability can be thought of as the proportion of total time that the vehicle is operational, i.e.,

$$A = \frac{Uptime}{Total_Time} \qquad (24\text{-}14)$$

where Total_Time is the sum of uptime and downtime. Thus, unavailability i.e., the probability that the system is not available is

$$U = 1 - A = \frac{Downtime}{Total_Time} \qquad (24\text{-}15)$$

If both failure and recovery times are exponentially distributed, it can be shown that

$$A = \frac{\lambda}{\lambda + \mu} \qquad (24\text{-}16)$$

where λ is the failure rate and μ is the reciprocal of the average recovery time (i.e., the "recovery rate"). Like reliability, availability is a probability, and hence availability analogs exist for the series, parallel, and m out of n system level reliability, see Eq. (24-4) through Eq. (24-8). Thus, for example, the availability of an n-redundant system (similar to the reliability equation shown in Eq. (24-6) is

$$A_{s,p} = 1 - \prod_{i=1}^{n} (1 - A_i) \tag{24-17}$$

and the availability of an m out of n system similar to the reliability equation shown in Eq. (24-8), is

$$A = \sum_{k=m}^{n} \frac{n!}{k!(n-k)!} A_i{}^k (1 - A_i)^{(n-k)} \tag{24-18}$$

24.1.2 Recent Spacecraft Reliability Experience

In a study of 1,584 Earth-orbiting satellites successfully launched between January 1990 and October 2008, Castet and Saleh [2009] found that at the vehicle level, there was a non-constant failure rate over the vehicle life. During the first year, there was a 3% loss with a 6% loss of those surviving after 6 years, and 8% loss of the survivors after 10 years. The authors reported the contributions of failures by subsystem as a function of time as seen in Table 24-3.

Over 15 years of operation, the control processor (CP) contributes approximately 6% to the total satellite failures, the attitude control system contributes 20% and the Telemetry, Tracking, and Command (TTC) subsystem contributes 16%. Functions related to electrical power (batteries, solar arrays, and electrical distribution) account for between 30% and 40% of the failures throughout the vehicle lifetime. For the first 10 years, TTC is the lead contributor to satellite failures.

Table 24-3. Frequency of Spacecraft Failures.

Subsystem	30 days	1 year	5 years	10 years
Attitude Control (Gyro/Sensor/Reaction Wheel)	8%	12%	11%	18%
Thruster/Fuel	13%	20%	16%	13%
Control Processor	0%	0%	4%	5%
Mechanisms/Structures/ Thermal	8%	10%	11%	7%
Payload Instrument	0%	2%	3%	4%
Battery/Cell	6%	2%	10%	6%
Electrical Distribution	12%	8%	10%	11%
Solar Array (Deployment and Operating)	25%	17%	12%	12%
Telemetry Tracking and Command	22%	23%	18%	19%
Unknown	8%	6%	5%	5%

24.1.3 Design Process for Reliability

The process for designing a reliable system should start at project conception. The reliability engineering function depends extensively on information and support from other project functions including system engineering, subsystem design, software development, parts material processes, quality assurance organizations, and system safety. There are multiple standards that describe the conduct of a reliability program in the space industry [DoD, 1988] or [ECSS, 2009a]. Table 24-4 describes the key reliability program activities by mission phase, and Table 24-5a and 24-5b shows the interactions, roles, and responsibilities among these project functions.

A reliability program plan should be written as part of the initial program planning function to define both the activities of the reliability engineering function and its interaction with the rest of the program. In large programs, this may be a separate document but in smaller programs, it can be a part of the system engineering management plan (SEMP). The plan should describe how reliability objectives are to be allocated (or reference a separate reliability allocation document), assign responsibility for achieving them, and establish milestones for evaluating the achievements. It also serves as an agreement with other spacecraft functions regarding their responsibilities in support of reliability.

Failure Modes and Effects Analysis

The Failure Modes and Effects Analysis (FMEA) is a qualitative analysis in which each component of a system is posited to fail, and the effects of that hypothetical failure on the surrounding components and the system level are evaluated. The benefits of the FMEA are:

- Identifying single points of failure and associated critical items
- Associating underlying failures with observable symptoms that can be used for health monitoring and diagnostics (if measurement and telemetry points are defined)
- Evaluating the effects of existing compensating provisions (e.g., on-board failure detection and recovery) physical, functional, or temporal redundancy

Several styles of FMEAs have been defined in standards such as DoD [1984], SAE [2001], and ECSS [2009b]. An FMEA can be performed at multiple levels of abstraction (called "indenture levels" in MIL STD 1629A). The lowest level is the part level (discrete parts and each pin of an IC). For a capacitor the failure modes are open and short (sometimes a change in capacitance may also be recorded). Failure effects are assessed at the part level and also at higher levels such as assembly and major component or subsystem. The failure effect of a shorted capacitor may be a bias shift on a transistor (part level) which in turn cuts off the output of a demodulator (assembly level) and causes loss of voice communications (subsystem level). In our

Table 24-4. Key Reliability Program Plan Activities.

Development Stage	Activities
Concept Development	Identification of reliability standards and methods; definition of quantitative requirements for reliability and availability based on operational concepts, definition of degraded modes and measures of effectiveness
Preliminary Design	Preliminary allocation of top level quantitative requirements to subsystems; definitions of operational modes and states; definitions of failure modes, effects, and compensations; definitions of reliability test plans; identification and justification (including mitigating actions) of single points of failure
Detailed Design	Detailed allocations of quantitative reliability and availability requirements to all elements of the space vehicle (to include both hardware, software, and data); verification of failure detection, recovery, and reporting provisions
Development and Fabrication	Ensuring conformance with the hardware and software design provisions; monitoring of problem and discrepancy reports for unanticipated failure behavior
Integration and Test	Review of test results for conformance or deviation from expected failure detection and recovery behavior; review of anomaly reports for unanticipated failure behavior; analysis of quantitative data for conformance with quantitative failure rate, recovery time, and recovery probability requirements
Operation	Review of data from on-orbit anomalies and failures for incorporation into future requirements

example, the detection may be by an output monitor included in the demodulator, and the failure mitigation may be use of an alternate voice channel. The FMEA may then be conducted at the function level, where functions are timers, counters, and shift registers, or at even higher levels. The advantage of performing an FMEA at higher levels is that there are usually fewer items that must be analyzed. The disadvantage is that failure modes may be less well defined, and effects may be more complicated to analyze.

Criticality Analysis can be added to the FMEA and it then becomes Failure Modes, Effects and Criticality Analysis (FMECA). Criticality here refers to the combination of frequency (or probability) of occurrence and the severity of the associated system effect. Thus, among two failure modes that produce the same system effect, the one having a higher probability is the more critical one. In the FMECA a probability is associated with each failure mode, and the probabilities of all failure modes that cause a given effect are added in a summary section of the FMEA. Loss of modulator output cited in the example may be caused by several failure modes of the demodulator components, and loss of the voice channel may be caused by failure modes other than those in the demodulator. Although there may be considerable error in the estimate of a given failure mode, these tend to be evened out when arriving at estimates of failure effects at the subsystem and higher levels. In digital microcircuits it is usually not possible to conduct FMECA at the level of primitive elements (gates or transistors) because there are too many of them and because causes of failure may affect multiple elements (e.g., voids in the oxide layer) in such cases, the FMECA must be performed at a higher level.

As noted above, failure modes at the function level are generally not as well known as those at the part level, and a single cause of failure may affect multiple functions. In assessing the effects of failure at the higher level it may therefore be advisable to be conservative.

Table 24-4, Fig. 24-2◀, Eq. 24-18◀

Other Reliability-Related Design Analyses

Other analyses that are commonly performed on space systems are:

- **Sneak Circuit Analysis (SCA),** which searches for unintended current flows. This analysis can be used to show that explosive or other one-shot devices will not be accidentally actuated, and that they will always be actuated when intended.
- **Worst Case Analysis (WCA),** which assumes that all components of the system are at their most pessimistic operational state (i.e., batteries at low voltage, resistors at either extreme of their tolerance range, signals at their lowest non-failed level) to assess whether the system as a whole will continue to function.
- **Thermal Analysis,** which determines the temperature at various points of the spacecraft. Thermal analysis is particularly important to piece part reliability prediction because failure rates can increase exponentially as a function of (absolute) temperature due to the *Arrhenius relation*.
- **Electromagnetic Interference / Electromagnetic Compatibility,** which analyzes for the presence of electromagnetic interference among the SV elements.

Failure Reporting

The *Failure Reporting and Corrective Action* (FRACAS) is a key element in any reliability program and it should cover both the pre-launch test and the on-orbit phases because:

1. It informs stakeholders that a failure has been observed

Table 24-5a. Interaction of Project Functions in Development of Reliable Space Systems—Concept Development through Detailed Design.

Project Function	Concept Development	Preliminary Design	Detailed Design
System Engineering	Requirements for reliability and dependability defined, complete, verifiable, consistent; definition of mission duration, size, weight, and power constraints; identification of space environment issues (e.g., radiation) based on orbit and operations constraints that affect reliability; identification of key reliability and availability issues connected to choices of technologies (e.g., propulsion, sensors, and power systems)	Requirements allocated to subsystems and disciplines (hardware, electronics, mechanical, thermal); derived requirements considering mission and operational environment defined. Trade-off studies performed where necessary on performance and redundancy requirements vs., size, weight and power constraints; definition of vehicle-level failure detection and recovery strategy and architecture including identification of failure containment and propagation boundaries	Design features traced to requirements; additional derived requirements for failure detection and recovery identified as design emerges; ensuring that proper margins are maintained for mechanical and electrical stresses and for computing and communication response time, error rate, and capacity factors
Subsystem Design	Extent of redundancy defined; major failure mechanisms and reliability design concerns identified; subsystem recovery strategies	Failure detection and reporting design provisions and telemetry defined; constraints for failure behavior at subsystem boundaries defined; diagnostic and built-in-test (BIT) provisions identified	Subsystem design incorporates requirements and architecture for failure detection and recovery; selected components are appropriate for electrical, thermal, and mechanical stresses; processors and communications lines have sufficient capacity; modular redundancy constraints are satisfied for switch over
Computer System Design and Software Development	Computer system architecture (including internal networks) accounts for recovery time and autonomous operational requirements of the mission; where multiple computer systems are used (e.g., payloads vs. bus), failure containment and recovery regions are defined; software development standards including testing at the unit and various stages of integration are defined; discrepancy reporting standards are defined; quantitative failure rate, recovery time, and recovery probabilities are allocated to major architectural components	Computer system and software top level accommodate quantitative and qualitative failure detection and recovery requirements; safe mode entry and recovery mechanisms are defined; redundancy and failure management designs are defined; mechanisms for maintaining consistency of replicated state data defined; software design and coding standards are defined for software components; generic purchased software components (e.g., operating systems, computer communications) conform to architecture and design; commands for vehicle control, recovery, and reconfiguration	Detailed design conforms to architecture and top level design features for failure detection and recovery; hardware infrastructure to accommodate failure detection and recovery is defined (e.g., interrupts, communications ports and connections, watchdog timers); exception handling strategies are defined; diagnostics and built-in-test mechanisms are defined; vehicle status and health monitoring telemetry is defined; all detailed design provisions conform with requirements
Parts, Materials, Processes	Standards for parts, materials, and processes are defined; requirements are consistent with mission reliability and availability requirements	Components (mechanical parts, solar cells, processors, FPGAs, memory chips, and other components) conform to reliability allocations and space qualification requirements; non-conforming or non-standard parts have appropriate documentation	Changes in the design to accommodate unavailability of qualified parts are analyzed for impact on reliability and reliability prediction models are revised as necessary
Test and Verification	Test strategies and test standards incorporate reliability, failure detection, and recovery	Verification and test program defined (including failure and recovery testing); identification of high cost and long lead testing facilities; definition of procedures for recording operating time and unplanned failures	Verification and test plans and analyses defined (including failure and recovery testing); design of test labs and special test equipment to include life testing and failure recovery testing where necessary
System Safety	Identification of top-level hazards that would result in death, injury, or catastrophic loss of mission capabilities; identification of program standards and requirements	Identification of hazards evident in the preliminary design, definition of requirements for mitigation, and probabilistic risk assessments; formulation of testing and verification requirements for defined mitigations	Review of design and analyses (e.g., FMECAs) to ensure that hazards have either been eliminated or that probabilities have been reduced to acceptable levels
Quality Assurance	Ensuring conformance with standards for reliability engineering, software development, parts selection, failure reporting, testing, and verification across all development phases		

Table 24-5a, Fig. 24-2◀, Eq. 24-18◀

Table 24-5b. Interaction of Project Functions in Development of Reliable Space Systems—Build through Operations

Project Function	Development and Fabrication	Integration and Test	Operations
System Engineering	Resolve cross-cutting issues and ensure interfaces preserve failure detection and recovery properties; addressing issues emerging from this and previous phases of the development	Verify that integrated system can be shown to meet requirements (including failure detection and recovery); resolution of critical issues (including those related to failure and recovery behavior)	Review on-orbit anomalies for system-wide issues; optimize operational and sustainability issues (for both space and ground segments)
Subsystem Design	Assemblies and units integrated into subsystems conform to expected failure and recovery behavior; diagnostic and BIT provisions across assemblies and units function as expected; electrical, mechanical, thermal stresses are within design margins	Anomalies discovered during testing related to design problems are addressed (corrected or mitigated)	Experience that would affect design of next generation subsystems is collected and analyzed
Computer System Design and Software Development	Computer system hardware infrastructure meets interface, capacity, response time, and failure containment requirements; software is designed and coded in accordance with project standards (especially those driven by reliability); unit testing addresses failure detection and recovery requirements for software components	Integration testing specifically addresses failure detection and recovery requirements allocated to the software and computing space vehicle elements. For software integration testing addresses failure containment, diagnostics, and reporting requirements; for system integration testing, system level failure detection and recovery behavior conforms to requirements; anomalies observed during any period of testing that are counter to assumptions and predictions are specially flagged and analyzed for their impact on the system design	Deficiencies and anomalies in on-orbit operation are analyzed for design impact; where possible, software changes are made and uploaded to the space vehicle; insights from on-orbit behavior are incorporated into future designs
Parts, Materials, Processes	Same as for detailed design	Test results are analyzed for failures or anomalies in conforming and specially qualified (or accepted non-conforming) components	Review of data from parts, materials and process on-orbit failures for incorporation into future requirements
Test and Verification	Verification and test procedures defined (including failure and recovery testing); development of test labs and equipment	Execution of test procedures; recording of results; analysis of results and identification of critical issues (including those related to failure and recovery behavior)	Acquisition of lessons learned from anomalies for future test plans and procedures
System Safety	Monitoring of development and fabrication activities to ensure that development meets requirements of mitigations identified in earlier design stages	Assessing test and verification results for hazard mitigation effectiveness; review of unanticipated failures for safety significance	Review of data from on-orbit anomalies and failures for incorporation into future requirements
Quality Assurance	Ensuring conformance with standards for reliability engineering, software development, parts selection, failure reporting, testing, and verification across all development phases		

2. It furnishes a record through which trends and correlations can be evaluated at a future time (an example of a trend is that the probability of failure increases after x hours of use; an example of a correlation is that part y fails during a particular step in the test sequence).
3. It permits reassessment of the predicted failure rates and is the basis for consequent modifications of the fault avoidance or fault tolerance provisions
4. It identifies unanticipated failure modes and effects and can inform on the effectiveness of recovery after a failure

Table 24-5b, Fig. 24-2◀, Eq. 24-18◀

Elements 2 and 3 require recording the operating time of all units in service; usually by an operating log maintained for each part number, with separate records for serial number. Corrective action is typically also recorded as part of the failure reporting. This facilitates configuration management in that it establishes at what point a component or subsystem has been returned to operational status. Reporting for each individual part also facilitates future investigation of the effectiveness of the corrective action. Corrective action frequently involves two steps: in the first, a failed part is replaced by a good one of the same design; the second step, addresses the

root cause, e.g., by tightening limits for the incoming test of this part. The results of retest are detailed in the corrective action report.

To establish a FRACAS the following must be identified:

- Scope of the activities (e.g., system test, field test, normal usage)
- Responsibility for cost and for report initiation
- Method and frequency of reporting (e.g., each incident or by time interval)

The format used for reporting of failures and corrective actions is not standardized. However, the following are the most essential data:

- Incident identification (e.g., report identification number)
- Date, time and locale of the incident
- Identification of the failed component (Part no., name, and serial number for hardware; task name, software configuration identification, version for software)
- Higher level part or system identifiers (subsystem or major component for hardware and software)
- Lower level part or system identifiers (usually available only after diagnosis)
- Operation in progress and environmental conditions when failure was detected
- Immediate and higher level effects of failure
- Names of individuals responsible for detection, verification and analysis
- Diagnosis of immediate, contributory and root causes of the failure
- Corrective action: Dates and nature of repair and results of retest

Parts Selection and Screening

The space operating environment includes cosmic rays, ionizing radiation, temperature extremes, vacuum conditions, and microgravity. In addition, parts and assemblies must withstand the forces of a rocket launch and explosive separation from the launch vehicle. Therefore, space qualified parts have unique requirements including: rigorous procedures for parts selection, evaluation and approval on both main spacecraft developer and its lower tier vendors and suppliers; parts and assembly handling and storage; quality assurance programs and documentation; and operating and failure data acquisition and dissemination. While these requirements apply to all aspects of the spacecraft—including structural components, thrusters, lubricants, adhesives, coatings, and propellants—reliability engineering is usually most concerned with electrical, electronic, and electromechanical (EEE) parts because they dominate the materials lists for a space vehicle. Requirements for space qualified EEE parts include [Hamiter, 1990]:

- Junction temperature ranges of –55 C to 125 C or wider
- Hermetic packaging for solid state devices and relays (plastic encapsulation is sometimes accepted when hermetic packaged parts are not available)
- Higher vibration capability
- Control of outgassing & flammability
- Extremely low defect levels (10 ppm)
- Extremely long reliability (1 to 15 × 10^{-9}/hr)
- Conservative derating & application practices (discussed in the next section).

There are multiple standards existing for parts specification, selection, and screening. Examples include ECSS-Q-60B [2007] (Space Components Steering Board in partnership with the European Space Agency), and (MSFC-STD-3012, prepared by NASA's Marshall Space Flight Center), MIL-PRF-38535J [DoD, 2010c] for integrated circuits, MIL-PRF-38534H [DoD, 2010b] for hybrids, and MIL-PRF-19500 [DoD, 2010d] for discrete parts), MIL-STD-202G [DoD, 2002], MIL-STD 883H [DoD, 2010a], and MIL STD 1580 [DoD, 2003]. However, because most are derived from DoD and were developed in the 1960s and 1970s, they share common elements including specification documents, standardized inspections, test methods (e.g., measurement of the leakage current at elevated temperature for semiconductors under specified conditions and for specified times), and destructive physical analysis (DPA) procedures.

Screening is testing intended to remove nonconforming parts (parts with random defects that are likely to result in early failures, known as infant mortality) from an otherwise acceptable lot and thus increase confidence in the reliability of the parts selected for use [Sahu, 2003]. This is graphically depicted in Fig. 24-3. For some component classes, inspection is part of the screening process. Screening can be conducted at several levels of acceptance, for example, MIL STD 883H (method 5004.11) specifies higher thresholds for space qualified parts (Class S) than for other military qualified parts (Class B). Higher screening thresholds reduce the probability of failure by rejecting the tail of the distribution but increase the cost of the remaining parts. The cost effectiveness of screening is high when (1) parts have an initially high failure rate, (2) the rejected fraction is low (not more than 20% of the lot), (3) where the cost of test is small when compared to the unit cost of the product under test (not over 10%), and (4) specially manufactured qualified parts would be bulkier or heavier. Screening can be applied to assemblies, (e.g., by subjecting them to combined thermal and vibration

Table 24-5b◀, Fig. 24-2◀, Eq. 24-18◀

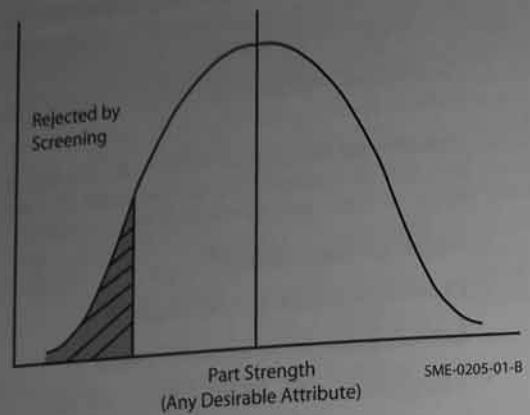


Fig. 24-3. Attribute Control by Screening. Screening rejects parts likely to fail in service.

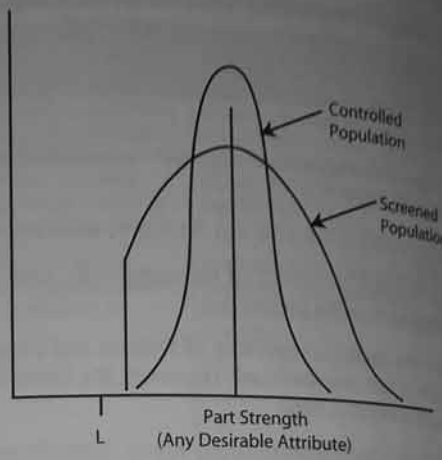


Fig. 24-4. Attribute Control by Process Control. In a controlled population fewer parts are near the acceptance limit than in a screened population.

environments and is thus more versatile than derating). Screening at the assembly level is also likely to result in a lower ratio of test cost to cost and thus is more cost effective. On the other hand, screening is not very effective for lowering the failure probability for components that have been derated—the failure probability due to external stresses in that mode is already very low.

Use of screening by itself is often criticized because it permits the use of products that do not, as delivered, meet all of the requirements of any given application. The alternative is the use of *qualified parts lists* (QPLs) and *qualified manufacturing lines* (QMLs) together with screening. QML and QPL parts are designed more conservatively, use more durable and superior quality materials, are built to tighter tolerances, and have more inspections. The net result is higher rigor in the control of the component manufacturing process so that it can be relied on not to produce the outliers that must be screened out. As shown in Fig. 24-4 both the screened population and the controlled population meet the acceptance criteria in that no parts fall below the lower limit, denoted as L in the figure. However, in the screened population a much larger fraction of the total population is near the lower limit than in the controlled population. Environmental effects and aging cause a dispersion of the attributes, and therefore will cause a larger fraction of the screened population to fall below the lower limit than in the controlled population. This effect can be compensated for by selecting an initial acceptance limit that is higher than the lowest value that can be tolerated in service. Where articles are procured specifically for space applications, costs will be reduced if the process can be improved so that only a small portion of the product will be rejected in screening.

Derating

The term *derating* is primarily applied to electrical and electronic components and involves the specification of a component that carries a higher rating than is needed for the application. The term *design margin* is mostly used in structure and thermal subsystems and means that a component is designed to carry more than the expected load and is defined by

$$M = \frac{\overline{S} - \overline{L}}{\sigma_S^2 - \sigma_L^2} \tag{24-19}$$

where M is the design margin, and S and L refer to the mean and standard deviations of the strength and load distributions [O'Connor, 2002]. Design margins and derating accomplish the same goal: prevention of component failure due to higher than expected external stresses or other deviations from the nominal conditions. The stress factors that are derated vary by component class. For example, for capacitors, stress factors may be voltage, for resistors, it might be power, for microcircuits, they may include supply voltage, power dissipation, junction temperature, output current, and clock frequency. There is no generally accepted method for determining which stress factors to derate and by how much. It is largely governed by the collective engineering judgment at the purchasing or developing organization. Examples are the NASA Goddard Space Flight Center PD-ED-1201 or Aerospace Corp. TOR-2006(8583)-5236 (used by the US Air Force).

Environmental protection can take the form of shock mounting, cooling or heating provisions, and shielding against radiation effects. Where derating reduces the failure probability by increasing the strength of the components, environmental protection reduces the failure probability by reducing the stress levels. In many cases, environmental protection adds considerable weight, and this, rather than cost, limits the amount of protection that can be provided.

24.1.4 Software Considerations

Spacecraft operations are becoming increasingly dependent on software, in the spacecraft and in the ground segment. Indeed, software is involved in nearly all functions of the spacecraft, and nearly all system functional requirements are in part allocated to software. Although there have been spectacular launch and on-orbit failures due to software defects [Lions, 1996]; [MCO, 1999]; [JPL, 2000], many more missions have been saved by software by means of work-arounds than have been lost due to it. This section considers software reliability issues from two perspectives: how software should be developed (specified, design, implemented, and tested) to support space vehicle and mission reliability, and reliability prediction of systems that include software.

Table 24-6 is a partial list of software specifications, designs, and development practices to enhance system reliability and availability by development phase. The most important distinction between hardware and software in space systems is that software failures can be recoverable if (1) the system is properly designed and implemented and (2) the health and telemetry data sent to ground control is sufficient for failures where ground intervention is necessary. The importance of defining correct and complete requirements to ensure mission success has been frequently noted (e.g., [Leveson, 1995]), and such definitions are critical for software requirements in space systems. The affected requirements concern not only that the software perform its functions correctly, but also that it manage the computer and system configurations properly and that it detect and respond appropriately to failures. Within the design, the most important considerations are the robustness of the interfaces that act as the failure containment boundaries and the completeness and correctness in the design of the recovery mechanisms (both functionality and timing). For implementation, the key issues are that proper coding standards are defined and followed, and that correctness of the actual implementations can be verified through both peer reviews and the definition of test cases.

Table 24-6. Software Design and Development Practices to Enhance System Reliability and Availability.

Phase	
Requirements	Requirements clearly identify and fully define operational modes (normal mode, graceful degradation, safe mode, or other contingency modes)
	Failure rates, recovery times, and recovery probabilities allocated to the software components that will exist at runtime
	Automated failure detection and recovery requirements defined, including explicit criteria for what failures classes will handled (For example, are only single-event failures handled)
	Timing requirements defined for both steady state and recovery—in particular, isolation, recovery and repair durations (average and maximum) have been identified and allocated
	Specified capabilities for monitoring and control by operators
Design	Existence of identifiable architecture components to manage failures in the systems
	Requirements for failure rates, failure detection, and recovery are derived as lower level design (not known at the beginning of the design) details are defined
	Fault containment boundaries are explicitly and unambiguously defined, and the failure containment mechanisms at these boundaries are dependable
	Protocols for processor health messages, heartbeats, and network traffic account for steady state and failure conditions
Detailed Design and Coding	Design standards exist to enforce architectural provisions for failure containment, failure detection, and recovery (including interfaces, data types, exception handling, input and output validation)
	Software development procedures emphasize testing and recording of results of testing for both normal and off-normal cases (including automated regression testing)
Verification (Test and Analysis)	Verify that all credible failure modes in lower level SW, HW, and networks been identified
	Verify traceability from the design to the system availability/maintainability requirements
	Verify that detection and recovery all identified failure modes addressed in the architecture
	Verify that failure detection/recovery durations meet the operational availability requirements
	Verify that Software architecture, when combined with the hardware architecture, will meet system reliability and availability requirements (including maximum outage duration)
	Test program should include prototype/benchmark/analysis on the system failure management—Provide analysis and results.
	Verify that the COTS/reuse products meet the reliability, availability, and fault management requirements (e.g., past service history, integration testing)

Table 24-6, Fig. 24-4◀, Eq. 24-19◀

Unlike electronic hardware, there is no widely accepted method of predicting software reliability in the absence of measurement of the actual system performance. This does not mean that the finite probability of software failures can be disregarded, instead, software failures should be recorded, analyzed and classified throughout the software development process—preferably, as part of a FRACAS. Software failures can be classified in two general categories: *Deterministic* and *Random*. Deterministic software failures occur consistently given a certain input data set and are nearly always due to some identifiable coding defect traceable to a logic or algorithmic error, design problem (e.g., interfaces), or a requirements flaw (omitted, vague, or incomplete). It may be difficult to find data sets that reproduce deterministic failures, but once found, such data sets will consistently produce the same unwanted result. *Random* failures, on the other hand, occur due to a timing problem, momentary buffer overflow, or other transient condition that make them difficult to reproduce. These conditions are frequently related to conditions external to the software itself, and because such conditions occur non-deterministically, the consequent failures can also be regarded as non-deterministic. Previous research on high integrity systems has shown that after software defects related to deterministic failures have been removed, the residual software defects lead to a failure behavior which can be characterized by a MTBF [Hsueh and Iyer, 1988] and hence can be characterized as random and amenable to stochastic modeling. A majority of such failures could be recovered by the use of physical redundancy as in the case of hardware ([Gray, 1990]; [Lee and Iyer, 1995]; [Tang and Hecht, 1995]). Such defects are often referred to as "Heisenbugs" after Werner Heisenburg's uncertainty principle [Bourne, 2004].

It can be argued that such software failures could ultimately be traced to a design implementation flaw and that they are therefore not truly random. However, the same argument would also apply to electronic hardware components for which parts-level reliability prediction methods discussed above are accepted practice, (i.e., that a detailed root cause analysis would reveal manufacturing flaws or design flaws). Nevertheless, the part-level reliability prediction methods described above that consider electronic component failures to be random are well accepted. In prediction of both hardware and software element and system reliability, we are less interested in the underlying causes of failures as we are in the probability of their occurrence.

Deterministic failures can be found and removed through a disciplined software development process in which requirements are carefully formulated and validated, the design is properly analyzed for both functional consistency and performance (capacity and response time), source code is checked for conformance to well established coding standards, peer reviews are conducted to find classes of defects readily identified by qualified people, and testing is performed consistently and thoroughly at multiple levels of integration (unit, subsystem, and system).

Table 24-6◀, Fig. 24-4◀, Eq. 24-19◀

Random failures, in software, can be addressed stochastically. No generally accepted methods exist for predicting software reliability without empirical data. Thus, software reliability can not be predicted, it can only be *measured*. This does not mean that the entire catellite must be completed before the quantitative software reliability can be determined. Software is composed of separate runtime components called *processes* that execute on a processor under the control of an *operating system kernel*. As the separate processes are developed and tested, their failure rates can be measured. Similarly, the failure rate of the operating system kernel, which may be a commercial product, can also be measured. Such software failure rates can be combined with hardware elements into a system level failure rate just as is the case for hardware components alone. Methods for doing so are described by Kececioglu [1993] and in IEC [1998], DoD [1996], and the NIST/SEMATECH online Handbook [2010].

24.1.5 Redundancy and Fault Tolerance

Fault tolerance protects against a wider spectrum of failure modes than fault avoidance. In most cases, it also requires more resources. At the system level, redundancy is the primary method of achieving fault tolerance. This section describes five types of redundancy strategies that are summarized in Table 24-7.

In practice the switching or voting provisions that are required for each partition add considerably to the cost of the implementation, and since they are not likely 100% reliable they may also reduce the reliability benefits. System test can also be adversely impacted by redundancy of small scope. For example, in Fig. 24-1, the system redundancy (middle line in the figure) requires only two tests. The quarter redundancy (top line) requires 16 test combinations. Selecting the scope of redundancy we must consider detection of the cause of failures, the ease of output switching or combining, the reliability that can be achieved, and the cost of implementation and test.

Replication

Replication involves installation of two or more identical components with switching to make one of them active. This type of redundancy offers very high protection against random failures, but not failures due to design deficiencies: if one component fails due to insufficient radiation hardness, the redundant one is very likely to fail soon thereafter. Because of its high cost, replication is used only sparingly in low value satellites. The cost of replication can sometimes be reduced by employing *k-out-of-n replacement*, in which a pool of spares can be assigned to replace any one of the pool of active units. An example is nonvolatile storage which usually consists of multiple physically identical modules. Providing one or more spare modules on the same data bus permits replacement of any one failed module. The same scheme can be used for multi-cell batteries, solar panels, and other elements of the electric power supply.

Table 24-7. Redundancy Strategies for Fault Tolerance.

Strategy	Protection Against	Disadvantages
Replication of the Same Design	Random failures	Higher acquisition cost, weight, power
Diverse Design for Each Channel	Random failures and failures caused by design deficiencies	Higher acquisition cost, weight, power, design, and logistics costs
Functional Redundancy	Random failures and failures caused by design deficiencies	May not always be feasible—existence of diverse method is necessary
Temporal Redundancy (Restart and Retry)	Transient and intermittent failures; some classes of software failures	Not effective against permanent failures; failure will persist until system is restarted
Information Encoding	Single Event Upsets and digital transmission errors	Correction capabilities are usually limited to 1 or 2 bits per event

In some cases (e.g., power supplies) the outputs can be combined so that the redundancy is completely passive. However, in most cases, some type of redundancy management such as voting or switching is necessary. Voting combines outputs of redundant units but carries a high cost because at least three identical units have to be installed to make it practicable. Switching is used in active/standby systems in which a switching mechanism configures the standby element into the circuit to replace the failed previously active elements. The effectiveness depends on the dependability of both the monitoring provisions that detect failure of the active component and the switching mechanism.

Diverse Design Redundancy

Installation of two or more components of different design (called *diverse design redundancy*) to furnish the same service has two advantages: it offers high protection against failures due to design deficiencies, and it can offer lower cost if the back-up unit is a "lifeboat," with lower accuracy and functionality, but still adequate for the minimum mission needs. The installation of diverse units usually adds to logistic cost because of additional test specifications, fixtures. and spare parts. This form of redundancy is economical primarily where the back-up unit comes from a previous satellite design, or where there is experience with it from another source. Where there is concern about the design integrity of a primary component, diverse design redundancy may have to be employed regardless of cost.

Functional and Temporal Redundancy

Functional redundancy (sometimes called *analytic redundancy)* involves furnishing a service by diverse means. An example is the determination of attitude rate from a rate gyro assembly (direct), and from observation of celestial bodies (indirect). It is particularly advantageous when the alternate is already installed for another service, (e.g., if a star sensor is provided for navigation). In these cases the only cost incurred is for the switching provisions and for data conversion. Both of these can frequently be achieved in an *existing* on-board computer thus further minimizing the cost.

Functional redundancy can also take the form of a ground back-up for functions preferably performed autonomously, (e.g., navigation, thermal control, or furnishing commands for sensor operation). In the communication subsystem there are frequently omni-directional and directional antenna systems. For some satellite missions these may also be considered a form of functional redundancy.

The chief benefits of functional redundancy are: it avoids the cost and weight penalties of physical redundancy, and it is inherently diverse, thus providing protection against design faults. The major limitation is that the back-up provisions usually entail lower performance.

Temporal redundancy involves repetition of an unsuccessful operation. A common example is a retry after a failure within the computing process. The same technique is applicable to acquisition of a star, firing of a pyrotechnic device, or communication with the ground. This is obviously a low cost technique. It is most effective when the design incorporates an analysis of the optimum retry interval and of changes that may improve the success of later operations, (e.g., switching a power supply, reducing loads on the power supply, or re-orientation of the satellite). The most important step is to plan ahead for retry of operations and to incorporate "hooks" that permit automatic or ground-initiated retry. "Safe mode" is a combination of both functional and temporal redundancy. Safe mode is entered upon detection of a major failure condition. In response, the satellite deactivates nonessential elements (i.e., all those not related to minimal attitude control, solar power, and ground communications). A ground control crew is then responsible for diagnosing the cause of the failure, taking corrective action, and reconfiguring the satellite to a normal or degraded mode.

Coding Techniques

Data coding provides robustness of data transmission and storage by enabling continued operation in the presence of a wide spectrum of errors, primarily in memory and data transmission. Coding techniques are also available for detection or toleration of errors in arithmetic processors.

The important coding techniques are error detecting code and error correcting code. The former is intended primarily for fault isolation (i.e., preventing an incorrect result from being used in subsequent operations). The latter is a fault tolerance mechanism that corrects a class of errors and permits operations to continue normally.

Table 24-7, Fig. 24-4◀, Eq. 24-19◀

All such measures require the addition of check bits to the bit pattern that represents the basic information. If there is agreement between the check and information bits, the data is accepted. If there is no agreement, the data is rejected (for error detecting code) or corrected (for error correcting code).

24.1.6 Human Factors Considerations

Operators are frequently ignored in reliability analysis and design but their action (or inaction) can drastically affect the mission duration and availability of space vehicles. A full discussion of human factors issues in ground control systems is beyond the scope of this chapter, but it should be noted that vehicle status monitoring (by means of telemetry and monitoring of payload signals) is the first line of defense in mitigating the effects of anomalies. Operators who act in time may be able to prevent a vehicle outage or ensure that an outage is recoverable. In order for operators to intervene in a manner that is timely and effective, they need adequate telemetry. The best means of defining telemetry requirements is through the FMEA. When failure mode compensating provisions depend on operators, an evaluation should be made to ensure that design information is provided.

24.1.7 Reliability Testing

Reliability testing assesses whether an element has sufficient durability for the mission (whether measured in cycles, time, or other relevant units). Reliability testing is costly and therefore performed selectively—usually on elements where wear out or material degradation is the primary failure mechanism. The least expensive reliability test is one that is not run at all as a reliability test, but rather as a part of a qualification test, lot acceptance test for purchased parts, or as an acceptance test on the spacecraft as a whole or on a major subsystem. To use these activities for reliability assessment may require some additional instrumentation and sometimes an extension of the test time, but these are very small resource expenditures compared to those required for even a modest separately run reliability test.

Other alternatives to obtaining reliability data by test are:

- Using test data obtained by others (including vendors) on the same component
- Using test or experience data on similar components
- Stress-strength analysis (particularly for mechanical components)
- Reliability prediction by MIL-HDBK-217 or similar sources

Once it has been decided that a reliability test is necessary, a suitable scope and test environment must be selected. Table 24-8 and Table 24-9 list considerations to be used in selecting these attributes. The attributes of the approaches lead to the test recommendations shown in Table 24-10.

Table 24-8. Considerations in Selection of Testing Scope.

Small Scope (parts and circuit boards)	Large Scope (assemblies, subsystems, and systems)
• Low cost and small size of individual test articles permit testing of multiple items • Inputs and outputs are easily accessible • The test environment can be tailored to the requirements or the unit under test • The test can be conducted early because it does not require integration	• Interactions between components can be observed • Test results are easily translated to effects on the mission

Table 24-9. Considerations in Selection of Testing Environment.

Quiescent (room ambient)	Stressed
• Low cost (no or only simple environmental chambers required) • Articles under test are easily accessible • No inducing of failures due to unusually high stress	• Increased probability of failure (less test time required) • Can identify an Environmental vulnerability of the unit under test

Table 24-10. Typical Uses of Reliability Testing.

Environment	Scope	
	Small	Large
Quiescent	Suitability test	High risk subsystems
	Critical components when vulnerabilities to stresses are not known	
Stressed	Critical components where vulnerabilities to stress are known	"Thermal vacuum" testing (expensive)

Table 24-10, Fig. 24-4◀, Eq. 24-19◀

The time required for any reliability test can be significantly reduced if *testing by variables* is employed. This means that the value of significant attributes is numerically recorded (as contrasted with the commonly used pass/fail procedure). From these data, the distribution of parameters can be plotted, and the probability of dropping below an acceptance criterion can be assessed. The general technique is similar to the one described in Sec. 24.1.3. For reliability assessment the parameter distributions are of interest, whereas in screening, the attributes of individual units are the chief criterion.

The small component populations that are typical of space procurements preclude the use of conventional reliability demonstration methods. Even a modest subsystem requirement, such as reliability of 0.95 for 17,000 hours (approximately 2 years), corresponds to an MTBF of over 330,000 hours and will require over 1 million component-hours of test to arrive at a statistically meaningful assessment by conventional methods. Yet, experimental verification of the claimed reliability of a component or subsystem is frequently desirable and sometimes required. So let's explore some low cost methods of accomplishing this.

The major causes of failures are workmanship and design. The first of these can be addressed by quality assurance. Design failures occur primarily because the strength of the component is not adequate for the environment in which it is used, or because the manufacturing process permits too much variability in component characteristics. This can more easily be seen in mechanical spacecraft components where reliability depends on (a) the margin between the nominal (mean) strength of the component and the maximum service load and (b) the variability of strength about the mean in the manufactured product. Since test can characterize the strength of mechanical components easily, strength-load margins (design margins) have always played a major role in their reliability assessment. For electrical and electronic components the same relation holds in principle, but it is usually much harder to define a single failure causing stress or load. Nevertheless, test data can give valuable insights into potential reliability problems. Important requirements are (a) recording of test numerical results and (b) statistical evaluation of the probability of failure from the numerical test results, e.g., by applying the 6-sigma criterion (the mean of a parameter is at least six standard deviations above the specified minimum of the specified maximum) [Harry, 1997]. The tests from which the required data is obtained can be qualification and acceptance tests.

24.2 Space System Risk Analysis

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The majority of the design process will focus on how to get a particular design to work in the way it is planned to achieve the mission objectives. By necessity, these designs become quite complex and quite expensive in order to achieve astounding accomplishments. The complexities, environments, and challenges inherent in any space mission mean that a large number of things need to work in order for the mission as a whole to be a success. If anything does not work as planned, or a failure occurs, the entire mission can be lost, and a very large amount of dollars, time, and opportunity or even human life can be lost with it. However, these same challenges and complexities also mean that getting the system to function properly even in the event of no anomalies is an expensive challenge, so spending more money, time, and effort to protect against anomalies is difficult. Therefore, we must figure out how to best apply our limited resources to maximize our chance of success. To do this, we have formalized the idea of risk.

Risk, R, is defined as the product of the probability of a negative event occurring, P, and the impact, or consequence, of that event, I.

$$R = P \times I \tag{24-20}$$

A risk analysis needs to consider both terms in the above equation (probability and impact). Risk is often confused with reliability. As discussed in Sec. 24.1 reliability is the probability of a failure not occurring. This is only half of the risk equation. What separates a risk analysis from a reliability analysis is the inclusion of a measure of the impact of the negative event, which can range from reduced performance to total mission failure.

Two other terms that are often confused are low risk and robust design. A mission with a *robust design* is one which is designed for uncertainty in the nominal conditions. This is often used in the context of unknown environmental conditions, but could be true for any nominal condition used during design. Unknowns in nominal conditions is one risk factor, and therefore a mission with a robust design has lowered the risk in that area, but may still have many other risk factors in other areas.

A third distinction that is important to make is the distinction between a risk and a problem or *anomaly*. For our definitions, a *risk* is a negative event which **may** occur in the future. A problem or *anomaly* is something which is occurring right now or has already occurred. Risks turn into problems or anomalies if the negative event that they are tracking occurs.

There are two basic types of *risk*: mission risk and implementation risk. The difference in the two types of risk is in the definition of the impact. *Mission risks* affect the total return from the mission and the impact is defined in terms of the amount of mission objectives that would be lost (or would remain) if the negative event were to occur. Mission risks can be thought of as the risk of a failure or anomaly occurring during operations. *Implementation risks* affect the budgets and margins of various resources and the impact is defined in terms of the percent of reserves or margin used to recover from the negative event. Implementation risks can be thought of as a risk prior to operations beginning that affects the cost, schedule, or engineering resources available. Clear-

Table 24-10◀, Fig. 24-4◀, Eq. 24-20

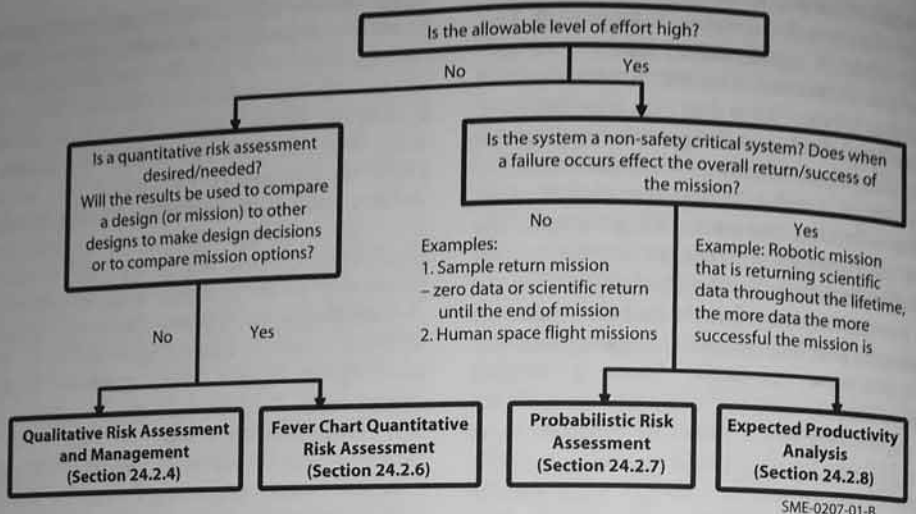


Fig. 24-5. Summary of various Risk Assessment Techniques. Where you fall in this chart may change throughout the design process.

ly the impact definitions are different for mission versus implementation risks. The probability values associated with the same "level" of risk can also be different if desired (i.e., a "low" probability may be 5% for implementation risks, but < 1% for mission risks). Neither type of risk is meant to imply more or less importance—either one can lead to mission failure. However, it is possible to be more or less risk adverse to one type of risk over the other.

24.2.1 Risk Assessment Throughout The Design Process

Many consider risk analysis to be a practice which examines finalized designs for areas that are the most risky, and adds extra redundancy or testing to those areas. This is expensive in terms of mass, cost, and schedule. In addition, this does not help an engineer choose between various designs or architectures with risk in mind. By turning risk assessment into a process that is ongoing throughout the design process, it is possible to include risk as a factor in design decisions, and to reduce the cost of risk mitigation, both of which will lead to lower risk designs.

To make risk a factor in design decisions throughout the design process it needs to be:

- Practical for designs in which little information is known
- Capable of maturing with the design process
- Easily understood by designers and engineers
- Tradable (quantifiable)

Several types of risk analysis can meet the above criteria. A risk analysis can range from very simple to very complex and can change throughout the design process. See Fig. 24-5 for a summary of various risk assessment techniques and when each type of technique may be most appropriate.

Table 24-10◀, Fig. 24-5, Eq. 24-20◀

As Fig. 24-5 shows, there are several questions that can be used to determine the appropriate risk assessment technique. If the level of effort needs to be low, you probably want to do only fever-chart level risk analysis. This can however, be either qualitative or quantitative, depending on what the goal of the analysis is. If the goal of the analysis is to manage and track risks only, then a qualitative fever chart analysis is appropriate. If the goal of the analysis is to both manage and track risks and to aid in design decisions based on risk, then a quantitative fever chart assessment may be more appropriate. If the available level of effort is, or will eventually be, moderate to high (e.g., a high-cost mission) then you can do more complex analyses. In these cases the type of analysis that is most appropriate depends on type of system you are analyzing. If the mission return is not affected by when a failure occurs, a Probabilistic Risk Assessment (PRA) is appropriate. If when a failure occurs affects the overall return of the mission, Expected Productivity Analysis (EPA) may be more appropriate.

Where you fall in Fig. 24-5, and therefore what type of analysis is most appropriate for your situation, can and often will change as the design matures and more information is available. It is also possible to do any of the four types of analysis shown in Fig. 24-5 for any phase of design, by simply adjusting the fidelity level of the analysis.

When to Use Expected Productivity vs. Reliability

We will use a simple example problem in order to show the differences between a PRA analysis and an EPA analysis. For this problem, the system of interest has only two failure modes, both of which are single point failures. The first failure mode is deployment failure. The deployment only occurs once and needs to be completed in order for the system to enter operations. The second failure mode is from a moving component. The component is required once operations begin and can fail at any point

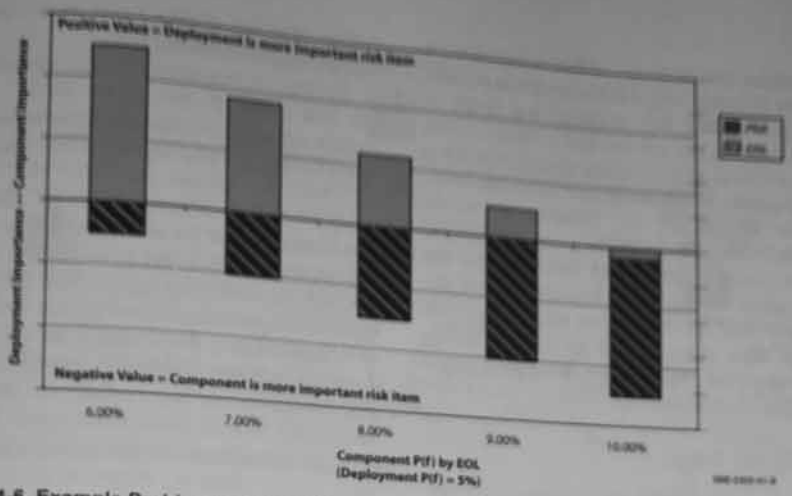


Fig. 24-6. Example Problem to Show Difference Between EPA and Reliability Analysis Techniques.

throughout operations. With this system there are two basic failure events (deployment failure and component failure) and if either event occurs the system is in a failed state. The importance of each of the two events was calculated using both PRA importance measures (Birnbaum Importance Measure—see Sec. 24.2.7) and EPA importance measures (see Sec. 24.2.8) for varying values for the probability of each event occurring. Additionally, the two events were ranked using each importance measure to determine which failure mode is the more critical mode and should have priority in terms of mitigation actions. The results are shown in Fig. 24-6.

In Figure 24-6, the probability of deployment failure is set at a constant value of 5%. An exponential failure rate was assumed for the component failure with a probability of failure by the end of the mission lifetime varying along the x-axis from 6% to 10%. The y-axis is the importance measure for the deployment failure minus the importance measure for the component failure. Therefore, a positive value implies that the deployment failure is the more important failure mode. Similarly, a negative value implies that the component failure is the more important failure mode. The first thing to note about this figure is that in four of the five cases tested the EPA importance measure ranked the two failure modes in opposite order from the PRA importance measure. These discrepancies mean that using PRA risk analysis techniques, an engineer or designer would assign more importance, and therefore more resources, to mitigating the component failure mode. However, if the key metric is actually the expected return of the mission and not the probability of failure by the end of the lifetime, more importance and therefore more resources should be focused on the deployment failure. The reasons for this are clear. The component has a higher probability of failure by the end of the mission lifetime. Therefore, measures that use probability of failure as the risk metric rank the component failure mode higher than the deployment failure mode. The component may fail at any point throughout the mission lifetime, however, including only a short time before the scheduled end of the mission. If the component fails after any significant amount of time has passed then the mission will still have returned some useful data. The deployment failure occurs at only one point in time, before operations. If the deployment fails, no data will be returned from the mission. This is not dependant on the timing of the failure since the event must occur before operations even begin. Therefore the deployment has a larger impact on the system return than the component failure. This higher impact outweighs the lower probability in ten of the eleven cases shown. This effect is not captured with the PRA risk metrics and importance measures.

In the above example we assume that the mission under consideration is returning data of some sort throughout time, and therefore the timing of when a failure occurs becomes important. In many cases the timing of when the failure occurs would not matter, such as a sample return mission, a manned mission, or a commercial satellite that needs to operate for X number of years to fulfill contract requirements. In these cases a failure is a failure, and when that failure occurs does not matter because the only thing being tracked is whether the end goal (returning the sample, returning the astronauts safely, or operating for X years) is met. In these cases a reliability-based metric can be used for a risk analysis, since the probability of meeting that end goal is what should be optimized.

24.2.2 Risk Management

According to NASA Continuous Risk Management definitions [JPL, 2003], Qualitative Risk Management should plan and organize to identify, analyze, plan, track, and control risks throughout the project life cycle. Risks should be identified and captured as described below. Next, they should be analyzed (either qualitatively or quantitatively) to assess those risks that should be accepted versus watched versus mitigated. For risks that

are accepted, nothing else is done—either it happens or it doesn't and you will deal with it if the problem occurs. For risks that are watched, no resources are spent to actively mitigate the risk, but it is tracked to keep a close eye on it, perhaps with a plan of action should it reach a particular trigger point. The most severe risks are mitigated. Mitigation approaches are planned to spend resources to reduce either the probability of occurrence, the impact of the event, or both. Both the risks and the mitigation approaches should be tracked throughout the lifecycle to ensure that the mitigation approaches are working and that other risks are not growing. Finally, these risks are controlled through the mitigation plans and through margin expenditure.

24.2.3 Identifying the Risks

One of the best ways to identify the risk items in a system is to ask the people who are designing the system. What the designers are worried about (what "keeps them up at night") are likely the biggest risk items. The difficulty in this approach is getting designers to admit their worries. One approach that helps with this is to first discuss with the designers the way the system works under nominal conditions. One of the best ways to learn how a system can break is to learn how it works. It can also be useful to interview operators of similar systems that have flown in the past or are currently flying, as these are the people who have to live with the consequences of a designer's choice. In any interview, don't feel the need to get numbers (probabilities, impacts) right away—start the conversation qualitatively by just discussing the system, how it works, and how the designer feels it might fail. You can go back to specific failure modes and discuss them quantitatively after this initial qualitative discussion.

Another way to identify risk items is to learn from the experience of others. Organizations will often keep risk databases, lessons learned, and even design checklists and rules which can be invaluable in determining risks in a design. The NASA lessons learned database and the military standards are two additional useful databases to search. A common way to reduce risk is to use a heritage system. It is important to realize that heritage systems will have lower risk than unique systems—but some unsolved risk items may still be present. In addition, it is easy to claim heritage, and therefore have very low margins, even if the heritage is not as direct as claimed. This can lead to large implementation risks, since modifications can lead to major increases in budget and schedule. Be especially careful of heritage claims if the new system will not be used in the same environment or with the same process as the inherited system—this can lead to both mission and implementation risks.

Even if not direct heritage, similar systems will have similar risks and problem areas, so researching these systems can be very useful. Along these same lines, if in doing this research you find an area of your system that is very unique, even just to your organization, you have also found a likely risk item. New technology or new developments are always a risk item—both a mission risk and an implementation risk.

Several more analytical tools can also be used to identify risk items. A few of the more common analytical tools are Failure Modes and Effects and Criticality Analyses (FMECAs) and fault or event trees. For information on FMECAs, please see Sec. 24.1.3. Fault trees and event trees help identify risks by identifying the steps that need to occur for either a negative or a positive event to happen. In a fault tree you identify what the main fault you will be analyzing is, and then proceed to identify all the various potential causes of that fault. In an Event tree or a Success tree, you identify the main event that needs to occur and then proceed to identify all the critical steps that need to occur successfully for the main event to be successful. In both cases, events, and therefore the tree itself, can be as detailed or as general as desired (e.g., mission success/failure, return a piece of data, loss of attitude knowledge or control, failure of a subsystem). The trees use "and" gates and "or" gates, along with several other variations to show logic flow. See Fig. 24-7 for some commonly used symbols in fault and event trees and see Fig. 24-8 for an example of a fault tree [Wertz & Larson, 1999].

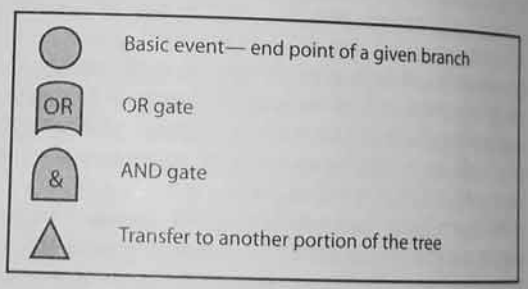


Fig. 24-7. Commonly Used Symbols in Fault Tree Analysis to Show Logic Flow.

Once risks are identified, they need to be documented in a risk data sheet, a risk list, or a risk database. Some of the important information to capture includes the risk statement, mitigation options, a point of contact, and any information available on the probability and the impact should the negative event occur, even if this information is qualitative. The risk statement should be in the form of "if event, then consequence." Examples of risk statements are: "If we have underestimated the scope of work for Technology A, then we will exceed the budget" (implementation risk), or "If we cannot develop Technology A to the level we expect, then we will not have the resources to meet the performance objectives", or "If a thruster fails, the mission will be lost."

24.2.4 Qualitative Risk Assessment

One of the most common tools used to both capture and track risks in qualitative risk assessment is the fever chart. *Fever charts* categorize risks into bins based on a relative scale for both probability and impact. The num-

Table 24-10◀, Fig. 24-7, Eq. 24-20◀

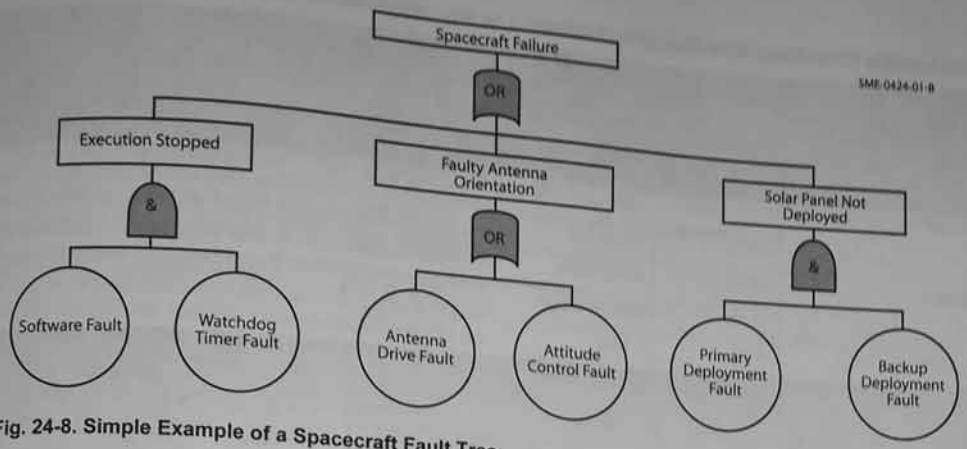


Fig. 24-8. Simple Example of a Spacecraft Fault Tree.

ber of bins defines the size of the fever chart—typical sizes include 3 × 3, 4 × 4, or 5 × 5. The definition of the bins can be qualitative—"High, Medium, Low" or "Very Likely, Somewhat likely, Not Likely, Very unlikely". Risks can then be characterized as "red" or high, "yellow" or medium, or "green" or low based on both their probability and impact scores. An example of a typical 5 × 5 fever chart is shown in Fig. 24-9.

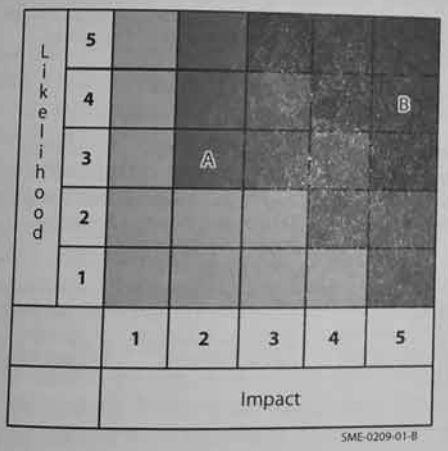


Fig. 24-9. Example Fever Chart. A and B are risks, placed in the appropriate row and column based on likelihood and impact. Light grey regions would be colored green in a typical fever chart. Similarly, dark grey would be red and medium grey would be yellow.

Fever charts can also be semi-quantitative by assigning a range of values to each bin (i.e., 1–5% probability). See Table 24-11 for some example values for fever chart bins. Assigning example values to the fever chart bins can greatly improve the consistency in the scoring across various team members. A 1% probability might be "Very high" for one person and "Low" for another depending on their personality and frame of reference—however, if they are given examples that show that 1% should be categorized as "Low", then both will score the risk more consistently. It can also be very useful to describe the various impact ratings as explicitly as possible. If the impact ratings are described as various percentages of the mission objectives being returned, then explicitly describe what those mission objectives are. If the impact ratings are described as whether or not full vs. minimum mission success is achieved, then explicitly describe what is required to meet full vs. minimum mission success. As an example a score of a 3 could mean: <10 images are returned; the mission lasts for <1 year; or instrument X and Y function, but instrument Z does not return any data. Describing the various impact levels in as much detail as possible, and specific to the particular mission being designed, will again greatly help in the consistency in scoring across various team members. It is important to also ensure that team members pay attention to these definitions and focus on scoring risks according to the definitions you have laid out for both impact and likelihood, and to not jump to the reaction of "this is clearly a yellow risk" or "I don't want to score it a 2 because then it will only be a green risk". Again, focusing on the specific definitions, and letting the color fall out of that definition, will add more consistency across the team. It should be noted that, especially in early design, even the quantitative estimates of what the probability or impact is will be very subjective, and therefore risk scores are still difficult to get consistent. However, using quantitative, explicit values or ranges that are specific to your mission will certainly help with this problem.

Once all the risks are identified and captured, you can produce a summary fever chart to show the number of risk items that fall into each bin to give a good visual snapshot of the number of risks and whether they are high, medium, or low risks. Fever charts can also be customized to show a particular organization's or a particular project's risk stance and preferences by adjusting which bins are categorized as red, yellow, or green. Note that the chart does not have to be symmetrical. Often those risks that are high impact but low probability are judged to be more severe than those that are high probability but low impact (yellow vs. green in Fig. 24-9).

Table 24-10◀, Fig. 24-9, Eq. 24-20◀

Table 24-11. Example Fever Chart Bins. Examples of definitions for fever chart bins.

Probabilities

		Example 1		Example 2		Example 3	
Bin	Qualitative Definition	Mission	Implementation	Mission	Implementation	Mission	Implementation
1	Very low	>0.1%	>1%	Less than 1%	Less than 1%	$0.1\% < P \leq 2\%$	$P \leq 10\%$
2	Low	>0.5%	>10%	1% to 5%	1% to 5%	$2\% < P \leq 15\%$	$10\% < P \leq 25\%$
3	Moderate	>1%	>30%	5% to 15%	5% to 50%	$15\% < P \leq 25\%$	$25\% < P \leq 50\%$
4	High	>5%	>50%	15% to 25%	50% to 80%	$25 < P \leq 50\%$	$50\% < P \leq 75\%$
5	Very High	>10%	>70%	> 25%	80 to 100%	$P > 50\%$	$P > 75\%$

Impacts

		Example 1		Example 2		Example 3	
Bin	Qualitative Definition	Mission	Implementation	Mission	Implementation	Mission	Implementation
1	Very low	Minimal (1%) loss of mission objectives	Minimal (1%) loss of reserves	Minimal or no impact to mission	Cost/schedule impact within subsystem purview	No impact to full mission success criteria	Consumes ≤ 1% of remaining Project reserves
2	Low	Small (10%) loss of mission objectives	Small (10%) loss of reserves	Small reduction in mission return	Cost/schedule impact within system purview	Minor impact to full mission success criteria	Consumes 1%–5% of remaining Project reserves
3	Moderate	Moderate (50%) loss of mission objectives	Significant (50%) loss of reserves	Cannot meet full mission success	Cost/schedule impact may require Project support	Moderate impact to full mission success criteria. Minimum mission success criteria, some margin will remain	Consumes 5%–20% of remaining Project reserves
4	High	Significant (90%) loss of mission objectives	Complete consumption of reserves (100%)	Cannot meet minimum mission success	Cost/schedule impact will certainly require Project support	Major impact to full mission success criteria. Minimum mission success criteria is achievable	Consumes 20%–50% of remaining Project reserves
5	Very High	Mission failure (100% loss of mission objectives)	Overrun (120% of reserves consumed)	Mission catastrophic	Cost/schedule impact severely jeopardizes the Project	Minimum mission success criteria is not achievable	Consumes > 50% of remaining Project reserves

24.2.5 Quantitative Risk Assessment

Quantitative Risk Assessment can be done in all design phases. You can estimate probabilities and impacts to the same level you estimate cost, schedule, or performance metrics early in the design phases. However, it is important to keep results relative if the analysis is done in early design phases as the absolute value of the probabilities and impacts is typically very uncertain. If the uncertainty in these values is extremely high, sensitivity studies can be done to determine for which areas of the design space various risks are the most important. Analyses can also be used to make relative comparisons between various designs or mission options. Without getting quantitative, it is impossible to determine which design is higher risk overall—a design with 1 red and 2 green risks or a design with 4 yellow risks. Quantitative analyses, even if done in a relative sense, can also help with questions like which yellow risk should I focus more on mitigating or would it be better to mitigate this set of green risks or this single yellow risk?

It should be noted that the results of quantitative models, even if used in a relative sense, are only as good as their inputs. While it is certainly possible to get very useful results out of quantitative models, even if the results are uncertain, there are some limitations. Results should be taken in context of the uncertainty on the inputs. If two designs are compared there is often a desire to come to a conclusion at the end of the analysis that one design is clearly superior to the other. However, if the analysis is done in the early design phases when the inputs are

Table 24-11, Fig. 24-9◀, Eq. 24-20◀

quite uncertain, and the final risk scores are separated by only 0.01%, it should be clear that this is not the case. Note that while this does not fulfill our initial desire to determine the more superior design, it is a valid result none the less. The result that two designs are equal, within the uncertainty of the analysis, is often just as important as a result that one design is superior to the other. As mentioned above, one of the best ways to deal with quantitative results in cases with large uncertainties is to do sensitivity studies. As an example, it can often be useful to determine at what probability of occurrence one risk becomes more severe than another, or what probability of failure is required for various components to meet an overall probability of failure requirement for the mission as a whole.

Several analysis techniques exist for adding quantitative aspects to risk analysis. The analysis techniques discussed below are presented from the simplest option, which is often useful to use in early design phases, to the most complex, which is most useful in the later design phases.

24.2.6 Fever Chart Quantitative Risk Assessment

It is easy to add quantitative assessment to fever charts by assigning a value to various levels of probability and impact. Note that you can choose to assign different values for probability depending on whether you are talking about a mission risk or an implementation risk. Impact values will be different by definition. You can choose either extreme bounding cases or the average of the range provided if you previously used fever charts during qualitative risk assessment. In fact, using a quantitative aspect to fever charts can often make defining where a risk belongs on the fever chart easier. Depending on the personality of the individual, it is sometimes easier to get a designer or engineer to choose a fever chart location for a risk item if they are given quantitative options. This allows you to ask "Is the probability more likely to be 0.1%, 1%, 5%, or 10%?" instead of an open-ended "What is the probability?" or a purely qualitative "High, medium, or low?" As mentioned above, this type of quantitative approach also greatly helps with consistency between the scoring for various team members.

Once risk items are scored on a fever chart with quantitative definitions you can combine the risk items to produce an overall risk "score" for the design. For mission risks, the impacts are defined in terms of the percentage of return you get from the mission, so the overall risk score is the expected percentage of objectives (scientific or otherwise) returned by the end of the mission lifetime. This can be calculated explicitly, since for each item you have defined (through the definitions of the bins) the probability of the event occurring and the percent of the mission objectives you will lose if the event occurs. While it is simplest to assume that all events are independent you can add calculations to include dependency if the increase in accuracy is warranted. To do this calculation you will also need to decide on a strategy of how to handle multiple events occurring, in terms of percent of objectives returned. The simplest approach is to assume that percentages can be added until you reach 100% of the objectives lost (be sure not to allow values greater than 100%). Equation 24-21 shows an example of this calculation for risks A and B shown in Fig. 24-9, using the definitions of impact and likelihood from the mission column in Table 24-11 Example 1.

$$
\begin{aligned}
E[\%Objectives] &= 1-\begin{pmatrix} AB\times P(AB)+\overline{AB}\times P(\overline{AB})+A\overline{B} \\ \times P(A\overline{B})+\overline{AB}\times P(\overline{AB}) \end{pmatrix} \\
AB &= 100\%+10\% = 110\% \Rightarrow 100\% \\
P(AB) &= 1\%\times 5\% \\
AB\times P(AB) &= 1\times(0.01\times 0.05) \\
E[\%Objectives] &= 1-\begin{pmatrix} 1\times 0.01\times 0.05+1\times 0.99\times .05+ \\ 0.1\times 0.01\times 0.95+0\times 0.99\times 0.95 \end{pmatrix} \\
E[\%Objectives] &= 94.9\%
\end{aligned}
\tag{24-21}
$$

To get more complicated, but also more accurate, you can determine for each combination of events whether or not the objectives that would be lost would be overlapping or not. An example would be a multi-element mission, such as an orbiter that will drop off a probe. The probe constitutes 25% of the mission objectives. All risks relating to probe failure would result in 25% of the objectives lost, but three probe failure risks should not result in 75% of the mission objectives being lost. There would be no difference between a single probe failure event occurring and multiple probe failure events occurring in terms of percentage of mission objectives lost. This approach obviously takes more effort to determine the impact of both individual risk events as well as various combinations of risk events, but will lead to a more accurate representation of the expected percent of mission objectives that will be returned.

Implementation risk impacts are measured in terms of percentage of resource reserves used, and therefore the overall risk score in this case is the probability that the mission reserves will be exceeded. For this metric, it does not matter if you use 50% of your reserves or 75%—it only matters if you use more or less than 100%. In other words, did you run over budget or not? This can also be explicitly calculated since for each item you have defined (through the definitions of the bins) the probability of the event occurring and the percent of the reserves you will use if the event occurs. Again, the simplest approach is to add percentages and assume independence. In this case you simply need to identify those combinations of events that result in over 100% of the reserves being used, and calculate the probability of any one of those combinations of events occurring. Equation (24-22) shows an example of this calculation for risks A and B shown in Fig. 24-9, using the definitions of impact and likelihood from the implementation column in Table 24-11 Example 1.

Table 24-11◀, Fig. 24-9◀, Eq. 24-21

$$
\begin{aligned}
&Resources(A,\overline{B}) = 10\% \\
&Resources(\overline{A},B) = 120\% \\
&Resources(A,B) = 130\% \\
&Resources(\overline{A},\overline{B}) = 0\% \\
&P[ExceedResources] = P[\overline{A},B] + P[A,B] \\
&P[ExceedResources] = 0.7 \times 0.5 + 0.3 \times 0.5 \\
&P[ExceedResources] = 50\%
\end{aligned}
\qquad (24\text{-}22)
$$

To get more complicated, but also more accurate, you can keep track of various resources—i.e., percentage of mass reserves used vs. cost reserves vs. schedule reserves, for example.

24.2.7 Probabilistic Risk Assessment

Probabilistic Risk Assessment (PRA) is the state-of-the-art risk assessment technique used by NASA in the later design phases. This method of quantitative risk analysis is used in the nuclear industry, and has been used at NASA since the Apollo program. Currently, the NASA policy states that a full scope PRA is required for all missions with a planetary protection requirement, all missions with nuclear payloads, all human space flight missions, all missions of high strategic importance, and all missions with high-schedule criticality. In addition, a limited scope PRA is required for any mission costing over $100M and is recommended for all missions [Stamatelatos, 2002].

A PRA is basically a quantifiable fault tree and event tree analysis. The systems engineer identifies major events that could affect the state of the system. These events are then placed in an event tree. Following each path through the event tree produces one sequence of events and results in one of a set of pre-determined end-states. The most common end-states are "ok" and "failed." While other end states can be included, each sequence needs to be laid out and assigned an end state, so a large number of end-state possibilities will lead to a very complicated and tedious analysis. The next step is to create fault trees for each event. These fault trees combine basic events using "and/or" gates among other logic to find the probability of each event in the event tree occurring. The probability of each end-state occurring can then be calculated by working through the logic in the fault and event trees and using the probabilities of the basic events (i.e., the probability of a failure of a given component). Those sequences of basic events that have the largest impact on the total probability of failure can then be identified. See Fig. 24-10 for an example of a basic Event Tree for the FireSat II Example. Each of the Events listed along the top will then be connected to a fault tree to determine the probability of taking each path. As an example, the Fault Tree shown in Fig. 24-8 might be linked to the Spacecraft (1 or 2) operation events.

Each of the bottom events in a fault tree is called a basic event, and is the level at which a probability value is

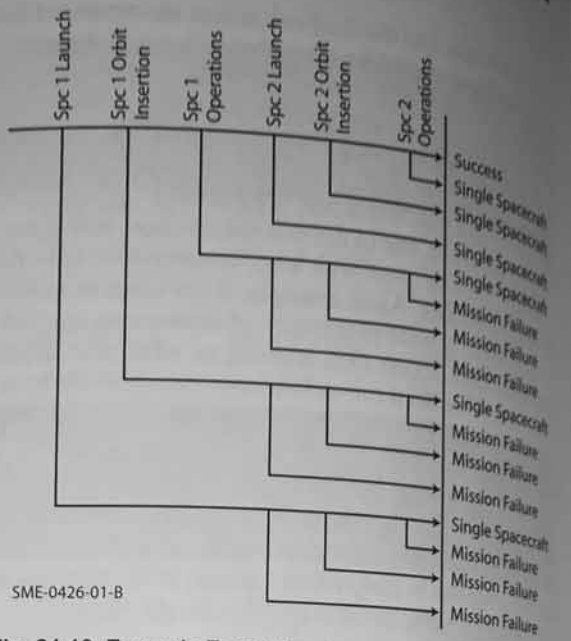


Fig. 24-10. Example Event Tree for the FireSat II Mission.

assigned. Depending on the complexity of the fault tree, a basic event can be the failure of a particular device, the failure of an entire subsystem, or anything in-between. A *cut-set* is a set of basic events that leads to a particular end state of the system. A cut-set in which all events are required to guarantee that particular end state is called a minimal cut-set [Stamatelatos, 2002].

As an example, look at the fault tree given in Fig. 24-8. In this case three cut-sets that lead to Mission Failure would be:

- Cut-set 1: Primary solar panel deployment fault and Secondary solar panel deployment fault
- Cut-set 2: Primary solar panel deployment fault and Attitude Control fault
- Cut-set 3: Attitude Control fault

These are just three example cut-sets—there are many other cut-sets that lead to mission failure. However, of these three cut-sets only cut-set 1 and cut-set 3 are minimal cut-sets. Cut-set 2 would lead to mission failure, but in this case the fact that the primary solar panel deployment failed did not affect the end state of the system—and therefore this is not considered a minimal cut-set. All of the *minimal cut-sets (MCSs)* for this fault tree are:

- MCS1: Primary solar panel deployment fault and Secondary solar panel deployment fault
- MCS2: Attitude Control fault
- MCS3: Antenna Drive fault
- MCS4: Software fault and Watch dog timer fault

We can use Boolean logic to determine the probability of mission failure using the minimal cut-sets and the probability of each basic event. Remember that in

Table 24-11◀, Fig. 24-10, Eq. 24-22

this case the basic events do not have to be mutually exclusive, so OR statements are not simple addition. When events are not mutually exclusive, we need to ensure not to double-count the possibility of both events occurring, as shown in Fig. 24-12.

$$P(A \text{ or } B) = P(A) + P(B) - P(AB) \quad (24\text{-}23)$$

If we assume the probability of each basic event in this example is 1%, then the probability of mission failure due to a spacecraft not operating correctly is:

$$\begin{aligned} PMF &= PMCS1 \text{ or } PMCS2 \text{ or } PMCS3 \text{ or } PMCS4 \\ PMF &= PPDF \times PSDF \text{ or } PACF \text{ or } PADF \text{ or } PSF \\ &\quad \times PWDTF \\ PMF &= 0.0001 \text{ or } 0.01 \text{ or } 0.01 \text{ or } 0.0001 \\ PMF &= 2.02\% \end{aligned} \quad (24\text{-}24)$$

Importance Measures

Importance measures give you a ranking of the importance of mitigating each individual risk, or reducing the probability of individual basic events. Note that some basic events could belong in more than one cut-set. In addition, some minimal cut-sets will have more than one basic event while others will have only one. Both of these situations mean that you cannot simply look at the probability of individual basic events to determine the most important events to mitigate—you need to follow the Boolean logic of the cut-sets to determine their importance. There are many valid importance measures that can be used to assign an importance to each cut-set, but the three most widely used are Birnbaum, Improvement Potential, and Fussell-Vesely. Each of these three importance measures are defined and discussed in Table 24-12, along with some suggested applications.

One of the most popular importance measures used in PRAs is the *Birnbaum importance measure* that calculates the sensitivity of the total probability of failure to the probability of a particular event or risk item. The equation for the Birnbaum importance measure, $I_{Birnbaum}$, is given by [Wang et al., 2004] [Hoyland & Rausand, 1994].

$$I_{Birnbaum}(A) = \frac{\partial P_{Failure}(total)}{\partial P_A} \quad (24\text{-}25)$$

where A is the event risk item of interest, $P_{Failure}(total)$ is the total probability of failure, and P_A is the probability of event A occurring. While Eq. (24-25) is the general form of this measure, a different form is often used to ease calculations. Equation (24-25), is the sensitivity of the total probability of failure to a change in the probability of the event of interest. This equation is greatly simplified if the change in the probability of the event is set equal to 1. In this case the denominator of Eq. (24-25) is simply 1 and the numerator is the difference in the system probability of failure with the probability of the event of interest set to 1 and 0. This form of the Birnbaum equation is given in Eq. (24-26) by [Smith et al., 2002] [Relex, 2005].

$$\begin{aligned} I_{Birnbaum}(A) = &\left(P_{Failure}(total) \middle| P_A = 1\right) \\ &- \left(P_{Failure}(total) \middle| P_A = 0\right) \end{aligned} \quad (24\text{-}26)$$

For our example above, the Birnbaum importance measure for Primary Deployment Failure and the Attitude Control Failure are shown below:

$$\begin{aligned} BIPDF &= P(Failure/PDF) - P(Failure/No\ PDF) \\ &= (PSDF \text{ or } PACF \text{ or } PADF \text{ or } PSF \times PWDTF) \\ &\quad - (PACF \text{ or } PADF \text{ or } PSF \times PWDTF) \\ &= 3.01\% - 2.01\% = 1.00\% \\ BIACF &= P(Failure|ACF) - P(Failure/No\ ACF) \\ &= 1 - (PPDF \times PSDF \text{ or } PADF \text{ or } PSF \times PWDTF) \\ &= 100\% - 1.02\% = 98.98\% \end{aligned} \quad (24\text{-}27)$$

As you can see from above, according to the Birnbaum importance measure, the Attitude Control Failure is a much more risky event than the primary deployment failure, even though both start with the same probability of occurrence. Therefore, when making investment decisions, one should mitigate an attitude control failure before being concerned with a primary solar array deployment failure. This makes sense since a primary so-

Table 24-12. Comparison of Importance Measures. A summary of three of the commonly used importance measures for PRA and when each may be the most appropriate.

Importance Measure	Largest Effect on Importance Measure	When to Use This Measure
Birnbaum	Impact of risk	Focus is on reducing the largest impact risks (i.e., single points of failure); Failure is not acceptable, even if very low probability (i.e., human missions)
Improvement Potential	Probability and impact of risk	Focus is on determining where to apply resources for mitigation approaches—which risks will give me the most bang for my buck if I mitigate them
Fussell-Vesely	Number of failure modes affected by the risk	Focus is on finding the few risks that I can mitigate that will improve multiple failure modes—similar to Birnbaum but most failures are not single point failures anymore, so I want to find those failures that combine with lots of other, different failures to cause overall mission failure

lar array deployment failure has already been mitigated in this system by adding a secondary deployment option.

One interesting thing to note about the Birnbaum Importance measure is that it is independent of the actual probability of failure of the component or subsystem that you are calculating the importance of. The Birnbaum importance measure is based more on the structure of the system. A second importance measure that can be used is called the *Improvement Potential* and is defined as the difference between the total system probability of failure given the current probability of event A occurring and the total system probability of failure if risk item A is completely mitigated [Rausand and Hoyland, 2004]. This is very similar to the Birnbaum importance metric, but uses the current value for P(A) instead of assuming that A has occurred.

$$\begin{aligned} \text{IPPDF} &= \text{PMF} - \text{P(Failure/No PDF)} \\ &= 2.02\% - 2.00\% = 0.02\% \end{aligned}$$

$$\begin{aligned} \text{IPACF} &= \text{PMF} - \text{P(Failure/No ACF)} \\ &= 2.02\% - 1.02\% = 1\% \end{aligned} \qquad (24\text{-}28)$$

Variations on the improvement potential include reporting the percentage improvement relative to the original probability of failure or using a mitigated probability of failure instead of completely mitigating the event in question.

The *Fussell-Vesely* importance measure calculates the probability of the union of all minimal cut-sets that contain the event of interest divided by the probability of the union of all minimal cut-sets [Relex, 2005]. This is shown in Eq. (24-29).

$$I_{Fussell-Vesely}(A) = \frac{P\left(\bigcup\{AllCutSetsContainingA\}\right)}{P\left(\bigcup\{AllCutSets\}\right)} \times 100\% \qquad (24\text{-}29)$$

It may be difficult for software programs to automatically determine all the cut-sets containing the specific event of interest. Since this is required for the numerator of Eq. (24-29), it is often useful to put the Fussell-Veseley importance measure in a different form. The probability of the union of all minimal cut-sets containing A can be thought of as the probability of the union of all minimal cut-sets minus the probability of the union of all minimal cut-sets not containing the event A. The latter can be easily found by setting the probability of event A to zero and recalculating the probability of the union of all minimum cut-sets. This form of the Fussell-Veseley importance measure is given below [Smith et al., 2002]; [Wolfram, 2005].

$$I_{Fussell-Vesely}(A) = \frac{P\left(\bigcup\{AllCutSets\}\middle|P(A)=P_A\right) - P\left(\bigcup\{AllCutSets\}\middle|P(A)=0\right)}{P\left(\bigcup\{AllCutSets\}\middle|P(A)=P_A\right)} \times 100\% \qquad (24\text{-}30)$$

Table 24-12◀, Fig. 24-10◀, Eq. 24-30

In Eq. (24-30), P(A) is the probability of event A occurring and P_A is the original probability of event A occurring.

While these importance measures are some of the more commonly used, there are numerous others to choose from, including: Risk achievement worth, Risk reduction worth, and criticality measures [Borgonovo, 2007] [Rausand and Hoyland, 2004]. See Table 24-12 for a summary of when it makes sense to consider each of the three importance measures described above.

Implementation Risks in PRA

Implementation risks can be treated in two ways using the PRA approach. First, a separate PRA can be done for implementation risks. In this case, the events that need to occur are meeting milestones or delivering technologies or subsystems that meet performance requirements and the end states of the PRA are whether or not the system was delivered for launch on schedule, within budget, and meeting all requirements. This type of analysis can be very complicated but also very useful, as there are many fewer single point failures in terms cost or schedule overruns compared to technical risk items, but many more combinations of events that are independent events (i.e., two independent subsystems both overrunning on mass budgets) that combine to create a minimal cut-set that could be mission critical.

Whether or not a PRA-type of analysis is done for implementation risks prior to a mission risk PRA, implementation risks can still be introduced into a PRA for mission risks by adjusting the probability that the system will begin in various initial states. The most common strategy is to simply adjust the probability of ever making it to the launch pad before getting canceled. However, implementation risks could lead to a "degraded state" of the system prior to operations if technology development items don't quite meet performance criteria or if mass overruns lead to the removal of redundant hardware. This type of analysis requires much more planning to determine what appropriate alternate or degraded initial states could be, but can lead to some interesting analyses.

Complex PRA Techniques

The above description of PRAs covers only the very basic concepts of a PRA analysis. PRAs can be as simple or as complex as we choose. They can begin very simply with high level events and full subsystem or major event failures in the fault trees, and can evolve into very complex analyses going down to component level failures as the design life progresses. In addition to simply adding more complexity, PRAs can also be modified to include more complex modeling cases. Improvements to the PRA process to include dynamic aspects are ongoing. While the dynamic aspect of systems is particularly important to the aerospace industry, researchers in the other industries have determined a need to include dynamic aspects in the PRAs analyzing nuclear systems as well. Methods to include dynamic aspects into a PRA-type of analysis include Phased Mission Systems and Dynamic Fault Trees. [Siu, 1994]; [Devooght and Smidts, 1996]; [Cojazzi, 1996]; [Meshkat et al., 2003]; [Chhikara et al., 2003].

24.2.8 Expected Productivity Analysis

The *expected productivity* of a system is the product of the probability of being in each functional state and the productivity in that state, summed over all states and all time. This could also be thought of as the expected value of the utility function or return for the mission. This is also called *Performability* in some areas [Ciciani & Vincenzo, 1987].

Combining the systems performance aspect with the probabilistic aspect of risk analysis leads to the need for much more customized tools for each individual mission. In the same sense that each mission needs a system performance model that is custom built for that mission, it will also need a risk assessment model that is custom built for that mission (utilizing the already custom built performance model). Engineers running these analyses should understand both the system performance and the probabilistic aspect of risk.

Calculating the expected productivity of a system is done in 3 basic steps:

1. Determine the probability of being in each state at each time
2. Determine the productivity in each state
3. Combine to produce the expected value of the total productivity by the end of mission life

For the first step, we can use *Markov Modeling* to determine the probability that the system is in any state at any given time. The process described below is only valid for *Markov systems*—defined as a system in which the future states of the system depend only on the current state, and not on any previous states. In other words, if my system has 2 actuators which can fail, it does not matter whether my system started with 10 actuators and 8 have already failed or if my system started with only the two—all that matters is that from this point on I have 2 actuators and either one can fail at a given rate.

The first step in determining the probability of being in any state at any time is to define what the possible states of the system are, and how the system would transition from one state to the next. This is done using a *Markov model*, or *state diagram*. The state of the system changes as failures occur in the system. The transition matrix, or A matrix, is defined as:

$$\frac{d\mathbf{P}(t)}{dt} = \mathbf{A}\mathbf{P}(t) \tag{24-31}$$

An A matrix can be created by looking at each state individually. Each row and column of the matrix corresponds to a different state. Fig. 24-11 shows a very simple Markov model, and the corresponding A matrix.

In Fig. 24-11 *d*, *m*, and *l* are the failure rates of three different components—component "D", component "M", and component "L". Note that these are failure rates, and not probabilities of failure. Assuming an exponential failure rate, λ, failure rates are related to the probability of failure by the end of the mission lifetime, P_f, through the equation:

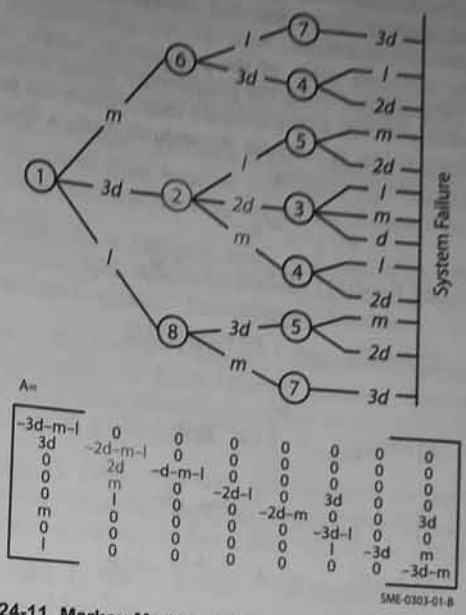


Fig. 24-11. Markov Model and Corresponding A Matrix for a sample system of three "D" components, 1 "M" component, and 1 "L" component. For this system to function: $d + m + l \geq 3$, $d + m \geq 1$ and $d + l \geq 2$.

$$\lambda = \frac{-\log\left(1 - P_f\right)}{life} \tag{24-32}$$

when *life* is the mission design lifetime.

Once the A matrix is calculated we can use it to determine the probability of being in any given state at any given time. From the definition of the A matrix [Eq. (24-31)], we can use Laplace Transforms to determine the probability vector (the probability of being in each state) for any time t.

$$\begin{aligned} s\mathbf{P}(s) - \mathbf{P}_o &= \mathbf{A}\mathbf{P}(s) \\ (s\mathbf{I} - \mathbf{A})\mathbf{P}(s) &= \mathbf{P}_o \\ \mathbf{P}(s) &= (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{P}_o \\ \mathbf{P}(t) &= e^{\mathbf{A}t}\mathbf{P}_o \end{aligned} \tag{24-33}$$

Next, we need to define the productivity vector, **C**. *Productivity* in a given state is defined as the fraction of "objects" that can be completed per unit time. In this case, an "object" is simply a single unit of the performance metric for the given mission. Examples of objects include an image, a measurement, a Kb of data, and so forth. The time to complete an object in a given state is calculated using a productivity model. This productivity

Table 24-12◀, Fig. 24-11, Eq. 24-33

model will be mission-specific and should be the same model used to determine the "value" of particular architectures or designs.

The expected value of the productivity metric can then be computed using the productivity vector and the probability vector. In the simplest cases, where the productivity vector is constant throughout time, a closed form solution is available.

$$\mathbf{P}(t) = e^{\mathbf{A}t}\mathbf{P}_o \tag{24-34}$$

$$E\left[Prod(t)\right] = \mathbf{C}\mathbf{P}(t) = \mathbf{C}e^{\mathbf{A}t}\mathbf{P}_o$$

$$E[Prod]_{Total} = \int_0^{life} E\left[Prod(t)\right]dt$$

$$E[Prod]_{Total} = \int_0^{life} \mathbf{C}e^{\mathbf{A}t}\mathbf{P}_o dt$$

$$E[Prod]_{Total} = \mathbf{C}\frac{1}{\mathbf{A}}e^{\mathbf{A}t}\mathbf{P}_o\Big|_0^{life}$$

$$E[Prod]_{Total} = \mathbf{C}\mathbf{A}^{-1}\left(e^{\mathbf{A}\cdot life} - \mathbf{I}\right)\mathbf{P}_o$$

For cases where the productivity vector is time-varying, it is easiest to use brute-force and step through time calculating both $\mathbf{P}(t)$ and $\mathbf{C}(t)$ for each time step. The expected productivity is then simply the product of the two vectors summed over all time-steps.

$$E\left[\text{Prod}\right]_{Total} = \sum_{t=0}^{life}\mathbf{C}(t)\mathbf{P}(t) \tag{24-35}$$

Implementation Risks and Early Mission Risks in Expected Productivity

Many of the risk items identified for missions are single events that occur before any objects are completed. Examples of these include a launch failure or a deployment failure. These risks result in a change to the initial conditions used to find the probability of being in each state. The probability that the system is in a completely failed state before objects are returned is the probability that at least one critical failure event occurred prior to this time. Care needs to be taken not to double count the probability of multiple critical failure events occurring, as shown by the overlap of events A and B in Fig. 24-12.

If both A and B are critical failure events that occur prior to objects being returned, then the probability that the mission is in a functional state, and therefore that neither A nor B has occurred, is given by:

$$\begin{aligned} &P(\text{neither A nor B}) = 1 - [P(\text{A}) + P(\text{B}) - P(\text{A and B})] \\ &P(\text{neither A nor B}) = 1 - [P(A) + P(B) - P(\text{A}) \times P(B)] \\ &P(\text{neither A nor B}) = 1 - P(A) - P(B) + P(A) \times P(B) \end{aligned} \tag{24-36}$$

In addition to risk elements that decrease the probability of beginning operations in the nominal state, other risk elements increase the probability that the system will

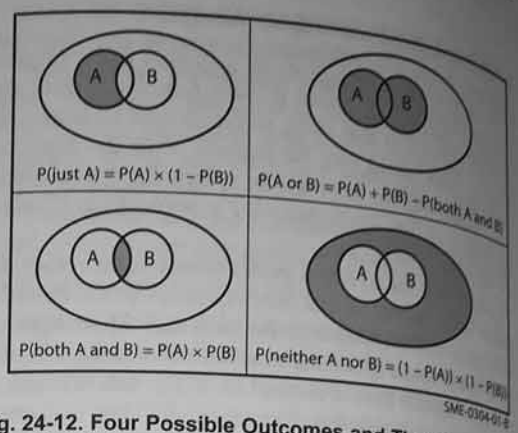


Fig. 24-12. Four Possible Outcomes and Their Probabilities from Two Independent, Probabilistic Events.

begin operations with degraded performance. Examples of these types of risk elements include a subsystem or technology not being developed to the required performance level or a partial deployment failure with which the system could still function, but at a reduced throughput rate or for a reduced time. These failure modes occur at or before the beginning of operations. Therefore, the initial system performance metrics, such as failure rates, lifetime, or efficiency, need to be adjusted before being used in conjunction with a dynamic failure model to find the overall expected total productivity. Using an observatory system example, a nominal system may have an observational efficiency of 1 (defined as equal time observing and non-observing). However, if control algorithms are not developed to the required level, the observational efficiency may be reduced to 0.5 (half the time taking observations is spent on overhead). These algorithms will not mature to the required level with some probability, *Pc*. As an example, let us take the probability of these algorithms not being developed to be 0.1. As shown in Eq. (24-37), the initial starting condition of this system is an expected observational efficiency value of 0.95.

$$\begin{aligned} &E\left[\text{Initial Observational Efficiency}\right] = \\ &\quad 0.5 \times Pc + 1 \times (1 - Pc) = 0.05 + 0.9 \\ &E\left[\text{Initial Observational Efficiency}\right] = 0.95 \end{aligned} \tag{24-37}$$

Once in operations, a single actuator may then fail. If this occurs, we assume for this example that the observational efficiency is once again cut in half. The productivity in the state in which a single actuator has failed should be based on an initial observational efficiency of 0.95:

$$\begin{aligned} &\text{Obervational Efficiency in Degraded State} = \\ &\quad \frac{1}{2} \times E\left[\text{Initial Observational Efficiency}\right] \\ &\text{Obervational Efficiency in Degraded State} = \\ &\quad \frac{1}{2} \times 0.95 = 0.475 \end{aligned} \tag{24-38}$$

Table 24-12◀, Fig. 24-12, Eq. 24-38

A separate, PRA-type analysis can be done for other implementation risks in Expected Productivity Analysis. See the Implementation risks portion of the PRA section for more information.

Importance Measures

An example of an expected productivity importance measure can be modeled after the concepts from both the Fussell-Veseley and Birnbaum importance measures. The impact of an event or risk item is determined by finding the expected productivity of the nominal system, in addition to the expected productivity given a zero probability of the given risk item occurring. Since the negative event would have lowered the expected productivity, the expected productivity should increase when the probability of the event occurring is set to zero. The impact of the event or risk item is then simply the expected productivity with the probability set to zero minus the expected productivity with the probability set to the original value. This is then normalized by the expected productivity with all probabilities set at the original values. The equation for the expected productivity, I_{EP}, importance measure is:

$$I_{EP}(A)=\left(\frac{\left(E[p]\middle|(P(A)=0)\right)-\left(E[p]\middle|P(A)=P_A\right)}{\left(E[p]\middle|P(A)=P_A\right)}\right)\times 100\% \tag{24-39}$$

when E[p] is the expected productivity, A is the risk item or event of interest, P(A) is the probability of event A occurring used in a given calculation, and P_A is the original input for the probability of event A occurring. Other importance measures could also be defined, using these concepts or more custom concepts, to create the importance measure that best fits the need for a particular mission or study.

24.2.9 Summary

No matter what type of analysis is done and how good that analysis is, risk will always be present—the question is which risks do you accept and which risks do you mitigate? Even when a risk is mitigated, it is often still present. Mitigating a risk simply means to improve it—by reducing either the consequence or the probability. Every manager will have a different risk posture, or level of risk that he is willing to accept. Some managers may even have different risk postures for different types of risks. The goal of a risk analysis is to convey what level of risk a project is currently taking, and to identify possible mitigation strategies and areas of a design that could make the largest impact on the overall risk level of the project.

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Table 24-12◀, Fig. 24-12◀, Eq. 24-39◀

25 Alternative Spacecraft Designs

25.1 Space Tethers
Space Tether Fundamentals; Space Tether Applications; Implementation and Integration

25.2 Inflatable Structures
Background; Engineering Inflatable-Only Structures; The Design Process; Engineering Inflatable-Rigidizable Structures; Reflector Mass and Stowage Volume vs. Diameter

25.3 SmallSats
We Were There First; Bigger is Better?; Smaller is Better?; Why Small is Different; Small Satellite Project Management; SmallSat Mission Examples; The Future

25.4 CubeSats
History of CubeSats; The CubeSat Philosophy from a University Student Perspective; Building CubeSats; Permits and Licenses; Past CubeSat Missions; CubeSat Missions—Applications; Launch Vehicles Currently Used and Proposed; Space Operations and Ground Stations; New Trends in Manufacturing; The Future of CubeSats of the Next Decade; List of CubeSat Builders and Component Suppliers

25.5 Differences Between International Approaches to Space
Comparative Perspective; Philosophy and Approach; Approaches to Organization; Conclusions and Lessons Arising

25.1 Space Tethers

Robert P. Hoyt, *Tethers Unlimited, Inc.*

A space tether is essentially a long wire or cable deployed from a spacecraft. Space tethers can provide a number of different capabilities for space missions, including attitude stabilization of spacecraft, generation of artificial gravity, formation flight of multiple spacecraft, and propellantless propulsion for orbital transfer, de-orbit, or drag-makeup. The principal advantage of space tethers is their ability to provide very large total delta V for certain missions with mass requirements far lower than other propulsion technologies. This advantage, however, comes with a trade-off in that the complex dynamics and physical extent of space tether systems can have a significant impact on system design and operations.

In this section, we will first review the basic physics and techniques involved in the use of space tethers. We will then summarize several of the primary applications of space tether technologies. Finally, we will discuss the issues that must be considered when integrating a space tether system into a space mission.

25.1.1 Space Tether Fundamentals

Gravity Gradient Forces on a Tethered System

A tethered system in orbit around a planetary body will experience forces due to the gradient of the body's gravitational field, and these "gravity gradient" forces play a dominant role in the behavior of tethered systems. As illustrated in Fig. 25-1, a tethered system in orbit will be in a stable equilibrium configuration when the tether is aligned along the local vertical, and the *tension, T*, on the tether balances the *gravity gradient force*, F_{gg}, experienced by both endmasses. For a system where the endmass masses are much greater than the tether mass, the tension is:

$$T = F_{gg} = 3GM\,mL/r^3, \qquad (25\text{-}1)$$

where G is the gravitational constant, M is the mass of the planetary body, m is the mass of one of the endmasses, L is the distance from the tethered system's center of gravity to that endmass, and r is the orbital radius from the system's center of gravity.

Tether Dynamics

The flexibility and tensile behavior of a tether and its interaction with the gravity gradient can result in the dynamical behavior of tethered systems being quite complex, exhibiting pendulum-like librations, transverse oscillations like a skip-rope, and longitudinal oscillations like a bungee cord. The design and operation of a tethered system must account for this dynamic complexity to ensure proper operation. As illustrated in Fig. 25-1, if a tethered system is rotated away from a local vertical orientation, the resulting gravity gradient forces will provide a restoring torque that will cause it to swing back towards vertical and librate around the local vertical orientation like a pendulum. The libration motions are more complicated than a simple pendulum because libration motions in the plane of the orbit couple into the orbital angular momentum of the system, whereas motion in the out-of-plane direction does not. As a result, librations in

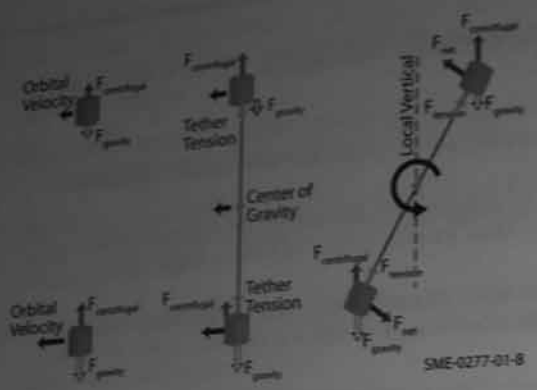


Fig. 25-1. Simplified Diagram of Gravity Gradient Effects on Space Tether Systems. On the left are shown two free-floating spacecraft in circular orbits. In circular orbits, the centrifugal force on each spacecraft must be balanced by the gravitational force towards the Earth, so the spacecraft at a lower altitude must orbit at a greater velocity than the higher altitude spacecraft. In the center we show the balance of forces when the two spacecraft are tied together with a tether. The tether forces the spacecraft to co-orbit at the velocity of the center of gravity of the system. The lower spacecraft, travelling slower than the circular-orbit velocity at its altitude, experiences a higher gravitational force than centrifugal force, and the difference is balanced by the tether tension. If the tethered system is rotated away from the local vertical, as illustrated on the right, the gravitational, centrifugal, and tether tension forces result in a restoring torque that will cause the system to librate around the local vertical.

the plane of the orbit have a period equal to $1/\sqrt{3}$ times the orbit period, and librations in the out-of-plane direction have a period equal to 1/2 of the orbit period. The in-plane librations of a tethered system can be pumped or damped by reeling the tether in or out, but such reeling maneuvers do not significantly influence the out-of-plane librations (see Sec. 25.1.3).

Momentum Exchange

A tether can act like a "sling" to provide a means to transfer orbital momentum and energy from one spacecraft to another. This technique of "momentum exchange" was demonstrated by the SEDS-1 tether mission, which flew as a secondary payload on a Delta-II upper stage. The SEDS-1 experiment deployed a small payload below the upper stage at the end of a 20 km long tether. Once the tether reached a vertical orientation, the upper stage, payload, and tether moved together in an orbit determined by their collective center of gravity, and in this configuration, the payload at the bottom end of the tether was moving at a speed less than orbital velocity at its altitude. The payload was then released from the end of the tether, injecting it into a sub-orbital trajectory that re-entered the Earth's atmosphere half an orbit later. After releasing the payload, the upper stage and tether entered an orbit with a slightly higher apogee. The maneuver is called 'momentum exchange' because the tether provided a mechanical path to transfer orbital momentum from the payload to the upper stage.

The delta V a momentum exchange tether imparts to a payload released from the tip of the tether can be calculated simply as the product of the angular velocity of the tether, measured in inertial coordinates, times the distance between the payload and the system's center of rotation prior to release. A simple rule of thumb for "hanging tether" payload releases such as SEDS-1 performed is that the payload will be injected into an orbit with a perigee reduced by a height roughly 7 times L, the distance between the payload and the tethered system's center of gravity. If the payload is released from the top of the tether, its apogee will be raised by an amount approximately $7L$. Larger orbital changes can be provided to payloads by spinning up the tether so that it is rotating in the orbital frame of reference. An important consideration in the design of tethers with tip velocities greater than several hundred meters per second is that each section of tether must support the centrifugal weight of not only the payload, but also of all of the tether between it and the payload. As a result, the minimum mass of a momentum exchange tether scales exponentially with the square of the delta V [Moravec, 1977]:

$$M_t = M_p \sqrt{\pi} \frac{\Delta V}{V_c} e^{\left(\frac{\Delta V}{V_c}\right)^2} erf\left[\frac{\Delta V}{V_c}\right] \qquad (25\text{-}2)$$

where *erf* is the error function, M_p is the payload mass at the tether tip, and V_c is the "critical velocity" of the tether material, which in turn depends upon the tensile strength, T, and density, d, of the tether material and the design safety factor, F as:

$$V_c = \sqrt{\frac{2T}{Fd}} \qquad (25\text{-}3)$$

With currently available high strength materials such as Spectra, Dyneema, and Zylon, this scaling of the required tether mass with tip velocity results in the system mass of momentum exchange tethers being roughly comparable to the mass of a chemical rocket stage for delta V of several hundred meters per second, but much larger than rocket masses for delta V greater than 1 km/s. However, unlike a chemical rocket, which consumes propellant, a momentum exchange tether can be re-used many times. Consequently, momentum exchange tethers will be most advantageous serving as a component of a system that transports many payloads over its operational lifetime.

Electrodynamics

If the tether contains conducting wires, it can enable interactions with the Earth's magnetic field to generate *Lorentz forces* that can be used to raise, lower, or change the plane of a spacecraft's orbit. As illustrated in Fig. 25-2, in an electrodynamic tether system, the orbital motion of the conducting wire across the Earth's magnetic field induces an electric field along the length of the tether. In a direct orbit, this electric field biases the top end of the

tether positive with respect to the ambient space environment, and the bottom end negative with respect to the ambient environment. In LEO, the ionospheric plasma can provide a conducting medium in which the tether can exchange charge carriers. If the tether system has a means for collecting electrons from the ionosphere at the top end of the tether and emitting them back into the ionosphere from the bottom end of the tether, the motion-induced electric field can drive a current up the tether, as illustrated on the left in Fig. 25-2. The current will interact with the Earth's magnetic field to induce a $\mathbf{J} \times \mathbf{B}$ Lorentz force that opposes the motion of the tether, lowering the orbit of the system. In this drag, or "deboost" mode, the electrodynamic tether converts the orbital energy of the system into electrical power, which is then dissipated through ohmic losses in the tether. If a load is placed in series with the tether, the system can be used to provide power for that load. Because that power is drawn from the system's orbital energy, an electrodynamic tether can be useful for providing occasional high peak power levels to a payload, but if used for frequent or continual power generation, the tether will cause hastened orbital decay of the system. Conversely, if a power source is used to apply a voltage to the tether to overwhelm the motion induced voltage so as to drive current in the opposite direction, the resulting Lorentz forces will provide a positive thrust that will raise the orbit of the system. In either the de-boost or boost mode, current is driven in only one direction along the tether. The current "loop" in the system is effectively closed by a system of waves in the ionospheric plasma that carry current away along magnetic field as the tether moves across them.

The feasibility of using conducting tethers to interact with the Earth's magnetic field and ionospheric plasma were demonstrated in the Plasma Motor Generator (PMG) experiment flown as a secondary payload on a Delta-II upper stage in 1993. The PMG experiment deployed a 500-meter long insulated conducting wire between the upper stage and a small endmass. Hollow cathode plasma contactors at both ends of the tether provided a low-impedance electrical connection to the ionospheric plasma. Over the course of approximately two orbits, the experiment demonstrated the feasibility of interacting with the ionospheric plasma to enable currents to be driven up or down the tether. The Tethered Satellite System 1-Reflight (TSS-1R) mission carried out in 1996 aboard the Shuttle Orbiter also investigated electrodynamic tether physics, testing different means of exchanging current with the ionosphere, including electron guns and passive collecting surfaces, and demonstrated the ability to drive collecting surfaces, and demonstrated the ability to drive ampere-level currents along the tether [Stone, Raitt, Wright, 1999].

25.1.2 Space Tether Applications

The ability of tethers to connect spacecraft over long distances, transfer momentum and energy between spacecraft, and generate thrust through interactions with the Earth's magnetic field can enable spacecraft applications that are impractical or impossible to perform with conventional spacecraft technologies.

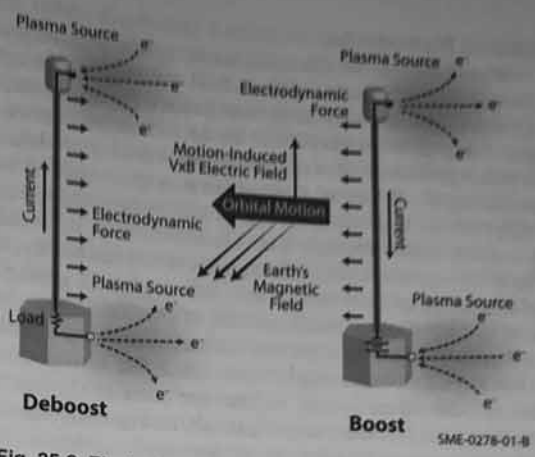


Fig. 25-2. Electrodynamic Tether Propulsion. In an electrodynamic tether, currents flowing along the tether interact with the Earth's magnetic field to induce Lorentz forces on the tether. The tether can be operated in either a deboost/power-generation mode, where the electric field induced by the motion of the tether across the magnetic field is allowed to drive current along the tether, converting orbital energy into electrical power and inducing a drag force on the system, or in a boost mode, in which power is applied to drive current against the motion-induced electric field and generate Lorentz forces that raise the orbit of the system.

Formation Flight

For a number of applications, including long-baseline interferometric astronomy, geolocation, and ionospheric science, it is desirable to fly two or more spacecraft in the same orbit, separated by a controllable distance. Although some formation configurations, such as trailing orbits or Hill's orbits, can be maintained using conventional propulsion systems, other configurations, such as co-orbiting spacecraft with constant separation along the local vertical and formations that rotate around a common center with a continuously changing radius, would require impractically large amounts of propellant to maintain for extended durations. A tether can provide a mechanical link to maintain such formations without consuming propellant. If the tether contains optical fibers or conducting wires, it can also provide a means for high-bandwidth communications between the spacecraft.

Because a tether provides strength in tension but little or no strength in compression, ensuring predictable behavior of a tethered formation requires keeping the tether in a taut state. To do so, a formation can rely on gravity gradient forces, centrifugal forces in a rotating system, or repulsive forces between the spacecraft. When evaluating tethered formation configurations in orbit around the Earth or other planetary body, it is important to consider the interaction between the gravity gradient forces on the system and the angular momentum of the tethered system. Generally, tethered systems that rotate in the plane of the orbit will have the least complicated dynamics, and will continue to rotate within that plane. Tethered systems that rotate in the out-of-plane direction will experience varying torques due to the gravity gradient that will result in precession and nutation of their angular momentum vector.

Table 25-0, Fig. 25-2, Eq. 25-3◀

Orbital Maneuvering

The ability of an electrodynamic tether to generate thrust without consuming propellant can enable a tether system to provide very large total delta V for orbital maneuvering of a spacecraft and do so with mass requirements significantly lower than conventional propulsion technologies. Whereas a rocket uses a gas expelled at high velocity to produce thrust, an electrodynamic tether generates thrust through Lorentz interactions with the Earth's magnetic field. Although some implementations of an electrodynamic tether may utilize plasma generators that consume a small amount of expellant, this expellant is used not as a reaction mass, but rather to create a low impedance electrical connection to the ionosphere. Instead, the tether system utilizes the mass of the Earth itself as its reaction mass. As a result, an electrodynamic tether system can generate thrust for very long durations with little or no consumption of expendables. Fig. 25-3 shows plots of the thrust-to-power ratio of a typical several kilometer long electrodynamic tether in a mid-LEO orbit. The thrust-to-power performance is significantly greater in the de-boost mode because in this mode the motion induced *electromotive force (EMF)* contributes roughly a kW per km of tether length to drive current along the tether, whereas in boost mode the applied power must overcome the motion induced *EMF*. In both cases, the thrust in the plane of the orbit decreases with inclination because the Lorentz force is always perpendicular to both the tether and the magnetic field vector, thus in high inclination orbits, the electrodynamic forces are primarily in the cross-plane direction.

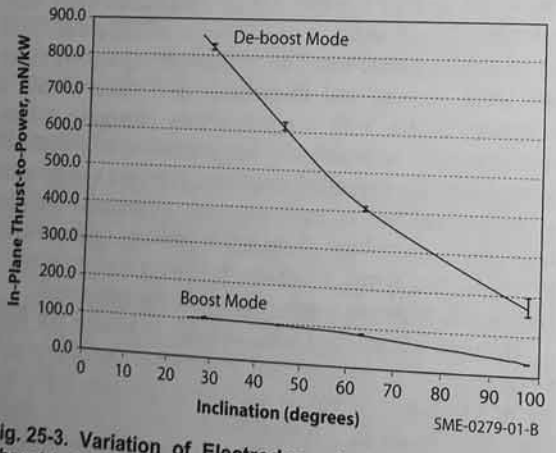


Fig. 25-3. Variation of Electrodynamic Tether In-Plane Thrust-to-Power Ratio with Inclination. Curves shown are for a 2.5 km tether at an altitude of 500 km.

Although an electrodynamic tether does not utilize propellant, we can provide a means for comparing the performance of a tether to other propulsion technologies by computing an "effective specific impulse" using the mass flow of expellant in place of the propellant mass flow rate (See Eq. (18-16) in Sec. 16.1.). Figure 25-4 shows values for the effective specific impulse of a typical several-kilometer long electrodynamic tether in a mid-LEO orbit. In both boost and de-boost modes, the effective specific impulse of the tether system is extremely large, ranging from 10,000 sec at high-inclination in boost mode up to over 100,000 sec in low-inclination de-boost mode. This very high effective specific impulse translates into a capability to provide multiple km/s delta V to a spacecraft with very low total mass requirements. Moreover, because an electrodynamic tether does not utilize an expelled gas to generate thrust, it is liberated from the inverse relationship between thrust and specific impulse typical of rocket-based propulsion systems, and thus it is able to provide both relatively high thrust and high specific impulse, as shown in Fig. 25-5.

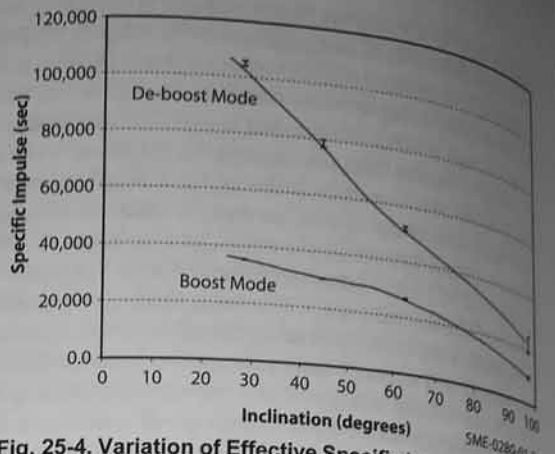


Fig. 25-4. Variation of Effective Specific Impulse with Inclination for an Electrodynamic Tether System. Curves shown are for a 2.5 km long tether at an altitude of 500 km, and the effective I_{sp} is calculated assuming the use of two plasma sources, each with an expellant flow rate of 0.1 mg/s.

Because the electrodynamic force on the tether has components both in the plane of the orbit and in the cross-plane direction, an electrodynamic tether can not only raise and lower the orbit of a spacecraft, but also change its orbit plane. With proper modulation of the current in the tether as it travels around its orbit, all six of the tether's orbital elements can be changed.

De-Orbit

The rapid growth of the space debris population over the past decade presents an increasing threat to operational spacecraft. As a result, many organizations that perform or regulate spaceflight have imposed requirements on new space missions to ensure that their spacecraft do not contribute to the debris population. The current default method for satisfying these end-of-mission disposal requirements is to reserve enough fuel on board the spacecraft to perform a controlled re-entry or de-boost the vehicle to an orbit low enough that aerodynamic drag will cause re-entry within 25 years. Unfortunately, relying upon thrusters for disposal requires reserving a significant fraction of the spacecraft's mass for disposal propellant, and it will not work if key subcomponents of the spacecraft, such as attitude control, command and control, or the thrusters have suffered fail-

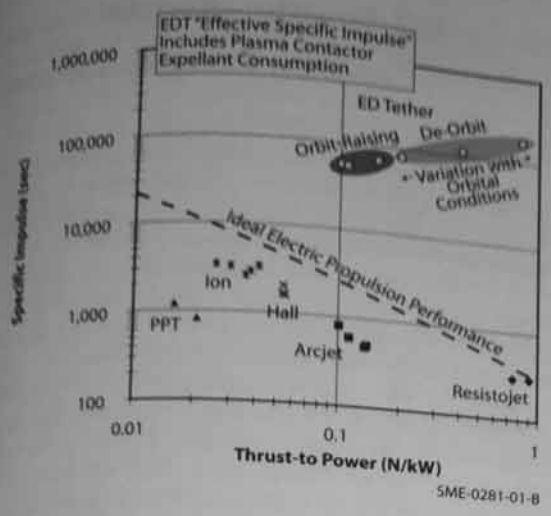


Fig. 25-5. Comparison of Thrust-to-Power vs. Specific Impulse for electric propulsion technologies and electrodynamic tethers.

ures. Because electrodynamic tethers can generate drag forces that are orders of magnitude greater than aerodynamic drag forces in LEO, they can provide a means of accomplishing rapid de-orbit of a spacecraft with significantly lower mass requirements [Forward et al., 2000]. Typically, an electrodynamic tether system can enable LEO spacecraft to meet the post-mission 25-year orbital lifetime restriction with mass impacts of less than 1% of the spacecraft mass. Furthermore, if properly implemented, they can do so even if the host spacecraft has become defunct. A drawback to this approach is the increase in collision probability described at the end of the next section.

25.1.3 Implementation and Integration

Although space tethers are a technology that sounds simple on the surface, like any other space technology the devil is in the details, and implementing a tether system properly to ensure mission success requires addressing a number of engineering challenges. Furthermore, the long tether lengths required for most applications as well as the dynamic characteristics of tethers in orbit present unique challenges for integrating the tether with a spacecraft's hardware and operations. Among the most important of the issues that must be addressed in implementing a space tether system and integrating it with its host spacecraft are the method of tether deployment, the design of the tether materials and structure to enable survival of the space environment, the impact of the tether system on the host spacecraft's behavior and operations, and the potential for collisions between the tether and other objects in orbit.

Tether Deployment

Of the nearly twenty space tether experiments that have been conducted on orbit, the handful that have failed to complete their missions have done so primarily due to problems experienced during the deployment of the tether. Deploying a tether from a spacecraft is challenging in part because a tether has essentially no compressive strength. If the tether becomes slack, its dynamic behavior is nearly impossible to predict with any accuracy, so it is critical that the tether be kept in a taut state during deployment. To keep the tether taut and well behaved, we must rely upon gravity gradient forces, centrifugal forces (in a spinning system), or thrusting by the tether's endmass to keep the tether under tension. As Eq. (25-1) indicates, the gravity gradient force depends linearly upon the tether length, so at the beginning of deployment there is no gravity gradient force to help pull the tether out of its deployer and keep it taut. Consequently, the most successful deployment schemes have accomplished deployment by impulsively ejecting an endmass away from the host spacecraft, and relying upon the endmass' momentum to pull the tether out of the deployer. In this method, the ejection impulse must be sufficient to overcome the drag the tether experiences coming out of the deployer until the gravity gradient force becomes large enough to keep deploying the tether.

An additional factor that must be considered in deployment of the tether is the dynamic behavior of the tethered system. The tether does not simply extend above or below the spacecraft as the tether length is increased. Rather, increasing the tether length tends to cause the object at the bottom of the tether to run ahead of the object at the top of the tether, resulting in librations in the orbit plane. Through proper modulation of the rate of tether deployment, however, a system can control the libration behavior and achieve a well-behaved vertical orientation or other desired end-state [Barkow et al., 2003]. Such controlled deployment of a tether was demonstrated successfully by the SEDS-2 mission, which deployed a 20 km long tether and achieved a final libration swing angle of less than 4 deg [Lorenzini et al., 1995].

Tether Materials and Design

A tether deployed in space must be designed to survive the space environment for the duration of its mission. In Earth orbit, a tether will be subjected to hypervelocity impacts by micrometeorites and orbital debris (MMOD). These impacts are so energetic that they will vaporize a portion of the tether regardless of what material it is made of, typically creating a crater with a diameter 3 to 5 times the diameter of the impactor. Consequently, to ensure that a tether will have a high probability of survival for its mission duration, it either needs to be constructed with a large diameter or with a structure that has redundant load bearing paths distributed over a large volume. The first method was successfully demonstrated by the TiPS tether experiment, which used an acrylic knitting yarn inserted inside a cylindrical braid of high-tensile strength fibers to expand the braid out to a larger diameter. The 4 km long TiPS tether survived for over ten years on orbit in an altitude region with a relatively high flux of small debris particles. The second method is used by approaches such as the Hoytether

[Forward and Hoyt, 1995] and the Carroll Caduceus, both of which distribute the tether material into multiple spatially-separated lines which are then periodically interconnected. The interconnections provide redundant load-bearing paths to enable tensile loads to be bridged around sections of tether that experience damage.

A tether deployed in space will also experience erosion and material degradation due to atomic oxygen, ultraviolet light, and energetic particle radiation. The high-tenacity polymers favored for momentum exchange or formation flight tethers and the polymeric insulators used on electrodynamic tethers can rapidly lose performance and fail due to these environmental effects. Several previous efforts have investigated the use of AO and UV-resistant coatings to protect tether materials. Coated tether samples tested on the MISSE-6 payload flown on the exterior of the ISS demonstrated the ability to survive for many months in the AO and UV environment.

Spacecraft Impacts

When integrating a tether subsystem into a spacecraft, the system design must account for a number of potential effects of the tether upon the spacecraft. The gravity gradient forces induced by the tether can serve as a simple means of stabilizing the satellite's orientation into a nadir- or zenith-facing orientation, but if the spacecraft's mission calls for it to point in other directions or perform slewing maneuvers, its ACS system must be designed to accommodate the torques induced by the tether tension. Furthermore, librations and other oscillations of the tether will result in attitude motion of the spacecraft that must also be accommodated by the ACS system design. Additionally, if the tether is conductive, there exists the potential for voltages induced along the tether to be transmitted to the spacecraft, and so the system design must ensure that the spacecraft will not experience adverse charging effects.

Collision Probabilities

The long lengths involved in most space tether systems raises the concern of the potential for collisions between the tether and other space objects. Although a tether may have a relatively small diameter, when calculating probability of collisions between the tether and other objects, the collision cross-section of a tether should be calculated using the length of the tether and an effective width defined by the characteristic size of the objects with which it could collide. In LEO altitudes, (the average characteristic size of the resident space object populations) on the order of 1.0 to 2.5 m, so the collisional cross sectional area of a tether system can be considerably larger than its physical cross sectional area. Consequently, tether systems should be flown in orbits in which close conjunctions with other objects will be relatively infrequent, or the mission design must enable the tether to be tracked and either maneuvered or retracted to avoid potential collisions.

25.2 Inflatable Structures

Constantine Cassapakis, Consultant

In this section, we consider the use of *deployable inflatable space structures* for space applications. It should be noted that this area of endeavor is large and this treatment is only meant to convey enough information on this class of structures to help the space mission analyst or engineer with some knowledge toward their use as payloads. It is also important to mention, that this section does not consider the type of inflatable structures used for Space Habitats, such as those conceived for the Lunar or Martian surfaces, or those that demonstrate the capability of LEO "man-rated" vehicles (the inflatable capsules of the "Genesis" type, successfully designed, built and flown by Bigelow Aerospace, are a good example of the latter). Rather, we will concentrate on the very lightweight "*Gossamer-type*" *structures*, as these truly stress the space inflatables technology to its limits. Should it be decided to delve into more detail towards possible mission implementation, the reader is directed to a variety of sources. Freeland et al. [1998]; Cassapakis [2008]; Satter and Freeland [1995]; and Cassapakis [1995] provide helpful overviews on the subject. Jenkins [1997] is a good source of information on most of the sub-technologies involved, providing a collection of writings by people working currently or recently in the field. An abundance of relevant technical papers on the subject matter can also be found on the websites of L'Garde, Inc., NeXolve, and ILC-Dover who have done extensive work in this area.

Plastic films, such as *polyesters* (e.g., Mylar®) or *polyimides* (e.g., Kapton®) and thin pliable laminates play an important role in this technology, and so do materials that become rigid upon inflation and deployment in the space environment. Much has been accomplished the last three decades in this fascinating technology, but most of the progress has been in its continued development rather than its frequent application in space missions. This technology still remains as one of the "Holy Grails" in the Space Structures field, as there still remain some serious risks to be retired. Yet, there are missions that will be difficult to accomplish without this technology, particularly those missions that would require either very large structures, or those in which the available stowage space is too small for a rigid deployable payload. When the stowage volume and mass requirements are too constricting for rigid deployable structures, the mission engineer should give *space inflatables technology* careful consideration.

25.2.1 Background

Mass and *stowage volume* have always been serious challenges to the aerospace engineer and more so in the engineering of spacecraft buses and their payloads. The expense of launching a 1 kg mass in space can reach beyond $18K (see Table 11-23) depending on orbit and does not seem to be getting much better with newer

booster technologies. In our experience, stowage volume is an even stricter constraint than mass, particularly for spacecraft that employ large deployable subsystems, such as solar arrays, dish antennas, solar concentrators, and solar sails. It is these constraints, along with the possible hang-ups of mechanically deployable structures comprised of a large number of rigid component parts (by comparison, inflatable bodies require much fewer parts), which pointed to the need for developing space inflatable structures. As early as the 1950s, aerospace engineers at Goodyear were experimenting with building inflatable antennas using elastomer-impregnated fabrics. These were fairly heavy, had low stiffness and did not attain the stowage volume possible with lighter materials.

The first inflatable structures that made it to space were the ECHO satellite series launched by NASA in the early 1960s, some of them being spherical shells of over 30 m in diameter and used as radio-wave reflectors (Fig. 25-6). The shell thickness was in the range of 38 microns, as Du-Pont-manufactured polyesters entered the plastic film market, offering a thickness as low as 6 microns and a respectable structural modulus in the few GP region.

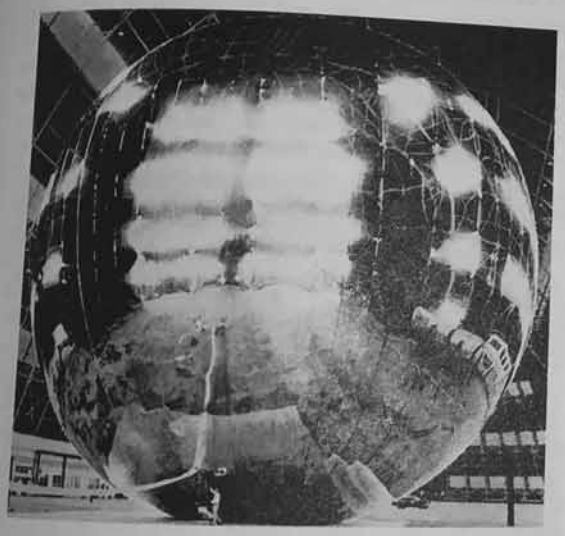


Fig. 25-6. NASA'S ECHO Satellite.

When the Soviet Union launched the Galosh ICBM interceptor in 1965, the US Air Force initiated a series of studies (ABRES) with several aerospace prime contractors of that time to establish the best recourse in countering this Soviet threat. Most of these multiyear studies and the concomitant experimentation resulted in favor of complementing the US Intercontinental Ballistic Missile (ICBM) fleet with inflatable missile decoys offering similar signatures in the radar and visible spectrum. The ease of manufacturing, folding, efficient stowage during launch and relatively light weight, meant low manufacturing and launch costs, thus allowing a significant number of them to be deployed around an ICBM, making it difficult, if not impossible, for an enemy to discriminate between it and the decoys. Ford Aeroneutronic (now a part of Loral), L'Garde, Inc., Tracor (now part of BAE), and AVCO (now Textron) among others, designed and built a number of these inflatable objects (Fig. 25-7). Lincoln and Sandia National Laboratories were also involved in the effort.

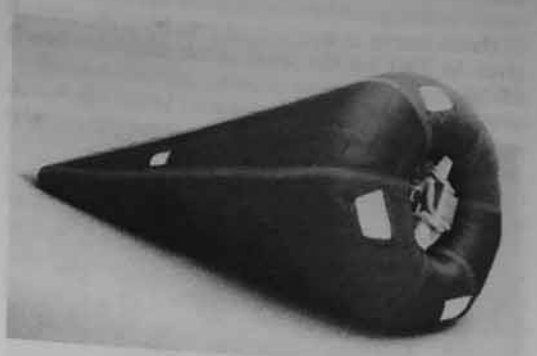


Fig. 25-7. An Early Example of a Re-entry Vehicle Inflatable Replica.

Though these space inflatable objects were never fielded as part of the US strategic defense, a large number of them have flown for testing purposes in the last 4 decades by the various incarnations of the US missile defense (BMO, SDIO, BMDO and presently, MDA). Much has been learned about the behavior of these materials and structures from these space tests. The latter were mainly sub-orbital, but there were some that were conducted orbitally. In the discussion that follows, we separate the engineering of this structural class into Inflatable-Only and Inflatable-Rigidizable Structures. Since for a long-term orbital mission, the latter will have a better chance of working, a table will be presented in Sec. 25.2.4 that will allow an approximate calculation of the mass and stowage volume of an inflatable-rigidizable parabolic antenna (Table 25-1).

25.2.2 Engineering Inflatable-Only Structures

We use the term "*Inflatable-only*" or "*purely inflatable*" structures for those that require constant pressurization in space to preserve their shape and structural properties. There are numerous examples, such as a *membrane lenticular* (lens-shaped) reflector or *light concentrator* (the large inflatable antenna in the IAE space shuttle mission of 1996 [Freeland et al., 1996] shown in Fig. 25-8); *conical structures*, such as those of Re-entry Vehicle replicas, *spherical target balloons*, simple or heavily instrumented), and a variety of other space targets; in general, any structure which is a body of revolution designed and fabricated with films or laminates, metalized or not, and reinforced with fabric fibers or not.

For closed structures, e.g., a sphere, there exists an important relationship between skin stress and inflation pressure:

Table 25-0, Fig. 25-7, Eq. 25-3◀